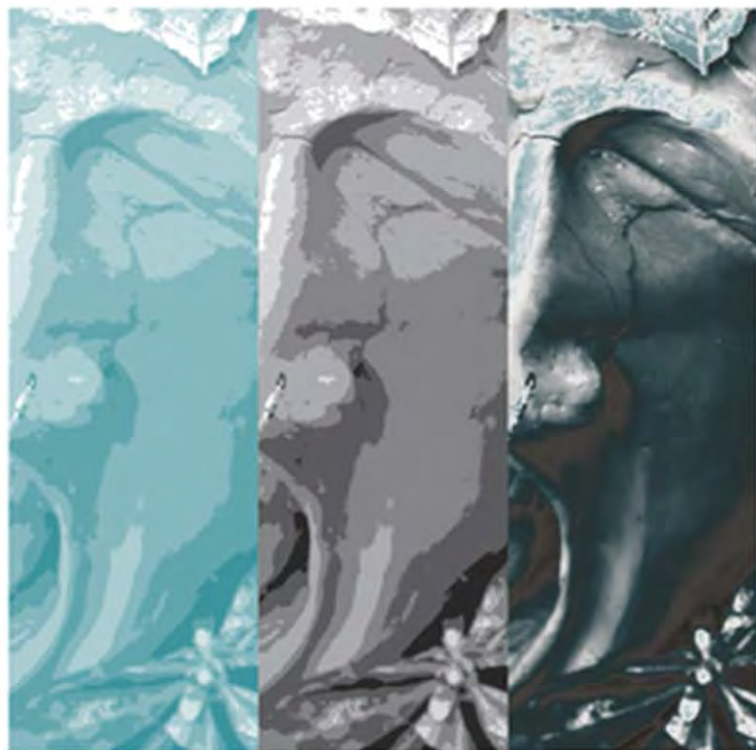


EDITED BY
JOHN READ AND JACQUI DILLON

Models of Madness

SECOND
EDITION



PSYCHOLOGICAL, SOCIAL AND
BIOLOGICAL APPROACHES TO PSYCHOSIS

PUBLISHED FOR

ISPS

THE INTERNATIONAL SOCIETY
FOR PSYCHOLOGICAL
AND SOCIAL APPROACHES TO PSYCHOSIS

ROUTLEDGE

MODELS OF MADNESS

This second edition of *Models of Madness* challenges the simplistic, pessimistic and often damaging theories and treatments of the 'medical model' of madness. Psychiatric diagnoses and medications are based on the false premise that human misery and distress are caused by chemical imbalances and genetic predispositions, and ignore the social causes of psychosis and what psychiatrists call 'schizophrenia'. This edition updates the now extensive body of research showing that hallucinations, delusions etc. are best understood as reactions to adverse life events and that psychological and social approaches to helping are more effective and far safer than psychiatric drugs and electroshock treatment. A new final chapter discusses why such a damaging ideology has come to dominate mental health and, most importantly, how to change that.

Models of Madness is divided into three sections:

- Section One provides a history of madness, including examples of violence against the 'mentally ill', before critiquing the theories and treatments of contemporary biological psychiatry and documenting the corrupting influence of drug companies.
- Section Two summarises the research showing that hallucinations, delusions etc. are primarily caused by adverse life events (eg. parental loss, bullying, abuse and neglect in childhood, poverty, etc) and can be understood using psychological models ranging from cognitive to psychodynamic.
- Section Three presents the evidence for a range of effective psychological and social approaches to treatment, from cognitive and family therapy to primary prevention.

This book brings together thirty-seven contributors from ten countries and a wide range of scientific disciplines. It provides an evidence-based, optimistic antidote to the pessimism of biological psychiatry. *Models of Madness* will be essential reading for all involved in mental health, including service users, family members, service managers, policy makers, nurses, clinical psychologists, psychiatrists, psychotherapists, counsellors, psychoanalysts, social workers, occupational therapists and art therapists.

John Read is a Professor of Clinical Psychology at the University of Liverpool and is Editor of the scientific journal *Psychosis: Psychological, Social and Integrative Approaches*. He is author of numerous books and over 100 research articles. In 2010 Professor Read was awarded the New Zealand Psychological Society's Hunter Award, presented, every three years, for excellence in scholarship and contribution to the profession.

Jacqui Dillon is a campaigner, writer, international speaker and trainer specialising in hearing voices, psychosis, dissociation, trauma, abuse, healing and recovery. She is the national Chair of the Hearing Voices Network in England, a Board member of Intervoice – the International Network for Training, Education and Research into Hearing Voices. Jacqui is an Honorary Lecturer in Clinical Psychology at the University of East London. This is her third co-edited book about psychosis.

The Book Series of The International Society for Psychological and Social Approaches to Psychosis

Series editors: Brian Martindale and Alison Summers

ISPS (The International Society for Psychological and Social Approaches to Psychosis) has a history stretching back more than fifty years during which it has witnessed the relentless pursuit of biological explanations for psychosis. The tide has been turning in recent years and there is a welcome international resurgence of interest in a range of psychological factors that have considerable explanatory power and therapeutic possibilities. Governments, professional groups, people with personal experience of psychosis and family members are increasingly expecting interventions that involve more talking and listening. Many now regard practitioners skilled in psychological therapies as an essential component of the care of people with psychosis.

ISPS is a global society. It aims to promote psychological and social approaches both to understanding and to treating psychosis. It also aims to bring together different perspectives on these issues. ISPS is composed of individuals, networks and institutional members from a wide range of backgrounds and is especially concerned that those with personal experience of psychosis and their family members are fully involved in our activities alongside practitioners and researchers, and that all benefit from this. Our members recognise the potential humanitarian and therapeutic potential of skilled psychological understanding and therapy in the field of psychosis and ISPS embraces a wide spectrum of approaches from psychodynamic, systemic, cognitive, and arts therapies to the need-adapted approaches, family and group therapies and residential therapeutic communities.

We are also most interested in establishing meaningful dialogue with those practitioners and researchers who are more familiar with biological-based approaches. There is increasing empirical evidence for the interaction of genes and biology with the emotional and social environment, and there are important examples of such interactions in the fields of trauma, attachment relationships in the family and in social settings and with professionals.

ISPS activities include regular international and national conferences, newsletters and email discussion groups. Routledge has recognised the importance of our field in publishing both the book series and the ISPS journal: *Psychosis – Psychological, Social and Integrative Approaches* with the two

complementing one another. The book series started in 2004 and by 2013 had 15 volumes with several more in preparation. A wide range of topics are covered and we hope this reflects some success in our aim of bringing together a rich range of perspectives.

The book series is intended as a resource for a broad range of mental health professionals as well as those developing and implementing policy and people whose interest in psychosis is at a personal level. We aim for rigorous academic standards and at the same time accessibility to a wide range of readers, and for the books to promote the ideas of clinicians and researchers who may be well known in some countries but not so familiar in others. Our overall intention is to encourage the dissemination of existing knowledge and ideas, promote productive debate, and encourage more research in a most important field whose secrets certainly do not all reside in the neurosciences.

For more information about ISPS, email isps@isps.org or visit our website, www.isps.org.

For more information about the journal *Psychosis* visit www.isps.org/index.php/publications/journal

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First edition

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Pamela Fuller

MODELS OF MADNESS: PSYCHOLOGICAL, SOCIAL AND
BIOLOGICAL APPROACHES TO PSYCHOSIS

Second edition

Edited by John Read & Jacqui Dillon

MODELS OF MADNESS

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approaches to psychosis

Second edition

*Edited by John Read and
Jacqui Dillon*

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In 1994 John joined the Psychology Department of the University of Auckland where, apart from undergraduate teaching and the professional training of postgraduate clinical psychology students, he has published numerous research papers, with psychiatrists and other psychologists, documenting the link between childhood adversity and psychosis. He has also published research about the infrequency of abuse inquiry by mental health professionals and the reasons for that, leading to a training programme on how and when to take trauma histories and how to respond to abuse disclosures. His other research publications include studies demonstrating that, contrary to the approach taken by most destigmatisation programmes, biological causal beliefs about mental health problems are related to increased fear and prejudice. He has also published research about the influence of the pharmaceutical industry.

In 2010 Professor Read was awarded the New Zealand Psychological Society's Hunter Award, presented, every three years, for excellence in scholarship and contribution to the profession. John is on the Executive Committee of the International Society for Psychological and Social Approaches to Psychosis [www.isps.org] and is the Editor of the ISPS journal *Psychosis: Psychological, Social and Integrative Approaches* [www.informaworld.com/psychosis].

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PREFACE TO THE FIRST EDITION (2004)

In June 2000 I was sitting outside my hotel in Stavanger, Norway, preparing the seminar I was to give later in the day, about child abuse and schizophrenia, to the 800 or so people who had come to the *13th International Symposium for the Psychological Treatments of Schizophrenia and other psychoses*. I will be forever grateful to Jan Olav Johannessen (the ISPS President) for inviting me to Norway, not least because of what happened next. At a nearby table three people were talking about how hard it is to get psychiatrists to realise the obvious fact that people are driven crazy by bad things happening to them. I overcame my usual shyness to introduce myself with some comment about 'Now I won't need to give my seminar – I'll be preaching to the converted' or some such nonsense. The three were to become good friends of mine before the conference was over. One was Volkmar Aderhold, a German psychiatrist with whom, a year later, I would be exchanging Dad stories in his Hamburg apartment (his was in the weapons branch of the S.S., mine was a British fighter pilot – so we both understand the intergenerational effects of bad things happening). The second was Petra Hohn, who gave a remarkable talk the next day – full of warmth, humanity and common sense – about her work in Sweden with people diagnosed 'schizophrenic'. The third, by far the shortest despite his hat, was Loren Mosher. When he said his name I had to stifle one of those awful 'not *the* Loren Mosher?' comments (although Loren would have loved it!). For 20 years I had been citing his work to any poor sod who would listen as evidence that people who go crazy need other people more than they need medical-sounding labels and tranquillizing drugs.

The point of this story is that after four days in Stavanger I plummeted from an enormous high (partly jet-lag, partly short Norwegian nights, but mostly the excitement of finding so many kindred spirits in such a short time) to an exhausted low in my one-night Oslo hotel, missing my new friends already. (I remember thinking I was too old to get bi-polar disorder.) Anyway, the next morning I found the solution to my 'low affect'. I decided to try to capture the spirit of Stavanger in a book and laid out, over breakfast, the first of many subsequent outlines and possible authors.

Soon after Routledge contracted the book I met Richard Bentall in Manchester: another hero from my younger days. His *Reconstructing Schizophrenia* was a gem. Richard was coming to the end of writing *Madness Explained: Psychosis and Human Nature*, a magnificent wide-ranging and scholarly treatise – a draft of which he let me read. I was delighted when, a few months later, in New Zealand, he agreed to come on board as a third editor, joining Loren Mosher and myself: a cooler head than either of ours, for sure. Now we had a Brit, a Yank and a Kiwi.

I am exhausted now, writing this last little bit of what has taken three years – mostly in two- or three-hour stretches between work (thank you, Fred and co. in the clinical programme, for being so understanding for so long) and being with my partner, Emma, and our two young children, Jessica and Ben. Well, they were young when I started. They have missed out at times. And so have I. Thank you, Emma, for all the ways you make my work possible and my life meaningful.

In 1994 I had left behind twenty years of trying to help people through madness in the face of a system that often didn't want to know and entered the no less bizarre world – to me – of a University. On a good day I really enjoy preparing the next generation of helpers. On a bad day it seems utterly futile. I know what they will be facing. So this book, in a way, is for them. The purpose of all these seemingly endless hours is to try to weaken the awful stranglehold that simplistic biological ideology has in the world of mental health, so that the thousands of people around the world trying to use their humanness to reach out, in the face of madness, to the humanness of others might be valued and nurtured in their work, rather than marginalised or scorned.

I end with three stories, from my very first mental health job, thirty years ago, as a nursing aide in a New York psychiatric hospital. I tell them to first-year undergraduate students, so they understand where my 'biases' come from (and probably to second- and third-year students as well – but they are too polite to tell me).

He had been on the ward for three days without opening his eyes. No mean feat. The doctors had been trying to figure this out, but 'eye-closed behaviour' just wasn't anywhere to be found in the *D.S.M.* One night, at about 2 a.m. I asked him, more from boredom than clinical judgment, why he kept his eyes shut. He immediately opened them, put his face uncomfortably close to mine, and said, 'It's about fucking time one of you idiots asked me that! I was put in here to get "insight", so that's what I'm fucking well doing!'

Just before my very first chance to help run the weekly group therapy session, an old woman approached me to explain that she wouldn't be talking in the group because she thought that whatever she had said in the past in the group had always been turned in to a 'symptom' and used against her. She didn't want me to be offended by her silence, she explained (I think she could see I was a little nervous). After the group I retreated to the staff team where

the patients' performance in the group was evaluated. The woman's not speaking was taken as a sure sign of her paranoia.

I was 'specialing' a teenage girl. This meant being locked in with her in the 'quiet room' (usually the noisiest place on the ward) to make sure she didn't try to harm herself. She hadn't spoken for weeks. A 'catatonic schizophrenic'. Having had no training, I tried: 'It's OK if you don't want to talk, but if you want to, I will listen.' Nothing. The next day she said one word: 'My'. The next day she said 'father'. The next day she didn't speak. The next day she said 'me'. The missing word, I learned later, was raped.

Thank you to all the people psychiatrists call 'schizophrenic' that I have known, for trusting me enough to teach me what being mad is like and about its many causes. Thank you to the many people ('patients', 'carers' and 'mental health staff'), including the wonderful contributors to this book, who, against all the odds, have not given up on the truth that when we are mad, we need other people to be people.

John Read
Auckland, June 2003.

PREFACE TO THE SECOND EDITION

The 23 of us who put together the first edition of *Models of Madness* could not have predicted how popular it would be. It has sold over 11,000 copies and, thanks to kindred spirits around the world, been translated into four other languages. Thank you, Jorge Tizon and Raimund Herder (Spain), Zeng Yong (China), Christer Skoog (Sweden) and everyone at PRO-PSY (Russia). I have been very fortunate to have been invited to talk about the ideas and research in the book at conferences in 16 countries (counting them just now produced some lovely memories). The most common reaction, everywhere, has been 'We sort of knew all that, but we didn't know there was so much research supporting our beliefs.' And that, really, is the main point of all the hours we have all put into this book – to provide people with the knowledge to be able to respond authoritatively to silly claims about it all being biological and everyone needing drugs.

The fact that a second edition is necessary is a mark of the progress that is being made – at least in the research world. The knowledge base around the social causes of madness, and the most effective social and psychological responses, is growing fast, including in areas, like childhood adversity, that were virtually taboo until recently. Indeed, some of the contributors who had got their chapter in early had to update the research again a year later.

Thank you so much to the 37 contributors to this edition, from ten countries and a range of disciplines (including 12 psychiatrists), for sharing your knowledge and experience. It was a labour of love (and one free copy!). Thanks also to everyone at Routledge for supporting the book and the rest of the wonderful ISPS book series.

I especially want to express my gratitude to Jacqui Dillon for her help, as co-editor, and for all her truly inspiring work (and that of all her colleagues in the international Hearing Voices network) helping the world not to be scared of voices or the people who hear them and educating the world about the origins and meanings of 'psychotic' experiences'. I will always remember sitting on stage behind Jacqui, in London, in 2008, at the first training day on abuse enquiry for British psychiatrists after the launch of the new NHS guidelines. I had bombarded the 100 or so psychiatrists with endless research

findings and techniques for asking about, and responding to, child abuse. Then Jacqui told the story of her life. From the tears in almost every eye, I think I know which part of the training they will remember most.

Sometimes I focus too much on the barriers to change and don't see the progress we *have* made. I was sitting with friends in a Copenhagen restaurant during the 2009 ISPS Congress moaning about the 'hypocrisy' of a prominent geneticist who had initially been among the most scathing deniers of the trauma-psychosis hypothesis but who had just put his name on the latest of the many papers confirming the hypothesis. My good friend Richard Bentall replied, 'For god's sake, John, you've won and you're still bitching!'

Thanks to everyone in ISPS for providing me with an international family, however fractious at times – like most families – within which I feel supported and nurtured in my efforts.

Thanks, again, to Emma and my two beautiful children, Jessica and Ben, just for being who you are and for sharing your lives with me.

Just before completing this edition, I was signed up by Liverpool (the University, not the football club). So, as Emma and I prepare to bring our 20 years in New Zealand to a close, it is time to acknowledge just a few (at the risk of offending others) of the many Kiwis who have inspired and sustained me: David Semp, Jim Geekie, Debra Lampshire, Vanessa Beavan, Jeremy Clark, Judi Clements, Thom Rudegeair, David Codyre, Ingo Lambrecht, Leigh Murray, Fred Seymour, John Thorburn, Arana Pearson and Jeffrey Masson.

This second edition of *Models of Madness* is dedicated to Loren Mosher, a co-editor on the first edition, who died shortly after its publication. You are still remembered and missed, Loren. It is also dedicated to everyone engaged in the international struggle to free madness from the frightening and pessimistic ideology promoted by biological psychiatry and the drug companies and to create more humane, evidence-based approaches. Remember, there are thousands of us!

John Read
Auckland, January 2013

FOREWORDS

As a practicing psychiatrist who has been working full-time for a number of years in a psychosis service, the second edition of this popular and most important book is a source of both considerable pleasure and some discomfort to me.

The pleasure is that so many chapters resonate with and validate many of my – often private – observations and beliefs about the limitations and sometimes even harmfulness of some of the ideology and practice of contemporary mental health services. The discomfort comes from the profound difficulties, and sometimes loneliness, in trying to effect change for the better and the exacerbation of personal guilt when some of the problems I encounter are so starkly portrayed, as they are in this book.

In the *Structure of Scientific Revolutions* (1962) Kuhn caused something of an outrage by suggesting that paradigm shifts result from a mixture of social forces, enthusiasm and the availability of an alternative paradigm. It is to be profoundly hoped that mental health services in some parts of the Western world do undergo a ‘scientific’ revolution in the not too distant future. This book contains many of the building blocks needed for such a revolution. It certainly takes a social perspective, and it contains many chapters that express enthusiasm and passion as well as many challenges to long-held ‘scientific’ beliefs and practices that hold centre stage in mainstream psychiatry. The theories that are challenged include genetic theories and the very concept of ‘schizophrenia’, as well as practices that involve the domination of powerful medications and the sidelining of involvement with the family in therapeutic services.

It is very satisfying that the questioning of well-established ideas or even beliefs is followed by alternative ideas and explanations that focus mainly on giving much greater emphasis to formative experiences, including difficult family problems and communication disturbances and trauma. The fact that there is personal meaning and history disguised to varying degrees in the symptoms of psychosis sadly still needs stressing, as many chapters impress on the reader.

A profound change is gradually taking place in the increasing exposure of the often corrupt science and marketing that leads to the excessive dominance

of the pharmaceutical industry in the mental health field. However, I think that many psychiatrists will only fight these pressures if during their training they have acquired other useful effective therapeutic tools that they are confident in using and if they have access to staff who also have good therapeutic skills and the time to use them.

The greatest leverage for change will probably be the increasing maturity and sophistication of organisations of families and experts-by-experience working in close cooperation with, and within, organisations like ISPS, which is composed of many members who are very sympathetic to many of the challenges raised in this book.

The book is written in a style that is fully accessible to professionals, service users and family members. I anticipate that it will play a part in generating an insistence that all members of staff teams working with psychosis have regularly evaluated competence in the psychological approaches to psychosis and that each team has available a range of psychological approaches, relevant to needs. This contrasts with the sad fact that at present such availability is often exceptional. One would not tolerate it in a surgical service: why do we do so in the mental health field?

We need a radical but common-sense shift to services that match patient and family needs, rather than the latter having to accommodate to inadequate service provision. These goals are at the heart of the objectives of the International Society for Psychological and Social Approaches to Psychosis (www.isps.org).

Some will say that it is easy enough to challenge existing practice and beliefs, but another thing to be sure one has better solutions. There is plenty in this book – and the other books in this ISPS series – to point the way. As well as those chapters challenging the conceptual basis for much practice in psychosis, I would like to highlight the intervention chapters, especially those that include reference to much more sophisticated thinking and practice in working with families, the importance of developing further the early intervention in psychosis frameworks, the need to retain the insights of the psychodynamics of psychosis and their incorporation into everyday practice, as well as retaining specialist practitioners, and the increasing evidence base for CBT approaches. If I were to forecast the future, I would envisage a rapprochement between CBT, psychodynamic and systemic approaches. This is already hinted at in the CBT chapter that emphasises the importance of *core* beliefs formed as a result of life experience. If core beliefs really are important, then what is needed is a move towards skills in providing longer-term interventions and therapies that tackle developmental disturbances and traumatic experiences that have deeply affected the security of the 'self'.

I am delighted that these days there is a much greater emphasis on competence in the professions. I hope that part of the revolution that is on its way will be for all mental health practitioners to be able to demonstrate expertise in areas determined by the needs of service users and their families. It is my

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experience that this is a difficult area to speak about but that the variations in standards and competencies are very worrying.

I am confident that this edition will be as popular as the first. Congratulations to John Read and Jacqui Dillon for their leadership in bringing this second edition of *Models of Madness* to completion.

Brian Martindale

* * *

It was with some sadness that, because of the burden of other commitments, I was unable to accept John's invitation to contribute as an editor for this second edition of *Models of Madness*, but I have followed its development closely and am delighted to see it finally appear in print. Loren Mosher was also unable to contribute, for more compelling reasons. He is sorely missed, not least because it will be difficult to find anyone able to fill his role as a forceful advocate of a more humane psychiatry and a challenging voice in psychosis research.

Happily, the first edition of *Models of Madness* proved to be more influential than the original editorial team had expected. Perhaps this was not only because the contributors were experts in their fields, but also because they argued passionately against the conventional approach to understanding madness which has so poorly served patients and their carers (although it has sometimes served professionals and the pharmaceutical industry very well). The first edition chimed well with the changing times, capturing the moment when the limitation of outdated, naively reductionist models of psychosis were becoming obvious, even to many working within the field of biological psychiatry.

The fact that a second edition is necessary at this stage reflects the exciting developments in our understanding and treatment of psychosis that have continued over the past few years. So many widely accepted assumptions about the nature of madness have been shown to be wanting that it is difficult to list them all. Whereas clinicians once saw patients with psychosis as almost a separate species, we now know that psychosis exists on a continuum with healthy functioning. Whereas it was once assumed that genetic factors dominate in the causation of psychosis, the past decade has yielded a massive amount of research on social factors (exposure to urban environments, social deprivation, migration and discrimination, and, of course, childhood trauma) that are, without doubt, causal. Whereas psychosis was once seen as an almost life-ending condition, shattering hopes and ambitions for the future, we now know that many people can lead happy, productive lives with persisting psychotic experiences and without the need for either psychiatric or psychological intervention. Truly, a revolution is occurring in our understanding of severe mental illness.

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This time round, John and Jacqui have done a wonderful job in bringing together most of the original contributors, persuading them to update their original material, and integrating their work into a coherent whole. This volume will serve as an inspiration, not only to established clinicians and researchers, but to the young people who will develop better services for people with psychosis in the future. It will also offer further hope to those people who matter most when we debate the nature of mental illness – the millions of experts by experience living in the world today, many of whom have contributed to our understanding, not only by participating in our studies, but also by sharing their stories of triumph over adversity.

Richard Bentall

Part I

THE ILLNESS MODEL OF PSYCHOSIS

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‘SCHIZOPHRENIA’ IS NOT AN ILLNESS

John Read, Loren Mosher and Richard Bentall

‘Schizophrenia is a chronic, severe, and disabling brain disease.’ In June 2003, this was the opening statement of the US government agency, the National Institute for Mental Health, on its public information website (www.nimh.nih.gov/publicat/schizoph.pdf) about the topic of our book. Such an opinion can still be found, in 2013, in most ‘educational’ material, from psychiatric textbooks to drug-company-sponsored pamphlets.

We disagree.

The heightened sensitivity, unusual experiences, distress, despair, confusion and disorganization that are currently labelled ‘schizophrenic’ are *not* symptoms of a medical illness. The notion that ‘mental illness is an illness like any other’, promulgated by biological psychiatry and the pharmaceutical industry, is not supported by research and is extremely damaging to those with this most stigmatizing of psychiatric labels. The ‘medical model’ of schizophrenia has dominated efforts to understand and assist distressed and distressing people for far too long. It is responsible for unwarranted and destructive pessimism about the chances of ‘recovery’ and has ignored – or even actively discouraged discussion of – what is actually going on in these people’s lives, in their families and in the societies in which they live. Simplistic and reductionist genetic and biological theories have led, despite the high risks involved and the paucity of sound research proving effectiveness, to the lobotomizing, electroshocking or drugging of millions of people.

The research we have gathered together in this book supports our belief that our efforts to understand and assist people experiencing the ‘symptoms of schizophrenia’ will benefit greatly from a fundamental shift away from unsubstantiated bio-genetic ideologies and technologies to a more down-to-earth focus on asking people what has happened and what they need.

We have not attempted an even-handed, ‘objective’ approach. What is required, after a hundred years or more of the dominance of an approach that is unsupported scientifically and unhelpful in practice, is a *balancing* stance rather than a balanced one. The traditional viewpoint is omnipresent in textbooks, research journals and the media. Other views have had difficulty being heard. As responsible critics, we have included discussion of bio-genetic

theories and the treatments they have spawned to show the methodological flaws involved in the research that purports to support those theories and treatments. Our main purpose, however, is to show that there are sound research-based psychosocial alternatives to both understanding the origins of and intervening with what is currently called 'schizophrenia'.

The three of us have worked with hundreds of people unfortunate enough to be labelled 'schizophrenic'. We have been saddened, and sometimes angered, by the readiness of some mental health professionals to believe that the complexities of human experience can be reduced to, and understood in terms of, a single, contextless, medical-sounding word. We have too often seen that instead of trying to find out how the individual – and their loved ones – understands what is going on and what might help, the only solution offered is a chemical or electrical one, usually delivered in a dehumanizing, prison-like 'hospital'. Together with the other contributors to this book, from six countries and a range of disciplines – including users of mental health services – we want to see a less ideological approach to understanding and treating 'madness'. We want to see a truly evidence-based range of possible explanations discussed when individuals and families come into contact with us, and a wider range of options offered in terms of what might be helpful.

Since the 1970s, the illusion of a balance, of an integration of models, has been created by the so-called 'bio-psychosocial' approach. An integral part of this has been the 'vulnerability-stress' idea that acknowledges a role for social stressors but only in those who already have a supposed genetic predisposition. Life events have been relegated to the role of 'triggers' of an underlying genetic time bomb. This is not an integration of models, it is a colonization of the psychological and social by the biological. The colonization has involved the ignoring, or vilification, of research showing the role of contextual factors such as stress, trauma (inside and beyond the family), poverty, racism, sexism, and so on in the aetiology of madness. The colonization even went so far as to invent the euphemism 'psycho-education' for programmes promulgating the illness ideology to individuals and families.

Perhaps most cruelly of all, the belief that because 'schizophrenia' is an illness and, therefore, life events and circumstances can play no role in its causation has led to the awful conclusion that nothing can be done to prevent it. Rather than lobby governments to fund primary prevention programmes that could improve the quality of life for children, adolescents and their families, biological psychiatry gives politicians a perfect excuse for doing nothing.

The majority of the public, many members of the mental health professions and most people labelled 'schizophrenic' understand that mental health difficulties originate in the life circumstances – past and present – of the individuals and families concerned. Only one very powerful branch of one profession, biological psychiatry, insists on overemphasizing biology and genetics. Its power is compounded by the support of the pharmaceutical industry. This book, then, has been written largely to bring together the body

of evidence that will increase the confidence of the majority when faced with that misguided but powerful minority who proclaim with all the trappings of scientific and professional expertness: 'it's an illness – so you must take the drugs', by force, if necessary.

This book is not only about the one in a hundred of us who attract the label 'schizophrenic'. Few families have no contact with someone who has been through the mental health system. The increasing medicalizing of human distress, ably abetted by drug company propaganda, knew no bounds in the latter part of the twentieth century. To market tranquillizers and antidepressants, what used to be called worrying and feeling sad are now 'anxiety disorders' and 'depressive illnesses'. Millions of our children are now on amphetamines to treat their difficulty concentrating and sitting still. Millions of older people sit tranquillized in 'homes'. Tens of thousands are still having electric shocks applied to their brains to cause convulsions in the name of 'psychiatric treatment'.

Beyond offering some alternatives and the evidence supporting those alternatives, this book examines the historical, economic and political contexts in which such simplistic bio-genetic ideology gained such a damaging supremacy. We do this lest we suffer illusions about the uphill struggle facing all of us determined to return mental health services to a more humane and effective pathway. Overcoming the obstacles will require that everyone – mental health professionals (including psychiatrists), people currently called 'schizophrenic', their loved ones, researchers, policy makers, service funders and managers and politicians – play their part. For our part, we have gathered together the research evidence that can be presented, in that struggle, to those whose minds are not forever closed to the rather simple ideas that human misery is largely inflicted by other people and that the solutions are best based on human – rather than chemical or electrical – interventions.

Our book is not the first to undertake such a task. For over 50 years, many others have documented the sterility and futility of trying to explain and treat 'madness' with the crude concepts and tools of biological psychiatry, and have offered alternatives. A selection of these is listed below. Our own contribution is partly an updating of the evidence, partly a resurrecting of forgotten or taboo research findings, and partly an introducing of newer approaches (such as understanding the role of childhood trauma, documenting that biological causal beliefs exacerbate stigma and explaining how cognitive psychology can render the seemingly incomprehensible understandable). It is also a straightforward, unashamed wake-up call. Everyone involved should act, in whatever way your circumstances allow, to end this madness.

NOTE

The text of this chapter is a slightly edited version of the first chapter of the first edition. The Further Readings have been updated.

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A HISTORY OF MADNESS

John Read

Sometimes we can understand the present better by reviewing the past. Others have done this in greater detail than is possible here (Appignanesi 2008; Cromby et al. 2013; Foucault and Khalfa 2006; McCarthy-Jones 2012; Porter 2002; Scull 2006). Historical vignettes were selected to illustrate four themes, so that we might be less surprised to discover those themes in the present:

- 1 One of the functions of ‘treatments’ for people considered mad is to suppress behaviours, thoughts and feelings disturbing to those with the power to determine and enforce social norms.
- 2 The ‘treatments’ are often unhelpful and sometimes damaging or even violent.
- 3 The experts of the day often camouflage, even from themselves, the sociopolitical function, and the damaging nature, of the ‘treatments’, behind theories about people having personal defects that are ameliorated by the ‘treatments’.
- 4 The targets of the treatments are typically relatively powerless groups such as the poor, ‘ethnic minorities’ and women (Chapters 14 and 15).

Past centuries have, of course, seen innumerable acts of kindness, by laypeople and experts, to those who experience extreme despair and disorientation or are sufficiently odd to be rejected by the majority. It is a central contention of this book, however, that until we achieve a better integration of psychosocial perspectives into our theories and practice, the themes of social control, unhelpful or damaging treatments targeted disproportionately at powerless groups, and the camouflaging of all this by simplistic theories focused on defective individuals rather than on their social context, will continue to distort our understanding of, and efforts to assist, people considered mad.

GODS AND DEVILS

Four thousand years ago, personal intention was attributed to almost everything. Madness resulted from the evil influence of some enemy, real or imagined. Approaches to 'treatment' were the same as those used for other disturbing events: threats, bribery or submission in the case of human ill-wishers; and prayer or sacrifice in the case of more lofty adversaries (Alexander and Selesnick 1996: 9). According to the prevailing worldview in Mesopotamia around 2000 BC, evil demons were responsible for ailments. The source of recovery was the appropriate good god. To understand 'madness', this world-view was invoked in the form of *Idta*, the demon responsible. *Idta* was assisted by a collection of lesser demons, and the whole department was answerable to *Ishtar*, the female deity supervising all witchcraft and evil (Alexander and Selesnick 1996). The Persians and Hindus also held evil gods, *Ahriman* and *Siva*, accountable for strange behaviours. The treatment of choice was prayers to the adversaries of these divine mischief-makers, *Ormuzd* and *Vishnu* (Gordon 1949).

In perhaps the earliest recorded attempt of the Hebrews to understand insanity, Moses was clear about one cause: 'The Lord will smite thee with madness' (Deuteronomy 28: 28). Moses came close to providing an early example of one of the central themes of this book – that madness is caused by adverse life experiences: 'You shall be only oppressed and crushed continually; so that you shall be driven mad by the sight which your eyes shall see' (Deuteronomy 28: 34). The theory remains, however, one of divine punishment, since it is the Lord who does the oppressing and crushing. He does so, moreover, for breaking the code of social norms – the Commandments. So in this instance madness is a result of divine punishment for breaking norms rather than a term applied to people who break norms to justify their punishment.

Shemia, however, invoked today's prevailing attitude, that breaking norms is a consequence rather than a cause of madness. Faced with *Jeremiah's* prophecy that the Israelites would be exiled, *Shemia* decreed that only a madman could utter such nonsense and ordered a public beating (Rosen 1964). The world-view of the ancient Hebrews, in the Old Testament, included the notion that a good way to enforce conformity to social norms was to eliminate people rather than just their behaviours: 'A man or woman who is a medium or a wizard shall be put to death; they shall be stoned with stones' (Leviticus 20: 27).

The reason Moses gave for punishing numerous deviant behaviours by death was 'So you shall purge the evil from Israel.' If 'Israel' refers to race rather than country, this represents an early version of the genetic theory of unacceptable behaviour and the eugenic approach to eradicating madness via the creation of a genetically untainted race (Chapter 4).

The Old Testament was used for centuries as an expert theory justifying violence against those considered mad. By the time of the first drafts of the

Talmud (AD 70–200), the Hebrew world-view had changed. Now, to be punished for one's madness one must be considered responsible for it. While the Old Testament clearly held madness to be the direct result of sinning, the Talmud says the 'mad' are 'not responsible for the damage or the shame they produce'. They must, therefore, be helped rather than stoned or purged (Rosen 1964).

Greece, too, had indulged in magical and religious theories. Homer, writing about 1000 BC, provides an explanation consistent with the world-view at that time. Whether it was Ajax killing sheep, believing them to be the enemy, or the daughters of Proteus believing they were cows and acting accordingly, there was always a god to blame. If divine intervention was the cause, it must be the cure. Accordingly, Aesculapius, originally a mere priest-physician, was promoted to godhood (Alexander and Selesnick 1996).

THE INTRODUCTION OF REASON

There followed in Greece what many describe as the most significant change in world-view that humanity has experienced. The 'Classical Era' was marked by the replacement of supernatural explanations with observation and reason. Priests were gradually replaced by physicians. The most famous doctor of all time, Hippocrates (460–377 BC), was among the first to promulgate the 'medical model' of madness. He produced the first of its endless futile attempts to reduce the complexities of unusual, distressed or distressing human behaviours to a set of categories of illness, each with its own physiological cause:

Men ought to know that from the brain and from the brain only arise our pleasures, joys, laughter and jests. . . . Those who are mad through phlegm are quiet, and neither shout nor make a disturbance; those maddened through bile are noisy, evil-doers, and restless, always doing something inopportune.

(Hippocrates 1931: 9)

Terms such as 'phlegmatic' survive today. So do simplistic biological theories about why some people are too noisy or too quiet. Hippocrates was just as convinced as biological psychiatry is today that he was discovering the physical causes of illnesses, rather than promoting a simplistic theory that justifies and camouflages the social control of unacceptable or disturbing behaviour. The Dogmatist sect was certainly convinced. In the first of many such enterprises (which continue today in the field of mental health), the Dogmatists tried to prohibit any investigation beyond the biological ideology of the day, because Hippocrates had already discovered everything worth discovering.

Ducey and Simon (1975) offer an insight, still valuable today, into one motivation for simplistic biological theories. The Greeks were becoming

aware, more than 2,000 years before Freud, of human conflicts: not only internal conflicts, such as the 'reasoning' versus 'desiring' dilemma identified by Plato (427–347 BC), but also conflicts between the individual and society. In the absence of religious instructions regarding such dilemmas, there was a need to identify other forces over which individuals had little control and for which they could not be held accountable:

The Hippocratic model of mental illness and of its treatment may have provided an important source of relief of anxiety, guilt and responsibility, by translating inner conflicts and dissonances into physiological and physicalistic terms.

(Ducey and Simon 1975: 19)

Hippocrates' treatments were probably effective in altering behaviour. I, for one, would have been less quiet or noisy rather than face being forced to vomit or defecate uncontrollably, or to swallow hellebore – a poison now used in insecticides.

During this period there was a fading of an extreme male dominance that had included the feeding of boys at the expense of girls (Lacey 1968). When Plato wrote 'within the woman there is a wild, animalistic, Bicchantic, frenzied creature, who must be gratified, or else she goes berserk' (Ducey and Simon 1975: 21), he was not arguing that generations of oppression had led to pent-up anger, nor suggesting that such oppression can drive women mad. He was laying the foundations of a different type of theory of female madness, a theory in keeping with the physiological explanations of the time, and that would be used not to liberate women but to continue their suppression.

The invention of 'hysteria' is an early example of the use of male experts' theories of madness to force women into the roles determined for them by men.

Whenever the womb – which is an indwelling creature desirous of child-bearing – remains without fruits long beyond the due season, it is vexed and takes it ill; and by straying all ways through the body and blocking up the passages of the breath and preventing the respiration it casts the body into the uttermost distress, and causes, moreover, all kinds of maladies until the desires and love of the two sexes unite them.

(Plato 1904: 916-c)

The treatments of choice for 'hysteria' were marriage and, if that failed, forced fumigation of the vagina – a form of mechanical rape. Although the male experts told their female 'patients' that the purpose of the 'treatment' was to attract the wandering uterus back to its rightful place, we can see with hindsight that it was the wandering of women from their subservient role that was the real concern.

A RETURN TO RELIGION

The social structure of the Roman Empire crumbled before epidemics and invasions. The resulting insecurity facilitated a return to supernatural beliefs. By the fourth century AD Constantine had made Christianity the official religion, the Church and State having become an inseparable, immense power. In the field of madness, observation, reason and physiological theories were replaced with variations on the old religious themes. In the New Testament Christ was venerated as the healer of the sick and the caster out of demons. As the Church became protector of the infirm and insane, images of Aesculapius were worshipped as images of Jesus.

In the search for some external force to explain frightening behaviour, it was thought that such behaviour was worse when people were alone at night. Obviously, the moon was responsible for craziness. Mad people have, ever since, been called 'lunatics'.

Another concept, more consistent with Christianity, explained almost any anxiety-provoking event, including epidemics, conflagrations and madness. The phenomenon of witchcraft is one of the most horrifying examples of the extent of the violence that can be justified by theories about groups of people considered defective.

The extent to which cultural changes affect theories about who or what is to blame for madness is well illustrated by dividing the Middle Ages into two periods. In the earlier period causal theories were influenced by the charitable aspects of Christianity. Care was relatively humane. Patients of the original Bethlem Hospital in London, later to degenerate into the infamous Bedlam 'snake-pit' (Russell 1996), wore arm badges on pass. The response was so positive that some people forged badges for themselves. At this time people thought to be possessed by the Devil were not held responsible. The treatment of choice – exorcism – was not, therefore, a punitive process aimed at the individual, but a benevolent one directed against the Devil.

From the eleventh century on, however, the breakdown of feudal structures, accompanied by insurrections, often aimed at the Church, demanded a set of scapegoats. Executions might deter other 'heretics'. Possessed women could not serve this purpose if they were not responsible for their actions. By the time Kraemer and Sprenger published their witch-hunting manual, *Malleus Maleficarum* ([1486] 1941), it had been discovered that witches invite the Devil into their lives.

Religious and social factors during this latter period of the Middle Ages had created an intense fear and hatred of women and a fiercely misogynist Church. Women were officially described as 'a foe to friendship, an unescapable punishment, a necessary evil, a natural temptation, a desirable calamity . . . an evil of nature painted with fair colours' (Kraemer and Sprenger [1486] 1941: 43). The vast majority of people tortured and murdered for witchcraft were indeed women (Michelet 1939). Many were guilty of nothing more than

having some healing skills. Their herbs and brews were a threat to the male priests' monopoly on healing. Others were guilty of no more than being female. The symptoms were listed in the diagnostic manual of the day:

All witchcraft comes from carnal lust which is in women insatiable. . . . Three general vices appear to have special dominion over wicked women, namely, infidelity, ambition and lust. Therefore they are more than others inclined toward witchcraft who more than others are given to these vices.

(Kraemer and Sprenger [1486] 1941: 47)

The possession of too much phlegm or bile had been replaced with possession by the Devil. While the various theories come and go, their purpose remains constant. Along with the mass murder of mental patients in Germany in the 1940s (Chapter 4) and the political oppression (often using the 'schizophrenia' label) of dissidents in KGB-managed Soviet 'psychiatric hospitals' (Bloch and Reddaway 1977, 1984; van Voren 2002; van Voren 2010), the Inquisition is a horrifically clear example of a well-organized programme of cruelty directed against those engaging in behaviours unacceptable to those in power. These programmes are always justified by theories disguising the violence as help for defective individuals. In the case of the murder of women diagnosed as witches, the 'help' took the form of purifying their souls so they could enter heaven.

THE RETURN TO SCIENCE

The return to observation and reason known as the Renaissance was a gradual and patchy process. For instance, Johann Weyer (1515–61) used observation in his relentless battle against the notion of witchery. He demonstrated that objects supposedly implanted in the stomach of an accused woman carried no signs of gastric juices, earning himself the diagnosis 'Weirus Insanus'.

Increased emphasis on the psychosocial origins of madness was reflected in Shakespeare's *Hamlet* and Cervantes' *Don Quixote*. Thomas Sydenham, the famous English physician, contributed to psychological approaches by diagnosing on the basis of the individual's emotional circumstances. Robert Burton's *Anatomy of Melancholy* ([1621] 1961), based partly on his need to understand his own unhappiness, described one of the more popular diagnostic categories of this period. His theories were a blend of demonological and physiological. His treatments focused, like those of his contemporaries, on exercise, diets, drugs and purgatives. He added, however, a moving description of one nonmedical treatment for his own personal malady:

It is the best thing in the world . . . to get a trusted friend, to whom we may freely and sincerely pour out our secrets; nothing so delighteth

and pleaseth the mind, as when we have a prepared bosom, to which our secrets may descend, of whose conscience we are assured as our own, whose speech may ease our succourless estate, counsel relief, mirth expel our mourning, and whose very sight may be acceptable unto us.

(Burton [1621] 1961: 108–9)

Such a therapeutic experience was not available to the thousands of Europeans who, under the guise of medical treatment, were being locked away, worked, purged and drugged. Soon after its 1656 opening, the Hôpital Général in Paris was ‘treating’ 6,000 people. Typical of the view taken of this period by historians of psychiatry is the opinion that:

The Age of Reason marked a great leap forward. Through the efforts of the great scientists, philosophers, men of letters, and artists of the 17th century mental illness was further extricated from superstition and authoritarian error.

(Alexander and Selesnick 1996: 104)

The direction being adopted, however, was a transition from error based on one form of authority (religion), to error based on an equally dangerous form of authority (simplistic and reductionist ‘medical science’). The violence continued, masked by a new rationale.

CATEGORIZATION AND CONFINEMENT

From the latter part of the seventeenth century until well into the eighteenth, the ‘scientific’ approach established the ascendancy it maintains today. During this ‘Enlightenment’ explanations of madness became dominated by categorization and physiology. Psychological and social factors were buried under a relentless drive to discover illnesses. This would eventually lead to the invention of ‘schizophrenia’ (Chapter 3). The hope driving this quest, then as now, was that correct categorization would lead to the discovery of physiological causes. Actually, those rare instances where physiological origins of madness have been found – such as the effects of advanced syphilis – have been the cause, rather than the result, of a new classification. The Enlightenment’s many attempts to find the imagined true categories of ‘mental illness’ led to no useful advances in physiological explanations. Meanwhile, the violent treatments continued.

Herman Boerhaave (1668–1738), for instance, justified blood-lettings and purgatives by discovering that ‘melancholia’ was caused by black juices. His invention of the rotating chair was more original. The experts’ theories used to portray this particular cruelty as a medical treatment varied. In Britain,

Erasmus Darwin discovered that it reintroduced harmony to the 'disordered motions' of the nerves. Benjamin Rush, meanwhile, was convincing patients in North America that they were being spun into unconsciousness to unclog their congested blood, which, he had discovered, was the true cause of madness. William Cullen (1812) added blistering of the forehead and cold dousings to the blood-letting, vomiting, purging and spinning.

The quest for localized brain pathology, still popular today, was well under way. At London's Bethlem asylum, John Haslem was already conducting autopsies (Alexander and Selesnick 1996). One wonders how many of these people had died as a result of the neglect and violence that was being justified by theories about disordered brains.

Who were these people? The 1656 Edict that founded the Hôpital Général defined its population as: 'the poor of Paris, of both sexes, of all ages and from all localities, of whatever breeding and birth, in whatever state they may be, able-bodied or invalid, sick or convalescent, curable or incurable' (Foucault 1965: 39). The mad were locked away not for being mad but for being poor. The 'Great Confinement', as Foucault describes this period, served the economic function of forcing inmates to work for very little pay, under the guise of exercise or 'occupational therapy'. It served the political function of suppressing, under the guise of help for the poor and sick, the increasing number of uprisings among the unemployed. It also bolstered a moral belief in hard work. The Edict established begging and idleness as the major causes of all disorders. The mad, Foucault argues, were being locked away all over Europe along with anyone else considered inconvenient or threatening to those in power, precisely because they were inconvenient or threatening. Calling prisons hospitals conceals this.

The treatments behind the locked doors 'were physical cures whose meaning had been borrowed from a moral perception' (Foucault 1965: 159). Many were based on purification of the body, symbolizing the notion that mental illness was caused by biological moral contamination. Doublet suggested, in 1785, that if the blood-lettings and purges failed, one should try abscesses and inoculation of scabies (Foucault 1965). This quest for the best purifying agent led to force-feeding tartar, chimney soot, woodlice and soap.

MORAL TREATMENT

Pinel in France, Tuke in England, Chiarugi in Italy and Muller and Reil in Germany are remembered for their humane treatment of mad people. Their various approaches have come to be known, accurately, as 'Moral Treatment' (Scull 1981). Pinel, renowned for unchaining the insane, found it impossible to differentiate the effects of madness from the effects of cruel treatments. Unlike many of the experts before them, Pinel and his fellow reformers were honest about their being in the business of imposing society's moral code on

deviant individuals. Pinel was particularly concerned to eradicate celibacy, promiscuity, apathy and laziness. Such a morality was a special goal for women because, in line with the efforts of the Greek priest-physicians to combat hysteria, 'marriage constitutes for women a kind of preservative against the two sorts of insanity which are most inveterate and most often incurable' (Foucault 1965: 258).

William Tuke founded the York retreat in 1792, on Quaker values. His son Samuel, who succeeded him, believed that 'to encourage the influence of religious principles over the mind of the insane is considered of great consequence as a means of cure' (Tuke 1813: 121). Patients would be invited to tea parties with the staff to assess their manners.

If the goals, for once, were less couched in medical terms (William was opposed to the medical management of madness), the means for imposing conformity remained the tools of social control. Samuel admitted what his predecessors had concealed behind the language of medicine when he wrote: 'the principle of fear, which is rarely decreased by insanity, is considered of great importance in the management of patients' (Tuke 1813: 141). Pinel argued that:

Considered as a means of repression, it often suffices to subject to the general law of manual labour a madman who is susceptible to it, in order to conquer an obstinate refusal to take nourishment, and to subjugate insane persons.

(Foucault 1965: 266–7)

It is important to our understanding of the political function of contemporary psychiatry that we realize *why* the medic was invited into this new version of the asylum. The medic, who had played little part in the huge confining hospitals, entered now not as a medical expert but as an authority figure to add weight to the efforts of the attendants:

The doctor's intervention is not made by virtue of a medical skill or power that he possesses in himself and that would be justified by a body of objective knowledge. It is not as a scientist that homo medicus has authority in the asylum, but as a wise man. If the medical profession is required, it is as a juridical and moral guarantee, not in the name of science.

(Foucault 1965: 270)

The authority of psychiatrists today still originates more from the political power we all bestow upon them than from any scientific expertise. Perhaps the source of this power is well disguised, even from those who wield it:

In the time of Pinel and Tuke, this power had nothing extraordinary about it; it was explained and demonstrated in the efficacy, simply,

of moral behaviour. . . . But very soon the meaning of this moral practice escaped the physician to the very extent that he enclosed his knowledge in the norms of positivism; from the beginning of the 19th century, the psychiatrist no longer quite knew what was the nature of the power he had inherited from the great reformers.

(Foucault 1965: 274)

Pinel clearly understood the reason for introducing the medic into his asylums, and the difficulty in distinguishing medical treatment from social control:

The doctrine in ethics of balancing the passions of men by others of equal or superior force, is not less applicable to the practice of medicine, than to the science of politics, and is probably not the only point of resemblance between the art of governing mankind and that of healing their diseases.

(Pinel 1801: 228)

Others offer a more positive perspective on Moral Treatment (Gale *et al.* 2008; Scull 1981). Whatever one's evaluation of it, it had little impact on the stranglehold of biological theories about madness. The categorizations and the quest for physical explanations continued. Despite producing an important treatise on psychotherapy in 1803, Johann Reil, for instance, lumped together all silent patients into yet another new category, for which he prescribed the treatment of standing next to cannon fire – an early version of shock therapy. Eventually the long list of useless categories would come to include 'dementia praecox' and, soon thereafter, 'schizophrenia'.

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THE INVENTION OF 'SCHIZOPHRENIA'

Kraepelin and Bleuler

John Read

PSYCHIATRY IN CRISIS

Emil Kraepelin and Eugen Bleuler, two 'grandfathers' of modern psychiatry, were jointly responsible for the invention of 'schizophrenia'. Their contribution has been critiqued many times before (e.g. Bentall 2003, 2009; Boyle 2002). My intention here is to demonstrate how their invention represents a continuation of (a) the futile search for categories of unusual behaviours in the hope of discovering illnesses, and (b) the historical themes of social control and harmful treatments disguised by theories about help for defective individuals (Chapter 2).

Throughout the nineteenth century, progress in medicine was impressive. The search for similar successes in relation to madness is understandable. The new discipline of psychiatry generated many new categorizations of madness, but each one was soon abandoned. The fragile new profession could not afford, however, to abandon the quest for mental 'illnesses' altogether. Its survival as a medical science depended on it.

In 1822, one physiological cause for one form of madness *was* found. Bayle's discovery of brain damage in '*dementia paralytica*' was later identified as syphilis. This fuelled hope that other mental 'illnesses' with physiological causes might exist. However, as Bleuler himself admitted: 'After Paresis was excluded . . . for 70 years theoretical psychiatry stood entirely helpless before the chaos of the most frequent mental diseases' (Bleuler 1924: 372). By the end of the century, psychiatry's need for a second discovery had become urgent.

EMIL KRAEPELIN

Those who knew Kraepelin described him as aloof and obsessional. He was a lifelong campaigner against the evils of alcoholism, a

fierce German nationalist and admirer of Bismarck, and wrote dreadful poetry in support of his social and political opinions. . . . As his detailed case studies reveal, patients were considered as bearers of symptoms rather than as people with histories and stories to tell.

(Bentall 2009: 28, 32)

One biographer wrote: 'Kraepelin's supremely controlling object was the furtherance of psychiatry' (Jelliffe 1932: 764), noting that he found financial backers for his mission:

Kraepelin succeeded in obtaining from the Rockefeller Foundation the sum of \$250,000 for his project, the first gift to be made by this organization to research work in Europe. A further gift from the Rockefeller Foundation increased the sum to \$325,000 under the stipulation that there should be strict adherence in the institute to Kraepelin's fundamental idea.

(Jelliffe 1932: 768)

We will probably never know whether there was any connection between the Rockefellers' owning the petroleum industry, the discovery that petroleum by-products could be used to produce medicines and the stipulation in Kraepelin's contract. (Sometimes paranoia is a state of heightened awareness; sometimes it's paranoia).

Outcomes

The task facing Kraepelin at Heidelberg University was the same as that faced by all those classifiers who had tried and failed before him: to group together behaviours all caused by some physical pathology.

In 1893 he claimed to have discovered a group of people in whom deterioration begins in adolescence and continues inevitably into a permanent dementia. Hence the name 'dementia praecox' ('praecox' meaning early). Many of the behaviours he decided were symptoms of dementia praecox had already been claimed by the inventors of two other illnesses, 'catatonia' and 'dementia paranoids'. Kraepelin created a conglomerate illness by incorporating and re-branding the other two. He did this despite the fact that the problems of many of the people labelled catatonic or paranoid neither began in adolescence nor resulted in dementia.

Despite one of the major defining characteristics of Kraepelin's new illness being an inevitable deterioration into dementia, some got better. Kraepelin made no claim to have cured them. Either his invention did not exist, or the people who got better didn't have dementia praecox. He argued:

The undoubted inadequacy of my former classification has led me once more to undertake the attempt to make a more natural grouping. . . . Recovered cases were not taken into account because of the uncertainty which still exists, but only such cases as had led to profound dementia.
(Kraepelin [1913] 1919: 89, 90)

Kraepelin claimed he had discovered an incurable degenerative illness. When people who he says have the illness get better, he says they haven't got the illness. He is then left with a group of people who don't get better and uses them as evidence that the illness exists. (This circular logic is still invoked today. When people diagnosed 'schizophrenic' go on to lead productive lives, they are often told that the diagnosis must have been wrong.) American psychiatrist Harry Stack Sullivan complained, in 1927, that: 'The Kraepelinian diagnosis by outcome has been a great handicap, leading to much retrospective distortion of data, instead of careful observation and induction' (p. 760).

Symptoms

Kraepelin couldn't, of course, wait for outcomes before making his diagnoses. All he could do, like his predecessors, was make a list of behaviours and apply a medical-sounding label. Then he claimed to have proved the existence of dementia praecox by showing that people who had it were exhibiting its symptoms; more circular logic.

His takeover of catatonia and paranoia had netted Kraepelin 36 groups of 'psychic' symptoms and 19 types of bodily symptoms. One patient could have symptoms entirely different from a second and totally the opposite to those of a third. They all supposedly had the same illness. Such constructs are known as 'dysjunctive' and are considered scientifically meaningless (Chapter 5).

Kraepelin ([1913] 1919: 3) asserted: 'We are justified in regarding the majority at least of the clinical pictures which are brought together here as the expression of a single morbid process, though outwardly they often diverge very far from one another.' Having listed the most important features of his illness, he admitted that some patients did not show any of these:

These features are certainly not demonstrable with full clearness in each individual case but still the general view over a great number of complete observations teaches that nowhere can a state be discovered which is not connected by imperceptible transitions with all the others.

([1913] 1919: 255)

So, if you can't see the features, you're just not looking hard enough. Even if after strenuous efforts you still can't perceive 'the imperceptible', it doesn't mean it isn't there. He even admitted that it can 'be very difficult, not only to

judge correctly of the diagnostic significance of such deviations, but even to recognize their very existence' (p. 257).

Causes

Kraepelin needed to identify some form of physical pathology that might be linked to the symptoms. Autopsies, conducted by the famous Alois Alzheimer, revealed nothing abnormal (Bentall 2009: 31). By 1913 Kraepelin admitted that the causes 'are at the present time still wrapped in impenetrable darkness' ([1913] 1919: 224). The key words are 'at the present time'. These words have been used ever since by researchers forever on the verge of finding the biological cause of schizophrenia.

Kraepelin ([1913] 1919: 1) began the eighth edition of his *Psychiatria* (in which dementia praecox was not the only illness described) thus: 'A series of morbid pictures are here brought together under the term "endogenous dementias" merely for the purpose of preliminary enquiry'. Psychiatry, however, urgently needed something more than 'preliminary enquiry' or 'impenetrable darkness'. Kraepelin suggested another, rather ingenious, re-branding exercise:

If therefore the name . . . is to be replaced by another, it is to be hoped that it will not soon share the same fate of so many names of the kind, and of dementia praecox itself, in giving a view of the nature of the disease which will turn out to be doubtful or wrong.

(Kraepelin [1913] 1919: 4)

He even recommended that 'a name that as far as possible said nothing would be preferable' (p. 4). To claim to have discovered an illness without identifying any consistent symptoms or outcome, and with no observable cause, stretches the basic rules of medical science. To propose a meaningless name to avoid assumptions that might be tested positions Kraepelin's invention beyond the realms of science altogether.

EUGEN BLEULER

Bleuler was influenced as much by Freud as by the dominant biomedical approach to madness. Unlike Kraepelin, Bleuler was 'an empathetic and workaholic clinician' who 'spent many hours getting to know his patients, and struggled to understand their inner life' (Bentall 2009: 93).

In 1911 Bleuler, working in Switzerland, published his famous *Dementia Praecox or the Group of Schizophrenias*. He divided Kraepelin's many-headed monster into several smaller beasts. He made explicit the criteria by which one knows whether a disease has been discovered:

We must ask ourselves the questions: in what connection with other symptoms and anatomical findings, in what sort of course, as the result of which causes does the symptom appear? . . . Only the answers to these questions can provide us with the concept of the disease.

([1911] 1950: 273)

Outcomes

Bleuler abandoned the idea that everyone with the disease ended up with 'dementia'. What did his search for some other consistent course reveal? 'It is impossible to describe all the variations which the course of schizophrenia may take' ([1911] 1950: 328). What about the relationship between outcome and symptoms? 'We have not discovered any correlation between the initial disease symptoms and the severity of the outcome or the illness' (p. 261).

Symptoms

Bleuler understood that discovering an illness requires more than listing symptoms.

Such terms as acute paranoia, acute hallucinatory insanity, confusion mentale, as well as mania and melancholia . . . do not designate 'diseases'. . . . Not only the names but the entire concepts were arbitrarily constructed by this or that observer, depending on which symptom he considered the most striking one. . . . A symptom, regardless of whether it is psychic or physical is never a disease; neither is a symptom-complex.

(Bleuler [1911] 1950: 272–3)

Since a symptom complex is not a disease, Bleuler had to demonstrate some connection between symptoms and causes. He admitted, however, that 'The anatomical findings do not correspond with the severity of the manifest symptoms' (p. 462). This is unsurprising, since 'We do not as yet know with certainty the primary symptoms of the schizophrenic cerebral disease' (p. 349).

Causes

The pathology of schizophrenia gives us no indication as to where we should look for the causes of the disease. Direct investigation for specific cause or factors has also left us stranded.

([1911] 1950: 337)

We do not know what the schizophrenic process actually is.
([1911] 1950: 466)

Bleuler had failed to answer any of the questions he stated must be answered before the discovery of a disease can be claimed.

SYMPTOMS OF AN ILLNESS OR BROKEN SOCIAL NORMS?

If Kraepelin and Bleuler's invention is not a disease, what is it? A primary function of treatments for the mad has always been to control behaviour unacceptable to those with the power to determine and enforce social norms (Chapter 2). Here is a tiny sample of the hundreds of 'symptoms' of their 'illness'.

Behaviours

They conduct themselves in a free and easy way, laugh on serious occasions, are rude and impertinent towards their superiors, challenge them to duels, lose their deportment and personal dignity; they go about in untidy and dirty clothes, unwashed, unkempt, go with a lighted cigar into church.

(Kraepelin [1913] 1919: 34)

The loss of taste often makes itself felt in their choice of extraordinary combinations of colour and peculiar forms.

(Kraepelin [1913] 1919: 55)

The want of a feeling of shame expresses itself in regardless uncovering of their persons, in making sexual experience public, in obscene talk, improper advances, and in shameless masturbation.

(Kraepelin [1913] 1919: 34)

More in girls, there is reported irritability, sensitiveness, excitability, nervousness, and along with these self-will and a tendency to bigotry . . . [and] those patients, belonging rather more to the male sex, who were conspicuous by docility, good nature, anxious conscientiousness and diligence, and as patterns of goodness held themselves aloof from all childhood naughtiness.

(Kraepelin [1913] 1919: 236)

The patients sit about idle, trouble themselves about nothing, do not go to their work.

(Kraepelin [1913] 1919: 37)

Patients are in love with a ward-mate with complete disregard of sex, ugliness, or even repulsiveness.

(Bleuler [1911] 1950: 52)

It hardly makes any difference to the patient whether he is addressing a person in authority or someone more humbly-placed whether a man or a woman.

(Bleuler [1911] 1950: 49)

A hebephrenic [a subtype of schizophrenia] whose very speech was confusion, held the cigar-holder to the mouth of another patient suffering from muscular atrophy. . . . He did this with a patience and indefatigability of which no normal person would ever be capable.

(Bleuler [1911] 1950: 48)

Perversions like homosexuality and similar anomalies are often indicated in the whole behaviour and in the dress of the patient.

(Bleuler 1924: 188)

Feelings

'Flat affect' and 'inappropriate affect' were 'primary symptoms'. Feeling two opposing emotions was also abnormal, such 'ambivalence' being a defining characteristic of 'schizophrenia'. Remembering too intensely was also a symptom:

Even decades later . . . nuances of sexual pleasure, embarrassment, pain or jealousy, may emerge in all their vividness which we never find in the healthy.

(Bleuler [1911] 1950: 46)

Thoughts

Unusual thought processes are 'wrong pathways of thought' carrying 'the schizophrenic mark of the bizarre' (Bleuler [1911] 1950: 25). They are wrong because they are 'pathways deviating from experience' (p. 354). Since a thought is, by definition, part of one's experience, whose experience is it from which these thoughts are deviating?

Kraepelin ([1913] 1919) defined 'incoherence of thought' in terms of whether psychiatrists can understand their patients, highlighting 'that peculiarly bewildering incomprehensibility' (p. 20). Instead of recommending that understanding unusual thinking be included in psychiatric training programmes, he added 'incomprehensibility' to his list of symptoms.

Bleuler ([1911] 1950) found two specific types of thinking particularly bewildering: 'Two ideas fortuitously encountered, are combined into one

thought' and 'The inclination to cling to one idea to which the patient then returns again and again' (p. 14). Bringing together ideas to form new ideas is the trademark of artists and philosophers. Holding steadfastly to one belief characterizes the religious or political enthusiast. Did Bleuler intend to pathologize these groups?

Poets and musicians must also be more sensitive than other people, a quality which is a hindrance to the daily tasks of life and often attains the significance of a disease.

(1924: 172)

Many schizophrenics display lively affect at least in certain directions. Among them are the active writers, the world improvers, the health fanatics, the founders of new religions.

([1911] 1950: 41)

Elsewhere, Bleuler appears to understand the difference between symptoms of an illness and deviance from social norms:

Nowhere is the question: 'sick or not sick?' put so often in such an inexorable manner and with such heavy consequences as in the judgment of mental conditions. But the given question is false. . . . Still more senseless is the question about well and sick where it is not a case of something added but of a plain deviation from the normal.

(1924: 170)

Nevertheless, as a list of unacceptable behaviours and experiences, the writings of Kraepelin and Bleuler are scary. What is left? What is considered sane? We must be heterosexual, abide by rigid roles for males and females, obey our superiors, want to work, like certain colour combinations, not feel too much or too little, not remember too vividly, not get too interested in new ideas, not show too much compassion for others, be well-behaved (but, if male, not too well-behaved), not write too much poetry or play too much music, not try too hard to change the world, and we must speak, even when very distressed, in a way that even psychiatrists can understand.

GENETICS

Kraepelin and Bleuler wanted to prove not only that all these behaviours were symptoms of a disease, but that the disease was genetically inherited. They searched for evidence in the families of 'schizophrenics', extending their diagnostic net to catch an even wider range of behaviours:

Among the brothers and sisters of the patients there are found striking personalities, criminals, queer individuals, prostitutes, suicides, vagrants, wrecked and ruined human beings, all being forms in which more or less well-developed dementia praecox may appear.

(Kraepelin [1913] 1919: 234)

Most frequently one finds in families schizoid characters, people who are shut-in, suspicious, incapable of discussion, people who are comfortably dull and at the same time sensitive, people who in a narrow manner pursue vague purposes, improvers of the universe, etc.

(Bleuler [1911] 1950: 391)

With this broadened range of behaviours Bleuler found 'hereditary tainting' in 90% of the families concerned. He found the same 'tainting' in 65% of people considered 'healthy'. Unlike genetic researchers of 'schizophrenia' today (Chapter 7), he understood that 'queer' parents having 'queer' children proves nothing about genetics.

If someone cares to assume that the modifications of the psychic or physical factors produced by communal living produces such accumulations of disease in a given family group, we would be unable to produce any proof to the contrary.

(Bleuler [1911] 1950: 337)

IGNORING THE SOCIAL CONTEXT

Any behaviour can be transformed into a symptom of 'mental illness' simply by an expert decreeing it so. It helps, however, if the behaviour is portrayed as meaningless or bizarre. An effective way to do this, then and today, is to ignore the social context.

Incarceration

One reality shared by most of the people who were supposed to have 'schizophrenia' was incarceration. Some people get upset when locked up. 'It is an everyday occurrence to see the patients soiling themselves or tearing their clothes, "because they won't let me go home"' (Bleuler [1911] 1950: 24). This is portrayed as paranoia: 'The "persecuted" are dissatisfied with their environment, considering it the source of the persecution' (p. 48). As an example of the symptom 'variability of attention', Bleuler (1924) cites 'Patients, who for a long time cannot give attention in any way, are suddenly capable of planning a complicated plan of escape' (p. 385). Only by ignoring

what may be the most important reality in one's life could such sensible behaviour be turned into a symptom of 'schizophrenia'.

To illustrate 'disconnected, confused babble', Bleuler cites a letter:

This is indeed no parish-house or even a poor-house but in this place there is noise, anger, grumbling-sunny-heavenly-knells all year round. Many a small and large land owner, wind-bag or poor drunk from Thalweil, Addisweil, from Albis, from Salz, from Seen, from Rorbach, from Rorbas have never again returned to their own homes, etc., etc., etc. Greetings to all who are still alive. My own relatives no longer exist.
(Bleuler [1911] 1950: 31)

If one remembers the social context, the letter seems to contain some understandable statements:

- The institution is not what most people think it is.
- Many people here are angry.
- Many different kinds of people are locked up here.
- Hope of release is minimal.
- One does not feel very alive when incarcerated.
- Since I have been here, my family have all died (or I feel deserted by my family).

The psychiatrist

Another reality in psychiatric hospitals is powerful doctors who ask questions to establish your insanity. Only by ignoring this, and the resentment it can cause, can Kraepelin include the following in his list of symptoms:

Often the patients refuse all information: 'That is their own affair', 'That is no one's business', 'People are cross-questioned in that way', 'They should not be there, did not require to give explanations'. One patient first asked the physician to show him the diplomas that he might know with whom he had to do.

(Kraepelin [1913] 1919: 49)

Resistance to those in power is also a symptom:

Neither on command, nor in accordance with the rules of the hospital do they want to get dressed or undressed or come to meals or leave their meals; but if they can perform the same acts outside of the required time or somehow contrary to the will of the environment, they often do them.

(Bleuler 1924: 406)

There is another form of resistance to outside influences, which comes into existence wholly by impulse that is without foundation on ideas or emotions. Imperative negativism, the carrying out of action exactly contrary to that wished, scarcely allows for any other interpretation. . . . When a patient begins to sing as soon as he is asked to be quiet, when he goes backward on being ordered to march, it is difficult to find an explanation in deliberation or emotional influences.

(Kraepelin [1913] 1919: 48)

One alternative interpretation is a deliberate, stubborn reaction to being told when to eat, sleep, get dressed, speak, be quiet and (one wonders what place this has in a hospital) to march. There are other examples of specific behaviours by psychiatrists actually eliciting 'mad' behaviour:

The patients do not note contradictions when we take their negative answers for positive ones. I asked a patient: 'Do you hear voices?' He definitely denied it. I continued: 'What do they say to you?' 'Oh, all sorts of things'.

(Kraepelin [1913] 1919: 54)

One might equally well conclude that the person asking the questions can't differentiate negative answers from positive ones.

Another example of apparently bizarre behaviour being explicable in terms of the actions of psychiatrists is the symptom 'waxy flexibility':

This very striking disorder is plainly seen in the group in figure no. 3, which brings together a series of patients suffering from dementia praecox. They were put without difficulty in the peculiar positions and kept them, some with a sly laugh, others with rigid seriousness.

(Kraepelin [1913] 1919: 53)

Despite admitting the patients were actually put in these positions by the psychiatrist (to see if they had the symptom of staying put), their maintaining those positions is perceived purely as a 'symptom' rather than a following of orders. Kraepelin asks them why they did it and is told 'I do it because you wish it so'; 'I place myself according to what is commanded' (Kraepelin [1913] 1919: 38–9).

Here is an example of the symptom 'automatic obedience':

He continues to put out his tongue when commanded to do so although one threatens to stab it, and causes him pain with a needle, as can be seen by the grimaces he makes.

(Kraepelin [1913] 1919: 38)

When presenting this material in undergraduate lectures I ask the students to imagine a Martian observing these two members of our species, and to guess which the Martian would consider mad. They usually answer 'both'.

Pressure to admit madness and deny social causation

Beyond the pressure – including specific instructions – to *act* crazy, there was also pressure to believe, or at least state, that you *are* mentally ill:

An often discussed criterion of cure is that of the patients' insight into the nature of illness. People who speak of their delusions and their weird behaviour during the attack as being pathological phenomena . . . are not without reason easily considered as cured; whereas the opposite is thought of as being a rather certain sign of continuing disease.

(Bleuler [1911] 1950: 257)

Furthermore, you had to admit that your problems were unrelated to your life history:

They do not always locate the illness where the observer sees it to be. They realise, for example, the poor state of their nerves, the senselessness of their behaviour, but they insist that both are quite understandable reactions to stimuli and irritations of their environment.

(Bleuler [1911] 1950: 257)

MEDICAL TREATMENT OR SOCIAL CONTROL?

A vital question, both before and since the invention of 'schizophrenia', is whether 'treatments' are best understood as attempts to cure an illness or as attempts to suppress socially unacceptable behaviour (Chapter 2). Kraepelin ([1913] 1919) devotes less than five of his 300 pages to *How to Combat It*. Apart from hospitalization, he offers little beyond 'occupation' and various restraints including bed-rest and prolonged baths. Some patients didn't think this was a good idea.

The patients do not remain in the bath, but always jump out again, perform neck-breaking gymnastics, roll about on the floor. The next thing now to be tried is to quiet the patients so far by a sedative, Nyoscine, Sulphonal, Trional, Veronal, that he may remain some hours in the bath.

(Kraepelin [1913] 1919: 257)

At this point in the history of schizophrenia, the use of drugs was not seen as a medical treatment. It was acknowledged that the drugs were to tranquillize or to get people to do as they were told. Leucocytosis consisted of injecting sodium nucleinate to cause fever. Kraepelin ([1913] 1919) admits that from a medical point of view 'nothing was achieved'. He adds, however: 'Under the influence of a fever . . . negativistic patients appear more docile and obey medical orders' (p. 281: 2).

Bleuler ([1911] 1950) also lists various forms of restraint, chemical and otherwise. He admits, however, that 'we know of no measures which will cure the disease, as such, or even bring it to a halt' and 'treatment of schizophrenia by medication does not exist' (p. 485). Nevertheless, he recommends sedatives: 'Patients, who otherwise would not get accustomed to work, to their bed, to bathing, and to normal conditions, may gradually be trained, while under the effect of mild sedation' (p. 486).

Apomorphine is a chemical restraint of a special type. Some acute agitated states can be interrupted immediately by an injection of an emetic dose. . . . I cannot recommend it on ethical grounds; but then, again, I must ask whether it is not more unethical to permit a whole roomful of patients to be annoyed by a single agitated patient than to cause the offender to vomit.

([1911] 1950: 486)

Not many psychiatrists today share Bleuler's understanding of the function of psychiatric drugs. Most believe that the drugs are treating an illness, rather than tranquillizing people into compliance, largely for the convenience of others.

Bleuler is remembered neither for delineating the criteria for discovering an illness nor for his honesty about the lack of evidence in relation to symptoms, outcome, cause or treatment. Long before his famous 1911 article was translated into English in 1950, the notion of a bio-genetically based illness called 'schizophrenia' had been unquestioningly accepted by psychiatrists all over the 'developed' world as if it were scientific fact. It quickly became the flagship of biological psychiatry.

Psychiatry survived its crisis. It did so, however, at an appalling cost for the millions who for a century thereafter have been branded with the scientifically meaningless and socially devastating label 'schizophrenia'.

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GENETICS, EUGENICS AND THE MASS MURDER OF ‘SCHIZOPHRENICS’

John Read and Jeffrey Masson

In 1941, the staff of the Hadamar Psychiatric Institution – psychiatrists, nurses and secretaries – attended a ceremony and were each given a bottle of beer. The occasion was the murder of the ten-thousandth mental patient (Proctor 1988).

Some might argue that what happened in Germany 70 years ago has nothing to do with how biological psychiatry operates today. We document these awful events, however, precisely because they so clearly illustrate themes present throughout the history of the treatment of people considered mad: (a) social control in the interests of the powerful, (b) damaging and violent ‘treatments’, and (c) the ability of expert theories to camouflage what is happening (see Chapter 2).

THE THEORY

The genetic theories that still dominate psychiatry today provided the motivation, the rationale and the camouflage for what happened. Psychiatrists had a leading role in developing the theory – that undesirable behaviour is genetically transmitted – that was used to justify compulsory sterilization and murder.

This theory, also promoted by famous psychologists Burt, Spearman and Cattell (Pilgrim 2008), provided the rationale for the eugenics movement. The inventors of schizophrenia were involved. Kraepelin stated, ‘Lomer has, it is true, proposed as a heroic prophylactic measure bilateral castration as early as possible, but scarcely anyone will be found who will have the courage to follow him’ ([1913] 1919: 278). Bleuler wrote, ‘Lomer and von Rohe have again recommended castration, which, of course, is of no benefit to the patients themselves. However, it is to be hoped that sterilisation will soon be employed on a larger scale . . . for eugenic reasons’ (Bleuler [1911] 1950: 473).

The eugenics movement, aimed at improving 'race hygiene' by eliminating tainted genes, was an international phenomenon (Strous 2010). The American 'scientific' journals *Eugenics Review* and *Eugenical News* gave the movement credibility for 20 years before the sterilization and murder of mental patients.

In 1920, Alfred Hoche, Professor of Psychiatry at Freiberg, and Rudolf Binding, a law professor, wrote, in *Release and Destruction of Lives Not Worth Living* (Binding and Hoche 1920), about 'those who are not capable of human feeling, those ballast lives, and empty human husks that fill our psychiatric institutions and can have no sense of the value of life. Theirs is not a life worth living; hence their destruction is not only tolerable, but humane' (p. 32).

In 1924, Bleuler recommended:

The more severely burdened should not propagate themselves. . . . If we do nothing but make mental and physical cripples capable of propagating themselves, and the healthy stocks have to limit the number of their children because so much has to be done for the maintenance of others, if natural selection is generally suppressed, then unless we will get new measures our race must rapidly deteriorate.

(1924: 214)

Bleuler's plea for action was answered, in Germany, by the 1933 law allowing compulsory sterilization in cases of 'congenital mental defect, schizophrenia, manic-depressive psychosis, hereditary epilepsy, hereditary chorea, hereditary blindness, heredity deafness, severe physical deformity and severe alcoholism' (Muller-Hill 1988: 30). The primary author of the legislation, working alongside Heinrich Himmler, was Ernst Rudin, Professor of Psychiatry at the Universities of Munich and Basel, Chair of the Association of German Neurologists and Psychiatrists, and President of the international Eugenics Federation. He had been recruited by Kraepelin to develop the new field of psychiatric genetics and succeeded Kraepelin as Director of the Kaiser Wilhelm Institute in Munich. Like Kraepelin, he was funded by the Rockefeller Institute (Pilgrim 2008). Rudin is still regarded as the 'father of psychiatric genetics' (Seeman 2005; Strous 2006).

THE FACTS

By 1939, about 350,000 had been sterilized (Strous 2006). Of the approximately 400,000 sterilized by the end of the war about one third (over 130,000) were 'schizophrenic' (Fuller Torrey 2010). In the 1930s, sterilization laws were passed in Norway, Denmark and Finland. In Sweden, 63,000 people, mostly women displaying 'antisocial behaviour', were sterilized under eugenic legislation, starting in 1934 (Muller-Hill 1988). The first country to translate

genetic theories into eugenic programmes had been the United States (Black 2003). Indiana passed the first compulsory sterilization law, in 1907. Winston Churchill wrote to Prime Minister Asquith urging that Britain emulate Indiana by compulsorily sterilizing the 'Feeble-Minded and Insane classes' (Gilbert 2009). By 1928, 20 other American states had followed Indiana's example (Fuller Torrey 2010). The laws were still valid in 19 states in 1985 (Seeman 2005). By 1932, 10,000 eugenic sterilizations had been perpetrated in California alone, two-thirds of which were cases of 'insanity' (Gosney 1937). In Canada, Alberta (in 1928) and British Columbia (in 1933) passed similar laws, also resulting in thousands of sterilizations (Seeman 2005).

By 1938, in Europe, sterilization began to be replaced with murder. The psychiatrists, and other doctors, started with between 5,000 and 10,000 children with psychological or physical abnormalities, in psychiatric institutions, at first starving them to death and later gassing them (Muller-Hill 1988). Despite the falsified death certificates, many people were aware of what was happening, including the local communities, who could smell the cremations and knew who was in the grey buses arriving at the hospitals (Muller-Hill 1988). The child victims knew. They played games with coffins (Dudley and Gale 2002).

In 1939, a plan to murder all mental patients was put into operation. The programme became known as 'Aktion T-4', based on the address of its headquarters at Tiergartenstrasse 4, Berlin. Those responsible for the plan and for 'selecting' who should die, on the basis of forms submitted by all German psychiatric institutions, included the Chairs of Psychiatry at Cologne and Berlin (Max de Crinis), Königsberg and Münster (Frederich Mauz), Marburg and Breslau (Werner Villinger), Würzburg (Werner Heyde), Düsseldorf (Friedrich Panse) and Bonn (Kurt Polisch), as well as Karl Schneider, Chair of Psychiatry at the University of Heidelberg (the same position held a generation earlier by Emil Kraepelin; Schneider had been Kraepelin's student).

About half of German physicians belonged to the Nazi party (Strous 2006) – seven times the rate for employed males (Seeman 2005) – with psychiatrists being the most involved (Dudley and Gale 2002; Strous 2006). Only a handful of psychiatrists refused to participate in the killings (Strous 2010). Fifty years later a president of the *German Society for Psychiatry and Neurology* acknowledged that 'The majority of psychiatrists involved in the infamous "euthanasia" did nothing to protect their patients, or try to protest, or stop the action. Generally the attitude varied between supporting assent, helpless giving in, and indifferent co-operation' (Meyer-Lindenberg 1991). For a tragic and moving account of one psychiatrist's attempts, after the war, to explain her involvement in mass murder, see Benedict and Chelouche (2008).

A tiny number of psychiatrists deserve to be remembered because they actively opposed the programme – notably John Rittmeister, Karsten Jaspersen and H-G. Creutzfeld (Strous 2010) and, during the planning stages, Karl Jaspers (Seeman 2005). Rittmeister is reported to be the only psychiatrist who lost his life for doing so (Meyer-Lindenberg 1991).

The murders were called 'euthanasia', 'mercy killing' or 'help for the dying'. By September 1941, over 70,000 mental patients had been killed, primarily with carbon monoxide (suggested by Professor Heyde, a psychiatrist), in six specially adapted psychiatric hospitals at Bernberg, Brandenburg, Grafeneck, Hartheim, Sonnenstein and Hadamar. Later, lethal injection became the preferred killing method. Of the 4,817 people arriving at Hadamar between August 1942 and March 1945, 4,422 (92%) died. The total figure for Germany alone has been estimated at about a quarter of a million (Fuller Torrey 2010; Wertham 1966). The number killed elsewhere is unknown. Wertham estimated that of all the patients in German and Austrian mental hospitals in 1939, fewer than 15% remained by 1945. In three months during 1940, over 4,000 were killed in Polish mental hospitals (Muller-Hill 1988). In French mental hospitals, 40,000 were starved to death (Koupernick 2001).

Of the approximately quarter of a million killed in Germany, at least half were diagnosed 'schizophrenic'. It has been estimated that at least 73% of Germans with this diagnosis (approximately 245,000) were either sterilized or murdered (Fuller Torrey 2010). Almost all Jewish people diagnosed with 'schizophrenia' (about 6,000) were murdered (Fuller Torrey 2010).

It is clear that individuals with this diagnosis were sterilized and killed disproportionately compared with individuals with other diagnoses. This was because of the strong belief among German psychiatrists that schizophrenia was genetically inherited.

(Fuller Torrey 2010: 29)

Professor Mauz had argued that there should be no exceptions 'as a matter of principle' for 'schizophrenics' (Muller-Hill 1988).

This annihilation of the majority of the people in one country with this supposedly genetically based 'illness' provided, with tragic irony, strong evidence against the theory used to justify the mass murders. While the prevalence, of course, had been drastically reduced, the incidence (new cases) was unaffected (Fuller Torrey 2010).

Towards the end of 1941, the gas chambers at psychiatric institutions were moved east to Belzec, Majdanek, Auschwitz, Treblinka and Sobibor to kill Jews. The doctors and nurses often accompanied the equipment (Muller-Hill 1988). For example, Dr Irmfried Eberl, a psychiatrist who had headed two hospitals (Bernberg and Brandenburg) where tens of thousands of patients had been murdered, was appointed Commandant of Treblinka (Strous 2009).

Thus the mass murder of mental patients by psychiatrists provided the 'scientific' rationale, the staff and the equipment for the Holocaust. One of the shamefully tiny number of papers on the killings published in psychiatric journals over the subsequent 70 years states: 'These programmes formed the template for the extension into concentration camps and the "Final Solution" which killed six million Jews' (Dudley and Gale 2002: 586).

After the dismantling of the gas chambers in the psychiatric hospitals, the killings continued with drug overdoses, starvation and injections of air into a vein. For a detailed and moving account of what has become known as this 'wild euthanasia' period, at Obrawalde hospital, see Benedict and Chelouche (2008).

PSYCHIATRY'S RESPONSE TO MASS MURDER

Although one psychiatrist, Hilde Wernicke (who supervised the killings at Obrawalde) was convicted and executed (Benedict and Chelouche 2008), almost all the psychiatrists involved in the killing escaped censure or punishment by the Allies (Dudley and Gale 2002; Muller-Hill 1988; Strous 2006; Wertham 1966). Ernst Rudin and Karl Schneider were investigated but were not prosecuted (Pilgrim 2008; Seidelman 1996). Schneider killed himself in 1946 (Seeman 2005).

Classic is the judgment of a Frankfurt court about a psychiatrist who not only killed many patients – adults and children – personally, but also watched their death agonies through the peep window of the gas chambers. 'We deal', said the court, 'with a certain weakness which does not as yet deserve moral condemnation'.

(Wertham 1966: 189)

A 1947 report by a German psychiatrist on 'Selection in Asylums 1939–1945' went unpublished because eminent psychiatrists Karl Jaspers and Kurt Schneider (to be distinguished from the deeply implicated Karl Schneider) were reluctant (Meyer-Lindenberg 1991). Many of the murderers returned to their careers. For example, Professor Heyde, the psychiatrist who had suggested carbon monoxide, practised from 1950 to 1959 in Flensburg (Muller-Hill 1988). After the War, of the first 12 presidents of the *German Society for Psychiatry and Neurology*, three had been organizers of the 'euthanasia' programme (Dudley and Gale 2002).

Psychiatry beyond Germany was equally unable to acknowledge what had happened. The profession cannot claim ignorance. In 1941, the 'euthanasia' programme had been described in *Reader's Digest* (Dudley and Gale 2002). In the same year, at a conference of the *American Psychiatric Association*, Dr Foster Kennedy presented a paper entitled 'The Problem of Social Control of the Congenital Defective: Education, Sterilization, Euthanasia'. He argued, about 5-year-old children, that 'It is a merciful and kindly thing to relieve that defective – often tortured and convulsed, grotesque and absurd, useless and foolish, and entirely undesirable – of the agony of living' (Kennedy 1942: 14). The *American Journal of Psychiatry* published the paper, with an Editorial that pathologized parents' objections to having their children killed (Joseph 2005).

After the War, Dr Werner Villinger, who was deeply implicated in the killing of children, became Chair of Psychiatry at the University of Marburg. He was invited to a White House conference on children.

For decades, histories of psychiatry omitted any mention of the murders (e.g. Alexander and Selesnick 1966; Schneck 1960). Howells's (1975) *World History of Psychiatry* had a chapter for each country, all but one of which was up to date. The chapter on Germany stopped at 1936. More recent histories have devoted less than a page (Freeman 1999; Shorter 1997). Stone's history (1997) cites the genetic work of Rudin, and his student Franz Kallmann (see below), with no mention whatsoever of the political context.

It is well documented (Pilgrim 2008; Seidelman 1996) that most psychiatric textbooks are silent on the issue of eugenics and psychiatry's role therein (e.g. Gelder *et al.* 2009; Kay and Tasman 2006). Some cite the work of Rudin and Kallmann uncritically or positively (e.g. Kirov and Owen 2009: 1463; Stone 2006: 13). *The American Psychiatric Publishing Textbook of Psychiatry* opens its section on the 'Genetics of Psychiatric Disorders', without context or critique, with: 'Beginning with the pioneering work of Rudin, Kallmann, and others in the Berlin school . . . ' (Choudary and Knowles 2008). The history chapter of the *New Oxford Textbook of Psychiatry* avoids the topic altogether (Pichot 2009). The parallel chapter in the *Comprehensive Textbook of Psychiatry* (Colp 2009) deploys a more active form of denial. There appears, in a table of 'Persons and Events in Psychiatry', an entry entitled 'Fuhrer Decree', which, the reader is informed, 'ordered doctors to kill patients' and which 'grew out of the Nazi doctrine of preserving racial purity'. The role of psychiatry is not mentioned. Four entries earlier, under 'Genetics of schizophrenia', the 'pioneering' work of Rudin and Kallman is lauded, with no link to the murders (pp. 4492–3).

A recent paper, 'Political Abuse of Psychiatry: An Historical Overview', covers past abuses in the Soviet Union and current abuses in China but makes no mention of Germany (van Voren 2010). Of the 771 papers in the journal *History of Psychiatry from 1997 to 2012*, only 11 address the murders in Germany (e.g. Benedict and Chelouche 2008; Joseph 2005 Westerman 2012).

In the United States, the 'scientific' journals *Eugenics Quarterly* and *Eugenics Review* had continued, into the 1960s, to provide credibility to the ideas that had led directly to the deaths of over a quarter of a million and indirectly to the deaths of a further six million. From 1947 to 1956, the *American Journal of Psychiatry* published annual updates of 'Psychiatric Progress' on the issue of 'Heredity and Eugenics', all written by Dr Franz Kallmann (e.g. Kallmann 1955).

Kallmann had argued, in Germany in the 1930s, that not only 'schizophrenics' but all their relatives should be sterilized. After the war he promulgated his beliefs about 'mental illness' and homosexuality in the United States, where he helped found the *American Society of Human Genetics*. His grossly flawed reports on the genetics of schizophrenia (Joseph 2004; Pilgrim 2008) still head the list of twin studies in many psychiatric and psychology

textbooks (e.g. Choudary and Knowles 2008; Kirov and Owen 2009). Unable to fulfil his dream of removing schizophrenic genes from the gene pool by sterilization, he advocated genetic counselling (e.g. Kallmann 1955).

This practice, of informing people diagnosed with 'schizophrenia' and their relatives that their offspring have a high probability of inheriting the 'illness', thereby discouraging reproduction, is still with us. Some have even argued that 'Access to genetic counselling should be available to all individuals with schizophrenia and is particularly important for family planning' (Hodgkinson *et al.* 2001: 123). Of 263 people attending a Canadian genetic counselling service in relation to schizophrenia, the majority (69%) were relatives who 'may indeed make childbearing decisions based on their perceptions of this risk' (Hunter *et al.* 2010: 147). In a recent survey US psychiatrists 'expressed a strongly positive view of genetic testing in psychiatry' (Hoop *et al.* 2008: 245). Some argue that psychology is as guilty as psychiatry of promoting simplistic genetic theories and, thereby, of leaving the door open for the continuation of eugenic theories and practices (Joseph 2005; Pilgrim 2008).

One of the most thorough documentations of the horrors perpetrated by psychiatry in Germany concludes: 'We are not dealing here with defects in the character of a few individuals, but rather with defects in psychiatry and anthropology as a whole' (Muller-Hill 1988: 109).

German psychiatry offered conducive conditions. From mid-19th century, a somatic approach dominated, the psychiatrist Griesinger asserting that 'mental disease is brain disease'. Emil Kraepelin's classification reflected therapeutic pessimism: For example, schizophrenia was organic, incurable and deteriorating. . . . Psychotherapy was separated from psychiatry, and regarded as suspect.

(Dudley and Gale 2002: 588–9)

Since the first edition of this book in 2004, a trickle of articles, mostly by psychiatrists, have broken the near-silence of the preceding 50 years (e.g. Benedict and Chelouche 2008; Fuller Torrey 2010; Pieczanski *et al.* 2004; Strous 2006, 2009, 2010; Westerman 2012). One of these asked:

Can this ever happen again? Can this happen today? . . . In one sense we do it now when we impose involuntary hospitalization on those with mental illness: will our motives for doing so be questioned by history? We currently prescribe large amounts of tranquilizing drugs that sometimes inadvertently impair our patients' health. Are we being hoodwinked into acquiescence by a profit-driven pharmaceutical industry. . . . Since Nuremberg, we have developed tight legislation to place constraints on human experimentation, but no such safeguards exist against biases that may influence our clinical decisions.

(Seeman 2005: 219)

The two paragraphs devoted to the murders in *A Century of Psychiatry* concluded: 'The facts are well known, but we still need an answer to the question – what theoretical or other ideas made so many psychiatrists who were not Nazis break their Hippocratic oath?' (Peters 1999: 89). Perhaps we need look no further than the opening words of the same article: 'Nazi psychiatry was not different in all respects from classical psychiatry. A shared belief between them was that endogenous psychoses were somatic, with mainly genetic causes, and they also shared a therapeutic nihilism'.

PSYCHIATRY'S RESPONSE TO HOLOCAUST SURVIVORS

In 1996 Yochevet Mark, a Holocaust survivor who, since making her way to Israel, had been repeatedly hospitalized as 'schizophrenic', was visited by her son in Geha Hospital. She believed he was an SS officer.

Her son was Dr Moti Mark, Israel's chief government psychiatrist. Dr Mark soon discovered that approximately two thirds of elderly patients (65 or over) in Israel's psychiatric hospitals were Holocaust survivors, including at his own Abarbanel, Israel's largest psychiatric facility (Yoram Barak, personal communication, 2011). With other psychiatrists, including Yoram Barak and Henry Szor, Mark campaigned to offer them treatment for Holocaust trauma rather than continue to be drugged for an illness called 'schizophrenia' (Rees 2002).

Following a 1999 Commission of Inquiry, the government closed three psychiatric institutions. Several hundred Holocaust survivors were moved into special facilities where their trauma was addressed. In 2001 Shosh Shlam's film *Last Journey into Silence* documented years of neglect and abuse.

Decades of using antipsychotic drugs like haloperidol and thorazine hadn't worked. In the lobby of the survivors' wards, patients still shake uncontrollably and grind their jaws grotesquely from the side effects of such drugs. Barak changed the diagnosis of schizophrenia attached to most of the 120 survivors in his wards to 'long-term post-traumatic psychosis'. . . . They had been kept heavily drugged and often in solitary confinement for decades. Many had lost the power of speech.

(Rees 2002: 41–2)

In the first edition of this book, we concluded only that this is an example of 'the awesome power of the current biogenetic ideology and classification system to distort reality and to destroy human beings'. In Israel, however, another dynamic was compounding psychiatry's reluctance to address the social causes of human distress.

Psychiatrists like Barak had to fight more than just a bad diagnosis made decades ago. They were up against a Zionist ideology that saw Holocaust victims as weaklings who had gone ‘like sheep to the slaughter’ – unlike the strong ‘new Jew’ Israel’s founders hoped to create. Holocaust survivors were treated with contempt in their new country.

(Rees 2002: 42)

AN ALTERNATIVE RESPONSE

As an antidote to the depressing nature of this chapter we include a moving example of what can happen when human beings do find the strength to break through the denial that we all so often use to shield us from the horrors that humans inflict on one another. In the new hostels to which the Holocaust survivors were moved, a team of psychiatrists (including Dori Laub and Rael Strous), social workers (including Baruch Greenwald and Ashrit Ben-Ari) and others attempted the daunting task of trying to reach the survivors. Beginning, in some instances, with helping them build relationships with animals that reminded them of childhood pets, the team moved onto ‘testimony therapy’ in which the survivors were gently encouraged to tell, for the first time in 50 years, what had happened in the ghettos and camps. Some of this work included filming interviews, not only for archival purposes but also so that staff and survivors could watch them together for therapeutic reasons (Greenwald *et al.* 2006; Laub 2006). Some of the survivors, many of whom had barely spoken for decades, did not recognize themselves when watching their videos for the first time. This testimonial therapy significantly reduced symptoms and improved social functioning (Strous *et al.* 2006).

Their trauma-related illness had been neglected in their decades-long treatment. Most of these patients had been diagnosed as having chronic schizophrenia, with no special attention given to the historical circumstances related to their psychiatric symptoms.

(Greenwald *et al.* 2006: 200)

After viewing Chana’s [diagnosed with ‘paranoid schizophrenia’] testimony in its entirety, the staff members can now more easily recognize the connection between Chana’s traumatic childhood, her suspicious behavior, lack of trust, self-neglect, and inability to make independent decisions.

(p. 213)

It was a communal bearing of a destiny that affected some – the patients – far more harshly than it had affected the others, the treaters.

Yet, as we discovered, everyone was affected, and the project provided a medium through which to express it, to experience it, and to begin to share it. . . . The treatment staff – after having lifted their own inner obstacles to speaking about the Holocaust – allowed for a homecoming for the patients.

(p. 264)

CONCLUSIONS

The sensitivity and courage required of psychiatrists and other staff when faced with trauma is not dependent on nationality. Denial is not limited to any one country or any one historical period. It is incumbent on all mental health workers to be constantly on the lookout for their own failures to perceive the myriad ways in which humans are harmed by other humans, including – perhaps the hardest of all to acknowledge – by mental health workers themselves.

Two comments, one on what happened during the Holocaust and one on what happened in Israel, are telling:

To remember the past requires an active effort and remembering is a prerequisite of mourning. All psychiatrists and every student of psychiatry should make this effort. . . . The 'scientific' psychiatrist does not console those in despair, he calls them depressed. He does not unravel the tangled thought-processes of the confused, he calls them schizophrenic. . . . This attitude reduces the person to a subservient depersonalised object. Such a process formed the bond which held the psychiatrists, anthropologists and Hitler together.

(Muller-Hill 1988: 110)

As a result of these meetings, the staff felt enriched by learning about and vicariously experiencing the patient's life experiences. Consequently, a new and deeper bond was created between the staff and the patients, based on a mutual understanding of the tragic events that played such a major role in the patient's life and pathology.

(Greenwald *et al.* 2006: 204)

In 2001 Israel's Health Minister, Nissam Dahan, met with some of the survivors 'to apologise that we did not treat you in the past as we should have' (Rees 2002: 41).

On 26 November 2010, 65 years after the killings ended, Professor Frank Schneider, President of the German Association for Psychiatry and Psychotherapy, addressed a commemorative event in Berlin. He began thus:

Ladies and gentlemen. Under National Socialism, psychiatrists showed contempt towards the patients in their care; they lied to them, and deceived them and their families. They forced them to be sterilised, arranged their deaths and even performed killings themselves. Patients were used as test subjects for unjustifiable research – research that left them traumatized or even dead. . . . For too long now we have been hiding, denying a crucial part of our past. For that, we are truly ashamed.

(Schneider 2011: 111)

Following a detailed documentation of the facts, and of psychiatry's subsequent denial and suppression of those facts, Professor Schneider added:

In the name of the German Association for Psychiatry and Psychotherapy, I ask you, the victims and relatives of the victims, for forgiveness for the pain and injustice you suffered in the name of German psychiatry and at the hands of German psychiatrists under National Socialism, and for the silence, trivialization and denial that for far too long characterised psychiatry in post-war Germany.

(p. 118)

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DOES 'SCHIZOPHRENIA' EXIST?

Reliability and validity

John Read

To say that 'schizophrenia' does not exist is not to say some people don't have experiences that are very unusual or difficult to understand, and who suffer extreme distress and confusion. The two questions addressed here are different: (a) Does 'schizophrenia' meet accepted scientific requirements for establishing that a concept exists? ('reliability'); and (b) Is 'schizophrenia' useful in terms of understanding or predicting anything? ('validity').

RELIABILITY: CAN WE AGREE WHO IS 'SCHIZOPHRENIC'?

The first fifty years

Ninety years ago, the textbook *Mental Diseases* stated: 'They [psychiatric diagnoses] do not contribute anything of value whatever to our knowledge of symptomatology, diagnosis or treatment' (May 1922: 246). By 1938, Boisen, finding huge variations in the application of the 'schizophrenia' diagnosis, concluded: 'It is clear that the Kraepelinian system is inadequate'. Nevertheless, it was accurately predicted that: 'There always will be many who will use the existing system of classification irrespective of whether or not it has any meaning, and even those who decry the orthodox classification will invoke it as an atheist when off guard invokes God' (Jellinek 1939: 161). By 1949, it was concluded that 'The psychiatric taxonomy which psychologists have been constrained to adopt is so inadequate, even for psychiatry, that no patching can fix it up' (Roe 1949: 38). In the same year clinicians were found to agree with others' diagnoses in only 33–50% of cases (Ash 1949). A large test–retest reliability study found that consistency for 'schizophrenia' was only 37% (Hunt *et al.* 1953).

A 'frontal attack' in search of schizophrenia

Researchers were urged to unleash 'a frontal attack' to discover why, 50 years after the invention of schizophrenia, experts couldn't agree who had it (Kreitman 1961). By employing artificial research conditions designed to maximize reliability, a group of diagnostic experts (Beck *et al.* 1962) increased agreement between pairs of clinicians, for schizophrenia, to 53%; or 42%, allowing for agreement by chance (Read 2004).

In 1974 prominent researchers announced that the average reliability was still only 0.57, that there 'appears to have been no essential change in diagnostic reliability over time', that 'there are no diagnostic categories for which reliability is uniformly high' and that reliability in 'routine clinical settings is even poorer' than in research studies (Spitzer and Fleiss 1974: 344). When 134 American and 194 British psychiatrists were given a description of a patient, 69% of the Americans gave a diagnosis of 'schizophrenia'. Only 2% of the British did so (Copeland *et al.* 1971).

The absurdity of all this was illustrated by the famous 'On Being Sane in Insane Places' study. Eight 'normal' people were admitted to psychiatric hospitals after saying they heard the words 'hollow', 'empty' and 'thud'. Seven were diagnosed 'schizophrenic'. Many of the patients, but none of the staff, recognized that the pseudo-patients were 'normal'. A follow-up study, in which the psychiatrists were told that pseudo-patients would be admitted, produced a 21% detection rate by staff. No 'pseudo-patients' had, however, been admitted (Rosenhan 1975). (The first part of the study was partially replicated 30 years later – Slater 2004.)

Redefining 'schizophrenia'

The 'frontal attack' had failed. However, another approach was available: redefinition. Following Kraepelin's repeated redefinitions of his own invention, Bleuler had further altered the definition (see Chapter 3). These redefinitions continued throughout the twentieth century, most famously based on Schneider's 'first-rank symptoms' such as hallucinations and delusions (Bentall 2009). The redefinitions were reflected in changes in the numerous official versions of 'schizophrenia', including the various editions of the *Diagnostic and Statistical Manual of Mental Disorders (DSM)*. The 'neo-Kraepelinians' (Bentall 2009) responsible for the third edition, in 1980, had overseen a massive research project to try to develop new definitions, and checklists of symptoms, which might finally produce adequate reliability. By 1987, however, the American Psychiatric Association (APA) admitted:

Despite extensive field testing of the *DSM-III* diagnostic criteria before their official adoption, experience with them since their publication had revealed, as expected, many instances in which the criteria

were not entirely clear, were inconsistent across categories, or were even contradictory.

(APA 1987: xvii)

This admission of the *DSM-III*'s failures was not the introduction to a paper arguing that psychiatry should finally recognize the futility of further redefinitions and abandon terms like 'schizophrenia'. It was the introduction to a 560-page book entitled *Diagnostic and Statistical Manual of Mental Disorders (3rd edition, Revised) (DSM-III-R)*.

A disjunctive category

As early as 1968, British psychologist Don Bannister had concluded that:

Schizophrenia as a concept, is a semantic Titanic, doomed before it sails, a concept so diffuse as to be unusable in a scientific context.

(Bannister 1968: 181)

We diagnose one person as schizophrenic because he manifests characteristics A and B and diagnose a second person as schizophrenic because he manifests characteristics C, D and E. The two people are now firmly grouped in the same category while not specifically possessing any common characteristic. . . . Disjunctive categories are logically too primitive for scientific use.

(pp. 181–2)

The category remains disjunctive today. To receive a diagnosis of schizophrenia using the *DSM-IV* (APA 1994), or the revised version thereof (APA 2000: 312), you need two out of five 'characteristic symptoms': hallucinations, delusions, disorganized speech, grossly disorganized or catatonic behaviour, and negative symptoms. There are 15 ways (1 + 2 vs. 3 + 4; 1 + 2 vs. 3 + 5, etc.) in which two people can meet the criteria without having any symptoms in common.

Furthermore, only one 'characteristic symptoms' is required if (a) a voice is 'keeping up a running commentary on the person's thoughts or behavior', (b) there are 'two or more voices conversing with each other', or (c) the 'delusions are bizarre'. Studies have found, unsurprisingly, that 'a reliable concept of bizarre delusion has yet to be achieved' (Bell *et al.* 2006), and that reliability is particularly poor in studies that most closely resemble real-life clinical situations (Cermolacce *et al.* 2010). Nevertheless, about 6% of 'schizophrenia' diagnoses are made solely on the basis of this symptom (Cermolacce *et al.* 2010).

To its credit, the APA has finally proposed, for *DSM-5* (due out in 2013), 'elimination of requirement that only one characteristic symptom be present'

because ‘no unique diagnostic specificity for these characteristic symptoms in comparison to others has been identified’ (APA 2011). The APA also proposes that at least one of the two required symptoms be hallucinations, delusions or disorganized speech. This would reduce the number of ways that two people can receive the same diagnosis without having any symptoms in common from 15 to 12. The construct will remain disjunctive and, therefore, scientifically meaningless.

The APA (2011) has recommended that a new diagnosis of ‘Psychosis Risk Syndrome’ be added in *DSM-5*. The diagnosis would have led to a further increase in the stigmatization and medicating of troubled youth who are not psychotic. This is already increasingly happening in the name of treating something called ‘prodromal’ psychosis in adolescents (Morrison *et al.* 2010), despite the inability of researchers to find a definition of ‘prodromal’ that adequately predicts psychosis (Larkin and Marshall 2010; Ross 2010). Fortunately, the proposal – which was described by Allen Frances, who chaired the production of *DSM-IV*, as the ‘single worst *DSM-5* proposal’ (Maxmen 2012) – was abandoned.

An alternative diagnostic system, the *ICD-10*, published by the World Health Organization (WHO 1992) and used by many European and developing countries, is even more problematic than the *DSM*. It doesn’t include the *DSM* requirement that ‘continuous signs of the disturbance persist for at least 6 months’, requiring only one month. It includes the subtype ‘Simple Schizophrenia’, which requires no symptoms of psychosis (requiring only ‘negative symptoms’). It is even more disjunctive than *DSM*, because it has a longer list of individual symptoms that, by themselves, warrant the diagnosis. *ICD-10* does not even include the *DSM* requirement that there be ‘Social/occupational dysfunction’, thereby making it possible to diagnose people whose ‘symptoms’ are not affecting their lives.

Bannister (1968) had recommended that ‘research into schizophrenia as such, should not be undertaken’. This recommendation has been made repeatedly (e.g. Bentall *et al.* 1988; Boyle 1990; Ross and Pam 1995). In 1992, after a lifetime studying diagnoses, British psychiatrist Ian Brockington concluded:

It is important to loosen the grip which the concept of ‘schizophrenia’ has on the minds of psychiatrists. Schizophrenia is an idea whose very essence is equivocal, a nosological category without natural boundaries, a barren hypothesis. Such a blurred concept is not a valid object of scientific enquiry.

(Brockington 1992: 207)

In 1996, US psychiatrist Howard James stated, again, that schizophrenia is ‘an unscientific and unprovable nosological construct’ (James 1996: 148).

In the same year that Brockington (1992) decried ‘a babble of precise but different formulations of the same concept’, researchers identified 16 systems of

classifying 'schizophrenia'. Of 248 patients, the number diagnosed as schizophrenic by these systems ranged from 1 to 203. This study was 'designed to maximise reliability and agreement' (Herron *et al.* 1992). Furthermore, within each of these 16 systems there are different ways of gathering and interpreting information. One study used four different recognized methods of assigning patients to eight *DSM-IV* diagnoses indicative of psychosis (including 'schizophrenia'). The agreement between pairs of systems ranged from 53% to 65% (McGorry *et al.* 1995). Another study found no statistically significant relationship between clinical diagnoses of schizophrenia and diagnoses based on *either* of two diagnostic processes used by researchers (Whaley 2001). In 2008 a renowned schizophrenia expert, British psychiatrist Robin Murray, acknowledged that:

Some psychiatrists react to such criticisms by totally denying their validity and by reasserting a simplistic medical disease model of schizophrenia. However, this is less than satisfactory because there remains widespread confusion regarding the meaning, boundaries and even value of the term schizophrenia.

(Murray and Dean 2008: 285)

In 2012, Dr Fibiger of the Psychiatry Department at the University of British Columbia, and ex-Vice President of Neuroscience at drug company Eli Lilly, acknowledged that 'Today, few would argue that syndromes such as schizophrenia and depression are single, homogeneous diseases', joined the chorus arguing for a focus on specific symptoms and predicted that 'concepts such as schizophrenia will surely be discarded and future generations will look back and might rightfully ask "What were they thinking?"' (Fibiger 2012: 50).

VALIDITY: IS 'SCHIZOPHRENIA' WHAT IT IS SUPPOSED TO BE?

As Brockington, Bannister, Boyle, Bentall, and others have been explaining for decades, without reliability investigating validity is meaningless. If researchers can't agree on who has 'schizophrenia', then the supposed properties of 'schizophrenia' cannot be evaluated. The people one researcher studies differ from those studied by other researchers.

If reliability *had* been established, we would investigate whether the construct was related to things that the theory underpinning the construct claims it is related to. In the case of the illness theory of schizophrenia, one would search – as Kraepelin and Bleuler had done, unsuccessfully – for: (a) a set of symptoms that occur together but not in other 'mental illnesses', (b) a predictable outcome, (c) a biological cause and (d) responsiveness to medical treatments.

Prominent diagnostic researchers Zigler and Phillips (1961) have explained that even when reliability *is* demonstrated, that does not guarantee validity.

Class membership conveys little information beyond the gross symptomatology of the patient and contributes little to the solution of the pressing problems of aetiology, treatment procedures, prognosis, etc.
(p. 612)

By 1988 a review of the research in each of the four validity areas (symptoms, aetiology, prognosis and treatment) concluded that “‘schizophrenia’ is not a useful scientific category and that for all these years researchers have been pursuing a ghost within the body of psychiatry’ (Bentall *et al.* 1988: 318).

In later chapters we analyse the attempts to discover a biological cause (Chapters 6 and 7) and effective medical treatments (Chapters 8 and 9). First, we consider the two other areas of validity: symptom specificity and outcome consistency.

Symptoms

There is no evidence that there is a set of ‘schizophrenic’ behaviours and experiences that occur together but do not occur in other psychiatric conditions. In 1952, Guertin used factor analysis to answer the question, ‘Is there such a thing as schizophrenia?’ He found no suggestion of a group factor that could be called ‘schizophrenia’. In 1973, the World Health Organization compared clusters of symptoms that occur together in real people with the groupings produced by diagnostic categories. It was found that ‘The clusters are defining different and more homogeneous groups than are the clinical diagnoses’ and ‘Patients diagnosed as schizophrenic are distributed in all clusters. No single “schizophrenic profile” was elicited’ (WHO 1973: 350).

‘Schizophrenic’ symptoms are frequently found in many other disorders. Most people diagnosed schizophrenic have sufficient symptoms of other disorders to earn additional diagnoses. This ‘co-morbidity’ has been found in relation to depression, obsessive-compulsive disorder, panic disorder, dissociative disorders, bipolar disorder, personality disorders, substance abuse, post-traumatic stress disorder and anxiety disorders (Craig and Hwang 2000; Crow 2010; Torgalsboen 1999). In one study people with dissociative identity disorder had more schizophrenic symptoms than people diagnosed schizophrenic (Ellason and Ross 1995).

Prognosis

There is no evidence that people receiving the diagnosis share a common prognosis (outcome). Eugen Bleuler ([1911] 1950) had acknowledged that it is impossible to describe all the variations that the course of schizophrenia may take. By the end of the century, a review concluded that ‘heterogeneity in outcome across individuals with schizophrenia remains the rule’ (Davidson and McGlashan 1997: 34).

Three findings characterize long-term outcome studies: (a) there are massive variations in outcomes, (b) many people labelled 'schizophrenic' recover, (c) the best predictors are psychosocial factors.

Estimates of long-term recovery rates vary, partly depending on definition, from 13% (Stephens *et al.* 1997) to 72% (Thara and Eaton 1996), with most falling between 30 and 55% (e.g. Harrison *et al.* 2001; Jobe and Harrow 2010; McGlashan 1988). From a 20-year study of hospitalized 'schizophrenics' at the Burghölzli Clinic (where his father Eugen had worked), Manfred Bleuler ([1972] 1978) found the following stable outcomes: severe, 15%; moderate, 17%; mild, 38%; recovery, 30%. Swiss psychiatrist Luc Ciompi (1980) found similar variability in a 37 year study: severe, 20%; intermediate, 26%; minor, 22%; recovery, 30%.

In Vermont, Harding and colleagues (1987) studied 'the sickest group in the hospital'. They had been hospitalized for ten years at the outset of the study. In the 12 months preceding the follow-up assessment, an average of 32 years later, 82% had not been in hospital, 40% were employed (despite two-thirds being 55 or older) and 68% had few or no symptoms. In 2001, a WHO review of 18 cities reported an average recovery rate, after 15 or 25 years, of 48% (Harrison *et al.* 2001). Ciompi concludes:

There is no such thing as a specific course for schizophrenia. Doubtless, the potential for improvement of schizophrenia has for a long time been grossly under-estimated. What is called 'the course of schizophrenia' more closely resembles a life process open to a great variety of influences of all kinds than an illness with a given course.

(Ciompi 1980: 420)

The myth of a consistent negative outcome was, after a century of pessimism that had damaged the lives of millions, eventually challenged. A 2001 review of the WHO International Study of Schizophrenia (ISoS) concluded:

Because expectations can be so powerful a factor in recovery, patients, families and clinicians need to hear this. . . . The ISoS joins others in relieving patients, carers and clinicians of the chronicity paradigm which dominated thinking throughout much of the 20th century.

(Harrison *et al.* 2001: 513)

Contrary to past and contemporary claims that 'schizophrenia' is a degenerative brain disease, a recent review concluded that 'there is overwhelming evidence that very few patients with schizophrenia show a progressive downhill course' (Jobe and Harrow 2010: 223). Leading researchers, however, recently bemoaned the fact that research in this field has had limited impact on either the pessimistic dominant paradigm or actual practice in the real world. Citing 11 studies that found that '48% to 68% of persons suffering

from schizophrenia can achieve significant improvement and often even full recovery', they point to the often made, but 'absurd' assumption, beginning in Kraepelin's time (see Chapter 3), that if someone recovers, they couldn't have had 'schizophrenia' in the first place (Ciompi *et al.* 2010).

Thus the diagnosis 'schizophrenia' has very little predictive validity. Furthermore, the huge variability in outcome between people with the same diagnosis is best predicted by psychosocial factors. Predictors of poor outcome include: poor social or cognitive skills, poor academic or work performance; high involvement with over-involved, critical families; being unmarried (especially in men); adverse life events; high anxiety; external locus of control beliefs and substance abuse (Harrison *et al.* 2001; Jobe and Harrow 2010; McGlashan 1988); and accepting a definition of oneself as 'mentally ill' (Yanos *et al.* 2010). British psychiatrist Richard Warner (2004) has demonstrated that people diagnosed 'schizophrenic' recover less well in periods of economic recession. Being male is a particularly powerful predictor of poor long-term outcome (Moller *et al.* 2010), but even this finding shows some variation between countries (Cohen *et al.* 2008).

There is one more example of both inconsistent outcome and psychosocial factors predicting outcome. It has been repeatedly demonstrated that outcomes in poorer countries (e.g. Nigeria and India) are far superior to those in wealthy industrialized societies (Harrison *et al.* 2001; Hopper *et al.* 2007).

Progress into a theoretical vacuum

The repeated redefinitions discussed earlier were not based on a solid theoretical framework but, rather, on the hope of increasing the chances that experts can agree about what they are looking at. One consequence of atheoretical redefinitions is to search for any behaviour about which clinicians can agree, regardless of whether it has anything to do with the original theory, and call it 'schizophrenic'. An example of this is the obsessive quest to find differences – any differences – between 'schizophrenics' and 'normals'. Here are some examples (reminiscent of Kraepelin's wide-ranging list of symptoms 100 years earlier) of the endless list of theoretically incoherent variables that have been researched as a result of psychiatry's reliability-at-any-price and any-variable-in-a-storm approach.

- reaction to 'body products' (Dimond and Hirt 1974);
- inadequate 'physical attractiveness' (Archer and Cash 1985);
- amount of DMPEA, a substance related to tannin, in the urine (Proctor *et al.* 1985) [*caused by the amount of tea drunk in psychiatric hospitals*];
- signs of homosexuality (Chaudhury and Jyothi 1996);
- handwriting (Agardi *et al.* 1997);
- inability to smell accurately (Striebel *et al.* 1999) [*an effect of antipsychotics*];

- inappropriate masturbation (Brooks and Waikar 2000);
- ear shape, position of eyes, size of gap between toes, finger curvature, snout reflex, upper lip width (Lawrie *et al.* 2001);
- absence of rheumatoid arthritis (Torrey and Yolken 2001);
- proximity to cats during childhood (Torrey and Yolken 2001);
- head circumference (too big or small), soft ears, third toe longer than second (Schiffman *et al.* 2002);
- amount of rainfall three months prior to birth (Messias *et al.* 2006);
- refusal to agree with psychiatrists that you have an illness (Arango and Amador 2011; Bleuler [1911] 1950 – see Chapter 2).

By 1973, the World Health Organization admitted that 'Kraepelin's attempt to create a concept of dementia praecox was unsuccessful as all later attempts in this direction have been', but offered an argument for carrying on regardless:

The word schizophrenia has come into such widespread use that it is necessary to have a practical definition of it in order to keep public discussion of schizophrenia within reasonable limits . . . for the benefit of non-professional contemporaries who enjoy talking about schizophrenia without knowing what it is.

(WHO 1973: 17)

Imagine a 'schizophrenic' claiming that the reason she keeps talking about her delusion is not because she still believes in it but because everyone else now believes in it, and she wants to make sure they get the facts right.

Diagnosis for social control

Sixty years ago, Zigler and Phillips, having acknowledged that diagnoses like schizophrenia contribute nothing to understanding causes or treatments, pointed out:

The present diagnostic system is quite useful when evaluated in terms of its administrative implications. . . . Examples would include legal determination of insanity, declaration of incompetence, type of ward required for custodial care, census figures and statistical data upon which considerable planning is based, screening devices for the military services, etc.

(Zigler and Phillips 1961: 615)

They seemed unaware of the implications of the fact that the only fields in which diagnoses have any validity are not medical but sociopolitical. Had they been writing today, they might have included health insurance – a powerful force perpetuating the use of unscientific diagnoses.

ALTERNATIVES

The need to stop using this unscientific, uninformative and unhelpful construct is being recognized, and acted on. In 2012, following the lead of other international organizations, members of the International Society for the Psychological Treatments of Schizophrenia (ISPS) voted overwhelmingly to remove 'Schizophrenia' from its name (Martindale 2012).

There *are* scientific and productive ways to understand 'madness'. They form the theoretical underpinning for much of what follows in this book and, hopefully, for clinical practice and research in the future.

Reliable constructs

Categorizing is not the problem. Categorizing is an inevitable human propensity, wired into our brains, without which we couldn't survive. The task is to find categories about which we can agree (reliable) and which are meaningful and useful (valid). This is not difficult.

The heterogeneity of the schizophrenia construct has been acknowledged for over a hundred years. Bleuler highlighted it by entitling his famous 1911 monograph *Dementia Praecox or the Group of Schizophrenias*. There have been many attempts to break the construct into smaller, more reliable, pieces. The latest official version, in *DSM-IV*, is: 'Paranoid', 'Disorganized', 'Catatonic', 'Undifferentiated' and 'Residual'. The APA has, thankfully, proposed that all these be removed from *DSM-5* (APA 2011).

The best way to break the construct into subtypes is to research which symptoms actually occur together. The first such evidence-based division was into 'positive' symptoms (additions to 'normal' experience, such as hallucinations and delusions) and 'negative' symptoms (deficits, such as blunted affect and social withdrawal) (Andreasen and Olson 1982). One example of the gains made when reliable constructs are used is the finding that people without negative symptoms are particularly likely to recover (Strauss *et al.* 2010). The positive symptoms were soon found to be reliably separable into 'psychotic' (delusions and hallucinations) and 'disorganization' (thought disorder, inappropriate affect and bizarre behaviour), producing, with negative symptoms, a three-factor model (Liddle 1987). Others have found four or five factors (Davidson and McGlashan 1997; Demjaha *et al.* 2009; Murphy *et al.* 2010; van Os and Kapur 2009).

Beyond these valuable efforts to discover reliable *groupings* of behaviours and experiences, there is an increasing focus on specific behaviours and experiences. We will see later (Chapters 16 and 22) that once we forget about the non-existent ghost of 'schizophrenia' and focus instead on definable and measurable constructs like hallucinations and delusions, significant progress is made in both understanding and helping. Constructs like hallucinations and delusions can themselves be broken down into clinically valuable and reliable

dimensional variables such as duration, intensity, frequency, conviction, disruption and distress.

Dimensions

Besides using only reliable variables, a second important principle is thinking in terms of dimensions rather than categories (McGorry 1998). 'Normal' human behaviours, thoughts and feelings vary over time and between differing social contexts. So do those considered 'abnormal' (Pilgrim 2000). The *DSM-IV* devotes just three of its 886 pages to its only dimensional measure (pp. 32–4). The Global Assessment of Functioning Scale (GAF) records clinicians' judgment of overall functioning from 0 to 100. Its inclusion was an acknowledgement that mental health is a matter of degree. Dimensional measures of mental health are more reliable than diagnoses. When asked to assess 258 patients in terms of *degree* of psychopathology, clinicians reported high agreement (Rosenthal *et al.* 1975). It was concluded that independent judges may not agree regarding what a person has, but they show remarkable agreement in deciding how severe it is. The *DSM-IV*'s GAF (and its two trial dimensions measuring relational and social/occupational functioning – pp. 758–60) have 'very high levels of inter-rater reliability' (Hilsenroth *et al.* 2000).

This dimensional approach is also helpful in showing that many of the 'normal' population have psychotic or 'psychotic-like' experiences (Johns and van Os 2001), and in researching links between psychosis and risk factors (Demjaha *et al.* 2009). Members of the public who believe there is a continuum between 'schizophrenia' and 'normality' hold fewer stereotypes about people with the diagnosis (Wiesjahn *et al.* 2013). A review of studies comparing dimensional and categorical representations of behaviours considered indicative of schizophrenia concluded that the dimensional approach appeared to be more useful in terms of yielding information on patients' needs and outcome (van Os and Verdoux 2003). It was encouraging to see that, largely because of the efforts of Dutch psychiatrist Jim van Os, dimensional measures of specific psychotic symptoms were considered by APA for inclusion in *DSM-5*. Sadly, they were not accepted into the main text and are due to be published only as an appendix, meaning that psychiatrists and other professionals are likely to continue to act as if 'schizophrenia' and 'psychosis' are categorical constructs that some of us have and some of us do not.

Combining the specific/measurable and the dimensional approaches produces descriptions of us as real people engaging in certain behaviours, thoughts or feelings to various extents at various times, depending largely on what is happening around us.

Focusing on needs and recovery

Questions like 'What would recovery mean to you?' and 'What do you need?' should demand at least as much attention as 'What is wrong with you?' and

'What should we call that?' Quality of Life measures are increasingly in use, covering domains such as self-esteem, family relationships, friendships, housing, income, autonomy, empowerment, physical health, emotional life and so on. Several tools with good reliability have been developed specifically for people diagnosed 'schizophrenic' (e.g. Boyer *et al.* 2010), although why people with this diagnosis should have different life needs from everyone else is not readily apparent. When the people with the diagnosis are themselves asked about recovery goals, for example, they go far beyond symptom reduction (Pitt *et al.* 2007), focusing more on the quality of their social lives (Byrne *et al.* 2010). A questionnaire recently developed in collaboration with service users has both 'intrapersonal' and 'interpersonal' items. Unlike the stigmatizing and unscientific construct 'schizophrenia', this measure has good reliability and validity (Neil *et al.* 2009) and is useful.

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BIOLOGICAL PSYCHIATRY'S LOST CAUSE

The 'schizophrenic' brain

John Read

We saw, in Chapter 5, that there is no relationship between 'schizophrenia' and symptomatology or outcome, two of the four variables to which 'schizophrenia' must be shown to be related if it is a scientifically valid construct. Here, and in Chapter 7, we consider the evidence pertaining to the third variable, aetiology, before discussing the fourth, treatment (Chapters 8 and 9).

The inventors of schizophrenia admitted that they could find no evidence of a biological cause (Chapter 3). For the past few decades three 'facts' have been repeatedly presented in textbooks, on websites and in 'educational' pamphlets funded by drug companies, to show that 'schizophrenia' is a biologically based illness:

- 1 'Schizophrenia' occurs with equally frequency in all countries.
- 2 The brains of 'schizophrenics' are abnormal.
- 3 There is a genetic predisposition to 'schizophrenia' (Chapter 7).

THE UNIFORM FREQUENCY MYTH

The first of the three 'proofs' that schizophrenia is a medical illness is that it occurs in the same proportion (about 1%) everywhere in the world. This is supposed to prove that environmental factors are irrelevant, and the cause must therefore be biological. However, many real medical illnesses, such as diabetes and multiple sclerosis, show huge variations between populations and locations, prompting Robin Murray to emphasize that if uniform prevalence *had* been found, 'such an occurrence would have made schizophrenia unique among diseases!' (Murray and Dean 2008: 292).

Furthermore the 'facts' on which this silly idea was based are untrue. Neither studies of prevalence (the number *with* an illness at a given point in time), nor studies of incidence (the number who *develop* an illness over a certain period) support the notion.

By 1987, a review of over 70 prevalence studies had reported that the highest rate (1.7%, in part of Sweden) was 55 times greater than the lowest rate (0.03%, among Amish people in the United States) (Torrey 1987). (There were no cases of 'schizophrenia' among 3,000 Kwaio adults in Melanesia.) A 2002 review found a 13-fold difference between highest (Puerto Rico) and lowest (Hong Kong) lifetime prevalence rates, and a 55-fold difference between highest (United States) and lowest (Canada) one-year incidence rates (Goldner *et al.* 2002). Saha *et al.* (2005), reporting on 132 prevalence estimates, found, even after excluding the highest and lowest 10%, that the highest estimates were (depending on type of prevalence estimate) between 5.3 and 6.3 times greater than the lowest.

It must also be remembered that 'Incidence rates of schizophrenia vary widely within the same country, between urban and rural populations and between sociocultural majority and minority populations' (van Os and McGuffin 2003: 291).

A scan of current psychiatric textbooks reveals that the uniform prevalence myth, promoted by most of them at the end of the twentieth century, has finally been abandoned.

A DISEASE OF THE BRAIN?

A more sensible way to prove that 'schizophrenia' is a medical illness is to show that 'schizophrenic' and 'normal' brains differ. However, there are three important issues that are frequently ignored when drawing conclusions about causation from brain research. The first is the lack of reliability of the construct under investigation (Chapter 5). The people studied by one researcher may have little in common with those being studied by another researcher.

Second, if 'schizophrenic' brains are different, does that mean we have found the cause of the 'symptoms'? When we are grieving the loss of someone we loved, our brains act differently from usual. Is our sadness caused by the brain's slower functioning or by our loss? This logical flaw has been raised repeatedly. For example:

Biological psychiatry looks for pathological changes, and most authors uncritically assume that such changes as they find are necessarily primary. There is no reason to believe this. . . . For the brain, since mental function has a physical substrate, the initiating event may well be the subject's own mental state.

(Mclaren 2000: 131)

The initiating event may also be external. Most studies of the brains of 'schizophrenics' completely ignore external events. It is as if psychiatric brain researchers fail to grasp what the brain is for. They ignore the fact that the brain is designed to respond to the environment. What use would a brain be that did not do so?

The third issue ignored in this quest for brain differences is the fact that one of the external initiating events that can change the brains of 'schizophrenics' is antipsychotic medication.

Biochemistry

The term 'chemical imbalance' is nowadays promoted as an explanation for a whole range of mental health problems (Read and Sanders 2010). Depression, for example, is supposedly caused by insufficient serotonin. Biological psychiatry's lead theory of schizophrenia, for the past 40 years, is that it is caused by too much dopamine.

Dopamine and serotonin are neurotransmitters. Neurotransmitters are chemicals that are released at the synapse, the point where two nerve cells meet, transmitting an impulse from one cell to receptors in the next cell. The receptors are sensitive to specific neurotransmitters. The dopamine theory of schizophrenia claims that in the 'schizophrenic' brain, those groups of nerve cells that communicate using dopamine are over-active. The theory was not developed on the basis of any direct evidence of over-activity. One source of *indirect* evidence is the effect on the brain of the drugs used to treat schizophrenia. These drugs, originally used for pre-surgery sedation, were first used on 'schizophrenics' in the 1950s. It was only later that it was discovered that one of their many effects was to block the dopamine receptor known as D₂. Psychiatry jumped to the conclusion that if these drugs cure 'schizophrenia', and they also block the dopamine system, then the cause of 'schizophrenia' must be over-activity of the dopamine system. This is as logical as proposing that headaches are caused by a lack of aspirin in the body (Jackson 1986).

Later, some postmortems on 'schizophrenics' did report signs of increased dopamine concentrations. By then, however, it had been discovered that the drugs not only cause a blockade of the dopamine system, but also initiate an attempt to compensate for the blockade. This was explained by Solomon Snyder, Professor of Psychiatry and Pharmacology at Johns Hopkins University:

Something within the neurons recognises this sudden absence of neurotransmitter molecules at their appropriate receptor site and one way or another transmits a message back to the dopamine neurons saying something like the following: 'We don't have enough dopamine. Please send us some more!' Where upon the dopamine neuron in question proceeds to fire at a more rapid rate.

(Snyder 1974: 241)

So, over-activity in the dopamine system can be caused by the drugs that are supposedly treating the illness, which is supposed to be caused by over-activity in the dopamine system! It is essential, therefore, that studies investigating whether 'schizophrenia' is caused by an over-active dopamine system use participants who have not received antipsychotic drugs. A 1982 review pointed out that most of the 'schizophrenics' in the post-mortem studies that had found elevated dopamine concentrations had received antipsychotics shortly before death, and that in drug-free 'schizophrenics' dopamine concentrations were normal. It was concluded that 'these findings do not support the presence of elevated dopamine turnover in the brains of schizophrenics' (Haracz 1982: 440). A subsequent review confirmed that 'no consistent differences between drug-free schizophrenics and normals have been found in terms of dopamine levels' (Jackson 1986: 129).

An alternative approach is to measure the dopamine metabolite, HVA, in cerebrospinal fluid (CSF). This is an indirect and unreliable approach but allows the use of live subjects, thereby making it possible to withhold antipsychotics. Again, no differences were found (Haracz 1982). The latest edition of the popular *Comprehensive Textbook of Psychiatry* admits that 'The measurement of dopamine and metabolites in CSF as a reflection of neurotransmitter activity in the brain has not contributed to the understanding of schizophrenia pathophysiology' (Saddock *et al.* 2009).

Another approach focuses on dopamine *receptors*. Heightened sensitivity, or increased numbers, of receptors would support the theory of an over-active dopamine system. By 1982, one laboratory had found no differences, but four had (Haracz 1982). However, it had just been reported, in the prestigious journal *Nature*, that antipsychotic drugs cause super-sensitivity in dopamine receptors (Murugaiah *et al.* 1982). Haracz acknowledged that the four studies in question included only a small number of patients who were drug-free for one month or more before death. Mackay *et al.* (1982) found no increased receptor sensitivity in those individuals who had not received antipsychotic drugs for a month before death. They also found that sensitivity was reduced to normal by washing out any remaining drug from the receptors. They concluded that 'the results suggest that these increases may be entirely explained by long-term neuroleptic medication' (p. 995). The previously mentioned textbook now acknowledges that: 'Increases in D₂ receptors is a consistent finding . . . but these findings have been largely attributed to a medication effect' (Saddock *et al.* 2009: 1491).

The same textbook adds the more general comment that 'Although the dopamine hypothesis has been the prevailing hypothesis for decades, evidence of dopamine dysfunction in human postmortem studies has not been forthcoming' (p. 1491). From the 1980s on, researchers have had increased access to sophisticated neuroimaging technology. Nevertheless, at the beginning of the twenty-first century, two reviews had reached similar conclusions. One acknowledged that 'few studies have provided convincing evidence of altered

dopaminergic activity' (Copolov and Crook 2000: 110) and reported that 11 of the 13 studies using the new neuroimaging techniques had found no significant differences in dopamine receptor density. The second found that 'In spite of decades of extensive research, the causes and exact sites of the presumed dopamine-mediated hyperactivity remain elusive' (Gainetdinov *et al.* 2001: 527).

This continued failure to confirm biological psychiatry's lead hypothesis led to the investigation of other neurotransmitters. As noted by one reviewer, however, these studies were no more productive than those involving dopamine (Dean 2000). Serotonin, for instance, is another neurotransmitter that is 'blocked' by antipsychotic medication and is therefore another suspect in the search for the cause. Several studies found no evidence of serotonin abnormalities in 'schizophrenics' (Dean 2000).

While most researchers continue to look for a purely bio-chemical solution, with the most recent candidates including glutamate, aminobutyric acid (GABA) and acetyl choline (Saddock *et al.* 2009: 1491–2), others are exploring the more productive path of linking biochemistry to the psychological and social domains. Kapur (2003), focusing on the role dopamine plays in rendering external stimuli 'salient' to the individual, seeks to explain hallucinations and delusions in terms of how events that seem insignificant to most of us are transformed into events of hugely significant personal meaning for others. The Traumagenic Neurodevelopmental Model points out that dysregulation of the dopamine system is a common finding in traumatized children, thereby demonstrating that finding biological differences between groups of people does *not* imply a purely biological aetiology (Read *et al.* 2001). Bentall (2009: 175), in explaining the development of paranoia, reminds us that one of the functions of dopamine involves the anticipation of threat and argues that 'the dopamine system becomes sensitized as a consequence of adverse experiences that predate the onset of the illness'.

Meanwhile, however, the original, simplistic, and unsubstantiated dopamine theory remains very popular. The dopamine theory has become the cornerstone of biological psychiatry's claim that 'schizophrenia' is an illness and of the pharmaceutical industry's marketing of antipsychotics (Chapter 10). The lack of evidence for the theory has only slightly dented biological psychiatry's enthusiasm. Entering 'dopamine' and 'schizophrenia' into the computerized database MEDLINE reveals that of the 90,207 papers ever published on 'schizophrenia' 6,521 (7.2%) are related to dopamine. In the decade 2001–10, 8.9% (3,130 papers) were related to dopamine.

Understandably, drug-company-sponsored organizations like the National Alliance on Mental Illness are particularly fond of the general belief that 'the brains of people with schizophrenia are different from the brains of people without the illness' and, more specifically, that antipsychotics 'rebalance dopamine' (www.nami.org). Drug company advertisements and websites promote the unproven theory, with no reference to the research showing that

adverse life events, including taking their products, are causing the 'imbalance'. For example, in 2012, Janssen, the manufacturer of the antipsychotic Risperidone, asserted:

Schizophrenia is a biologically based brain disease. The most recent advances in brain imaging have confirmed imbalances of two brain chemicals – dopamine and serotonin.

(www.mentalwellness.com)

Without the dopamine theory there would be no way to justify the use of the drugs. Without the drugs there may be no 'chemical imbalance' to support the theory.

Brain anatomy

Another key component of the quest to prove 'schizophrenia' to be a medical illness is the finding that some 'schizophrenics' (a minority) have larger spaces within their brains than 'normals'. The enlargement of these fluid-filled gaps, called ventricles, is related to a reduction in brain tissue in general (and, in particular, reductions in the frontal cortex, amygdala, hippocampus and thalamus). Despite the three major problems with this kind of research identified above, 'For many biological psychiatrists, this has been taken as incontestable evidence of a disease process at work' (Bentall 2009: 153).

An early review (Reveley 1985) had found that between 6% and 60% of 'schizophrenics' have enlarged ventricles ranges. A more recent analysis, of over 90 computerized tomography studies, revealed 'a substantial overlap (approximately 60%) between the schizophrenic population and the control population' (Copolov and Crook 2000: 109). What does it mean that a certain percentage of people labelled 'schizophrenic' have enlarged ventricles? Because they are found in many other disorders, such as depression and alcoholism (Copolov and Crook 2000), they certainly do not constitute a specific cause of schizophrenia. All we really know is that somewhere between 6 and 60% of people labelled 'schizophrenic' have experienced one or more of the factors that cause enlarged ventricles.

The finding that when one identical twin has 'schizophrenia' and the other doesn't, the one with 'schizophrenia' has larger ventricles raised 'the possibility that presently unidentified environmental factors result in increased ventricular volume' (Copolov and Crook 2000: 109). One of these unidentified factors, as was the case for dopamine dysregulation, is trauma – especially in childhood, when the brain is developing. Early childhood trauma can cause ventricular enlargement, cerebral atrophy and dysfunction of the limbic system (hippocampus, amygdala and septum) (Nemeroff *et al.* 2006; Read *et al.* 2001; Teicher *et al.* 2006). Given that the majority of people diagnosed with 'schizophrenia' had been abused or neglected as children (Chapter 18),

this would account for much of the ‘evidence’ that ‘schizophrenia’ is a ‘brain disease’. This simple biological fact is still ignored by psychiatry textbooks (e.g. Black and Andreasen 2011; Gelder *et al.* 2009; Kay and Tasman 2006; Lieberman *et al.* 2006; Saddock *et al.* 2009).

The only environmental factors that *are* considered by brain researchers, or discussed in textbooks, are those occurring at, or around, birth. The reason for permitting discussion of these factors, but not others, seems to be that they are themselves biological and do not, therefore, represent a threat to the ‘medical model’. A recent textbook puts it this way:

Individuals may carry a genetic predisposition, but this vulnerability is not ‘released’ unless other factors also intervene. Although most of these factors are considered environmental . . . most are also biological rather than psychological and include factors such as birth injuries, poor maternal nutrition, or maternal substance abuse.

(Black and Andreasen 2011: 120)

Even the deficits in cognitive functioning found in ‘schizophrenia’, and cited unquestioningly as evidence of a biological disease process, can be caused by adverse events in childhood (Read *et al.* 2001). The researchers studying these children, however, rarely take an interest in what is happening in the lives of the children. Studies of the very earliest ‘signs’ of ‘schizophrenia’, such as minor physical abnormalities that develop during pregnancy, are assumed to be evidence of a genetically based illness. These, however, are by no means specific to ‘schizophrenia’ and are related to prenatal factors (such as mother’s exposure to stress) (Cannon *et al.* 2004). Furthermore, research into ‘epigenetic’ processes has recently established that even the earliest ‘evidence’, including during pregnancy, need not be an inevitable consequence of faulty genes, because genes can be turned on and off by the environment (Read *et al.* 2009).

The second type of environmental factor that causes ventricular enlargement and cerebral atrophy is, once again, antipsychotic medication. This possibility had been ignored by all but one of the first 60 studies (Reveley 1985). The exception (Lyon *et al.* 1981) had found that as the dose of antipsychotics increases, the density of brain tissue decreases.

Recent reviews find numerous studies confirming that antipsychotics are related to reductions in brain volume and increases in ventricles (Navari and Dazan 2009; Weinmann and Aderhold 2010). While industry-sponsored studies (e.g. Lieberman *et al.* 2005) sought to prove that this is only the case for the first-generation ‘typical’ antipsychotics rather than for the newer ‘atypicals’, others argued that these brain changes are not an effect of the drugs at all but are part of the disease process. Two recent developments have proved otherwise. The first was the discovery that antipsychotics (‘typical’ and ‘atypical’) cause reduced brain volume in monkeys (Dorph-Petersen *et al.* 2005). This cannot be blamed on a disease process.

The second was the largest MRI study to date, published in the most prestigious psychiatry journal, *Archives of General Psychiatry*. In September 2008, the esteemed psychiatrist and neuroscientist, Nancy Andreasen (famous for her separation of positive and negative symptoms – Chapter 5) announced the findings to the media, saying: ‘The more drugs you’ve been given, the more brain tissue you lose. . . . The prefrontal cortex doesn’t get the input it needs and is being shut down by drugs. That reduces the psychotic symptoms. It also causes the prefrontal cortex to slowly atrophy’ (New York Times 2008). Curiously, the paper, which answered one of the most pressing problems in psychiatry, was not published until 2011 (Ho *et al.* 2011). (Nor were its findings mentioned in any psychiatric textbooks published in the interim period, including two for which Andreasen was an editor (Black and Andreasen 2011; Gelder *et al.* 2009).) Not only did the study involve 674 brain scans, on 211 patients (far more than most previous studies), covering an average of 7.2 years (the longest to date), it controlled for substance abuse and, of crucial importance to the question at hand, illness severity and duration.

Biological psychiatry and the pharmaceutical industry had had three years to prepare their response to this proof that antipsychotics (‘typical’ and ‘atypical’), rather than the ‘illness’ for which the drugs are prescribed, is causing the brain to atrophy. The first page of the paper directs the reader – in large bold print – to an ‘Editorial Comment’. Here, the commentator (who discloses income from eight drug companies) argues that ‘The idea that strategic reductions in brain volume can be functionally beneficial is supported by the improvements in cognitive capacity that accompany cortical gray matter volume reductions during adolescence’ (Lewis 2011).

Researchers of a less biological bent, and with no drug company sponsorship, point out, sarcastically: In other words, psychotic patients are hypothesized to have too much brain in the first place (Bentall and Morrison 2011). [It is, in fact, akin to the original justification for electroshock therapy 70 years ago (Chapter 8): that it works *because* it causes brain damage.] Bentall and Morrison conclude, instead, as any independent observer would, that ‘the effect is causal, and not some artifact of an underlying schizophrenia disease process’. They describe the Editorial Comment as a last-ditch defence of the bio-bio-bio model of psychiatric disorder.

The defence is likely to continue, regardless of how much evidence is produced to the contrary. Drug company sites fail to inform the public about the causes of the brain differences. For example, Eli Lilly, the maker of olanzapine, simply claims that:

Many studies of people with schizophrenia have found small differences in brain structure, such as an enlargement of the fluid-filled cavities, called the ventricles, in the interior of the brain.

(www.zyprexa.com/Schizophrenia)

It is not only the industry, and industry-sponsored groups like NAMI, that continue to assert that we are dealing with a ‘brain disease’, a ‘medical condition’ (www.nami.org). In 2012 the psychiatrists at the US Government’s National Institute of Mental Health were still trying to persuade us, with no reference to the studies cited above, that:

Schizophrenia is a chronic, severe, and disabling brain disorder. . . . Fluid-filled cavities at the center of the brain, called ventricles, are larger in some people with schizophrenia. The brains of people with the illness also tend to have less gray matter.

(www.nimh.nih.gov/health/publications/schizophrenia)

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‘SCHIZOPHRENIA’ AND HEREDITY

Why the emperor (still) has no genes

Jay Joseph

The consensus position in psychiatry during the past few decades has been that schizophrenia is caused by an inherited genetic predisposition in combination with environmental events or triggers. This is known as the ‘predisposition-stress’ or ‘diathesis-stress’ theory. The most frequently cited evidence in support of the genetic portion of this theory comes from family, twin, and adoption studies, and has compelled researchers to search for predisposing genes at the molecular level. However, the 40-year search for ‘schizophrenia genes’ has failed to bear fruit (Akil *et al.* 2010; Collins *et al.* 2012; Gilmore 2010; Turkheimer 2011). In this brief review we will see that the available evidence provides little if any support for a genetic basis or predisposition for schizophrenia, which suggests that the ongoing inability to identify genes is the result of psychiatry’s failure to critically reexamine the research upon which genetic theories are based.

FAMILY STUDIES

Psychiatric genetic family studies identify people diagnosed with a given disorder, and then determine whether their biological relatives are diagnosed with the disorder more often than are members of the general population or a control group. If a disorder is found to cluster or ‘run’ in families, the disorder is familial. This is seen in schizophrenia family studies, where, in the more methodologically sound studies, the first-degree biological relatives of people diagnosed with schizophrenia are diagnosed with the same disorder roughly four times more often than the 1% rate in the general population (although some studies have found no significant elevation; Joseph 2006).

The key point here is, to the neutral observer, obvious: ‘Familial’ is not the same as ‘genetic.’ Although the familiarity of schizophrenia was once seen as

conclusive proof of its genetic basis, most investigators now realize that disorders and behavioral traits can run in families for non-genetic reasons such as exposure to common rearing patterns and other aspects of the physical and social environment. As behavioral geneticist Robert Plomin and his colleagues point out, ‘Many behaviors “run in families,” but family resemblance can be due to either nature or nurture’ (Plomin *et al.* 2008: 70). They concluded, ‘Family studies by themselves cannot disentangle genetic and environmental influences’ (p. 151).

Therefore, although some authors continue to mistakenly interpret family data in support of genetic transmission (e.g. Gilmore 2010), psychiatric genetic and behavioral genetic researchers have turned to twin and adoption studies, which, they believe, *are* able to disentangle potential genetic and environmental factors.

TWIN STUDIES

The logic of schizophrenia twin studies, which use a technique called the ‘twin method,’ seems straightforward: If reared-together MZ (monozygotic, identical) twin pairs, who share 100% of the same genes, resemble each other more for schizophrenia than do reared-together same-sex DZ twin pairs (dizygotic, fraternal), who share on average only 50% of the same genes, the genetic position is confirmed. (When both members of a twin pair are diagnosed with schizophrenia, the pair is said to be *concordant* for the condition; when one twin is diagnosed but the other is not, they are *discordant* for schizophrenia.) However, we will see that genetic interpretations of twin method data are based on the acceptance of an implausible theoretical assumption.

As seen in Table 7.1, the pooled pairwise concordance rates for schizophrenia are MZ 39.9%, and DZ 7.5%. Reviewers sometimes divide these investigations into the ‘classical’ studies published before 1962, and the more methodologically sound ‘contemporary’ studies published after this date (Gottesman 1991). The older, more methodologically suspect classical studies, which used non-blinded, poorly defined diagnoses and potentially biased resident hospital samples (e.g., Kallmann 1946; Rosanoff *et al.* 1934; Slater 1953), reported higher rates than the more recent studies based on registers or consecutive hospital admissions. We see in Table 7.1 that the rounded pooled pairwise concordance rates in the contemporary studies are MZ = 23%, and DZ = 5%. Moreover, MZ concordance is 18% or less in three contemporary studies, which include two studies with large MZ samples. Thus in the more methodologically sound studies, when one member of an MZ pair is diagnosed with schizophrenia, nearly 80% of the time his or her genetically identical co-twin is *not* diagnosed.

Table 7.1 Results of schizophrenia twin studies

Researcher(s)	Year	Country	Pairwise concordance rates					
			MZ pairs			Same-sex DZ pairs		
			N	C	%	N	C	%
Classical Studies								
Luxenburger ^a	1928	Germany	17	10	59	13	0	0
Rosanoff <i>et al.</i>	1934	USA	41	25	61	53	7	13
Essen-Möller ^b	1941, 1970	Sweden	7	2	29	24	2	8
Kallmann	1946	USA	174	120	69	296	34	11
Slater	1953	UK	41	28	68	61	11	18
Inouye	1961	Japan	55	20	36	17	1	6
Contemporary Studies								
Tienari	1963, 1975	Finland	20	3	15	42	3	7
Gottesman and Shields	1966b	UK	24	10	42	33	3	9
Kringlen ^c	1967	Norway	45	12	27	69	3	4
NAS–NRC ^d	1970, 1983	USA	164	30	18	268	9	3
Fischer	1973	Denmark	25	9	36	45	8	18
Koskenvuo <i>et al.</i> ^e	1984	Finland	73	8	11	225	4	2
Onstad <i>et al.</i>	1991	Norway	24	8	33	28	1	4
Franzek and Beckmann	1998	Germany	9	6	67	12	2	17
Cannon <i>et al.</i> ^f	1998	Finland	—	—	—	—	—	—
Cardno <i>et al.</i> ^g	1999	UK	42	13	31	56	5	4
Pooled rates			761	304	39.9	1,242	93	7.5
‘Classical’			335	205	61.2	464	55	11.9
‘Contemporary’			426	99	23.2	778	38	4.9

Notes

N = Number of twin pairs; C = Number of concordant twin pairs. Concordance rates based on the researchers’ ‘strict’ definition of schizophrenia; age correction factors not included. Unless otherwise noted, when two dates are stated, the first indicates the year results were first published, the second indicates the final publication, whose figures are reported in the table.

^aReported by Gottesman and Shields 1966a.

^bMZ figures from Essen-Möller 1970. DZ concordance rate based on (1941) definite cases among co-twins, as reported by Gottesman and Shields 1966a: 28.

^cBased on a strict diagnosis of schizophrenia; hospitalized and registered cases.

^dNational Academy of Sciences–National Research Council. Original report by Hoffer and Pollin 1970; final report by Kendler and Robinette 1983.

^eThe Koskenvuo *et al.* study is rarely mentioned in textbooks or reviews.

^fCannon *et al.* 1998 reported probandwise concordance rates of MZ = 46% and DZ = 9%. The pairwise equivalents are not listed because Cannon *et al.* did not give the number of pairs in each group.

^gPairwise data supplied by A. Cardno (personal communication, 19 March 2010).

The equal environment assumption: the Achilles’ heel of the twin method

There are several important methodological problems found in schizophrenia twin research. These include:

- the lack of an adequate and consistent definition of schizophrenia;
- the questionable reliability and validity of schizophrenia (Chapters 5 and 6);
- the use of non-blinded diagnoses, and diagnoses made on the basis of inadequate information;
- the use of unreliable methods of zygosity determination (whether a pair is MZ or DZ);
- the unnecessary, potentially biasing, use of age-correction formulas;
- the use of non-representative or small sample populations;
- the lack of an adequate description of the methods;
- investigator bias in favor of genetic conclusions.

These important problems, explained more fully elsewhere (Joseph 2004, 2006), have undoubtedly inflated concordance rates. Nevertheless, there is little doubt that MZ pairs resemble each other more than do DZ pairs for schizophrenia, and for most other psychiatric disorders and behavioral traits as well. The decisive question, however, is: ‘Why?’

The twin method’s critical theoretical assumption is that MZ and DZ pairs experience roughly the same childhood and adult environments. However, this ‘equal environment assumption’ (EEA) is not supported by the evidence, since MZs clearly experience much more similar environments than DZs (Joseph 2004; 2006; Kringle 1967). Because it is widely understood that MZ pairs are treated more similarly, encounter more similar environments, and experience a greater level of ‘identity confusion’ (Jackson 1960) than DZ pairs, some critics have argued that MZ–DZ comparisons measure nothing more than the *environmental* differences distinguishing the two types of twin pairs.

Although most twin researchers now recognize that MZ pairs experience more similar environments than DZ pairs (e.g., Bouchard and McGue 2003; Kendler 1983), they continue to uphold the validity of the twin method and the equal environment assumption on the basis of two main arguments. The first is that, although MZ and DZ twin pair environments are different, these environments must differ on ‘trait-relevant’ factors that have been shown to have a causal influence on the disorder in question (e.g. Bouchard and McGue 2003). However, twin researchers do not make this ‘trait relevant’ argument when addressing potential environmental confounds in *family* studies. In this case they correctly recognize that, because family members share a common environment (‘trait-relevant’ or not), one cannot draw valid conclusions in

favor of genetics on the basis of finding a family resemblance for behavioral traits. Many critics have argued that a similar evaluation holds true for the twin method as well.

The second argument contemporary twin researchers make in defense of the EEA is that, although it is true that MZ environments are more similar than DZ environments, MZ pairs ‘create’ or ‘elicit’ more similar environments for themselves by virtue of their greater genetically caused similarity of behavior (e.g. Kendler 1983; see also Joseph 2012). Twin researchers fail, however, to understand that the *reason* MZ pairs experience more similar environments than DZ pairs is irrelevant in assessing the validity of the EEA. The only relevant question is whether – not why – MZ environments are more similar (Joseph 2004, 2010a).

Moreover, this ‘twins create their own environment’ argument rests on the logical fallacy of circular reasoning, as twin researchers simultaneously and circularly *assume and conclude* that MZ pairs’ greater behavioral resemblance is caused by their greater genetic similarity. In the process, modern twin researchers must assume the validity of genetic interpretations of previous twin studies in order to validate subsequent twin studies (Joseph 2010a, 2012).

Finally, the circular ‘twins create their own environment’ argument renders the ‘trait-relevant’ argument irrelevant because, even if critics demonstrate that MZs experience more similar trait-relevant environments than DZs, twin researchers could still argue in favor of the twin method’s validity on the basis of MZ pairs having ‘created’ more similar ‘trait-relevant’ environments for themselves. (For an example of psychiatric genetic twin researchers making this argument, see True *et al.* 1993.)

Thus, there are two main conclusions that one can reach on the basis of schizophrenia twin method data (Joseph 2010b):

- 1 *Twin researchers’ and contemporary psychiatry’s conclusion, based on the acceptance of the equal environment assumption:* The greater resemblance of MZ vs. same-sex DZ twin pairs provides solid evidence that a sizable portion of the schizophrenia population variance can be explained by genetic factors, or
- 2 *Twin method critics’ conclusion, based on the rejection of the equal environment assumption:* The twin method is a faulty instrument for assessing the role of genetics, given the likelihood that MZ vs. same-sex DZ comparisons measure environmental rather than genetic influences. Therefore, it is likely that previous interpretations of the twin method’s results in support of genetics are wrong.

I argue here, and in detail elsewhere (Joseph 2004, 2006, 2010a), that the evidence supports the acceptance of Conclusion #2. Therefore, we can assume that the equal environment assumption (EEA) is false, and that the results of

schizophrenia twin studies can be completely explained on the basis of methodological problems and the more similar treatment, socialization resemblance, environment, and emotional bond experienced by MZ vs. DZ twin pairs. Like schizophrenia family studies, schizophrenia twin studies cannot disentangle the potential impact of genetic and environmental influences, and all previous interpretations of these studies in favor of genetics should be rejected outright.

Thus, it is likely that twin studies of psychosis have revealed little more than MZ pairs' more similar environments, and their greater propensity to experience *folie à deux* (shared psychotic disorder) than DZ pairs (Jackson 1960). As schizophrenia genetic researcher David Rosenthal (1970: 247) concluded, due to MZ pairs' greater 'identificatory bond', the 'difference in concordance rates between the two types of twins could have a psychological explanation.'

Other twin studies

Two other types of schizophrenia twin studies should be mentioned. The first (Fischer 1973; Gottesman and Bertelsen 1989) studied schizophrenia rates among the biological offspring of discordant MZ pairs. Genetic theories predict that we would find comparable schizophrenia rates among the offspring of the diagnosed twins, versus the offspring of their non-diagnosed co-twins. However, these studies are greatly flawed, and no valid conclusions about genetic factors can be drawn from them (Joseph 2004; Torrey 1990). Although no systematic reared-apart twin study of schizophrenia has ever been published, there have been a handful of single-case reports of purportedly 'reared-apart' MZ pairs concordant or discordant for schizophrenia. In Farber's (1981) review of these cases, she concluded that only nine MZ pairs qualified as having been truly reared apart. However, even in these cases the twins were aware of each other's existence and had periodic contact. Regardless of how many individual reared-apart MZ pairs are reported as being concordant for schizophrenia, however, they do not constitute scientific evidence in favor of genetics because, among other reasons, they were not ascertained in a systematic study.

ADOPTION STUDIES

Most schizophrenia researchers believe that adoption studies are able to make a clean separation between genetic and environmental influences, since adoptees inherit the genes of their biological (birth) parents but are reared in the environment of another (adoptive) family with whom they share no genetic relationship. Although adoption studies appear at first glance to be free from the environmental confounds plaguing family and twin studies and have

therefore played a crucial role in creating the perception that schizophrenia is a genetic disorder, a closer look at these studies reveals several invalidating flaws and biases.

In all six adoption studies of schizophrenia, the researchers compared the diagnostic rate of their index (experimental) group relatives or adoptees, versus the diagnostic rate of control group relatives or adoptees.

- 1 Heston (1966) assessed the rate of schizophrenia among 47 adopted-away biological offspring of women diagnosed with schizophrenia who were confined to Oregon state mental hospitals.
- 2 Rosenthal and colleagues (1968, 1971) studied the adopted-away biological offspring of Danish parents diagnosed with schizophrenia, 'schizophrenia spectrum disorders,' or manic depression.
- 3 Kety and colleagues (1968) began with the records of Danish adoptees from the greater Copenhagen area, identified those diagnosed with a 'schizophrenia spectrum disorder,' and recorded diagnoses among their adoptive and biological relatives. In a follow-up (Kety *et al.* 1975), the investigators interviewed and rediagnosed many of the 1968 relatives. The study was then extended to the rest of Denmark, and the final results were published in 1994 (Kety *et al.* 1994).
- 4 The 'crossfostering' investigation by Wender and colleagues (1974) studied the adopted-away biological offspring of Danish parents not diagnosed with schizophrenia, but who were raised by an adoptive parent eventually diagnosed with schizophrenia.
- 5 In contrast to the earlier investigations, Tienari and colleagues' Finnish study (1987, 2003) took the important (but previously ignored) step of looking at adoptive family environments as well as adoptees' genetic background. Their index group consisted of the adopted-away biological offspring of mothers diagnosed with schizophrenia and 'schizophrenia spectrum disorders.' Tienari and associates concluded that both genetic background *and* adoptive family rearing environment are 'predictor variables' for schizophrenia.
- 6 Lichtenstein and colleagues (2009) used hospital and population records of Swedish parents and children (including some adoptees) to assess whether schizophrenia and bipolar disorder are genetically distinct disorders. Based on these records, they found an elevated schizophrenia risk among the adopted-away biological offspring of parents diagnosed with schizophrenia. However, they did not state how many adoptees were studied.

Methodological problems

In all schizophrenia adoption studies, the investigators concluded that genetic factors play a major role, while Tienari added the important finding that disturbed family environments also play a role. However, these studies have

been the subject of several critical analyses (e.g., Benjamin 1976; Boyle 2002; Jackson 2003; Joseph 2004, 2006; Lidz and Blatt 1983; Lidz *et al.* 1981; Rose *et al.* 1984).

The Danish–American investigations by Kety *et al.* (1968, 1975, 1994), and Rosenthal *et al.* (1968, 1971) are the most frequently cited studies. The most important methodological problems in these studies, elaborated upon by the previously listed critics and others, are outlined below.

- The Danish–American investigators decided to expand the definition of schizophrenia to include non-psychotic 'schizophrenia spectrum disorders,' and they would not have found statistically significant results without such an expansion. The 1968 Kety study found *zero* cases of chronic schizophrenia among the 65 identified first-degree biological relatives of adoptees diagnosed with a schizophrenia spectrum disorder, and Rosenthal *et al.* (1971) found that only one of the 76 adopted-away biological offspring of a parent diagnosed with a schizophrenia spectrum disorder had received a hospital diagnosis of schizophrenia.
- In the Kety *et al.* (1968) study there is evidence suggesting that the researchers decided to change the study's design after the initial relative group comparisons failed to obtain statistically significant results in the genetic direction (Joseph 2004: 220–22).
- The researchers failed to adequately define schizophrenia and 'schizophrenia spectrum disorders.'
- In the Rosenthal *et al.* (1971) study, the researchers counted 'manic-depression' as a 'schizophrenia spectrum disorder' despite their insistence elsewhere that this diagnosis is genetically *unrelated* to schizophrenia. Without these manic-depressive subjects, Rosenthal would not have been able to claim statistically significant results in the genetic direction (Joseph 2004; Lidz *et al.* 1981).
- In the Kety *et al.* (1968, 1975) studies, there were inconsistencies in the way that the researchers decided to count and diagnose dead or unavailable relatives (Joseph 2004; Rose *et al.* 1984).
- The researchers failed to provide case history information on adoptees or relatives and failed to study important environmental variables.
- In the Kety *et al.* Adoptees' Family study (1975), the 'procedure of counting up all the possible relatives of each index case and pooling them as if they were independent samples . . . would allow some families to disproportionately affect the results' (Benjamin 1976: 1130). Thus, the investigators' decision to emphasize the diagnostic rate among individual *relatives*, as opposed to individual *families*, violated the assumption of independent observations underlying the statistical comparisons they used.
- In the Kety *et al.* studies (1968, 1975, 1994), the researchers counted first-degree biological relatives (e.g. siblings) and second-degree biological relatives (e.g. half-siblings) with equal weighting.

- The researchers decided to include many late-separated and late-placed adoptees in their samples. Therefore, during sensitive developmental periods, these adoptees (a) were reared for a certain period of time by their biological parent(s), (b) suffered a disruption of attachment bonds with their biological parent(s), and/or (c) were placed in unstable environments between separation and adoption.
- The investigators used substandard interviews to make diagnoses. In the Kety *et al.* (1975) study, many of these ‘interviews’ never took place, but instead were *fabricated* by the investigators on the basis of hospital records (Kendler and Gruenberg 1984; Rose *et al.* 1984). In the raw data Kety called them ‘pseudointerviews,’ but no mention of their existence appeared in any Danish–American publication.

Table 7.2 summarizes the presence or absence of important methodological problems in all schizophrenia adoption studies published to date.

Selective placement: the Achilles’ heel of adoption studies

A critical component of psychiatric adoption research is the ‘no selective placement assumption,’ which requires investigators to assume that factors relating to the adoption process (including the policies of adoption agencies) did not lead certain groups of adoptees to be placed into environments contributing to a higher rate of the disorder in question. The investigators must assume that children were not systematically placed into adoptive homes correlated with the socioeconomic status – or presumed genetic status – of their biological family. In many psychiatric adoption studies, however, the evidence suggests that index adoptees did experience more psychologically harmful rearing environments than those experienced by control adoptees (Joseph 2004, 2006; Rose *et al.* 1984). This suggests that adoptees with a biological family history of mental disorders were seen as inferior potential adoptees and therefore were placed into more chaotic and harmful (and potentially ‘schizophrenogenic’) adoptive families.

Most adoptees were placed in the early- to middle part of the twentieth century in Denmark, the United States (Oregon), and Finland. The Swedish study index relatives were born between 1932 and 2002. All four regions had laws permitting the compulsory eugenic sterilization of people labeled ‘schizophrenic,’ ‘insane,’ ‘feeble-minded,’ and so on. Society and scientists alike viewed as axiomatic that the offspring of ‘insane’ people were the undesirable carriers of ‘hereditary taint.’

Denmark

In 1929 Denmark became the first European nation to pass a eugenics-inspired sterilization law (Hansen 1996). This law was in force until well after the last

Table 7.2 Important methodological problems in schizophrenia adoption research

Researchers	Year	Country	Significant chronic Schiz first-degree Bio relative rate vs. controls? ^a	Significant chronic Schiz first-degree Bio vs. controls? ^a	Used late separated adoptees?	Probable selective placement blindly? bias?	Diagnoses made placement blindly?	Adequate definition of Schiz?	Counted SSDs as adoptive family rearing environments?	Adequate case history information provided?	Used fabricated interviews to make diagnoses?	Counted first- and second-degree relatives equally?
Heston	1966	USA (Oregon)	Yes	N/A	No	Yes	No ^b	No	No	No	No	N/A
Kery <i>et al.</i>	1968	Denmark	No	No	N/A	Yes	Yes	No	No	No	N/A	Yes
Kery <i>et al.</i>	1975	Denmark	No	No	N/A	Yes	Yes	No	No	No	Yes ^d	Yes
Kery <i>et al.</i>	1994	Denmark	Yes	No	N/A	Yes	Yes	No	No	No	?	Yes
Rosenthal <i>et al.</i>	1968, 1971	Denmark	No	N/A	Yes	Yes	Yes	No	No	No	No	N/A
Wender <i>et al.</i>	1974	Denmark	No	N/A	Yes	Yes	Yes	No	No	No	No	N/A
Tienari <i>et al.</i>	1987, 2003	Finland	No	N/A	Yes	Yes	Yes	Yes ^c	Yes	No	No	N/A
Lichtenstein <i>et al.</i>	2009	Sweden	Not stated	N/A	Not stated	Yes	No ^e	Yes ^f	No	No	No	No

Notes

Schiz = Schizophrenia; Bio = Biological; SSD = Schizophrenia spectrum disorders (as the researchers defined them); N/A = Not applicable.

^aUsing the traditional one-tailed .05 level of statistical significance. Based on a comparison between either (a) the first-degree biological relatives of chronic schizophrenia index adoptees vs. the first-degree biological relatives of control adoptees (Kery); (b) the chronic schizophrenia rate among the adopted-away biological offspring of people diagnosed with chronic schizophrenia vs. the biological offspring of controls (Heston, Rosenthal, Tienari, Lichtenstein); or (c) the chronic schizophrenia rate of cross-fostered adoptees vs. controls (Wender).

^bAlthough two of the three raters made diagnoses while unaware of the adoptees' group status, the third rater, Heston, was not blind to their status.

^cUsed DSM-III-R criteria (APA 1987).

^dSee Rose *et al.* 1984, and Kendler and Gruenberg 1984.

^eLichtenstein *et al.* 2009 obtained diagnoses from hospital records. As they acknowledged (p. 238), 'diagnostic bias could have happened if diagnostic assessments by physicians were influenced by knowledge of family psychiatric history.'

^fUsed *International Classification of Diseases* (ICD 8, 9, 10) criteria (WHO 1967, 1978, 1992).

studied Danish adoptees were placed (placements were made between 1924 and 1947). The Danish adoption agencies checked potential adoptees' perceived genetic family backgrounds to determine their suitability (or desirability) for adoption (Mednick and Hutchings 1977). Clearly, the most qualified potential adoptive parents, who were usually informed of 'deviance' in the adoptee's family background, would not have selected children with a biological family history of mental disorders.

Oregon

Similar conditions existed in Oregon (Joseph 2004), where adoptees were placed between 1915 and 1945. Although Heston failed to mention it, Oregon passed a law in 1917 creating a 'State Board of Eugenics,' whose duty was to authorize the compulsory sterilization of 'all feeble-minded, insane, epileptic, habitual criminals, moral degenerates and sexual perverts,' because they might produce 'inferior' offspring (Olson 1920: 1487).

An additional Oregon law passed in 1919 stipulated that, if a person had been admitted to a mental hospital, this constituted 'prima facie evidence that procreation by any such person would produce children with an inherited tendency to feeble-mindedness, insanity, epilepsy, criminality or degeneracy' (Olson 1920: 3176). Because Heston's index adoptees were born to women hospitalized with schizophrenia, it is unlikely that these 'inferior' children were placed into the same types of adoptive homes as the control adoptees.

Finland and Sweden

In 1935 the Finnish parliament passed the Sterilization Act, which allowed the compulsory eugenic sterilization and castration of 'idiots,' 'imbeciles,' and the 'insane,' which included people diagnosed with schizophrenia and manic depression (Hietala 1996). In 1950, Finland passed the Castration Act, which permitted the compulsory castration of criminals, the mentally retarded, and the 'permanently mentally ill.' Compulsory eugenic sterilization was not legally abolished in Finland until 1970. Sweden also had a long history of eugenics and compulsory eugenic sterilization (Broberg and Tydén 1996).

Kety argued (Kety *et al.* 1994) that because his studies began with diagnosed adoptees, as opposed to diagnosed biological parents, they were less vulnerable to selective placement bias. However, in 8 of 33 index *adoptive* (rearing) families, a parent had been admitted to a Danish psychiatric facility, which was not true for any of the 34 control adoptive families (Rose *et al.* 1984). This finding suggests that index adoptees were placed into more psychologically harmful adoptive homes than the control adoptees. Thus, the higher rate of schizophrenia spectrum diagnoses among index versus control biological relatives

might reflect little more than the agencies’ placement of children with ‘hereditarily tainted’ biological relatives into these types of adoptive homes.

It appears that a violation of the crucial ‘no selective placement assumption’ has introduced a major environmental confound into all schizophrenia adoption studies. Thus, like family and twin studies, the results of these studies are explainable on nongenetic grounds. When we consider these studies’ other glaring methodological problems (Table 7.2), they clearly fail to provide scientifically acceptable evidence in favor of genetics.

THE FRUITLESS SEARCH FOR SCHIZOPHRENIA GENES

Based on the widespread yet mistaken belief that family, twin, and adoption studies have established schizophrenia as a genetically based disorder, psychiatric molecular genetic researchers have attempted to identify genes for schizophrenia. However, although gene-finding attempts go back to the 1970s (Elston *et al.* 1973) and earlier, to date they have been unable to do so (Collins *et al.* 2012; Gilmore 2010; Joseph and Ratner 2013; Plomin 2013; Turkheimer 2011). As the Nobel-Prize-winning coauthors of a ‘Policy Forum’ article published in *Science* recognized, there has been a ‘frustrating lack of progress’ in understanding the genetics of mental disorders (Akil *et al.* 2010: 1580).

Still, despite the publication of over 1,700 schizophrenia molecular genetic studies (www.szgene.org/), the search continues on the basis of researchers’ belief that it is beyond question that schizophrenia is a ‘highly heritable’ disorder. Researchers long ago abandoned the search for a major causative single gene and now view schizophrenia and other psychiatric conditions as ‘multifactorial complex disorders,’ which refers to a condition being caused by a complex interaction of multiple genes and multiple environmental risk factors. It is important to note, however, that the failure to discover genes is currently a *defining feature* of ‘multifactorial complex disorders’ in psychiatry (Joseph 2006, 2012).

Until the mid-2000s, the most common methods in molecular genetic research were linkage studies and association studies. These methods did not yield any genetic variants that were shown by replication studies to cause schizophrenia. Since 2005, researchers have pinned their hopes on genome-wide associations studies (GWAS), which Plomin and colleagues (2009: 873) defined as ‘A hypothesis-free genetic method that uses hundreds of thousands of DNA markers distributed throughout the chromosomes to identify alleles that are correlated with a trait.’ We must keep in mind, however, that *association* (correlation) is not the same thing as *cause*.

Although genome-wide association studies remain the current focus of attention, a seemingly more potentially fruitful research avenue concerns *epigenetics*, which refers to the effect of environmental factors on gene expression (Petronis 2004; Read *et al.* 2009).

Many authors of the 1,700+ studies published over the past 40 years claimed to have found a marker for a schizophrenia gene, with the various claims encompassing most of the 23 pairs of chromosomes. However, subsequent attempts failed to replicate the original findings. In a highly publicized case, Sherrington and colleagues believed they had identified a genetic marker for schizophrenia in their (1988) *Nature* publication, but this result was not replicated. Clearly, psychiatric molecular genetic research is massively plagued by false positive results, and systematic error appears to have been repeated year after year and decade after decade (Ioannidis 2005). In this context, leading psychiatric genetic researchers have publically asked funding sources not to 'give up' on schizophrenia GWAS research (Sullivan *et al.* 2012).

Nevertheless, because some researchers claim that schizophrenia genes have already been found (e.g., Cross-Disorder Group of the Psychiatric Genomics Consortium 2013; Kim *et al.* 2011), accompanied by sensationalized media accounts, it is still widely believed that gene variants or genetic markers for schizophrenia and other psychiatric disorders have been found. In fact, the past 40 years has been characterized by the stunning and unexpected *failure* to discover genes for these disorders (Joseph 2012). Attempts to identify genes presumed to underlie personality traits and IQ (general intelligence) have also failed (Joseph 2011; Plomin 2012; Wahlsten 2012).

In 2008, leading genetic researchers developed the 'missing heritability' position to explain the lack of gene findings for psychiatric disorders and common medical conditions (Gershon *et al.* 2011; Manolio *et al.* 2009; Plomin 2013; Turkheimer 2011). Regarding schizophrenia, proponents of this position argue that schizophrenia genes are 'missing' because researchers must find better ways to uncover them, as opposed to the critics' contention that these genes probably do not exist (see Joseph 2006, 2012; Latham and Wilson 2010).

A 2012 study (Collins *et al.* 2012) co-authored by many of the world's leading schizophrenia molecular genetic researchers taking part in the International Schizophrenia Consortium (ISC; <http://pngu.mgh.harvard.edu/isc>) compared the results of 732 previously identified 'hypothesis-driven candidate genes' for schizophrenia with genome-wide association study results from the ISC. The investigators found 'no notable ISC results' for the most studied candidate genes (Collins *et al.* 2012: 607). They concluded that these results 'suggest, but do not prove, that many traditional ideas about the genetic basis of SCZ [schizophrenia] may be incorrect,' and that 'it is possible that the next few years will lead to marked changes in major hypotheses about the genetic basis of SCZ' (p. 614).

In an earlier review of the 'genetics of schizophrenia' question, I concluded: 'Based on the weight of the evidence, it is predicted here that a gene for schizophrenia will not be found, because it does not exist' (Joseph 1999: 137). Some 14 years later, I see no reason to modify this prediction.

Thus, the time has come to institute a moratorium on schizophrenia molecular genetic research and to undertake a thorough reassessment of the original

family, twin, and adoption studies that inspired the fruitless search for genes in the first place. As the authors of an article on the 'facts' of schizophrenia concluded: 'A reconsideration of our basic strategies and fundamental assumptions may be in order' (Tandon *et al.* 2008: 12). Indeed, such a reconsideration is long overdue.

CONCLUSION

The fact that mainstream psychiatry regards the genetic basis of schizophrenia as a proven fact speaks volumes about the discipline's failure to critically analyze the methods and assumptions of its own research. In countless textbooks in psychiatry, psychology, and other behavior sciences, we find the same uncritical acceptance of the conclusions of twin and adoption researchers. Too often, these textbooks provide an inaccurate presentation of the evidence supporting the genetic position, and only a handful attempt some kind of critical analysis (Joseph 2006).

Psychiatry's uncritical acceptance of the conclusions of schizophrenia twin and adoption researchers is an appalling development in the history of scientific research. It can be understood much more by psychiatry's interest in maintaining itself as a viable profession than on the basis of a careful analysis of the original studies. Moreover, genetic theories aid the interests of the social and political elites, and the interests of the psychopharmaceutical industry, to locate the causes of psychological distress within people's bodies and brains, as opposed to their familial, social, and political environments (Read *et al.* 2009, and see Part II of this book). It is not surprising, therefore, that these groups continue to channel research funding in the genetic and biological direction. Focusing attention on genetic research, regardless of the massive flaws, biases, and untenable assumptions contained therein, successfully diverts attention from the social factors that contribute to people exhibiting behaviors given the schizophrenia label.

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ELECTROCONVULSIVE THERAPY

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A recent *British Journal of Psychiatry* editorial states that ‘ECT has saved and significantly improved the lives of tens of thousands of patients since the 1930s’ (Gazdag *et al.* 2009). Psychiatry textbooks (e.g. Black and Andreasen 2011; McGuffin and Farmer 2008), and a 2007 book, professing to be a ‘fair and comprehensive investigation of ECT’ (Shorter and Healy 2007), all claim ECT is safe and effective.

Electroconvulsive therapy (ECT) involves passing sufficient electricity through the brain to cause a grand mal seizure. It is usually performed 6–12 times over a three- or four-week period. Although nowadays used mostly for depression, ECT was invented to treat ‘schizophrenia’, which remains the second most common diagnostic indicator for ECT. Other factors, however, are more predictive than diagnosis. Women, internationally, are about twice as likely to get ECT as men. The average age of recipients is 61 (Read 2004).

THE FIRST ECT

The idea that schizophrenia could be cured by inducing epileptic seizures was based on the belief, in the 1920s and 1930s, that there was a ‘biological antagonism’ between schizophrenia and epilepsy (Fink and Sackeim 1996: 28). Some doctors treated epilepsy with injections of the blood of ‘schizophrenics’ (Kalinowsky 1986). Others used a range of substances, including insulin, to induce seizures in ‘schizophrenics’.

The first use of electricity to cause convulsions occurred in Italy in 1938. Ugo Cerletti experimented first with dogs, placing electrodes in the mouth and rectum. Many died (Kalinowsky 1986). The idea of bypassing the heart by placing the electrodes on the head came to him at a slaughterhouse where he saw hogs electrocuted via metallic tongs applied to their heads. He found his first human subject, a homeless man, at a railway station:

As soon as the current was introduced, the patient reacted with a jolt, and his body muscles stiffened; then he fell back on the bed without loss of consciousness. . . . It was proposed that we should allow the patient to have some rest, and repeat the experiment the next day. All at once, the patient, who evidently had been following our conversation, said clearly and solemnly, without his usual gibberish: 'Not another one! Its deadly!' (Impastato 1960: 1113–14)

DOES ECT WORK?

Neurologist John Friedberg (1976: 31) points out that there were no studies comparing recipients and non-recipients of ECT in the 1940s, and that 'the influence of ECT was on the minds of the psychiatrists, producing optimism and earlier discharges'. The first studies actually found no difference between (Scherer 1951), or *lower* recovery rates (Karagulla 1950) for, ECT recipients and non recipients.

However, because of frequent fractures of the spine, a disguisable placebo was impossible. In the early 1950s, muscle relaxants and general anaesthesia were introduced, making it possible to evaluate this new 'modified ECT' by comparing it to control groups rendered unconscious by general anaesthesia but not given ECT ('simulated-ECT' [SECT]). The first such study, in which neither psychiatrists nor patients knew who received ECT, involved people diagnosed 'schizophrenic'. It found no difference in outcome between the two groups (Miller *et al.* 1953). The second study found the same, for both depression and schizophrenia (Brill *et al.* 1959).

Comparisons with simulated ECT: depression

After the introduction of antipsychotics in the 1950s, the use of ECT for schizophrenia declined. Depression became the primary target. The vast majority of studies ever since have failed to use SECT, thereby failing to control for the placebo effects of hope and expectancy on the part of psychiatrists and patients. This has been justified in terms of the 'ethical difficulties' of withholding a treatment 'known' to be effective. The *assumption* that it is effective is used to justify not ascertaining whether it *is* effective.

Many of these non-SECT studies report high 'response rates'. However, by 2003 the UK ECT Review Group (Carney *et al.* 2003) found that only 73 of 624 studies (12%) met their standards for inclusion in their review, adding: 'The quality of reporting', of the 12%, 'was poor'. For example, a study in the *British Journal of Psychiatry* (Shukla 1981) claimed that the proportions showing at least 'moderate improvement' were: depression, 100%; schizophrenia, 97.6%. The description of how improvement was measured was: 'A record was kept of progress.'

There have only ever been 10 studies comparing ECT to SECT for depression, and none since 1985 (Read and Bentall 2010). None found any benefit beyond the treatment period. Five found no difference *during* treatment. Of the five that did, two had invalidated their findings by giving ECT to the SECT group after one (Freeman *et al.* 1978) or three weeks (West 1981). In the third study (Brandon *et al.* 1985), both groups improved, but some in the ECT group – the ‘deluded’ and ‘retarded’, but not the ‘neurotic’ – improved more than the SECT group. In the fourth study, both the ECT and the SECT groups improved during treatment, but the ECT group improved more than the SECT group. There were, however, no differences at one-, two- or six-month follow-ups (Gregory *et al.* 1985). The fifth was the famous Northwick Park study (Johnstone *et al.* 1980). During treatment there were no significant differences for two of the three subgroups – ‘agitated’ and ‘retarded’. The only positive finding, for the ‘deluded’ subgroup, was perceived only by psychiatrists. Ratings by nurses and patients produced no significant differences for any of the three subgroups. Even the one positive finding (for the one subgroup, perceived only by psychiatrists) had disappeared one month after the end of treatment. A recent reviewer commented:

Rigorously defined endogenously depressed patients did exceptionally well with sham ECT, just as well as with real ECT. This needs explaining because it is common wisdom that endogenous (melancholic) depressions are not supposed to be placebo responsive.

(Rasmussen 2009: 59)

Comparisons with simulated ECT: schizophrenia

By 1976, famous British psychiatrist Anthony Clare reviewed the evidence and concluded that ‘ECT in schizophrenia had little to offer’. Yet ECT advocates still argued that it is effective for schizophrenia (e.g. Fink and Sackeim 1996). Nevertheless, psychiatric textbooks still recommend ECT for schizophrenia today (Black and Andreasen 2011: 547; Girgis *et al.* 2008: 606).

An American Psychiatric Association report (2001) acknowledged that none of five pre-1980 studies comparing ECT with SECT found any differences, even in the short term. It claimed, however, that three later studies demonstrated ‘a substantial advantage’ for ECT. In all three, however, both groups were receiving antipsychotic medication, and the advantage was only short-term. An example of ‘substantial advantage’ is the Leicester ECT Trial (Brandon *et al.* 1985). Both the ECT and the SECT groups improved on all four measures used. The ECT group showed faster improvement on two of the four scales, but not on ‘global psychopathology’. Eight weeks after treatment, ECT recipients had deteriorated on three of the four measures, but the SECT group continued to improve, overtaking the ECT group on all four measures within six weeks.

Another of these studies (Taylor and Fleminger 1980) found that during treatment there was equal improvement in both groups but that ECT reduced general psychopathology faster than SECT. Again, psychiatrists were the only group to perceive the difference. Nurses and relatives did not. After four weeks, the ECT group deteriorated while the SECT group continued to improve – a pattern that was continuing 16 weeks after treatment. The third study cited as evidence of ‘substantial advantage’ found that after eight weeks ‘the groups were similar’ and that ‘the advantage was totally lost with the passage of time’ (Abraham and Kulhara 1987).

An Indian study of 36 ‘schizophrenics’, missed by the APA, found no differences in symptom reduction, even in the short term, between ECT and SECT (Sarita *et al.* 1998). A subsequent Nigerian study also failed to find significant differences between ECT and SECT, either at the end of treatment or 20 weeks later (Ukpong *et al.* 2002). None of the three meta-analyses of ECT for schizophrenia identified in a recent review (Read and Bentall 2010) found any evidence of long-term benefits.

It should be noted that people diagnosed ‘schizophrenic’ receive longer courses of treatments (Fink and Sackeim 1996; Girgis *et al.* 2008: 587), often in futile efforts to get a result. Seizure threshold increases after each individual shock, by an average of 269% within one course (Chanpattana *et al.* 2000). Therefore people diagnosed ‘schizophrenic’ are exposed to more frequent ECTs *and* to particularly dangerous dosages of electricity.

DOES ECT PREVENT SUICIDE?

A cornerstone of the case for ECT is the claim that it prevents suicide (e.g. Gazdag *et al.* 2009; Shorter and Healy 2007). There has, however, never been a single ECT vs. SECT study on suicide. A review commissioned by New Zealand’s Ministry of Health (2004) concluded that there is ‘No definitive randomized evidence that ECT prevents suicide’.

There have been some studies comparing groups who have, and have not, received ECT (in which any differences could be explained by placebo effects). Almost all these studies found no differences (Read and Bentall 2010). For example, a study frequently cited in support of the claim that ECT prevents suicide found, three years post-discharge, four suicides among the 257 who had received ECT (1.6%) and four (1.5%) among the 262 who had not (Avery and Winokur 1976). The authors later reported on the first six months (perhaps because all four non-ECT suicides occurred in that period). The difference, 0% vs. 1.5%, remained statistically insignificant (Avery and Winokur 1978).

Many other studies found no difference (Read and Bentall 2010). For example, a US study of 1,076 inpatients (Black *et al.* 1989) found no differences in suicide rates over two years between depressed people who received ECT (2.2%), antidepressants (2.6%) or neither (1.9%).

Some studies report an association between ECT and *increased* suicide risk (e.g. Sharma 1999). Among 149 inpatient suicides in Denmark, the inpatients who had completed ECT in the previous week were nearly five times more likely to kill themselves than those not treated with ECT (Munk-Olsen *et al.* 2007). Because such studies do not randomize patients to treatment, it is possible that highly suicidal patients were more likely to receive ECT. Nonetheless, such findings hardly testify to an anti-suicide effect.

DOES ECT CAUSE MEMORY DYSFUNCTION?

A 2003 review of studies of self-reported memory loss at least six months post-ECT found a range of 51–79%, and an average of 70% (Rose *et al.* 2003). The range for ‘persistent or permanent memory loss’ was 29–55%; average 38%. The New Zealand Government report (Ministry of Health 2004) concluded that ‘ECT may permanently affect memory’ and bemoaned the ‘slowness in acceptance by some professional groups that such outcomes are real and significant in people’s lives’. A current psychiatry textbook asserts, rightly, that ‘All patients should be informed that permanent memory loss may occur’ (Black and Andreasen 2011: 55).

Retrograde amnesia

Retrograde amnesia is the inability to recall past events. Three facts are generally accepted: (a) retrograde amnesia occurs to some extent in almost all ECT recipients; (b) memory of events closest to the treatment are most affected; and (c) some improvement occurs over time, with distant memories returning before recent ones (APA 2001: 71).

The APA report acknowledges, however: ‘In some patients the recovery from retrograde amnesia will be incomplete, and evidence has shown that ECT can result in persistent or permanent memory loss.’ A recent review confirmed that ‘ECT can cause persistent or permanent memory loss, especially autobiographical memory’ (Singhai 2011: 25).

All 25 ECT recipients in a recent study suffered deterioration in autobiographical memories (Merkl *et al.* 2011). An earlier study established that immediately after ECT, memory gaps had been caused for a period spanning 25 years, which reduced to a three-year span seven months after ECT (Squire *et al.* 1981). Three years after ECT, memory for events during the six months before treatment remained lost (Squire and Slater 1983).

Despite repeated claims, for 60 years, that ECT is safe, the first large-scale prospective study of cognitive outcomes did not occur until 2007. Six months after ECT, autobiographical memory was significantly worse ($p < .0001$) than pre-ECT levels (Sackeim *et al.* 2007). The degree of impairment was related

to the number of shocks. Twelve per cent were deemed to have suffered 'marked and persistent retrograde amnesia'. Women and older people (both of whom receive ECT more frequently) were particularly impaired. The impairment was greater among those who received bi-lateral ECT rather than unilateral ECT. Bilateral (in which the electrodes are placed on opposite sides of the head) remains the most common form of ECT despite multiple previous findings of greater cognitive dysfunction (McGuffin and Farmer 2008; Singhai 2011).

In an early study, in which ECT recipients performed worse than non-recipients on ability to recall 1960s celebrities and childhood memories, the average time since ECT was 8.4 years (Freeman *et al.* 1980).

Anterograde amnesia

Almost everyone receiving ECT suffers anterograde amnesia, the inability to retain new information. For many this ability returns over a period of a few weeks. The APA Report (2001) cited 11 studies demonstrating anterograde amnesia in the first few weeks after ECT, concluding that 'returning to work, making important financial or personal decisions, or driving may need to be restricted' (p. 70). The report claims that 'no study has documented anterograde amnesia effects of ECT for more than a few weeks'. Studies showing that it persists for one (Falconer *et al.* 2010), two (Porter *et al.* 2008; Squire and Slater 1983) and three months (Halliday *et al.* 1968) might be consistent with 'a few weeks'. Two other studies are not. One found that an average of 8.4 years after treatment, ECT recipients were impaired on retention of new information (Freeman *et al.* 1980). The other found that ECT recipients scored worse than a non-ECT control group on short-term memory tests, designed to detect brain damage, at 10 and 15 years after ECT. The researchers concluded that these findings 'suggest that ECT causes irreversible brain damage' (Goldman *et al.* 1972).

The majority of studies measure only verbal memory. In an attempt to limit the memory loss, unilateral ECT was introduced, in which both electrodes are placed next to the non-dominant hemisphere, rather than on opposite sides of the head, as in bilateral ECT. This may reduce the damage to verbal memory somewhat, thereby creating the impression that unilateral ECT is safer than bilateral. However, there is increased damage to the rarely measured visual memory (Johanson *et al.* 2005).

'Subjective' memory loss

ECT proponents typically respond to these findings by portraying the memory loss as 'subjective'. They argue that memory problems are found only in those who fail to respond to ECT, and that the memory loss is part of the unimproved depressive illness (e.g. McGuffin and Farmer 2008: 629).

These claims have been refuted empirically. McElhiney *et al.* (1995) identified five previous studies showing no significant relationship between anterograde or retrograde amnesia and clinical change after ECT. Their own study found that retrograde amnesia was related to the ECT and not to mood state before or after ECT. Another study found no correlation between changes in depression and changes in any of the 12 cognitive measures (Neylan *et al.* 2001). The only large-scale prospective study found no relationship between degree of depression and 19 out of 22 cognitive measures (Sackeim *et al.* 2007). A reviewer found ‘no evidence of a correlation between impaired memory/cognition after ECT and impaired mood, much less a causal relationship’ (Robertson and Pryor 2006).

Even if there was a relationship, causality might be in the other direction. Losing your memory is depressing. Just before killing himself, shortly after ECT, Ernest Hemingway asked: ‘What is the sense of ruining my head and erasing my memory, which is my capital, and putting me out of business? It was a brilliant cure, but we lost the patient’ (Hotchner 1967: 308).

BRAIN DAMAGE

Further evidence that the adverse effects of ECT are not imaginary is provided by the many studies documenting brain damage (Breggin 2008; Frank 1978; Friedberg 1976, 1977; Sterling 2000). In the 1940s and 1950s autopsies consistently provided evidence of brain damage, including necrosis (cell death). A *Lancet* review concluded that ‘all parts of the brain are vulnerable’ (Alpers 1946). A review of the first 20 years of autopsies concluded: ‘damage to the brain, sometimes reversible but often irreversible, occurred in the course of electric shock treatments’ (Allen 1959).

In 1974, Karl Pribram, head of Stanford University’s Neuropsychology Institute, wrote: ‘I’d rather have a small lobotomy than a series of electroconvulsive shock. . . . I just know what the brain looks like after a series of shock – and it’s not very pleasant to look at.’

CT scans subsequently confirmed frontal lobe atrophy following ECT (Calloway *et al.* 1981; Carney *et al.* 2003). An MRI study found that the number of ECTs correlated with reduced grey matter density, after controlling for level of depression (Shah *et al.* 2002). Besides this *structural* damage, EEG and brain imaging studies have documented *functional* brain impairments as well. Marked deactivation in several cortical regions has been found (Schmidt *et al.* 2008). ECT also strongly activates the HPA axis, leading to a marked rise in cortisol levels (Zis *et al.* 1996), which has been found to correlate with post-ECT verbal learning impairment (Aperia *et al.* 1985).

In fact, in the 1940s it had been argued that ECT worked *because* it causes brain damage and memory deficits. In 1941, Walter Freeman, who imported ECT from Europe to the United States, wrote: ‘The greater the damage, the

more likely the remission of psychotic symptoms. . . . Maybe it will be shown that a mentally ill patient can think more clearly and more constructively with less brain in actual operation' (p. 83). The paper was entitled 'Brain Damaging Therapeutics'. Another US psychiatrist explained:

There have to be organic changes or organic disturbances in the physiology of the brain for the cure to take place. I think the disturbance in memory is probably an integral part of the recovery process. I think that it may be true that these people have for the time being at any rate more intelligence than they can handle and that the reduction in intelligence is an important factor in the curative process. . . . Some of the very best cures that one gets are in those individuals whom one reduces almost to amnesia.

(Myerson 1942: 39)

This line of argument was recently resurrected by a Scottish research team, whose 2012 study found that ECT reduces the 'functional connectivity' of the brain. Instead of cautioning against ECT because of the finding, they celebrated their claim to have finally discovered how ECT works, by correcting a supposed 'hyperconnectivity' in depressed people (Perrin *et al.* 2012).

PSYCHOLOGICAL EFFECTS

There has been very little research asking people how they felt about having ECT (MIND 2001). One study described 'a complex range of emotional responses including feelings of humiliation, increased compliance, failure, worthlessness, betrayal, lack of confidence and degradation, and a sense of having been abused and assaulted' (Johnstone 1999: 76). The UK NICE guidelines (2003: 15) acknowledge that some people report 'feelings of terror, shame and distress, and found it positively harmful and an abusive invasion of personal autonomy, especially when it was administered without their consent'.

ECT-RELATED DEATHS

Psychiatric textbooks and official reports claim that the risk of death from ECT is inconsequential. One example is the Task Force Report of the APA (2001). Note, first, whom the profession of psychiatry selected to assess ECT: six of the eight Task Force members were Directors of ECT services. The group included the incoming and outgoing presidents of the Association for Convulsive Therapy. Three were financially involved with companies manufacturing ECT machines.

The APA report (2001) devotes just one of its 245 pages to deaths, which it calls 'General Issues' (this is one page more than Shorter and Healy's book).

The APA report claims, without citing any research, that ‘A reasonable current estimate’ is ‘1 per 10,000 patients or 1 per 80,000 treatments’ (on the assumption that the average course is eight treatments). Three years later this claim was repeated, verbatim, in a Report of the Royal College of Psychiatrists in the United Kingdom (Benbow 2004). It continues to be regurgitated in textbooks (e.g. McGuffin and Farmer 2008) and reviews (e.g. Singhai 2011). The only figures about deaths in a 154-page report by the Food and Drug Administration (2011: 70) were ‘mortality rate of 1:10,000 patients, or 1:80,000 treatments’.

The APA (2001) and FDA (2011) reports also repeat the claim, made for decades, that the ECT death rate is about the same as that associated with general anaesthesia for minor surgery (estimated at one per 13,000: Lagasse 2002). This ignores the obvious fact that even if this were true for an individual administration of ECT, the risk to an ECT recipient is significantly greater than that of minor surgery because they each receive an average of eight treatments.

Table 8.1 lists studies showing that the 2001 APA claim, which became a mantra for official statements, is a gross underestimation. In 1957, Impastato

Table 8.1 Studies finding significantly higher ECT-related death rates than official statements by the American Psychiatric Association, the Food and Drug Administration, and the Royal College of Psychiatrists

<i>Claims</i>	<i>Year</i>	<i>Country</i>	<i>Death rate/ patients</i>	<i>n</i>
APA	2001	USA	1/10,000	unstated
RCP	2004	UK	1/10,000	unstated
Textbook ^a	2008	UK	1/10,000	unstated
Journal ^b	2011	UK	1/10,000	unstated
FDA	2011	USA	1/10,000	unstated
<i>Studies</i>				
Impastato	1957	international	1/1,000 1/200 ^c	over 250,000
Strensrud	1958	Norway	1/298 ^d	893
Frank	1978	international	1/1,447	130,216
Freeman and Kendell	1980	Scotland	1/92	183
Pippard and Ellam	1981	UK	1/371	2,594
Shiwach <i>et al.</i>	2001	USA	1/627	8,148
Nuttall <i>et al.</i>	2004	USA	1/1,140	2,279

Notes

APA = American Psychiatric Association; FDA = Food and Drug Administration; RCP = Royal College of Psychiatrists.

^aMcGuffin and Farmer 2008: 628.

^bSinghai 2011: 25.

^cOver 60 years.

^dWomen.

reported 254 deaths caused by ECT and calculated a death rate of one per 1,000 patients overall and a death rate in people over 60 of one in 200, 50 times higher than the APA's claim. In 1978 Frank reviewed 28 articles in which psychiatrists had spontaneously self-reported ECT-related deaths. Of 130,216 patients there were 90 ECT-related deaths, one per 1,447 people.

A 1980 survey asked British psychiatrists to report ECT-related deaths (Pippard and Ellam 1981). Excluding deaths that occurred more than 72 hours after treatment, there were four in 2,594 patients, that is, one per 648.5 people. Of the additional six who died within a few weeks of ECT, two were from heart attacks and one from a stroke, the most common causes of death from ECT. Inclusion of these three deaths produces a mortality rate of one per 371 ECT recipients. This is roughly consistent with an earlier Norwegian survey in which three of 893 women (one in 298) died as a result of ECT (Strensrud 1958).

Of 2,279 ECT recipients at the Mayo Clinic in Minnesota, 18 died within 30 days (Nuttall *et al.* 2004). The paper reporting this claimed that 'all deaths appear to be unrelated to ECT'. One was a cardiac arrest (within two days of ECT) and one a stroke. Including only these two, the death rate is one per 1,140. Of 8,148 ECT recipients in Texas, seven died within 48 hours (Shiwach *et al.* 2001). Excluding the two 'unlikely to have been related to ECT', this is a rate of one per 1,630. Eight more died within two weeks of 'cardiac events'. If these are included the rate becomes one per 627.

It could not be determined whether the one death (four days post-ECT) among 75 French ECT recipients was ECT-related. This study, by anaesthetists, found one or more 'complications' for 51 (68%), 12 (16%) of which were 'potentially life-threatening' (Tecoult and Nathan 2001).

Many of these findings relied on the reporting of deaths by those responsible for giving the ECT. A more objective measure was accidentally provided by a study of patients' attitudes (Freeman and Kendell 1980). The researchers wanted to interview 183 people one year post-ECT. However, 12 had died (four had killed themselves). Counting only the two deaths occurring *during* ECT, the death rate was one per 91.5.

CONCLUSIONS

Every administration of ECT causes loss of memory for the period surrounding the treatment. For some people the loss is persistent and covers a greater time span. For others the damage is permanent. Besides the fear and other psychological effects (Johnstone 1999), short-term effects include 'headaches, muscular aches, drowsiness, weakness, nausea and anorexia' (Singhai 2011: 25). Those exposed to these short- and long-term risks, and to the slight but significant risk of death, receive little or no benefit. There is no evidence that ECT has any benefit lasting beyond a few days. It does not prevent suicide

and, for a minority, may precipitate it. The short-term benefit gained by a small minority does not warrant the risks involved (Read and Bentall 2010). In other branches of medicine, a treatment with such an imbalance between risk and benefit would be deemed unethical. Two recent books (Andre 2008; Breggin 2008), and a review (Ross 2006), have reached similar conclusions. A recently published review, by two of this chapter's authors, concluded:

The continued use of ECT therefore represents a failure to introduce the ideals of evidence-based medicine into psychiatry. This failure has occurred not only in the design and execution of research, but also in the translation of research findings into clinical practice. It seems there is resistance to the research data in the ECT community, and perhaps in psychiatry in general.

(Read and Bentall 2010: 344)

An entertaining example of this resistance occurred in the process of getting this review published. It was rejected by one psychiatry journal on the basis of reviews that included vitriolic statements such as 'seems to be part of an ongoing campaign against ECT' and 'authors are hardly (hopefully not!) familiar with the clinical realities regarding severely ill and painfully suffering depressed or psychotic patients'. Soon after the paper was finally published, prominent Australian psychiatrist Ian Hickie informed the media, without reading it, that it was 'ridiculous' (Stark 2010).

The dwindling number of psychiatrists who continue to use ECT genuinely believe ECT is a medical procedure treating an illness. Nevertheless, the research indicates that ECT is a contemporary example of the long history of punitive and damaging treatments (Chapters 2 and 4). Cerletti's obituary in the *American Journal of Psychiatry* placed his invention in the historical context of 'attempts carried out in all possible ways, from the most cruel to the most bizarre, to cause a major shake-up in the psychic economy of mental patients' including 'surprise baths, sudden showers, rotating machines, intimidation systems and blood-letting' (Mora 1964: 201).

We hope that ECT will soon be looked back on as another abandoned attempt to shock distressed people back to sanity. Since the 1980s, ECT rates have fallen steadily in many countries (Read 2004). In some, including Italy and the Netherlands, it is hardly ever used. In New Zealand numbers fell, partly because of a public campaign by ECT recipients, from over 900 patients per year in 1999 to 217 in 2008/2009 (Ministry of Health 2010: 27). In England the number of individual administrations has fallen steadily, from 159,600 in 1980, to 27,128 in 2006 (Singhai 2011).

In the United States, however, about 100,000 people are still given ECT every year (Breggin 2008; Fink 2000). However, January 28, 2011, may be remembered as a proud landmark in the history of how the United States treats its citizens when they are feeling at their lowest. After 60 years of ECT

being completely unregulated, the 18 members of the Food and Drug Administration's Neurological Devices Advisory Panel met to decide whether to classify ECT machines as 'high risk'. The Panel was presented with 3,045 written submissions and a 154-page 'FDA Executive Summary' of the research (FDA 2011) and listened to two days of verbal submissions. After all this, they voted 10 to 8 that ECT should be classified as 'high-risk' for people with depression, and (somewhat illogically) by greater margins for other diagnoses – e.g. 13 to 4 for schizophrenia.

The way to accelerate the process of adding ECT to the list of discarded ineffective and damaging treatments is mapped out by a famous ECT enthusiast. Max Fink (2001: 11) reminds us that it was only when psychotherapists, psychologists, psychoanalysts, nurses, social workers and clergy 'took public positions to restrict the availability of ECT' and 'lay groups, many led by former patients' demanded that 'ECT and lobotomy be outlawed', that ECT has been curbed in the past.

EXAMPLES OF PERSONAL ACCOUNTS OF ECT

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ANTIPSYCHOTIC DRUGS

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Antipsychotic drugs are regarded as the cornerstone of treatment for 'schizophrenia' and other psychoses. They are also frequently used for a range of other 'disorders' with children, adolescents, old people and prisoners (Ross and Read 2004). Their introduction has been credited with a reduction in long-stay hospital-based care; however, others have pointed out that this reduction began before the drugs were introduced (Ross and Read 2004).

Antipsychotics have been characterized as falling into two main groups: 'typical' or 'first-generation' antipsychotics (FGAs), developed in the 1950s, and 'atypical' or 'second-generation' (SGAs), which emerged in the 1990s. SGAs were thought to be more effective than FGAs, with fewer adverse effects. However, the results of two large effectiveness studies and comprehensive reviews have undermined this distinction (Jones *et al.* 2006; Leucht *et al.* 2009b, 2009c; Lieberman *et al.* 2005; Tyrer and Kendall 2009), with some calling for it to be abandoned (Kendall 2011). Most now agree that the perception of SGAs as a major breakthrough in the treatment of schizophrenia was largely a consequence of aggressive marketing by the pharmaceutical industry (Kendall 2011; Leucht *et al.* 2009d).

THE EFFECTIVENESS OF ANTIPSYCHOTIC MEDICATION FOR PSYCHOSIS

The first part of this chapter examines the evidence for effectiveness, focusing on systematic reviews of clinical trials where participants have been randomly allocated to receive antipsychotics or placebo under 'double-blind' conditions (when neither participant nor researcher is told what the participant is taking).

Antipsychotics for people at high risk of developing psychosis

Cochrane Systematic Reviews are independent integrations of high-quality studies. One of these recently concluded that there is no clear evidence that antipsychotics are effective in preventing psychosis in people judged to be at

high risk of developing this condition (Marshall and Rathbone 2011), despite three clinical trials. A subsequent study found no benefit of risperidone (a popular SGA) over placebo when both groups also received psychological therapy (Yung *et al.* 2011). Unfortunately all studies suffered from various limitations (small sample sizes, lack of blind raters, high drop-out rates, low rates of people becoming psychotic, or multiple interventions). Thus, there is currently no robust evidence that antipsychotics are beneficial for this group. Despite this, many of these young people continue to be prescribed antipsychotics (Nieman *et al.* 2009), meaning they are being exposed to their adverse effects (see below) without clear evidence of benefit.

Antipsychotics in early-episode psychoses

Practice guidelines universally recommend treating first episodes of psychosis (FEP) with antipsychotics for 6–24 months (Gaebel *et al.* 2005). A 1991 review found that immediate use of antipsychotics improved outcomes (Wyatt 1991), hence subsequent research that did not immediately medicate everyone was deemed unethical on the basis of withholding a proven treatment (Rothman and Michels 1994). In 2006, however, it was found that the majority of studies in the 1991 review had been uncontrolled (meaning that influences other than antipsychotics could be responsible for the effects) and that, when removing these studies from the analysis, short periods of medication-free research in controlled environments did not demonstrate harm to those later needing medications (Bola 2006). This finding was corroborated by direct investigation (Johnstone *et al.* 1999), and the ethical prohibition was removed (McGlashan 2006).

Approximately 40% of people with an early (first or second) episode of psychosis recover without antipsychotic medication (Bola and Mosher 2002; Bola *et al.* 2006). The current practice of medicating nearly everyone with FEP precludes research intended to distinguish those who benefit from antipsychotics from those who do not. This practice also unnecessarily exposes a large subgroup to the drugs' adverse effects.

A recent Cochrane review of randomized controlled trials (RCTs) found 'Data are too limited to assess outcomes from initial antipsychotic medication treatment for individuals with an early episode of schizophrenia' (Bola *et al.* 2011). When the evidence standard is relaxed slightly to include quasi-experimentally evaluated treatments, a common practice when there are few available RCTs, five studies of initial psychosocial treatments combined with time-limited (1–6 week) postponement of antipsychotic medications each demonstrated small to medium advantages, after two to three years, compared to immediate medication treatment (Bola *et al.* 2009). Furthermore, in each of these studies, 27–61% of subjects were not maintained on medications at the follow-up point, and most had never been medicated. These projects demonstrate potential for *both* better outcomes *and* reduced medication dependence from initial psychosocial treatment *without* medication.

The question of the overall effectiveness of antipsychotics in early episodes *and* the question of how to distinguish people needing antipsychotics from those not needing medication both remain unanswered. Rothman and Michels (1994) suggest that the ‘continuing unethical use of placebo controls’ might be rephrased as the ‘continuing unethical use of unproven antipsychotic treatment’ in early-episode psychosis. Neither the practice of medicating most people who develop psychosis nor the clinical guidelines that support this practice (e.g. Gaebel *et al.* 2005) are evidence-based.

Antipsychotics for people with persistent symptoms

First-generation antipsychotics

Cochrane reviews of the two most popular FGAs – chlorpromazine (Thorazine) and haloperidol (Haldol) – report favourable results (Adams *et al.* 2007; Joy *et al.* 2006). In the haloperidol review, the best-quality data indicates that 14% improved with placebo while 44% improved with haloperidol, suggesting that 30% of participants taking haloperidol benefitted (Joy *et al.* 2006). The best quality data in the chlorpromazine review suggested that 13% of people improved with placebo while 28% improved with chlorpromazine. Therefore around 15% of people taking chlorpromazine could be said to benefit (Adams *et al.* 2007). These reviews included many trials with relatively low rates of drop-out, so we can be relatively confident about the results. However, they are also quite old, with very few reporting usable data with modern rating scales.

Newer trials using reliable and valid scales are available for haloperidol, as this has been the comparator drug of choice when evaluating SGAs. These studies suggest much lower effectiveness. In the most recent review of trials carried out after 1990, haloperidol led to an average reduction of about 8–9 points on a commonly used assessment measure called the Positive and Negative Syndrome Scale (PANSS) (Kay and Opler 1987), which is considerably less than the 15 points thought to be required for minimal clinical improvement (Leucht *et al.* 2006). The proportion improving with placebo was around 19%, while 29% improved with haloperidol (Leucht *et al.* 2009a), suggesting only 10% benefitted. Interpreting this small effect is difficult. These newer trials may be more sophisticated, but most also suffered from very high rates of drop-out (>50%). Nonetheless, the results suggest considerable uncertainty about the effectiveness of this drug more than 50 years after its introduction.

Second-generation antipsychotics

The main finding of the most recent review of 38 clinical trials of SGAs was that people taking these drugs had a higher average degree of improvement in symptoms than people taking placebo (Leucht *et al.* 2009a). However, the improvement on the PANSS was only 10 points, and therefore did not meet

an empirically derived threshold for minimal clinical improvement (Leucht *et al.* 2006).

The review also looked at the proportion of people achieving an important clinical response. They found that 24% and 41% of participants achieved this degree of improvement when taking placebo and SGAs, respectively. This suggests that 17% of people taking SGAs could be said to benefit from them in the short- to medium term. The review also found that 17% of those taking SGAs over the long term met criteria for relapse, compared to 39% of those taking placebo: a difference of around 22%. The reviewers concluded that these results mean that antipsychotics are ‘no wonder-drugs’, and wondered whether the benefits of antipsychotics have been overestimated.

Supporting this view, another review (Lepping *et al.* 2011) examined 120 studies to determine whether the degree of symptomatic improvement associated with antipsychotics was clinically meaningful, as assessed by the PANSS and by another well-established rating scale, the Brief Psychiatric Rating Scale (BPRS) (Overall and Gorham 1962). BPRS data from over 9,500 participants suggested that antipsychotics as a class were associated with moderate ‘global improvement’. However, PANSS data from over 21,500 participants suggested that antipsychotics were associated with *less* than *minimal* global improvement. The authors suggest that the BPRS figures are more favourable to antipsychotics because negative symptoms (i.e., reduced motivation, reduced anticipatory pleasure), not normally responsive to antipsychotics, are measured more thoroughly by the PANSS.

*Pills vs. injections: is the limited efficacy of antipsychotics
a consequence of poor adherence?*

The discontinuation rates in trials of antipsychotics for psychosis are typically extremely high (Rabinowitz *et al.* 2009). Adherence is also poor in clinical practice. In the United Kingdom, more antipsychotic prescriptions go unredeemed by service-users than for any other drugs (NHS Information Centre 2011). In Norway, only 43% of those who redeemed an initial prescription returned for a second (Kjosavik *et al.* 2011).

Some argue that better adherence would dramatically improve outcomes (e.g., Velligan *et al.* 2010). If this were true, then we would expect to see greater improvement when people are given antipsychotics in a long-acting injectable or ‘depot’ form, where near-maximum adherence is guaranteed. A recent analysis of 10 studies comparing depot and oral antipsychotics for a year or longer reported results that seem, at first glance, to support this theory (Leucht *et al.* 2011). However, the favourable results for depot antipsychotics did not persist in the six studies where the researchers did not know which participants were receiving depot. When we pooled the results of these studies, there were no significant differences in rates of relapse or rehospitalization between the groups. Of those receiving depot preparations (the maximum-adherence group) 24.8% and 22.1%

relapsed or were hospitalized, respectively, compared to 29.3% and 21.2% of those receiving the traditional oral form. A large two-year study carried out by the Veterans Affairs Administration in the United States, published after the Leucht review, found no difference between a depot version of risperidone (a popular SGA) and oral antipsychotics on rates of re-hospitalization, symptoms, quality of life and functioning (Rosenheck *et al.* 2011). Those receiving depot simply experienced greater extra-pyramidal side-effects (see below) and injection-site-related pain. The most recent meta-analysis, based on data from 21 trials involving almost 5,000 patients, concluded that long-acting injections do not have a significant advantage over pills (Kishimoto *et al.* in press).

The role of unblinding

The placebo effect in antipsychotic trials seems substantial. It is also highly variable (Leucht *et al.* 2009a; Potkin *et al.* 2011) and has grown over time (Kemp *et al.* 2010). It has been suggested that the placebo effect in older trials of FGAs appeared to be smaller because researchers may have been able to guess who was and was not receiving the drug, because of the obvious adverse effects (Kinon *et al.* 2011). Only three trials (all chlorpromazine vs. placebo) have directly tested this (Engelhardt *et al.* 1964; Grygier and Waters 1958; Hall and Dunlap 1955). All three found that those assessing participants could guess above chance who was receiving drug or placebo. If this occurs in most trials it could bias results in favour of the active drug treatment (Perlis *et al.* 2005), particularly when those carrying out the assessments have a financial interest in demonstrating a particular drug to be effective (Heres *et al.* 2006).

Surprisingly, we could not find any trials that tested whether *participants* can tell if they are receiving antipsychotics or placebo, despite it being well established that antipsychotics have prominent subjective effects (Artaloytia *et al.* 2006; Kapur 2004; Moncrieff *et al.* 2009). This is a worrying omission, because there is evidence that people are more likely to improve when they *believe* they will receive an effective treatment, regardless of whether they actually do (Mallinckrodt *et al.* 2010).

Finally, a recent analysis by the US Food and Drug Administration found that antipsychotics had an advantage of only six points over placebo on the PANSS in trials submitted in support of new drug applications between 1998 and 2008, compared to 11 points for trials submitted between 1991 and 1998 (Khin *et al.* 2012). Possible explanations (Kinon *et al.* 2011) include improved methodology and reduced biased reporting (Moher *et al.* 2001).

Conclusions regarding efficacy

When considering all these findings, it is worth remembering that most of the studies have been carried out, or funded, by pharmaceutical companies

(Perlis *et al.* 2005). Trials with pharmaceutical industry support are nearly five times more likely to report positive results than those without (Perlis *et al.* 2005). Antipsychotics are worth many billions of dollars to these companies (see Chapter 10), which have a track record of not publishing data suggesting that these and other psychopharmacological agents are either ineffective or harmful (Spielmanns and Parry 2010; and Chapter 10). This is not an entirely new problem. Follow-up data from an influential early study of FGAs, suggesting that those receiving antipsychotics were more likely to relapse than those on placebo (Schooler *et al.* 1967), remain unpublished. Moreover, at least three systematic reviews found compelling evidence that trials of FGAs or SGAs with small or negative effects have not been published (Joy *et al.* 2006; Leucht *et al.* 2009a; Turner *et al.* 2012). This is likely to lead to overestimation of effectiveness and an incomplete understanding of adverse effects, prompting some to declare that this practice ‘is as fraudulent to medicine as Enron, Lehmann, and Madoff are to finance’ (Davis *et al.* 2011).

There is no robust evidence that antipsychotics prevent psychosis onset, and claims for efficacy for people experiencing early psychosis have been challenged by a recent Cochrane review (Bola *et al.* 2011). For those with persistent symptoms, around 10–15% taking FGAs and around 17–22% of those taking SGAs experience an improvement in symptoms attributable to the drugs rather than to placebo effect or natural recovery.

ADVERSE EFFECTS

First-generation antipsychotics (FGAs)

Both first-generation (FGAs) and second-generation antipsychotics (SGAs) carry the risk of severe adverse effects. Nearly everyone prescribed antipsychotics has some adverse effects (Chiang *et al.* 2011), with almost one in four describing them as intolerable and one in three citing them as the most important reason for stopping medication therapy (Borneo 2008). It is important to be aware of these effects in order to respond and prevent them from becoming permanent problems.

Adverse effects of FGAs include the ‘anticholinergic’ effects that result from blocking the neurotransmitter acetylcholine, such as dry mouth, blurring of vision, tachycardia (fast or irregular heart rate), constipation, urinary retention and delirium, with higher dosages carrying greater risks (Kane and Correll 2010). As a result of effects on the sympathetic nervous system, but also due to direct cardiac effects, hypotension, palpitations, tachycardia and sudden cardiac death (due to lengthening of the heart’s Q-T-interval) can occur. Many FGAs result in adverse sexual effects, including decreased libido and inability to sustain erections or ejaculate (Murthy and Wylie 2007).

Endocrine effects such as weight gain and hormonal changes differ between different antipsychotics. FGAs can cause neurological disorders ('extrapyramidal symptoms') resulting from dopamine blockade. These include a syndrome of restless legs ('akathisia'), tremor, rigidity of the muscles, involuntary movements, prolonged or uncontrolled muscle contraction and lack of coordination (dyskinesia), lack of movement, involuntary and spasmodic movements of the eyes and tardive dyskinesia. Both FGAs and SGAs can result in the symptoms of Parkinson's disease and abnormal movements and muscle tone that severely compromise daily living and lead to stigmatization. Among the worst effects are restlessness and agitation, with unpleasant emotional states and strange impulses, which are often not taken seriously because these drugs are expected to reduce irritability. Restless legs syndrome accounts for a substantial proportion of suicide attempts, which are most common among young people recently diagnosed and medicated (Drake and Ehrlich 1995).

Tardive dyskinesia

Tardive dyskinesia (TD) involves uncontrollable movements of the tongue, lips, face, hands and feet. The average prevalence of TD among people on antipsychotics is 30% (Llorca *et al.* 2002). Older studies, mainly of FGAs, found that it affects 10–50% of those in long-term treatment (Appelbaum *et al.* 1985). Among people over 45 years, 26% taking FGAs develop TD after one year and 60% after three years (Jeste *et al.* 1995), with 23% developing severe TD within three years (Caligiuri *et al.* 1997).

Severe TD is a serious, potentially disabling, movement disorder. Severe oral dyskinesia may result in dental and denture problems that can progress to ulceration and infection of the mouth, and muffled or unintelligible speech. Severe TD of the mouth and face can impair eating and swallowing, which can produce significant health problems. Gait disturbances due to limb dyskinesia can leave patients vulnerable to falls and injuries. Severe TD may impair mobility and often impacts on an individual's ability to work. Some patients with obvious TD experience shame, guilt, anxiety and depression (Caligiuri *et al.* 1997: 148). The pharmaceutical industry has admitted that TD is irreversible in 75% of cases (Hill 1986).

Tardive dyskinesia occurs because of the dopaminergic overactivity that develops in response to the blockade of the dopamine system by the antipsychotic drugs. This overactivity 'results in an imbalance between the dopamine and acetylcholine . . . which manifests itself in the symptoms of tardive dyskinesia' (Berger and Roxroth 1980: 105). Anti-cholinergic drugs prescribed to counteract the extrapyramidal effects of the antipsychotics can further upset the balance between the dopaminergic and cholinergic systems and thereby often exacerbate TD (Gerlach *et al.* 1974).

The antipsychotic drugs mask the symptoms of tardive dyskinesia in up to 40% of people. Indeed, the drugs that cause the brain damage underlying the

symptoms of TD are the best, or only, 'treatment' for the symptoms (Hill 1986). It also means that discovery of TD often occurs only following discontinuation of the drugs (Crane and Smith 1980). Therefore estimates of the prevalence of TD based on people still receiving the drugs are probably underestimates.

The response of psychiatry to TD has been shameful (Hill 1986). Antipsychotics were described as 'among the safest drugs available in medicine' (Baldessarini and Lipinski 1976: 48). The first book on the subject, published a quarter of a century after the introduction of the drugs, notes in its preface that 'The majority of psychiatrists either ignored the existence of the problem or made futile attempts to prove that these motor abnormalities were clinically unrelated to drug therapy' (Fann *et al.* 1980).

By the early 1980s, Roche still claimed that only between 3% and 6% of people on antipsychotics develop TD and, therefore, only 5–10 million of the 150 million people Roche estimated were on the drugs at the time had TD (Hill 1986). A conservative estimate based on the research available at the time was 33 million TD cases worldwide, for 21 million of whom the brain damage was irreversible (Hill 1986). A 1992 estimate, based on everyone who had ever received the drugs to that date, was 86 million TD cases, 57 million of which were irreversible (Hill 1992).

By 1983 TD had, justifiably, been described as 'one of the worst medically-induced disasters in history' (Breggin 1983) and has been the subject of legal action, particularly in the United States. A large out-of-court settlement was won by an American TD sufferer on the basis of Article 3 of the Human Rights Act 1988, which states 'no one shall be subjected to inhuman or degrading treatment or punishment' (Chari *et al.* 2002).

Neuroleptic malignant syndrome

Neuroleptic malignant syndrome involves muscle rigidity, fever, autonomic nervous system instability and reduced consciousness. It occurs in 0.02–1.9% of individuals taking antipsychotics (Yassa *et al.* 1997), with risk being greater with higher or rapidly escalating dosages, intramuscular injections and taking other psychotropic medications. Of those who develop the condition, between 4% and 25% die, although mortality has fallen in recent years due to earlier recognition and treatment (Ahuja and Cole 2009).

Atypical antipsychotics (SGAs)

Although the SGAs are marketed as being more effective and having a more favourable 'side-effect' profile than FGAs, these claims are largely drug company hype (see Chapter 10). The term 'atypical' is primarily an invention of the pharmaceutical industry. 'Atypicals' have nothing that clearly distinguishes them from 'typical' drugs. 'As a group they are no more efficacious, do

not improve specific symptoms, have no clearly different side-effect profiles than the first-generation antipsychotics, and are less cost effective' (Tyrer and Kendall 2009).

Cardiovascular effects and blood problems

Cardiac arrhythmia is an adverse effect of both FGAs and some SGAs and may be related to an increased mortality rate (Reilly *et al.* 2002). Lengthening of the Q-T interval of the heart can lead to ventricular fibrillation (severely abnormal heart rhythm) and cardiac death. Among the newer antipsychotics, sertindole and ziprasidone carry the highest risk. However, 'all currently available antipsychotics have electrophysiological properties that should increase the risk of sudden cardiac death' (Ray and Meador 2002). A major study found that there are no differences between FGAs and SGAs in sudden death rates and that combination therapy with other psychotropic drugs is particularly dangerous (Ray *et al.* 2009).

Some antipsychotics may cause substantial blood pressure falls, with cardiovascular instability, flushing and palpitations. This is one reason why clozapine has to be started at very low doses and increased slowly. Clozapine, marketed as especially useful in 'treatment-resistant' patients, was initially marketed as being safer than FGAs. It was soon revealed, however, that it can cause agranulocytosis, a reduction in white blood cells leading to a decreased ability to combat infections (Kane and Marder 1993) and may, therefore, result in death from infectious complications (Alvir *et al.* 1993).

Another effect is the higher risk of clots due to effects on blood coagulation. People on antipsychotics have a one-third greater risk of blood clots leading to deep vein thrombosis (Parker *et al.* 2010) and an increased risk of stroke, particularly in older people (Douglas and Smeeth 2008). The risk for blood clotting disorders seems to be higher for some SGAs than for FGAs (Douglas and Smeeth 2008).

Extrapyramidal effects

It is repeatedly claimed that SGAs have a lower risk of neurological (extrapyramidal) effects than FGAs. For instance, Jeste *et al.* (1999) found tardive dyskinesia rates, after nine months, of 30% for FGAs but only 5% for SGAs. However, many of the comparisons have been made, conveniently, with the 'high-potency' drug haloperidol, which has a particular high risk of extrapyramidal effects (Leucht *et al.* 1999). In fact, comparisons between SGAs and 'low-potency' FGAs (e.g. thioridazine) find no differences in extrapyramidal effects (Schillevoort *et al.* 2001). A recent independent study allowed doctors and patients to choose either an SGA (except clozapine) or an FGA. No differences were observed, after one year, in terms of symptoms, quality of life, adverse effects or subjective satisfaction (Jones *et al.* 2006). With regard to

TD, there does seem to be an advantage with some newer SGAs; however, the evidence is not definitive. One review of the TD research concluded that the SGAs are 'promising' but that 'further long-term studies and observations are still necessary before reaching any conclusion' (Llorca *et al.* 2002). Another review, based on three studies of only one year duration, suggested a lower TD risk with SGAs (Correll *et al.* 2004); however, the studies have substantial limitations, and it remains unclear whether SGAs really have a lower risk, or are used in lower dosages, or just delay the risk.

Metabolic effects

The medical community has for decades largely neglected metabolic and other non-neurological somatic effects of antipsychotics. The perceived necessity to take these drugs and a focus on the extrapyramidal effects seem to have decreased prescribers' sensitivity to these less striking, longer-term and highly significant metabolic effects. Furthermore, people diagnosed with 'schizophrenia' have, on average, many independent risk factors and may share a more unhealthy lifestyle compared to the general population, in which the prevalence of diabetes and metabolic syndrome has also been increasing in recent years (de Hert *et al.* 2009).

It is now well established that many SGAs, unlike most FGAs, produce an increased risk of glucose intolerance and diabetes, especially in men (Hedenmalm *et al.* 2002). Most SGAs are also more likely than FGAs to cause obesity, which, besides the social stigma involved, can lead to hypertension, cardiovascular disease, hyperlipidaemia (e.g. high cholesterol level) and, again, diabetes. A higher 'biologic' vulnerability to somatic disorders seems unlikely as unmedicated first-episode patients do not have differences in their glucose and lipid metabolism, or in the prevalence of diabetes, compared to the general population (Sengupta *et al.* 2008).

There is a high variability between different antipsychotics in causing metabolic effects. Olanzapine, clozapine and quetiapine carry the highest risks. In the CATIE study, sponsored by the US National Institute of Mental Health, weight gain greater than 7% after 18 months occurred among 30% of people treated with olanzapine, 16% of those with quetiapine, 15% with risperidone, 12% with perphenazine and 7% with ziprasidone (Meyer *et al.* 2005). This situation is concerning because metabolic syndrome doubles the 10-year risk of coronary heart disease (Daumit *et al.* 2008). Overall differences in cardiovascular risk between different drugs result largely from differential effects on cholesterol and other blood lipid levels.

These adverse effects, combined with sexual dysfunction (Murthy and Wylie 2007) and feelings of anxiety, restlessness and dissatisfaction (dysphoria), are a major reason why half of the people prescribed antipsychotics today 'do not fully comply with treatment' (Perkins 2002). Drop-out rates for SGAs are very high, with three quarters discontinuing their assigned

drug treatment within 18 months in the CATIE study (Lieberman *et al.* 2005). This also reflects low tolerability. It has to be taken into account that tolerability and preferences with regard to adverse effects differ between individuals. A British study found that the sexual dysfunction and weight gain 'are more troubling than the extra-pyramidal effects that the atypical compounds [SGAs] are designed to avoid' (Day *et al.* 1998). These effects and others, such as bad breath, acne, disturbances of vision, hair loss, voice changes and incontinence, contribute to stigmatization and compromise a person's attractiveness but are largely underestimated by clinicians and rarely taken into account when choosing antipsychotics (Seeman 2011).

Reductions in brain volume

There are many other adverse effects of antipsychotics that, together with the limited efficacy of these drugs, make them less than perfect interventions. Long-term prescription of antipsychotics is increasingly seen as a problem because of the significant influence of constant dopamine blockade on the lives of people but also due to the high risk of somatic disorders. A further concern is the effects of antipsychotics on brain morphology and the possible contribution of long-term treatment with FGAs to progressive brain tissue loss, as suggested by a review (Weinmann and Aderhold 2010) and a recent study (Ho *et al.* 2011). This study is important as it is a large longitudinal study using magnetic resonance imaging (MRI). It found that long-term use of antipsychotics, FGAs and SGAs – including clozapine – contributes to brain volume reductions in people diagnosed with schizophrenia. Longer duration of follow-up was significantly linked to greater reductions in total brain tissue. Even after taking into account possible confounding factors such as illness severity and substance misuse, antipsychotics were significantly associated with brain volume reductions. Although some suggested that these reductions may be evidence of the *benefit* of antipsychotics (Lewis 2011), we suggest that service-users be encouraged to make their own judgement.

A recent meta-analysis of first-episode studies found that 'grey matter volume in the anterior cingulate and left insular clusters was influenced by exposure to antipsychotics; patients receiving medication were more likely to show structural abnormalities in these regions' (Radua *et al.* 2012).

A recent Chinese study reported that previously unmedicated patients had acute alterations in white matter microstructure after only six weeks of antipsychotic treatment and concluded 'antipsychotic medication is likely to be the major factor' (Wang *et al.* in press). Together with the debunking of the myth of schizophrenia as a progressive brain disease, many authors admit that antipsychotics have a substantial role in producing those brain changes which are detectable in people under psychiatric treatment (Zipursky *et al.* in press).

Mortality

People with a diagnosis of schizophrenia have a life expectancy at least 20 years shorter than the general population. Although some of this elevated mortality risk is due to factors such as smoking, lack of exercise and poor diet, there is increasing evidence that antipsychotics also contribute to mortality (Weinmann and Aderhold 2010; Weinmann *et al.* 2009), primarily via their cardiovascular, pulmonary, neurological and metabolic effects. Prospective studies have found that both total dosage of antipsychotics (Loas *et al.* 2008) and number of different antipsychotics (Joukamaa *et al.* 2006) are related to mortality. One large Finnish study appeared, on superficial reading, to suggest that long-term treatment with antipsychotics *lowers* mortality (Tiihonen *et al.* 2009). In fact, this study demonstrates the difficulties in evaluating drug effects on mortality in a situation where most people with psychosis receive multiple drug regimes. The study actually showed that those with only a short period of medication have the lowest risk of death. In addition, there was evidence for neurodegenerative effects, effects on thinking, sedation and other adverse effects leading to negative symptoms, sensory deprivation and life-style changes associated with shortened life expectancy.

CONCLUSION

The facts about antipsychotic medication appear to be changing. For people with a schizophrenia-spectrum diagnosis, evidence is accumulating that their benefits have been overestimated and adverse effects underestimated. There is no clear evidence that antipsychotics prevents psychosis, but they are being prescribed for that supposed purpose anyway (Nieman *et al.* 2009). Around 10–22% of people with persistent difficulties who receive modern antipsychotics will experience a benefit that can be attributed to the drugs rather than to placebo effects or natural recovery. The clinical relevance of some of these benefits is, at best, uncertain (Lepping *et al.* 2011). This limited efficacy is unlikely to be a consequence of poor adherence, as injections are no better than pills.

There is, however, growing evidence that antipsychotics may be responsible for some of the problems commonly thought to be a sign or symptom of an underlying disease process (Artaloytia *et al.* 2006; Faber *et al.* 2011; Ho *et al.* 2011). Leading psychiatrists have expressed bitter distrust of drug companies for the promotion of SGAs as wonder-drugs (Kendall 2011; Tyrer and Kendall 2009), and their derisory attempts to hide, or spin away, the data on their adverse effects (Spielmans and Parry 2010). Modern antipsychotics may possibly cause less tardive dyskinesia, but they lead to considerable weight gain and increase the risk of diabetes and other problems. All antipsychotics may increase mortality due to their effects on physical

health, although this must be weighed against their potential for reducing suicide risk.

Doctors have a duty to ensure their patients have access to all the information they need to make decisions about their treatment. However, one study found that only 10% of people on these drugs knew about the risk of agranulocytosis (Angermeyer *et al.* 2001) which causes decreased ability to combat infections. Neither the *New Oxford Textbook of Psychiatry* (Gelder *et al.* 2009) nor the *Comprehensive Textbook of Psychiatry* (Saddock *et al.* 2009) make any mention of neurodegeneration or reduced life span. Recently the section on mortality in the ‘Schizophrenia; Just the Facts’ series in a leading journal made no mention of the possibility that antipsychotics might play a role in shortening life span (Tandon *et al.* 2009; Weinmann *et al.* 2011).

Bearing all this in mind, we propose:

- 1 To reduce the number of people unnecessarily exposed to antipsychotics, we urgently need to identify who does and who does not benefit from this treatment.
- 2 Psychiatrists should give patients the option of not taking antipsychotics, providing them with the evidence on their benefits and harms so they can make their own informed decisions (Hamann *et al.* 2003) in accordance with their own values and goals (Fulford 2011). Letting people choose whether to receive treatment may involve swimming against the tide (Burns *et al.* 2011; Morrison *et al.* 2012), but the reward may be greater autonomy and dignity for service users.
- 3 Policy-makers must fund more research into alternatives to antipsychotic medication.

The ethical prohibition on medication-free research appears to have been lifted (Nelson *et al.* 2011), and there is emerging evidence that psychological treatments might be effective for people who refuse antipsychotics (Morrison *et al.* 2011). It is clear that service-users want more than just drugs (Warner *et al.* 2006), and psychosocial treatments appear to be highly acceptable (Villeneuve *et al.* 2010). However, as leading US psychiatrist Thomas Insel recently concluded, psychosocial treatments for psychosis are ‘powerful’ yet ‘woefully underused’ (Insel 2010).

Finally, we agree with the authors of the recent Cochrane review of risperidone, who concluded:

People with schizophrenia or their advocates may want to lobby regulatory authorities to insist on better studies being available before wide release of a compound with the subsequent beguiling advertising.

(Ratthalli *et al.* 2010: 18)

POSTSCRIPT

Just before going to print, an Editorial appeared in the *Schizophrenia Bulletin*. The ex Vice President of Neuroscience at Eli Lilly wrote:

Given that there cannot be a coherent biology for syndromes as heterogeneous as schizophrenia, it is not surprising that the field has failed to validate distinct molecular targets for the purpose of developing mechanistically novel therapeutics. . . . Psychopharmacology is in crisis. The data are in, and it is clear that a massive experiment has failed: despite decades of research and billions of dollars invested, not a single mechanistically novel drug has reached the psychiatric market in more than 30 years. . . . Nearly every major pharmaceutical company has either reduced greatly or abandoned research and development of mechanistically novel psychiatric drugs.

(Fibiger 2012: 69)

Almost simultaneously the *British Journal of Psychiatry* also published an Editorial, entitled ‘Antipsychotics: Is it time to introduce patient choice?’:

In the context of emerging evidence regarding the overestimation of the effectiveness of antipsychotics and the underestimation of their toxicity, as well as emerging data regarding the possibility of alternative treatments, it may be time to reconsider the prevailing opinion that all service users with psychosis require antipsychotic medication in order to recover.

(Morrison *et al.* 2012: 84)

The Editor commented:

This is not a wild cry from the distant outback, but a considered opinion by influential researchers who help to formulate NICE guidelines. And the reasons for the change in view are not just, as some evidence suggests, a consequence of biased representation of drug treatment in the mass media, but an increasing body of evidence that the adverse effects of treatment are, to put it simply, not worth the candle.

(Tyrer 2012: 5)

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DRUG COMPANIES AND 'SCHIZOPHRENIA'

Unbridled capitalism meets madness

Loren Mosher, Richard Gosden and Sharon Beder

It seems important to provide a historical context in order to understand how the current domination of schizophrenia's 'treatment' by neuroleptic drugs (we use this term and antipsychotic interchangeably) came to be. Not only do they dictate practice, they also buttress the biomedical theorizing that dominates thinking about the problem.

Chlorpromazine ('Thorazine'), the first neuroleptic, arrived on the scene in the early 1950s. It received Food and Drug Administration [FDA] approval in 1955. By 1958 it was in almost universal use in American mental hospitals. Such rapid adoption of a new treatment was unheard of in psychiatry. How did this occur? In 1956 Smith Klein and French [SKF], Thorazine's manufacturer, assembled its American drug 'detailers' (salespersons) to instruct them to behave as 'assault troops' (Johnson 1990) to convince psychiatrists to use their new 'magic bullet'. This was the first massive public relations foray by a pharmaceutical company into a previously small market – institutional psychiatry. In its first year the drug made \$75 million for SKF. The rest, as they say, is history. Thorazine and its successors, despite their adverse effects, are widely viewed as the only 'real' treatment for 'schizophrenia'. The basic elements of this aggressive sales campaign, refined and expanded (detailed below), would be used time and again to sell new drugs to the psychiatric market. The introduction of chlorpromazine was a defining moment in psychiatry.

It is abundantly clear to any thoughtful observer of the psychiatric scene that drug company influence is now pervasive. Between 1993 and 2001 prescription drug spending tripled in the United States – from \$50 billion to 150 billion or more (Szegely-Marzak 2001). In 2000 psychotropic drug sales in the United States totaled \$23 billion. Between 1990 and 2000 spending on antidepressant drugs rose 800% (Tanouye 2001). Over the same period the availability of the new 'atypical' antipsychotic drugs caused spending on

neuroleptics to rise 600%, to 4 billion dollars in 2001 (Moukheiber 2001). The successful selling of 'atypical' antipsychotics, the reason for this 600% rise – to replace the older, no longer patented, neuroleptics, is the major focus of this chapter. One indication of the success of the selling of the 'atypicals' is the increase in use of antipsychotic drugs in youth (under 18). In 1992, 50,000 youth outpatients received them; by 2002, the figure had reached 530,000 (Thomas 2002).

The large pharmaceutical companies average between 30 and 40% of revenues spent on marketing and administration, 15–20% profit and 12–15% on research and development. In the United States they paid an average of 16% of revenues in taxes, whereas all other industries averaged 27% (Angell and Relman 2001). They have 625 paid lobbyists in Washington, DC – one per congressperson (Public Citizen 2001)! They spent nearly \$2.7 billion on direct-to-consumer [DTC] (TV and non-medical magazine) advertising in 2001. Roughly a third of the American Psychiatric Association's (APA) budget is derived from various drug sources (*Psychiatric News* 1997, 15 August, p. 4). APA meetings, like many psychiatric conferences around the world, are dominated by drug-company-sponsored exhibits and symposia providing attendees with a variety of enticements: music, food, drink, disc players etc.

Drug companies provide substantial support to nearly all of the mental health advocacy organizations like the National Alliance for the Mentally Ill (NAMI), the National Mental Health Association (NMHA), the National Alliance for Research on Schizophrenia and Affective Disorders (NARSAD), the Anxiety Disorders Association, the Global Alliance Mental Illness Advocacy Network, etc. (Harris 2009a). The only groups Big Pharma doesn't support are the true consumer advocacy organizations like the Support Coalition International (SCI), the National Empowerment Center (NEC), and the National Association for Rights Protection and Advocacy (NARPA).

Perhaps the industry's most successful marketing tool is direct personal contact with doctors by their 'sales representatives', who supply physicians with sanitized information, promotional materials and samples of their company's products. In 2001 there were, industry-wide, 83,000 such persons in the USA, twice the 1996 number. They cost about \$8 billion a year, and their samples an equivalent amount (Angell and Relman 2001).

The drug industry supports clinical trial research at universities to the extent that it is doubtful that many departments of psychiatry could survive without it. The industry owns the data from clinical trials it supports, decides which studies are published, chooses authors, ghost writes articles and revises them to present the best possible interpretation of the data (Angell 2000).

In 2001 John le Carre wrote, from the perspective of a non-scientist:

Of all the crimes of unbridled capitalism, it seemed to me, as I began to cast round for a story to illustrate this argument in my most recent novel, that the pharmaceutical industry offered me the most eloquent

example. Big Pharma is engaged in the deliberate seduction of the medical profession, country by country, worldwide. It is spending a fortune on influencing, hiring and purchasing academic judgment to a point where, in a few years' time, if Big Pharma continues unchecked on its present happy path, unbought medical opinion will be hard to find. And consider what happens to supposedly impartial academic medical research when giant pharmaceutical companies donate whole biotech buildings and endow professorships at universities and teaching hospitals where their products are tested and developed. There has been a steady flow of alarming cases in recent years where inconvenient scientific findings have been suppressed or rewritten, and those responsible for them hounded off their campuses with their professional and personal reputations systematically trashed by the machinations of public relations agencies in the pay of the pharmas. (le Carre 2001: 11)

In 'Is Academic Medicine for Sale?' the editor of the *New England Journal of Medicine* wrote, from her *professional* viewpoint:

The ties between clinical researchers and industry include not only grant support, but also a host of other financial arrangements. Researchers also serve as consultants to companies whose products they are studying, join advisory boards and speakers' bureaus, enter into patent and royalty arrangements, agree to be the listed authors of articles ghostwritten by interested companies, promote drugs and devices at company sponsored symposiums and allow themselves to be plied with expensive gifts and trips to luxurious settings. Many also have equity interest in the companies.

(Angell 2000: 1517)

In 2002 the *British Medical Journal* published an article entitled 'Selling Sickness: The Pharmaceutical Industry and Disease Mongering' (Moynihan *et al.*). The paper argued that 'Pharmaceutical companies are actively involved in sponsoring the definition of diseases and promoting them to both prescribers and consumers. The social construction of illness is being replaced by the corporate construction of disease' (p. 886). It concluded that 'A publicly funded and independently run program of "de-medicalization," based on respect for human dignity, rather than shareholder value or professional hubris, is overdue' (p. 890).

Psychiatrists Joanna Moncrieff and Phil Thomas (2002) replied:

The influence of the pharmaceutical industry is particularly pernicious in psychiatry where the possibilities for colonizing ever more aspects of human life are potentially limitless. Psychiatry is an area of

controversy, where different paradigms and approaches to treatment are hotly contested. The financial muscle of the pharmaceutical industry has helped to tip the scales in favor of a predominantly biological view of psychiatric disorder. This has submerged alternative therapeutic approaches, despite the fact that user-led research indicates that service users find a wide variety of non-medical approaches valuable in coping with emotional distress.

UNDERSTANDING BIG PHARMA'S MARKET TACTICS

The approval process

The American process is described here, but it is roughly similar in many other countries. For a new drug two studies must be submitted to the FDA that indicate the drug is better than placebo and is without serious adverse effects for condition X. The drug need not be more effective than those already available for X. Data from 'failed studies' – those showing no significant drug placebo differences – are not supplied to the FDA. If approved, the drug can be marketed for X. The companies can later apply for the drug's indications to be extended to new conditions or new populations (e.g., youth, the elderly). This opens a new market. The 'new indications' technique was highly successful with the SSRI's – first approved only for depression but now approved for obsessive compulsive disorder, post-traumatic stress disorder, and various anxiety disorders.

Once a drug is on the market, doctors can legally prescribe it for 'off-label' (unapproved) indications. Because doctors are often given a variety of drug company 'perks', beginning while in training, they are 'open' to listening to the detailers and may in fact be rewarded with more goodies if they prescribe enough of a new 'silver bullet' (Wazana 2000). The companies track prescribing practices very closely.

Sales campaigns

- 'Carpet bombing' of doctors' offices with visits from sales representatives distributing promotional materials and samples.
- Sponsorship of nationwide symposia keynoted by 'thought leaders' in the field who are on the company's payroll, including 'educational' forums for government, academia, and the public.
- Publication of selected (most positive) studies of the drug, authored by high-profile investigators for credibility. Up to 50% of these may be 'ghost authored' (Healy 2002a).
- Endorsements by various organizations with paid alliances with the drug company – NAMI, NMHA etc.

- Generation of 'media hype' via celebrity endorsements, articles placed by public relations agencies, and direct-to-consumer advertising.
- Expanding the prescriber base: The company will focus its detailers' efforts on getting primary care doctors comfortable with prescribing their new 'safe, effective and well tolerated' drug that has ordinarily been the province of a particular specialty group. This has been an extraordinarily successful technique with both the SSRI antidepressants and the 'atypical' antipsychotic drugs.
- Large market formulary publishers are lobbied to convince them the drug should be included in the formularies so it will be paid for by programs like the relevant National Health Care agency or Medicaid, managed care companies, insurers and the military in the United States.
- Substantial political campaign contributions are made, and legislators and government regulatory bodies are lobbied to be sure pharmaceutical company interests are advanced.
- Persons less suspect than drug corporation executives (prominent academicians and scientists) are paid to represent them in taking the lead to counter criticisms. If this fails, critics are systematically discredited, demonized or harassed by colleagues employed in some capacity by the company (Healy 2002a, 2002b). Funding will be withdrawn from journals or other outlets that publish material unfavorable to the pharmaceutical company's interests (Healy 2002a, 2002b). Investigators have even been sued by the company that paid for the research when negative results were published (Bodenheimer and Collins 2001).
- If sued because of a drug's adverse effect(s), companies settle out of court and 'seal' the case to avoid harmful publicity.
- As patents begin to run out, the companies engage in a number of strategies to keep generic versions of the drug off the market. They buy off the potential generic drug makers, tie them up in long court proceedings, produce a new form of the drug (e.g. long-acting) and threaten groups working to enhance the availability of generic drugs (Peterson and Abelson 2002).

CONFLICT OF INTEREST

It has become nearly impossible for the American drug regulatory authority, the FDA, to convene advisory groups that do not contain members with conflicts of interest in the discussion of drugs under consideration. *USA Today* (2000) analyzed the financial conflicts at 159 FDA advisory committee meetings from January 1, 1998 through June 30, 2000. At 92% of the meetings, at least one member had a financial conflict of interest. At 55% of the meetings, half or more of the FDA advisors had conflicts of interest. About half the FDA's budget is derived from drug company fees. Is the fox watching the chicken coop?

SELLING THE ATYPICAL ANTIPSYCHOTICS

Good drug/bad drug

In the 1990s a new type of drug, called 'atypical antipsychotics' were approved, beginning in 1990 with clozapine (Clozaril), followed by risperidone (Risperdal), olanzapine (Zyprexa) quetiapine (Seroquel) and ziprasidone (Geodon). The companies' task was to persuade purchasers (including governments) to purchase these newer drugs instead of the older ones, even though they cost up to ten times as much.

The older drugs – made by the same companies – were suddenly portrayed as not very good. It was as if the deficiencies and toxicities of the older agents (chlorpromazine, haloperidol, fluphenazine etc.) had just been discovered. They were ineffective in 30–40% of cases, had very high rates of unpleasant side effects and caused, over the longer term, the now 'terrible' iatrogenic diseases of tardive dyskinesia, tardive dementia and neuroleptic-induced deficiency syndrome. Before the pricey new drugs arrived, these problems were never sufficient to warrant questioning their use. Suddenly they became intolerable. All stops were pulled out as Big Pharma used its tried-and-true PR methods, which had been perfected in the selling of the SSRI's to replace tricyclic antidepressants.

Nevertheless, the FDA, in its final letter of approval to Janssen, made explicit its conclusions about the relative merits of risperidone and haloperidol. Robert Temple, Director of the FDA's Office of Drug Evaluation, told Janssen:

We would consider any advertisement or promotion labeling for RISPERDAL false, misleading, or lacking fair balance under section 502(a) and 502(n) of the ACT if there is presentation of data that conveys the impression that risperidone is superior to haloperidol or any other marketed antipsychotic drug product with regard to safety or effectiveness.

(Whitaker 2002: 277)

By 2000 Geddes *et al.* had reviewed 52 studies involving 12,649 patients and concluded, 'there is no clear evidence that atypical antipsychotics are more effective or are better tolerated than conventional antipsychotics.'

Setting the agenda

When the atypical neuroleptics became available in the 1990s, the drug companies decided that the best approach to maximizing profits was to expand the size of the market. Hitherto the market for schizophrenia drugs had been restricted by diagnostic conventions on the one hand, and civil liberties protections on the other. The drug companies set out to expand the market by

promoting the concept of an additional pre-psychotic phase of schizophrenia that requires 'preventive treatment' with their new drugs. To further expand the market, they also participated in campaigns to weaken civil liberties protections to increase the number of people who could be treated involuntarily. Finally, as they had done to promote the SSRI antidepressants, they expanded the prescriber base to primary care physicians, gerontologists and pediatricians. This was accomplished by sending sales representatives to these physicians' offices to promote the 'safety, effectiveness and well tolerated' mantra, this time about the atypical antipsychotics as compared with the 'problematic' older neuroleptics. So, the impression was left that the use of these newer agents would be safe for youth, non-psychotic persons and the elderly. The previously noted ten-fold increase in use of antipsychotic drugs in the 18-and-under population in the last decade is stark testimony to the success of this tactic.

The overall solution was the development of a two-fold public relations campaign. The first part involves harnessing support groups for relatives of people suffering from 'schizophrenia' as the driving force for an advocacy coalition (see NAMI's contribution to the media hype below). This has been achieved by carefully focused funding of these organizations (Gosden 2001: 94–97). Once they were made dependent on drug company 'sponsorship', they could then be used as public relations front-groups to assist with planting stories in the media about the efficacy and safety of the new drugs. They also supported claims that schizophrenia has been proven to be a brain disease requiring urgent drug treatment at the earliest signs. An example of this practice can be found at *Schizophrenia.com* (2001a) which purports to be 'A Not-for-Profit Information, Support and Education Center' representing consumers'. However, *Schizophrenia.com* acknowledges that it is funded by Janssen Pharmaceuticals (*Schizophrenia.com* 2001b).

Media hype

Another part of their strategy involved a media hype by 'experts'. Risperidone, the *Washington Post* reported (16 February 1993), 'represents a glimmer of hope for a disease that until recently had been considered hopeless.' George Simpson, a prominent psychopharmacologist (and long-term drug company 'consultant') at the Medical College of Pennsylvania, told the *Post*: 'The data is very convincing. It is a new hope, and at this moment it appears, like clozapine, to be different from all existing drugs.' The *New York Times* (15 January 1992), quoting Richard Meibach, Janssen's clinical research director (no conflict of interest here for sure), reported that 'no major side effects' had appeared in any of the 2,000+ patients who had been in the clinical trials.

Olanzapine, the *Wall Street Journal* (10 April 1996) announced, has 'substantial advantages' over other current therapies. 'Zyprexa is a wonderful drug for psychotic patients,' said John Zajecka, at Rush Medical College in

Chicago. Stanford University Psychiatrist Alan Schatzberg, meanwhile, informed the *New York Times* (14 April 1998) 'It's a potential breakthrough of tremendous magnitude' (Whitaker 2002: 260–1).

NAMI put it together into its full mythic glory. In 1999, it copyrighted *Breakthroughs in Antipsychotic Medications* (Weiden *et al.*). Inside the front cover were framed color photos of the new wonder pills. The NAMI authors wrote:

Conventional antipsychotics all do about the same job in the brain. They all correct brain chemistry by working on the dopamine systems in the brain . . . the newer medications seem to do a better job of balancing all of the brain chemicals, including dopamine and serotonin . . . just give the new medication plenty of time to do a good job!

(Weiden *et al.* 1999: 2)

These glowing endorsements should be contrasted with the review by Geddes *et al.* (2000) and with the research summarized in Chapter 9.

A question of ethics

One aspect of the campaign involves funding selected psychiatric researchers to promote the doubtful belief that schizophrenia must be detected and treated in a pre-psychotic stage to avoid brain deterioration (Gosden 2001: 224–47). This argument has the potential to vastly expand the market for schizophrenia drugs, but treating 'at risk' adolescents with these very powerful agents raises serious ethical issues that the drug companies fail to attend to in any meaningful way. The risk of serious adverse effects, treating large numbers of 'false positive' youth with these powerful drugs, the potential for stigmatization and the creation of a self-fulfilling prophecies are given scant consideration in this 'damn the torpedoes, full-speed ahead' endeavor.

Let the force be with you

A key element of the PR strategy involves funding from the drug company Eli Lilly being channeled through both the *World Psychiatric Association* (Rosen *et al.* 2000) and NAMI (Oaks 2000: 14; Silverstein 1999) to mount an anti-stigma campaign. The thrust of the anti-stigma campaign is to advocate for the elimination of discrimination against people diagnosed with schizophrenia, *so long as they are taking medication* – by force, if necessary. (See Chapter 12 for research showing that the campaign has actually *increased* stigma.)

Meanwhile, in what appears to be a coordinated strategy, the Treatment Advocacy Center (TAC), originally established as a branch of NAMI, has been feeding a different, but complimentary, line to the media and the public

about the dangerousness of *untreated* schizophrenia. This involves associating untreated schizophrenia with news stories about violent behavior (Torrey and Zdanowicz 1999: 27) and promoting wild hyperbole about the murderous intentions of untreated schizophrenics: 'Violent episodes by individuals with untreated schizophrenia and bipolar disorder have risen dramatically, now accounting for an estimated 1,000 homicides annually in the United States' (TAC 2001). This is intended to send an agenda-setting spin in the opposite direction, by scaring the public and impacting on governments as a law and order imperative. The policy intention of this counter-spin is to weaken civil liberties protections in mental health laws in order to increase the number of people eligible for involuntary treatment.

Involuntary treatment is an essential part of the market. A central objective of this hyper-stigmatizing law-and-order part of the campaign is the introduction of legislation that will allow the forced medicating of people in their own homes.

POLICY IMPLICATIONS

Public policy analysts often dissect the strategies and techniques of vested interests like Big Pharma using agenda-setting theory. Cobb *et al.* (1976) propose three models of agenda-building: an outside-initiative model where citizen groups gain broad public support and get an issue onto the formal agenda; a second model where the issues are initiatives that come from government and may need to be placed on the public agenda for successful implementation; and an inside-access model where the policy proposals come from policy communities with access to government, usually with support from particular interest groups, but with little public involvement.

The campaigns run by public relations consultants to distort the mental health agenda in favor of the pharmaceutical industry utilize all three of these models. They run coordinated campaigns that involve funding consumer advocacy groups to simulate outside-initiatives; they plant stories in the media that are designed to gain public acceptance of policies that are already on the government agenda; and they use the insider-access model when they utilize industry lobbying organizations to gain easy access to government. This access is facilitated by the millions of dollars drug companies donate to politicians and their parties (Mintz 2000; Public Citizen 2000).

The use of public relations techniques for setting political agendas has become standard practice in most advanced democracies. The consequences are slowly becoming apparent. The system of representative democracy is being reshaped into a new kind of 'managed corporatocracy' in which public opinion and government policy are custom-made products that can be shaped, packaged and sold by skilled PR experts. This example of a successful

campaign to sell a very expensive product – the atypical antipsychotic drugs – is chilling testimony to the power of Big Pharma to have its way with us.

The cynical manner in which this shaping, packaging and selling has been carried out in regard to mental health policy-making over the last three decades should serve as a warning to anyone who believes that the public good should come before corporate profits.

The vulnerability of ‘mental patients’ to exploitation, and the existence of mental health laws that provide for involuntary detention and treatment of certain classes of distressed people, creates conditions that require vigilant protection of civil liberties and human rights. To do so in the face of the power of a \$150-billion-a-year industry (in the United States) that has only the ‘bottom line’ to guide its activities is a formidable, and perhaps impossible, task without assertive governmental regulatory intervention.

However, Big Pharma’s power may be curbed by legal actions against the industry for failure to adequately disclose adverse drug effects and against doctors for overuse and misuse of (failure to adequately monitor) psychotropics. Professional associations (like the American Psychiatric Association) could implement strict disclosure rules about conflicts of interest that should at least highlight the degree to which they exist among its members.

Individual policy-makers, managers and professionals working in the mental health field, and psychiatrists especially, have an ethical and professional responsibility to consider the extent to which they, and their professions, are influenced by drug company money and propaganda and, having considered, to act. Refusing to take drug company money would be a start. If that proves difficult, you will at least have learned just how dependent your department, journal or professional organization has become. We have to start somewhere to gain back control.

POST 2004: SECOND EDITION UPDATE

Unfortunately, the problem of drug company influence seems to have grown even worse in recent years, although it is, at last, coming under increasing scrutiny. More peer-reviewed journals are now insisting on disclosures by researchers of their conflicts of interest arising from drug company funding. Organizations like the Alliance for Human Research Protection (AHRP 2010) have been active in distributing information about drug company practices. Leading figures from psychiatric professional organizations have, after decades of silence from such groups, spoken out. In 2005 the President of the American Psychiatric Association warned:

If we are seen as mere pill pushers and employees of the pharmaceutical industry, our credibility as a profession is compromised. As we address these Big Pharma issues, we must examine the fact that as a

profession, we have allowed the bio-psychosocial model to become the bio-bio-bio model.

(Sharfstein 2005: 3)

In the same year, the President of the Royal College of Psychiatrists – UK, put it this way:

I cannot be the only person to be sickened by the sight of parties of psychiatrists standing at the airport desk with so many perks about them that they might as well have the name of the company tattooed across their foreheads.

(Shooter 2005: 82)

Political over-sight has increased in some countries. For example, United States Senator Chuck Grassley has used his position as a senior member of the Senate Finance Committee to 'shed light on financial relationships between drug companies and doctors' (Grassley 2010). First, he revealed that academic psychiatrist Melissa DelBello had been paid \$238,000 by AstraZeneca, some of which she had failed to report to her employer, the University of Cincinnati. DelBello has published research claiming that children did well on the company's antipsychotic drug Seroquel (Goldstein 2008). Grassley used the DelBello case to argue for legislation requiring drug companies to disclose all payments made to physicians, including gifts and travel expenses. He was unsuccessful.

Grassley went on to expose drug-company-associated conflicts of interest involving a small Who's Who of American academic psychiatry. Not long after he became president-elect of the American Psychiatric Association early in 2008, Alan Schatzberg, chairman of the psychiatry department at Stanford University School of Medicine, was skewered by Grassley for failing to declare drug company payments of some \$70,000. He was also accused of failing to report an investment of \$6 million he held in a company that was seeking approval to market the drug Mifepristone for psychotic depression. At the time he was conducting clinical trials on the drug (Garber 2008).

Grassley also exposed Joseph Biederman, a 'world-renowned Harvard child psychiatrist whose work has helped fuel an explosion in the use of powerful antipsychotic medicines in children' (Harris and Carey 2008). Biederman was reported to have received fees of at least \$1.6 million from drug companies between 2000 and 2007 but failing to report most of it to Harvard authorities. Biederman's psychiatric colleague at Harvard, Timothy Wilens, also owned up to receiving more than \$1.6 million over the same period. Another, Thomas Spencer, admitted to receiving over \$1 million. Biederman and his colleagues are credited with playing a major part in the 'fortyfold increase from 1994 to 2003 in the diagnosis of pediatric bipolar disorder and a rapid rise in the use of powerful, risky and expensive antipsychotic medicines in children' (Harris 2008).

In March the following year it was reported that the Harvard three – Biederman, Wilens and Spencer – had been named in a subpoena relating to lawsuits claiming that ‘makers of antipsychotic drugs defrauded state Medicaid programs by improperly marketing their medicines’ (Harris 2009b).

On the drug company advocacy front, business is booming at NAMI. Grassley’s inquiry and the *New York Times* uncovered that ‘drug makers from 2006 to 2008 contributed nearly \$23 million to the alliance, about three-quarters of its donations’ (Harris 2009a). They also found that NAMI received directions from AstraZeneca on how to advocate on issues that boost drug profits (Harris 2009a). NAMI received about \$3.85 million from drug companies in 2009 (NAMI 2010), \$1.25 million from AstraZeneca alone (a tiny fraction of its \$4.5 billion plus sales of Seroquel).

While direct-to-consumer advertising remains illegal everywhere except in the United States and in New Zealand, the Internet provides drug companies with a way around this restriction (Read and Cain in press). A 2008 study found that more than half of websites on ‘schizophrenia’ are drug-company-funded and that these are significantly more likely than other websites to promote bio-genetic rather than psychosocial causal explanations of schizophrenia; to emphasize medication rather than psychosocial treatments; to portray ‘schizophrenia’ as a debilitating, devastating and long-term illness; and to link violence to coming off medication. ‘These results suggest that the documented influence of the pharmaceutical industry over research, professional organisations, teaching institutions, clinical practice and regulatory bodies may now extend to public promotion, via the Internet, of perspectives conducive to maximisation of sales’ (Read 2008: 99).

Back at psychiatry headquarters – the American Psychiatric Association – everything is also business as usual. APA experts are negotiating the contents of the new *DSM*—5. The majority of them, it seems, are in the pay of drug companies: ‘Of the 170 contributors to the most recent edition of the American Psychiatric Association’s Diagnostic and Statistical Manual of Mental Disorders (*DSM*), ninety-five had financial ties to drug companies, including all of the contributors to the sections on mood disorders and schizophrenia’ (Angell 2008). The working group for the section on psychosis probably had their sponsors in mind when they thought up a new diagnosis of ‘Psychosis Risk Syndrome’ (Chapter 5). This is the name they have given to the supposed pre-psychotic phase of schizophrenia introduced some years earlier by the drug companies to expand the market. American psychiatrist Colin Ross (2010: 109) has observed that ‘the main effect of Psychosis Risk Syndrome will be further acceleration of the already rapidly growing prescribing of antipsychotic medications to children’. British psychologists, meanwhile have argued: ‘Those with most to gain from the inclusion of this new diagnosis in *DSM*—5 would appear to be pharmaceutical companies, whereas those with the least to gain and most to lose would seem to be service users themselves’ (Morrison *et al.* 2010: 96).

There is one problem the drug companies can't overcome: the passage of time. Annual global sales of antipsychotics reached a high point in 2008 of about \$22 billion (Healthcare Finance News 2009) but have since been on the slide. This has been due to the progressive expiry of patents for a number of the atypicals. Companies are suffering big losses as a result. Eli Lilly, for example, lost US patent protection for its blockbuster Zyprexa in October 2011, and sales of the drug had already fallen by 56% in the first quarter of 2012. Sales of Zyprexa reached \$5 billion in 2010, but Lilly is expecting that figure to be down by over \$3 billion on 2012. (Murphy 2012).

But this isn't likely to herald the end of drug company influence on matters concerning 'schizophrenia'. Already there are a number of new schizophrenia drugs coming down the pipeline and onto the market. This new generation of antipsychotics will undoubtedly be hyped for their lack of side-effects and miraculous properties – just as the atypicals were.

NOTE

When we (RG and SB) were asked to update this chapter for this second edition, we decided to simply add a section at the end describing some significant developments since 2004, leaving intact (apart from some minor editing) the first version, which was written with our dear friend Loren Mosher, who died in 2004.

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Part II

SOCIAL AND
PSYCHOLOGICAL
APPROACHES TO
UNDERSTANDING
MADNESS

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PUBLIC BELIEFS ABOUT THE CAUSES OF 'SCHIZOPHRENIA'

Bad things happen and can drive you crazy

John Read, Lorenza Magliano and Vanessa Beavan

THE PUBLIC'S BELIEF IN PSYCHOSOCIAL CAUSES

The idea that bad things happen and can drive you crazy is not controversial to the public, including people diagnosed 'schizophrenic' and their relatives. Studies spanning 60 years show that while the public understands that many factors influence our mental health, they place far more emphasis on adverse life events than on biology or genetics. This belief has survived concerted efforts by biological psychiatry and drug companies to persuade us we are wrong (Read and Cain in press and see Chapter 12).

Table 11.1 updates previous reviews (Read 2007; Read *et al.* 2006). It presents studies of 84 samples of people, from 25 countries, comparing levels of belief in bio-genetic causes ('genetic', 'brain disease', 'chemical imbalance' etc.) and psychosocial causes ('stress', 'poverty' etc.). Some use vignettes – descriptions of a person exhibiting the symptoms of 'schizophrenia'. Some use the diagnostic label 'schizophrenia'. Some ask people about the causes of their own 'illness', or that of a relative. Of these 84 samples, 61 (73%) believe more strongly in psychosocial than bio-genetic causes, six (7%) apportion equal emphasis and 17 (20%) favour bio-genetic causes.

For example, in Japan the most frequently cited cause (65%) was 'problems in interpersonal relationships', with only 27% citing 'predisposition and genetics' (Tanaka *et al.* 2005). In South Africa 83% stated that schizophrenia is caused by 'psychosocial stress' (difficulties in work or family relationships, stressful life events), while only 43% thought it was a 'medical disorder' (brain disease, heredity, constitutional weakness) (Hugo *et al.* 2003).

Although 70% of Australians endorsed 'inherited or genetic', the most endorsed causes were 'problems from childhood such as being badly treated or abused, losing one or both parents when young or coming from a broken

Table 11.1 Relative strength of bio-genetic and psychosocial causal beliefs about 'schizophrenia' or psychosis among the public, patients and families

<i>Country and study</i>	<i>Sample</i>	<i>Stimulus</i>	<i>Preference for bio-genetic vs psychosocial causal beliefs</i>
<i>Australia</i>			
Jorm <i>et al.</i> 1997	public	vignette	PS
Jorm <i>et al.</i> 2005a	public	vignette	PS
<i>Austria</i>			
Angermeyer and Matschinger 1996	family [o]	vignette	BIO
Grausgruber <i>et al.</i> 2007	public	label	BIO=PS
	family	label	PS
<i>Bali</i>			
Kurihara <i>et al.</i> 2006	family	label	PS [S]
<i>Brazil</i>			
Monteiro <i>et al.</i> 2006	family	relative	PS [S]
Peluso <i>et al.</i> 2008	public	vignette	PS
<i>Canada</i>			
Norman and Malla 1983	public [a]	vignette	PS
Thompson <i>et al.</i> 2002	public	label	BIO
Stip <i>et al.</i> 2006	public	label	BIO
Gladstone <i>et al.</i> 2007	family	relative	PS
<i>China</i>			
Phillips <i>et al.</i> 2000	family	relative	PS
Furnham and Chan 2004	public	label	PS
<i>England</i>			
Furnham and Rees 1988	public	label	PS
Furnham and Bower 1992	public	label	PS
Pistrang and Barker 1992	patients	self	PS
Jones <i>et al.</i> 2003	patients [v]	self	PS[S]
Furnham and Chan 2004	public	label	BIO
McCabe and Priebe 2004	patients	self	PS
Dudley <i>et al.</i> 2009	patients	self	PS
Penny <i>et al.</i> 2009	family [p]	relative	PS
<i>Ethiopia</i>			
Shibre <i>et al.</i> 2001	family	label	PS [S]
<i>France</i>			
Durand-Zaleski <i>et al.</i> 2012	public	label	PS
<i>Germany</i>			
Angermeyer and Klusmann 1988	patients	self	PS
Angermeyer <i>et al.</i> 1988	family	relative	PS
Angermeyer and Matschinger 1994	public	vignette	PS
Angermeyer and Matschinger 1996	family [o]	vignette	BIO
Angermeyer and Matschinger 1999	public	vignette	PS
Gaebel <i>et al.</i> 2002	public	label	BIO

BELIEFS ABOUT THE CAUSES OF 'SCHIZOPHRENIA'

Holzinger <i>et al.</i> 2002	patients	self	PS
Gäbel <i>et al.</i> 2008	public	label	BIO
Angermeyer <i>et al.</i> 2009	public	vignette	BIO=PS
<i>Greece</i>			
Alivistasos and Lykestos 1964	family	relative	PS
Molvaer <i>et al.</i> 1992	patients	self	PS
Economou <i>et al.</i> 2009	public	label	PS
<i>India</i>			
Srinivasan and Thara 2001	family	relative	PS
Kermode <i>et al.</i> 2010	public	vignette	PS
<i>Ireland</i>			
Barry and Greene 1992	public	vignette	PS
<i>Italy</i>			
Magliano <i>et al.</i> 2004b	public	label	PS
Magliano <i>et al.</i> 2004a	family	label	PS
Pantusa <i>et al.</i> 2007	public	label	PS
Magliano <i>et al.</i> 2009	patients	self	PS
<i>Japan</i>			
Tanaka <i>et al.</i> 2005	public	vignette	PS
Nakane <i>et al.</i> 2005	public	vignette	PS
Kawanashi 2006	family	relative	PS
<i>Malaysia</i>			
Razali <i>et al.</i> 1996	patients	self	PS [S]
<i>Mongolia</i>			
Dietrich <i>et al.</i> 2004	public	vignette	PS
<i>New Zealand</i>			
Walker and Read 2002	public [s]	vignette	PS
Taitimu 2007 [m]	public	vignette	PS [S]
Beavan 2007	patients [v]	self	PS
Geekie and Read 2009	patients	self	PS
<i>Nigeria</i>			
Adebowale and Ogunlesi 1999	patients	self	PS [S]
	family	relative	BIO=PS [S]
<i>Pakistan</i>			
Zafar <i>et al.</i> 2008	public	vignette	BIO=PS=S
<i>Russia</i>			
Dietrich <i>et al.</i> 2004	public	vignette	PS
<i>South Africa</i>			
Hugo <i>et al.</i> 2003	public	vignette	PS
Samouilhan and Seabi 2010	public [s]	vignette	BIO
Sorsdahl <i>et al.</i> 2010	public	vignette	BIO=PS=S
<i>Switzerland</i>			
Buadze <i>et al.</i> 2010	patients	self	PS
<i>Turkey</i>			
Karanci 1995	family	relative	PS
Sagduyu <i>et al.</i> 2001	public	label	PS
Sagdayu <i>et al.</i> 2003	family	label	PS
Taskin <i>et al.</i> 2003	public	label	PS

(Continued overleaf)

Table 11.1 Continued

<i>Country and study</i>	<i>Sample</i>	<i>Stimulus</i>	<i>Preference for bio-genetic vs psychosocial causal beliefs</i>
<i>United States</i>			
Jones <i>et al.</i> 1963	patients	self	PS
Weinstein 1974	patients	self	PS
Wahl 1987	public	label	PS
McGill <i>et al.</i> 1983	patients	self	PS
	family	relative	PS
Lefley 1985	family [r]	label	BIO
Link <i>et al.</i> 1999	public	vignette	BIO=PS
Martin <i>et al.</i> 2000	public	vignette	BIO
Corcoran <i>et al.</i> 2003	family	relative	PS
van Dorn <i>et al.</i> 2005	patients	label	BIO
	family	label	BIO
	public	label	BIO
Esterberg and Compton 2006	family	label	BIO
Kawanashi 2006	family	relative	PS
Corcoran <i>et al.</i> 2007	family	relative	PS
Lopez <i>et al.</i> 2009	public [l]	vignette	PS
	family [l]	vignette	PS
Demir <i>et al.</i> 2010	public [z]	label	BIO
Pescosolido <i>et al.</i> 2010	public	vignette	BIO
Wasserman <i>et al.</i> 2010	family	relative	BIO

Notes

BIO = Biogenetic causal beliefs. PS = psychosocial causal beliefs. [S] = Greater belief in spiritual factors than in PS or BIO. [m] = See text for description of this study of Maori. [o] = Members of a family organisation. [a] = Adolescents. [v] = Voice-hearers, including some without contact with psychiatric services (<50%). [p] = Pakistani. [s] = Students. [r] = Relatives who were also mental health staff. [l] = Latinos. [z] = Police.

home' (91%), 'day-to-day-problems such as stress, family arguments, difficulties at work or financial difficulties' (90%) and 'recent death of a close friend or relative' (88%) (Jorm *et al.* 2005a).

In a London study the most endorsed causal model of schizophrenia was 'unusual or traumatic experiences or the failure to negotiate some critical stage of emotional development'; 'it seems that lay people have not been converted to the medical view and prefer psychosocial explanations'; 'subjects agreed that schizophrenic behaviour had some meaning and was neither random nor simply a symptom of an illness' (Furnham and Bower 1992: 206).

The most extensive series of surveys has been in Germany. In 2006 the researchers reviewed all their studies and concluded that, for both depression and schizophrenia, 'acute stress in the form of life events is the most frequently endorsed cause, followed by chronic stress in partnership and family' (Angermeyer and Dietrich 2006).

Table 11.1 also shows that in some countries and cultures spiritual explanations are held as strongly as, or more strongly than, either bio-genetic or psychosocial explanations. For example, among Māori in New Zealand the most common framework for understanding 'extraordinary experiences' such as hearing voices is a spiritual one, including messages from ancestors, *matakite* (gift), *makutu* (curse) and several types of *mate Māori* (spiritual disorders). For many Māori the question about *causes* made little sense. For example: 'For me hearing voices is like saying hello to your *whanau* [family] in the morning, it is nothing unusual' (Taitimu 2007).

Even where spiritual beliefs dominate, psychosocial explanations tend to be endorsed more than bio-genetic ones (see Table 11.1). Indeed, the finding is remarkably robust over time and across cultures. A recent study of 240 community members and 60 health workers in India found that 83% of both groups endorsed 'problems from childhood', financial difficulties', family conflict', 'recent death of relative or friend', 'difficulties at work' and 'having an addicted family member', whereas only 41% of the community members and 18% of the health workers endorsed 'inherited/genetic' (Kermode *et al.* 2010). There is one culture that differs. Of the 24 samples from North America (USA and Canada), 11 (46%) favour bio-genetic causes. Of the 60 samples from the 23 other countries, only six (10%) do so.

One other variation is important. Of the 31 samples stating causes about either themselves (16) or a relative (15) only one (3%) prioritized bio-genetic causes. Of the 26 responding to an unlabelled vignette five (19%) did so. However, of the 27 responding to the diagnostic label 'schizophrenia', 11 (41%) preferred bio-genetic causes. (We discuss the negative effects of labelling in Chapter 12.)

'Patients'

All 16 samples of people, from nine countries, who had been diagnosed with psychotic disorders understood the causes of their problems in terms of what has gone on in their lives more than in terms of genes and neurotransmitters. The causes espoused as 'likely/very likely' by German patients were: 'recent psychosocial factors', 88%; 'personality', 71%; family, 64%; and 'biology', 31% (Holzinger *et al.* 2002). A study in London found that only 5% of people diagnosed 'schizophrenic' believed that their problems were a result of having a 'mental illness', and only 13% cited other biological causes, but 43% cited social causes such as interpersonal problems, stress and childhood events. Members of ethnic minorities within the sample were even less likely to endorse biological causes (McCabe and Priebe 2004). In an Italian study, 76% of people diagnosed with 'schizophrenia' mentioned at least one social cause (family conflicts, psychological traumas and losses), with only 10% citing a biological cause (Magliano *et al.* 2009). In an international study, 297 of 306 'typical schizophrenics' (97%) did not believe they had an illness (Murray and Dean

2008: 285). (This was dismissed as 'lack of insight': a 'symptom' of the illness that psychiatrists believe their patients have got but which patients do not).

Family members

If one relied on the Internet and media to learn what relatives think, you might conclude that they strongly endorse the illness model. It seems, however, that 'family organizations' like SANE in the United Kingdom, NAMI (National Alliance on Mental Illness) in the United States, and the Schizophrenia Society of Canada, which receive so much media coverage (and so much drug company sponsorship), are not representative of family members in general.

Of the 26 samples of relatives, 19 (73%) agreed with the 'patients' and the majority of the public. If the Austrian and German samples drawn from unrepresentative 'family organizations' (Angermeyer and Matschinger 1996) are excluded, the proportion increases to 79% (19/24). Of the 15 family samples, 13 (87%) that were asked about their own 'schizophrenic' relative (rather than being presented with a label or vignette) concurred with the patients and the public. Even in the United States, five of the nine (57%) family samples (and four of the five when asked about their own relative) endorsed psychosocial causes more than bio-genetic ones.

A study of a 'psycho-education' programme designed to teach the illness model found that before the training only 14% of relatives believed 'genetic inheritance' was a cause, and only 24% believed this even after being 'educated' (McGill *et al.* 1983). (Only 6% of the patients believed it before, or after, the programme).

In an Italian survey of 709 relatives of people diagnosed 'schizophrenic', the most commonly endorsed causes were 'stress' and 'psychological traumas', with 68% believing that schizophrenia is caused entirely by psychosocial factors (Magliano *et al.* 2004a). Similarly, 55% of Indian relatives cited social stressors, whereas only 5% cited heredity and 14% brain disorder (Srinivasan and Thara 2001).

Mental health staff

There has been insufficient research into the causal beliefs of staff. An Italian study found that the cause most frequently endorsed by psychiatrists and nurses was 'heredity', but relatives were most likely to endorse 'stress' and 'psychological traumas' (Magliano *et al.* 2004a). A US study found that patients, family members and the public all endorsed 'a non-biomedical view of mental illness' more frequently than clinicians. Patients were the most likely (66%), and clinicians the least likely (18%), to cite 'the way he was raised'. Clinicians were significantly more likely to endorse 'genetics' as a cause than the other three groups. Patients were less likely than clinicians to endorse 'chemical imbalance'. 'Stressful circumstances' was cited significantly

more often by both patients and family members than by clinicians (van Dorn *et al.* 2005).

Of a random sample of 154 British psychiatrists 86% cited 'biochemical abnormalities' and 87% 'family history/genetics'; only 22% cited 'childhood factors' (Baillie *et al.* 2009). A larger survey, of 2,813 British psychiatrists, found that 0.4% thought the causes of schizophrenia are 'primarily social' while 46.1% thought they are 'primarily biological' (Kingdon *et al.* 2004). Thus for every psychiatrist who agrees with the majority of patients, family members and the public, 115 do not. The sad scenario of a distressed person trying to explain their difficulties, and the causes, to a psychiatrist trying to help but who has been trained to see only 'symptoms' of an 'illness' (including 'lack of insight') occurs thousands of times every day all over the world.

This is sad not only for patients and clinicians. There are numerous subjective accounts of the frustration of family members caused by purely bio-genetic theories and the over-reliance on medication and hospitalization that follows (Read and Magliano 2012).

THE PUBLIC'S REJECTION OF MEDICAL TREATMENTS

The public has an equally strong preference for psychosocial solutions over medical interventions for 'schizophrenia'. This has been found in 15 countries (Table 11.2). (The one exception, in Greece, used the label 'schizophrenia' (Economou *et al.* 2009). Most of the other studies used an unlabelled vignette).

For example, the most preferred treatment for schizophrenia in Germany is psychotherapy (65%), compared to psychiatric drugs (15%) and electroconvulsive therapy [ECT] (1%) (Riedel-Heller *et al.* 2005). In Canada, 90% or more endorse work/recreation opportunities, involvement of family/friends and group homes. Only 49% endorse drug treatment and 42% (only 18% of relatives) psychiatric hospitals (Thompson *et al.* 2002). When Japanese are asked what might help best, 7% mention medication, 44% talking to friends/family and 74% counselling (Jorm *et al.* 2005b).

In Australia, the public rate psychotherapy as 'helpful' and antipsychotic medication as 'harmful' (but not as harmful as psychiatric hospitals or ECT). They are also more likely than staff to rate counsellors and clinical psychologists as 'helpful' and to endorse: help from family and friends, becoming physically active, self-help books, stress management, meditation, relaxation and yoga (Jorm *et al.* 2005a). Similar differences between staff and the public regarding belief in medication vs. psychosocial approaches have been found in Switzerland (Lauber *et al.* 2005).

For Australians, the *reasons* for rejecting antipsychotic medication include that they 'have more risks than benefits', 'lack efficacy because they do not deal with the roots of the problem', and 'are prescribed for the wrong reasons

Table 11.2 Public preference for medication or psychosocial treatments for 'schizophrenia'

Country	Study	Stimulus	Preference for medicine vs psychosocial treatments
<i>Australia</i>	Jorm <i>et al.</i> 2005a	vignette	PS
	Wright <i>et al.</i> 2005	vignette	PS
	Wong and Xuesong 2011 [c]	vignette	PS
<i>Austria</i>	Katschnig <i>et al.</i> 1993	label	PS
	Jorm <i>et al.</i> 2000	vignette	PS
<i>Canada</i>	Norman and Malla 1983 [a]	vignette	PS
	Thompson <i>et al.</i> 2002	label	PS
	Stip <i>et al.</i> 2006	label	MD=PS
<i>China</i>	Wong and Xuesong 2011	vignette	PS
<i>England</i>	Furnham and Bower 1992	label	PS
<i>Germany</i>	Angermeyer and Matschinger 1994	vignette	PS
	Benkert <i>et al.</i> 1997	label	PS
	Angermeyer and Matschinger 2005	vignette	PS
	Riedel-Heller <i>et al.</i> 2005	vignette	PS
	Gaebel <i>et al.</i> 2008	label	PS
	Angermeyer <i>et al.</i> 2009	vignette	PS
	Economou <i>et al.</i> 2009	label	MD
<i>Hong Kong</i>	Wong & Xuesong 2011	vignette	PS
<i>India</i>	Kermode <i>et al.</i> 2009	vignette	PS
<i>Italy</i>	Magliano <i>et al.</i> 2004b	vignette	PS
	Pantusa <i>et al.</i> 2007	label	PS
<i>Japan</i>	Jorm <i>et al.</i> 2005b	vignette	PS
<i>Russia</i>	Angermeyer and Matschinger 2005	vignette	PS

<i>Slovakia</i>	Angermeyer and Matschinger 2005	vignette	PS
<i>South Africa</i>	Hugo <i>et al.</i> 2003	vignette	PS
	Sorsdahl and Stein 2010	vignette	PS
<i>Switzerland</i>	Lauber <i>et al.</i> 2005	vignette	PS
<i>Turkey</i>	Sagduyu <i>et al.</i> 2001	label	PS
<i>United States</i>	Kuppin & Carpiano 2006	vignette	PS=MD

Notes

MD = Medication. PS = Psychosocial treatments. [c] = Chinese. [a] adolescents.

(e.g., to avoid talking about problems, to make people believe things are better than they are, as a straitjacket' (Jorm *et al.* 2000: 404).

A 2006 review found that 'the public's beliefs about psychological interventions (e.g. psychotherapy and counselling) are predominantly favourable, while very negative views prevail about pharmacological treatments'. Surprisingly, they also found that 'the particular liking of psychotherapy is more developed for schizophrenia than for depression' (Angermeyer and Dietrich 2006). Even in the United States the public values 'talk to family/friends' (94%), 'therapist' (86%) and 'self-help group' (83%) as much as 'prescription medication' (85%) (Kuppin and Carpiano 2006).

When Austrians are asked what they would do if a relative were to become psychotic, the most common response is 'talk to them' (Jorm *et al.* 2000). Similarly, for South Africans the best treatment is 'talk it over with someone' (Hugo *et al.* 2003).

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PREJUDICE, STIGMA AND 'SCHIZOPHRENIA'

The role of bio-genetic ideology

John Read, Nick Haslam and Lorenza Magliano

Of the many psychosocial causes of the suffering of people who experience psychosis, one of the most pervasive and destructive is discrimination. Many people describe discrimination as more long-lasting and disabling than the psychosis itself (Schulze and Angermeyer 2003), and it is recognized as a major barrier to recovery (Corrigan and Kosyluk 2013; NICE 2009).

The stereotype of the 'schizophrenic', with the toxic combination of dangerousness and unpredictability at its core, is remarkably consistent over place and time. The stereotype of the violent 'madman' remains despite people diagnosed with psychosis actually being more likely to be assaulted than to assault others (Morgan *et al.* 2012). Additional components of the stereotype include: worthless, cold, foolish, dirty, insincere and not responsible for one's own behavior (Read *et al.* 2006). Attitudes seem to be worsening (Schomerus *et al.* 2012). In the United States, the perception of dangerousness, and unwillingness to have a 'schizophrenic' as a neighbour, increased between 1996 and 2006 (Pescosolido *et al.* 2010). The German public's desire for distance from people diagnosed 'schizophrenic' increased between 1990 and 2001 (Angermeyer and Matschinger 2005a).

These attitudes are acted on. Discrimination when seeking work, housing or insurance and rejection by friends and families are commonplace (Read *et al.* 2006; Schulze and Angermeyer 2003; Thornicroft *et al.* 2009). Prejudice is fuelled by the media, not only via sensationalist attributions of violence to 'schizophrenics' (Cain *et al.* 2013; Castiello and Magliano 2007) but also by metaphorical use of the term 'schizophrenic' to denigrate people as unpredictable and irrational (Magliano *et al.* 2011a).

Relatives also experience prejudice (Angermeyer *et al.* 2003). Negative attitudes exist, too, among some health and mental health staff (Magliano *et al.* 2002; Sartorius 2002; Uçok *et al.* 2004). A survey of Australian service users and their families found that 'they experience more stigma and

discrimination from mental health professionals than from any other sector of society' (Walter 1998: 70).

A recent study further suggests that, for people diagnosed 'schizophrenic', having been abused as a child leads to particularly high levels of experienced discrimination, internalized stigma and anger towards self (Outcalt and Lysaker 2012).

Discrimination, in combination with anticipated discrimination (Angermeyer *et al.* 2004; Thornicroft *et al.* 2009) and internalized stigma (Dinos *et al.* 2004), leads, unsurprisingly, to decreased life satisfaction and self-esteem, and to increased alcohol use, depression and suicidality (Read *et al.* 2006). A recent study of 732 people diagnosed with 'schizophrenia', from 27 countries, found that 64% had stopped themselves from applying for work, training or education, and 58% from looking for a close relationship, because of anticipated stigma (Ucok *et al.* 2012). Anticipated discrimination was strongly related to actual discrimination.

DESTIGMATIZATION PROGRAMMES

'Mental illness is an illness like any other'

The core assumption of most anti-stigma programmes is that the public should be taught to recognize the problems in question as illnesses or diseases, and to believe that they are caused by biological factors like chemical imbalance, brain disease and genetic heredity. The thinking behind this well intentioned 'mental illness is an illness like any other' approach is that if we can't control our behavior, we can't be held responsible and, therefore, can't be blamed. It is the 'mad not bad' argument, with the mad part portrayed as biologically based illnesses (Figure 12.1).

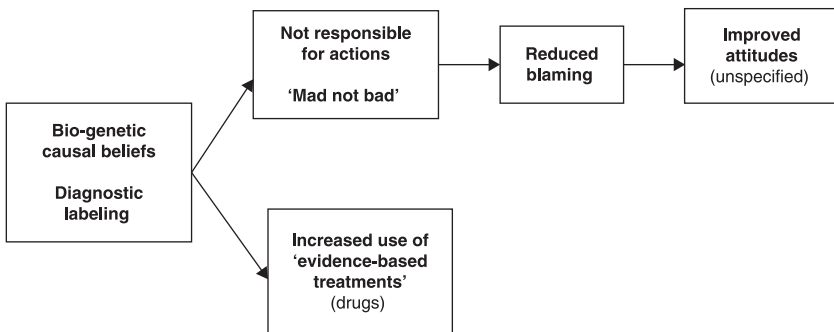


Figure 12.1 'Mental Health Literacy': The plan.

The goal is to get us all to view madness as biological psychiatry sees it. The public, however, is firmly convinced – internationally – that madness is caused more by bad things happening than by genetics or chemical imbalances (Read *et al.* 2006; Read 2007; and Chapter 11). This battle has been going on for a long time. By 1961, the US Joint Commission on Mental Illness and Health had concluded:

The principle of sameness as applied to the mentally sick versus the physically sick has become a cardinal tenet of mental health education. . . . Psychiatry has tried diligently to make society see the mentally ill in its way and has railed at the public's antipathy or indifference.

(JCMIH 1961: 59)

'Mental health literacy'

More recently, the battle has been fought under the banner of improving our 'mental health literacy'. In 1997 Australian psychologist Anthony Jorm introduced the term, defining it as 'knowledge and beliefs about mental disorders which aid their recognition, management or prevention' (Jorm *et al.*, 1997). This sounds innocent enough until you understand exactly what 'knowledge and beliefs', and what 'management' techniques, are considered 'literate'. We are 'literate' if we can 'recognize' that certain behaviours are examples of a particular psychiatric illness, if we 'know' that these are biologically caused, and if we 'believe' in psychiatric drugs. In 2000 Jorm wrote: 'What is most surprising is that psychological interventions are seen by the public as highly effective for psychotic disorders' (p. 397). 'If the public's mental health literacy is not improved, this may hinder public acceptance of evidence-based mental health care' (p. 396), meaning drugs.

Jorm is not alone in implying that the millions of us who understand that bad things happening can drive us crazy, and who prefer human to chemical solutions, are suffering from 'mental health illiteracy.' A study (Thompson *et al.* 2002) emanating from the World Psychiatric Association campaign to improve attitudes about schizophrenia (Sartorius 2002) portrayed the belief that schizophrenia is a 'debilitating disease', caused predominantly by biochemical imbalance as 'sophisticated' and 'knowledgeable'. The study was funded by drug company Eli Lilly. Another study measured 'knowledge' with items like 'good mental health is the absence of brain disease' (Aghanwa 2004). The South African public's belief in social causes, and their rejection of medication, are characterized as 'misinformation' demonstrating the 'need to address ignorance' (Hugo *et al.* 2003). One of the questions used to assess 'schizophrenia knowledge' is: 'Schizophrenia is caused by poor parenting'; the 'correct' answer is 'No' (Faulkner *et al.* 2010). 'Psychosis literacy' is measured by 'illness attributions' and rejection of the 'social world' as a relevant

construct (Lopez *et al.* 2009). The role of the drug companies in all this is evidenced by a study showing that more than half of websites about 'schizophrenia' are drug-company-funded and that these industry sponsored sites are significantly more likely than other sites to portray 'schizophrenia' as a biologically based illness (Read, 2008). While this has been found to be the case for websites about other mental health problems, the promotion of the illness model by drug company funded websites is particularly pronounced in 'schizophrenia' websites (Read and Cain, in press). A recent Canadian study of 'psychosis literacy' actually instructed participants that a 'psychotic disorder is a disease of the brain' and, unsurprisingly, found that 76% 'thought the main cause of a psychotic episode was a chemical imbalance in the brain', which the researchers then described as 'a high level of knowledge of aetiology' (Addington *et al.* 2012).

This has also been going on, for just as long, inside mental health services. Family members are sold the illness model of schizophrenia in what, paradoxically, have been called 'psycho-education' programmes (Chapter 26). Meanwhile, patients who, like the public, insist that their problems have resulted from adverse life events (the vast majority – Chapter 11) are told they are lacking 'insight' into the fact that they have an 'illness'. This is the exact opposite of the original psychodynamic meaning of the term (and the meaning used by many laypersons) – that is, our ability to relate our current problems to past events or unconscious motives. A drug-company-sponsored American psychiatrist has gone so far as to claim that this 'lack of insight' – which he calls 'anosognosia' – is 'a manifestation of the illness' of schizophrenia and has a biological cause (Amador 2010).

THE NEGATIVE EFFECTS OF 'MENTAL HEALTH LITERACY'

The relationship of causal beliefs to attitudes

These campaigns to improve our 'mental health literacy', and thereby reduce stigma, claim to be designed to increase our 'evidence-based' knowledge (that 'mental illnesses' are biologically caused) and to increase our use of 'evidence-based' treatments (drugs). The campaigns themselves, however, are far from evidence-based. They have, for decades, ignored the research about what increases and decreases prejudice.

Table 12.1 documents 57 findings about the relationships between causal beliefs and attitudes about 'schizophrenia', 'psychosis' or 'mental illness', generated by 39 studies from 16 countries. The studies are of two general types. The 22 correlational studies report the relationships between biogenetic or psychosocial causal beliefs and either attitudes (using semantic differentials such as 'safe/dangerous', 'predictable/ unpredictable' etc.) or

Table 12.1 Studies finding significant relationships between bio-genetic or psychosocial causal beliefs and positive or negative attitudes

<i>Studies</i>	<i>Sample</i>	<i>Country</i>	<i>Category</i>	<i>Causal belief</i>	<i>Attitude</i>
<i>Correlational studies</i>					
Golding <i>et al.</i> 1975	public adolescents	USA	mental illness	BIO	—
Norman and Malla 1983		Canada	schizophrenia	PS	+
Chou and Mack 1998	public	Hong Kong	mental illness	BIO	—
Kent and Read 1998	staff	New Zealand	mental illness	BIO	—
Martin <i>et al.</i> 2000	public	USA	mental illness	BIO	—
				BIO	+
Magliano <i>et al.</i> 2001	relatives	Italy	schizophrenia	PS	+
Read and Harre 2001	students (psyc)	New Zealand	mental illness	BIO	—
Angermeyer and Matschinger 2003	public	Germany	schizophrenia	BIO	—
				PS	+
Magliano <i>et al.</i> 2003	public	Italy	schizophrenia	BIO	—
				PS	+
Taskin <i>et al.</i> 2003	public	Turkey	schizophrenia	BIO	—
				PS	—
Dietrich <i>et al.</i> 2004	public	West Germany	schizophrenia	BIO	—
				PS	+
		East Germany		BIO	—
		Russia		BIO	—
		Mongolia		PS	+
				BIO	—
Magliano <i>et al.</i> 2004	public	Italy	schizophrenia	PS	+
Angermeyer and Matschinger 2005a*	public	Germany	schizophrenia	PS	—
Phelan <i>et al.</i> 2006	public	USA	schizophrenia and depression	BIO	—

(Continued overleaf)

Table 12.1 Continued

<i>Studies</i>	<i>Sample</i>	<i>Country</i>	<i>Category</i>	<i>Causal belief</i>	<i>Attitude</i>
Grausgruber <i>et al.</i> 2007	public	Austria	schizophrenia	PS	+
Jorm and Griffiths 2008	public	Australia	schizophrenia	BIO	—
Kernode <i>et al.</i> 2009	public and health workers	India	depression and psychosis	PS	+
Naanyu 2010	public	South Africa	schizophrenia and depression	BIO	—
Pescosolido <i>et al.</i> 2010*	public	USA	depression	BIO	—
Rüsch <i>et al.</i> 2010	public	USA	schizophrenia	BIO	—
	patients		schizophrenia	BIO	—
Magliano <i>et al.</i> 2011b	students (med)	Italy	schizophrenia	BIO	—
Peluso and Blay 2011	public	Brazil	schizophrenia	BIO	—
<i>Experimental studies</i>					
Rothaus <i>et al.</i> 1963	employers	USA	mental illness	PS	+
Morrison 1976	patients	USA	mental illness	PS	+
Morrison 1977	patients	USA	mental illness	PS	+
Morrison <i>et al.</i> 1978	students	USA	mental illness	PS	+
Fisher and Farina 1979	patients	USA	self	BIO	—
Morrison <i>et al.</i> 1979	students	USA	mental illness	PS	+
Morrison and Teta 1979	students	USA	mental illness	PS	+
Morrison and Teta 1980	students	USA	mental illness	PS	+
Mehra and Farina 1997	students (psy)	USA	mental illness	BIO	—
Read and Law 1999	students (psy)	New Zealand	mental illness	PS	+
Walker and Read 2002	students	New Zealand	schizophrenia	BIO	—
Schulze <i>et al.</i> 2003	adolescents	Germany	schizophrenia	PS	+
Lam <i>et al.</i> 2005	public	England	mental illness	PS	+

Breheny 2007	students	New Zealand	schizophrenia	BIO	+
Bennett <i>et al.</i> 2008	students	Wales	depression	BIO	-
Lincoln <i>et al.</i> 2008	students (psy)	Germany	schizophrenia	BIO	-
	students (med)		schizophrenia	PS	+
Campbell <i>et al.</i> 2011	adolescents	England	psychosis	BIO	+
				PS	+

Notes

BIO = bio-genetic causal beliefs. PS = psycho-social causal beliefs. +: positive attitudes. -: Negative attitudes (e.g. dangerousness, unpredictability, fear, desire for distance, pessimism about recovery, perceived severity of disturbance).

*Trend analysis over time period during which both bio-genetic beliefs and negative attitudes increased

desire for social distance (using 'behavioural intention' questions about willingness to be friends with, work with, etc.). The 17 experimental studies involve manipulations such as showing a video of the same person, but with different causal explanations, to two groups of participants.

Correlational studies

The 22 correlational studies, spanning 35 years and 14 countries, generated 37 significant findings (see Table 12.1). For 23 of the 24 significant findings about bio-genetic causal beliefs the relationship was with *negative* attitudes. For 12 of the 13 significant findings about psychosocial beliefs the relationship was with *positive* attitudes.

For example, a survey of 1,550 South Africans found that 'Biogenetic causes are associated with intolerance, while non-biogenetic attributes reveal tolerant attitudes' (Naanyu 2010). A study of 5,025 Germans found that belief in 'brain disease' and 'heredity' increased perceived dangerousness, fear and desire for distance (Dietrich *et al.* 2006). In Italy naming heredity as a cause of schizophrenia is associated with pessimism about recovery in medical students, relatives and lay people (Magliano *et al.* 2011b, 2001, 2003).

A German study comparing beliefs in 1990 and 2001 found that 'Although the endorsement of biological causes increased substantially, the public's rejection of people with schizophrenia increased in the same period'. In both 1990 and 2001 biological causal beliefs were related to greater desire for distance (Angermeyer and Matschinger 2005a: 333). Similarly the US public's belief in 'chemical imbalance' and 'genetic problem' as causes of 'schizophrenia' increased significantly between 1996 and 2006 (along with endorsement of psychiatric drugs), as did 'unwillingness to have as a neighbour' (Pescosolido *et al.* 2010).

Experimental studies

The 17 experiments, in six countries and spanning 50 years, produced 20 significant findings. In five of the seven findings about bio-genetic beliefs, they made attitudes worse. In all 13 cases psychosocial beliefs improved attitudes.

In the 1970s James Morrison conducted six studies, on students and patients, all finding that seminars critiquing a medical model and presenting psychosocial perspectives significantly improved attitudes (see Table 12.1). These findings were replicated with New Zealand students, using lectures with similar themes. Perceptions of 'dangerous', 'unpredictable' and 'antisocial' were all significantly reduced (Read and Law 1999). Presenting a psychosocial explanation to patients induces more efforts to change than a disease explanation (Fisher and Farina 1979).

A New Zealand study found that viewing a video of a person describing hallucinations and delusions followed by a bio-genetic explanation

significantly increased perceptions of dangerousness and unpredictability. A video of the identical behaviours explained as reactions to adverse life events led to a small (non-significant) improvement in attitudes (Walker and Read 2002). Similarly, students in Wales were significantly more likely to assume the person presented in a schizophrenia vignette was dangerous, and unlikely to recover, if the vignette was followed by a genetic explanation than an explanation that his wife had recently died (Bennett *et al.* 2008).

Only one experiment has measured actual behaviour change. Participants in a learning task increased the intensity of electric shocks more quickly if they understood their partner's difficulties in disease terms than if they believed they were a result of childhood events (Mehta and Farina 1997).

Combining the correlational and experimental studies, in 28 of 31 cases (90%) bio-genetic causal beliefs are related to negative attitudes. In 24 of 26 cases (92%) psychosocial beliefs are related to positive attitudes. In total, 52 of the 57 findings (91%) show that the vast expenditure of drug company and taxpayers' money on biologically oriented destigmatization programmes is not evidence-based.

A sociologist recently noted that 'Although the public increasingly endorses a biomedical model, including a growing appreciation of genetic causes, the stigma of mental illness remains' (Schnittker 2013). An article published by the Canadian Health Services Research Foundation (CHSRF 2012), in a series of papers 'using evidence to debunk common misconceptions in Canadian healthcare', concluded that 'evidence actually shows that anti-stigma campaigns emphasizing the biological nature of mental illness have not been effective, and have often made the problem worse'.

A review, by possibly the most prolific research team in this field, found that 'Generally, biogenetic causal attributions were not associated with more tolerant attitudes; they were related to stronger rejection in most studies examining schizophrenia' and concluded that 'Biogenetic causal models are an inappropriate means of reducing rejection of people with mental illness' (Angermeyer *et al.* 2011: 367). The same team reviewed the relatively few studies that have tracked attitudes and causal beliefs over time and found a trend 'towards a biological model of mental illness' but 'at the same time, attitudes towards persons with mental illness have not changed for the better, and have even deteriorated towards persons with schizophrenia' (Schomerus *et al.* 2012). They added: 'While the approach to depict mental disorder as an "illness like any other" and to emphasize its biological correlates seems useful to enhance the acceptance of professional medical treatment for mental disorders, it is not suitable to improve social tolerance. Dissemination of biological knowledge is not a solution to discrimination and stigmatization.' It can be concluded, then, that the outcome of spending millions of dollars (largely by drug companies) to teach the public to think more like biological psychiatrists has had two outcomes: more discrimination and more drugs.

Furthermore, a 1980 US National Institute of Mental Health report had found that treatment by medical professionals or modalities, or in medical settings, is more stigmatizing than other approaches. Medication for psychosis causes drooling, tremors, obesity, drowsiness, and sexual dysfunction (Chapter 9). An Italian study found that 68% of people diagnosed with 'schizophrenia' believe it is easy to recognize people who are taking these drugs (Magliano *et al.* 2008). A recent review found that 'stigma associated with antipsychotic medications is not only pernicious, but pervasive across an array of personal and social contexts' (Sajatovic and Jenkins 2007: 111).

The relationship of diagnostic labelling to attitudes

Those attempting to improve our 'mental health literacy' also bemoan our inability to recognize and name 'mental illnesses' like 'schizophrenia' (Ganasen *et al.* 2008; Jorm *et al.* 1997). Labelling does increase belief in bio-genetic causes (Angermeyer *et al.* 2009; Jorm *et al.* 1997; Magliano *et al.* 2003, 2011b; and Chapter 11). It also accomplishes another of the goals of illness-based destigmatization programmes; increased endorsement of psychiatric drugs and psychiatrists (Angermeyer and Dietrich 2006; Angermeyer *et al.* 2009).

In terms of stigma, however, labelling is just as damaging as bio-genetic causal beliefs. Studies of our ability to label certain behaviours 'schizophrenic', or to describe a vignette of a 'schizophrenic' person as 'mental illness' show that such labelling is related to:

- *perceived dangerousness* (Angermeyer and Matschinger 2005b; Angermeyer *et al.* 2004; Kingdon *et al.* 2008; Martinez *et al.* 2011; Peluso and Blay 2010; Wright *et al.* 2011);
- *perceived unpredictability* (Angermeyer and Matschinger 2005b; Kingdon *et al.* 2008; Magliano *et al.* 2003, 2004; Wright *et al.* 2011);
- *perceived lack of responsibility for own actions* (Penn and Nowlin-Drummond 2001);
- *perceived lack of 'humanity'* (Martinez *et al.* 2011);
- *perceived severity of the problem* (Cormack and Furnham 1998; Magliano *et al.* 2003);
- *perceived dependency* (Angermeyer *et al.* 2004);
- *pessimism about recovery* (Angermeyer and Matschinger 1996; Kingdon *et al.* 2008; Magliano *et al.* 2003, 2011b);
- *fear* (Angermeyer and Matschinger 2003);
- *rejection and desire for distance* (Angermeyer *et al.* 2004, 2009; Magliano *et al.* 2011b; Martin *et al.* 2000; Stilton *et al.* 2011; Taskin *et al.* 2003).

It is unsurprising, therefore, that the majority of people who are given a diagnosis of 'schizophrenia' conceal it from others (Ucok *et al.* 2012).

A study of 166 Scottish GPs shows that labelling also affects professionals and can impact the provision of care. The GPs read case vignettes that were identical apart from mention of a previous diagnosis of schizophrenia, depression, diabetes, or no illness. Those responding to the vignette mentioning 'schizophrenia' were more concerned about possible violence and less willing to treat that patient (Lawrie *et al.* 1998).

This relationship between labelling and negative attitudes also exists within people who experience psychosis. A British study found that 'patients who accepted their diagnosis reported a lower perceived control over illness' and that depression in psychotic patients is 'linked to patients' perception of controllability of their illness and absorption of cultural stereotypes of mental illness' (Birchwood *et al.* 1993: 387). Many psychiatrists, aware of the damage done by the 'schizophrenia' label, suggest that the diagnosis be kept secret (Sartorius 2002). In an Italian study, only 30% of people who had been diagnosed with schizophrenia had been told (Magliano *et al.* 2008). In another study psychiatrists told patients their diagnosis 85% of the time for depression but only 17% of the time for 'schizophrenia' (Ucok *et al.* 2004).

A recent independent inquiry into the 'schizophrenia' label in the United Kingdom surveyed over 500 people diagnosed 'schizophrenic' and found that over 80% believe the diagnosis is 'damaging and dangerous' (ISL 2012).

Imposing 'mental health literacy' on other cultures

Some of those promoting the illness model to their own citizens, thereby increasing prejudice and fear, also try to impose their beliefs on other countries and cultures (Taitimu and Read 2006). Psychiatrists have bemoaned the poor 'mental health literacy' of people in India (Kermode *et al.* 2009), Pakistan (Zafar *et al.* 2008), Bali (Kurihara *et al.* 2006), Nigeria (Ohaeri and Fido 2001), Malawi (Crabb *et al.* 2012) and of the Xhosa people of South Africa (Mbanga *et al.* 2002). The definition of being 'mental health illiterate' has been expanded to 'also mean that the knowledge and beliefs held may be derived from other sources, such as superstitions or cultural and personal beliefs' (Ganasen *et al.* 2008: 23). Terms like 'misinformation' and 'lack of knowledge' are used to describe spiritual and social causal beliefs. A typical conclusion is 'Interventions aimed at increasing the mental health literacy of traditional healers are essential' (Sorsdahl *et al.* 2010). All these researchers express concern that other cultures' beliefs limit the use of psychiatric drugs.

There seems to be no awareness that supplanting indigenous beliefs with those of a dominant culture is a cornerstone of colonization and is also a primary contributor to the poor mental health of colonized peoples (Johnstone and Read 2000). Also ignored is the fact that recovery rates for 'schizophrenia' are higher in 'developing' countries than in 'developed' countries (Chapter 5). Perhaps these countries have something to teach about how to reduce prejudice (Rosen 2003).

The same process occurs *within* nations. Atlanta researchers are alarmed that low-income African-Americans insist on believing in 'debunked theories of schizophrenia that focused on the family's effect on causing schizophrenia' (Broussard *et al.* 2010). Researchers in California bemoan the poor 'psychosis literacy' of Latinos, especially their failure to make 'illness attributions' and their insistence that the 'social world' is important in understanding psychosis (Lopez *et al.* 2009).

HOW DOES THE MEDICAL MODEL PRODUCE NEGATIVE ATTITUDES?

The evidence shows us that while the public may assign less blame to individuals for their biologically-determined mental illness, the very idea that their actions may be beyond their conscious control can create fear of their unpredictability and thus the perception that those with mental illness are dangerous, leading to avoidance. Biological explanations can also instill an 'us vs. them' attitude, defining individuals with mental illness as fundamentally different. . . . Similarly, mental illnesses are seen as less responsive to treatment and more persistent and serious when framed as biological diseases. This framing may suggest that people with mental illnesses will never recover, which contributes to stigmatizing attitudes. . . . Presenting mental illness in the context of psychological and social stressors normalizes symptoms, creating a healthier public perception of mental illness.

(Canadian Health Services Research Foundation 2012: 3)

Efforts to reduce our fear and prejudice by trying to teach us to think like biological psychiatrists, and to learn to diagnose unusual behavior, which have been going on for over 60 years now, have been an abject failure. Why has it not worked? Why do biological causal beliefs make things worse rather than better?

Recent research offers some answers to these questions, using the concepts of 'essentialism' and 'dehumanization'. Essentialist thinking involves seeing social groups as having fixed underlying essences that define who they are. This view of people as falling into deep-seated 'natural kinds', much as we see animals falling into distinct species, is widespread and can have dire consequences. For example, people who hold essentialist beliefs about race tend to show higher levels of racial prejudice and essentialist views of gender promote sexist attitudes (Keller 2005). These negative implications of essentialist thinking appear to be especially marked when the imagined essences are genetic or otherwise biological.

There is now ample evidence that many people conceive of mental health problems in essentialist ways (Ahn *et al.* 2006; Haslam and Ernst 2002) by endorsing some version of the biomedical disease model. On this understanding, people who experience problems are discrete kinds of person who have a fundamental, biological flaw that is essentially unalterable and beyond their personal control. This analysis suggests four ways in which essentialist thinking promotes stigma (Haslam 2011).

First, people who tend to essentialize human variation are especially likely to endorse social stereotypes (Bastian and Haslam 2006). Essentialist thinking about mental health problems should therefore promote agreement with society's prevailing negative images of 'mental illness'. A recent study has found that members of the public who believe that there is a continuum between 'schizophrenia' and 'normality' hold fewer stereotypes about people with the diagnosis (Wiesjahn *et al.* 2013). In short, essentialist thinking carries with it a tendency to accept uncritically the derogatory stereotypes that attach to mental health problems.

Second, because essentialist thinking leads people to view social groups as discrete categories, bio-genetic explanations encourage the view that people who display unusual behaviour are categorically and fundamentally different from normality, as if they are a distinct species (Mehta and Farina 1997). Benning *et al.* (2006: 89) note how psychiatric stigmatization involves 'an exaggerated attribution of 'other-ness'. This attribution, in turn, reduces motivation to interact with people seen as deviant: the 'social distance' dimension of stigma that has repeatedly been shown to correlate with bio-genetic explanations.

A third reason why essentialist thinking increases stigma is that it represents unusual behaviour as something that is caused by a biological aberration outside the person's control. Although this belief in uncontrollability may decrease moral blame, this does not ensure positive attitudes. Hinshaw and Stier (2008: 379) argue 'the belief that negative but uncontrollable traits are always viewed benignly is simply wrong'. Denying personal control implicitly removes people from the rights and responsibilities of the wider community and portrays them as untamed and uncontrollable. As Campbell *et al.* (2011: 42) note, the 'no-fault brain disease' model leads people to consider those with mental illness as not accountable for their actions, which is a key component of the unpredictable stereotype associated with psychosis. Similarly, viewing distressed people as lacking personal control may lead people to see the imposition of harsh and patronizing forms of control as appropriate (Mehta and Farina 1997; and Chapters 2 and 4). Although a recent review found no general population studies reporting on the relationship between bio-genetic causal beliefs about schizophrenia and perceived responsibility (Angermeyer *et al.* 2011), a small Chicago study found that genetic beliefs were associated with lower perceived responsibility and increased desire for distance (Rüsch *et al.* 2010).

Fourth, essentialist thinking involves a tendency to see social groups as having fixed, unalterable natures. As a result, it promotes a pessimistic view of mental health problems: the person experiencing them is essentially and unchangeably deviant. Essentialist beliefs about abnormality, exemplified by a causal explanation in which bio-genetic flaws underlie unusual behaviour and experience, therefore lead people to see problems as permanent, forever tarnishing the identity of the person.

Essentialist thinking is also linked to the concept of dehumanization (Haslam 2006). If essentialist thinking leads us to view those with mental health problems as categorically different, uncontrollable and untamed, it should be no surprise that they will also be seen as less fully human. As Martinez *et al.* (2011: 10) argue, ‘when people are confronted with a general label of mental illness, in the absence of other information, their default response is to accord the targeted individual with lowered human status’. These authors found that labelling people as having chronic mental illness led participants to ascribe lesser humanity to them, which in turn was associated with perceiving them as threatening and dangerous. Thus the combination of bio-genetic explanations and labelling can trigger essentialist and dehumanizing perceptions that can account for many of the key elements of psychiatric stigma.

Figure 12.2 attempts to portray the interactions between the research summarized in this chapter, and contrasts the claims made by proponents of the illness-model approach to destigmatization with what research shows is actually happening (Figure 12.1). Given that the only promise kept is that of increasing endorsement and use of psychiatric drugs, it is easy to see why drug companies are keen to fund these ‘destigmatization’ programmes.

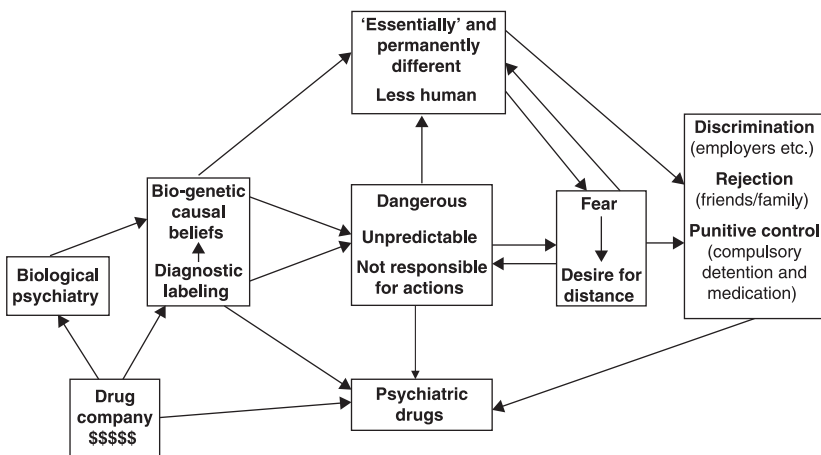


Figure 12.2 ‘Mental Health Literacy’: The outcomes.

ALTERNATIVES

Destigmatization programmes that combat prejudice by portraying mental health difficulties as understandable reactions to adverse events will not only be based on sound evidence that this approach works, they will also have solid evidence behind them to show that social factors are indeed causal (Chapters 14–19). They will also have the added bonus of 'speaking the same language' as the public rather than trying to promulgate a set of causal beliefs that the public everywhere rejects (Chapter 11).

It is not clear why social causal beliefs improve attitudes. Perhaps the 'social' is more important than the 'causal'. It seems *any* normal information about a person increases 'ascribed humanity' (Martinez *et al.* 2011). Indeed, the strongest predictor of low fear and discrimination is amount of contact with the people who are the object of the discrimination (Angermeyer and Dietrich 2006; Angermeyer *et al.* 2004; Read and Harre 2001; Read and Law 1999; Schulze *et al.* 2003). It seems unfortunate, therefore, that one goal of the illness model approach, that of increasing confidence in medical professionals and treatments, seems to reduce confidence in the helping capacities of ordinary members of the public (Riedel-Heller *et al.* 2005), thereby reducing precisely the sort of contact needed to combat prejudice.

Perhaps anti-discrimination work in the mental health field has been overly focused on causality, in ways that have not occurred in other areas. In challenging the discrimination faced by people who use wheelchairs, the cause of their impairment is irrelevant. What is relevant is their right to access social and economic life (Read *et al.* 2006; Sayce 2000, 2003; Sayce and Curran 2007).

An increasing number of community-based efforts to reduce discrimination against people labelled mentally ill either do not adopt a medical paradigm or ignore causality altogether. The principles on which many of these programmes are based include, in addition to the crucial one of increasing contact: acknowledging and valuing differences; promoting positive participation and contributions by people with mental health problems; addressing the power differentials that underpin discrimination; targeting different groups (gender, age, ethnicity, etc.) differently; tackling stereotypes about violence (including efforts to constrain the media in this area); legislation to decrease discrimination; and including the people on the receiving end of the discrimination in the design and management of programmes (Corrigan and Kosyluk 2013; Estroff *et al.* 2004; French *et al.* 2011; Harrison and Gill 2010; Hinshaw and Stier 2008; Laroie and van der Linden 2009; Pinfold *et al.* 2005; Sayce 2000, 2003; Sayce and Curran 2007; Schulze *et al.* 2003).

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LISTENING TO THE VOICES WE HEAR

Clients' understandings of psychotic experiences

Jim Geekie

Schizophrenia is the lynchpin of models of mental illness. Regardless of what we as individuals make of this term, it would be difficult to dispute that it is *the* central concept around which Western notions and practices of mental health care revolve. Rather than – or as well as – viewing schizophrenia as the ‘lost cause’ of psychiatry, we could view it as the ‘Holy Grail’ of mental health workers, including, but not limited to, psychiatrists. The term ‘schizophrenia’ also has a broader significance in Western culture, beyond the field of mental health. The common everyday translation of the term is, I propose, roughly equivalent to ‘madness’ or ‘insanity’. Thus, we could view our search for the Holy Grail of schizophrenia in broader terms as reflecting the efforts of our culture to come to grips with ‘madness’.

I discuss this search for the Grail, focusing not on what has been found – or, more accurately, not found – but on certain aspects of the nature of the search itself. In particular, my interest here is in who it is that has been deemed able to contribute to this search and who has been excluded. I argue that the very people who have, by definition, the closest relationship with experiences considered indicative of schizophrenia – clients of mental health services who happen to have that diagnosis – have been largely excluded from the search. I locate this discussion within the context of power relationships, drawing on the work of Foucault. I then discuss how mental health workers may contribute to remedying this situation by paying greater attention to the sense clients make of ‘schizophrenic experiences’. As part of this discussion, I outline my research looking at clients’ understandings of their psychotic experiences.

SCHIZOPHRENIA AS AN 'ESSENTIALLY
CONTESTED CONCEPT'

Even the most superficial glance at the schizophrenia literature reveals considerable controversy. As outlined earlier in this book (Chapters 3, 5–7) and elsewhere (Pilgrim 1990), this disagreement is as old as the concept itself. Disagreements about schizophrenia relate to some of the most fundamental aspects of the concept, such as whether it exists outside the minds of those who use the term, whether it should be viewed as a medical or social problem, and how best to assist those who may be troubled with schizophrenia (Bentall 2003, 2009; Boyle 1993). For example, in an extensive review of factors influencing prognosis for those diagnosed schizophrenic, Warner (1985) questions the received wisdom that the introduction of antipsychotic medications led to improved outcomes. Whether one accepts such arguments, it is impossible to dispute that they illustrate considerable disagreement in this field.

What, then, are we to make of this controversy, given that it seems to permeate the concept? Should we view it as an unfortunate – but temporary – stage in the development of our investigations into schizophrenia? On what grounds might we make such a claim, given that the term has been imbued with controversy since its inception, and, as books such as this demonstrate, the controversy shows no signs of abating? It appears, rather, to make more sense to see the controversy as being *integral* to the concept itself.

One possible response to this situation is to view 'schizophrenia' as what the philosopher Gallie (1955–56) referred to as an 'essentially contested concept'. In keeping with Wittgenstein's (1958) notion that meaning is determined by usage, Gallie (1955–56) suggests that there are some concepts that 'are essentially contested, concepts the proper use of which inevitably involves endless disputes about their proper use on the part of their users' (p. 169).

We could view schizophrenia as an 'essentially contested concept', implying that the disagreement surrounding the term is not incidental, accidental or temporary, but is an *essential* aspect of the concept itself. Like the search for the Holy Grail, efforts to resolve this debate are doomed to failure, as this debate constitutes the primary meaning of the term.

MARGINALIZATION OF THE CLIENT'S VOICE

Whether the notion of schizophrenia as an 'essentially contested concept' is accepted, the more general point that schizophrenia literature is characterized by disagreement is undeniable. This leads us now to consider *who* has a voice in this debate. A whole range of professionals and academics have made significant contributions to the vast literature. This is perhaps not surprising, given the amount of health care resources dedicated to providing treatment to those diagnosed schizophrenic (Knapp 1997). However, what is surprising is the almost

total absence of the voices of those who have most direct experience of schizophrenia: the person diagnosed 'schizophrenic'. An important qualification needs to be made at this point. I am not suggesting that those who have this diagnosis have not expressed their thoughts about the nature of this experience. One could argue that they do so at each and every meeting with a clinician. Furthermore, there is an extensive literature where those who experience mental health problems – including 'schizophrenia' – have expressed their thoughts and feelings on the experience (Beavan 2011; Beavan and Read 2010; Kaplan 1964; McCarthy-Jones 2013; Romme *et al.* 2009; Romme and Escher 2011; Geekie *et al.* 2012). I argue that those voices, both in the professional literature and in clinical settings, have been marginalized, and that this has been to the detriment of clients' interests, clinical practice and research efforts investigating the nature of the experience. This neglect of the clients' understandings of their own experience is reflected in the conclusion of a review article: 'research dealing with patients' own attributions for their illness has been virtually non-existent' (Molvaer *et al.* 1992).

One manifestation, and consequence, of this marginalization is the paucity of terms available to us to refer to the multitude of ways in which people with a diagnosis of schizophrenia relate to their experience. The term that first comes to mind in this regard is 'insight'. Unfortunately, as the term is now used, it usually refers to whether the client agrees with the clinician's position – namely, that the client is suffering from an 'illness' (Amador and Kronengold 1998; Kirmayer and Corin 1998) – rather than being truly concerned with the client's understanding.

Rare exceptions in this area are the related concepts of 'sealing over' and 'integration' as ways of conceptualizing clients' recovery styles following a psychotic episode (McGlashan *et al.* 1975). 'Sealing over' is the tendency to dismiss the experience as having little personal relevance, whereas 'integration' reflects a curiosity about the experience and its personal significance. Drayton *et al.* (1998) found that those who 'seal over' are more likely than 'integrators' to become depressed. Such research supports the notion that there are sound clinical grounds for attending to clients' responses to their psychotic experiences. Other variables such as distress (Romme and Escher 1989) and coping strategies (Carr 1988) have also been found to be associated with the sense the client makes of psychotic experiences. There is also a strong moral case to be made for attending to how the individual thinks about her or his experience. Shotter (1981) argues that 'authoring' one's own experience is a basic human right: 'In a moral world, no one but the person in question has the status, the authority, under normal conditions, to decide what his experience means to him.'

POWER AND KNOWLEDGE

How, then, are we to account for the 'paucity of research concerned with the individual's psychological adaptation to psychosis' (Drayton *et al.* 1998)? In

examining the relationship between power, knowledge and social practice, Foucault (1980) proposes that those with power act in such a way as to legitimize their own knowledge (and hence practice) while simultaneously discrediting other knowledges that may challenge their dominant knowledge. Foucault refers to the knowledge of those who are subject to the particular social practice or institution (such as prisons, mental health services, etc.) as 'subjugated knowledge' and proposes that simply giving voice to such knowledge is one way of challenging the hegemony. 'It is through the re-appearance of this knowledge, of these local popular knowledges, these disqualified knowledges, that criticism performs its work' (Foucault 1980: 82).

The knowledge that people with a diagnosis of schizophrenia have about their own experience (and hence about 'schizophrenia') is an example of 'subjugated knowledge', marginalized because it challenges the dominant discourses of professionals. This marginalization occurs in both research literature and in the face-to-face clinical encounters between client and clinician.

I want to make it clear that I am *not* claiming that the knowledge that people diagnosed schizophrenic have is necessarily superior to, or more accurate than, the professional, 'scientific' knowledge. I propose only that it is a legitimate form of knowledge and, as such, has an important contribution to make. Incorporating this knowledge within the debate will not magically lead to a resolution of all controversy. However, I believe that the debate will be richer if clients' voices are included rather than excluded, and that these voices will enhance understandings of 'schizophrenia' and efforts to provide effective support to people with that diagnosis (Geekie *et al.* 2012).

A STUDY OF THE SUBJECTIVE EXPERIENCE OF PSYCHOSIS

How, then, might we, in both research and clinical settings, attend more to the voices of those with this diagnosis? I outline one way in which this might be achieved by describing a research project that looked at how people make sense of their 'psychotic' experiences.

As a clinical psychologist working in a first-episode psychosis service, my primary role is to offer psychological support/therapy to the clients of this service. The exact nature of the work varies from client to client, but I have found that whatever the focus of therapy, the client's 'explanatory model' (Kleinman 1980) of his or her 'psychotic' experience is always of importance and sometimes the main focus. Here, I describe one aspect of this research project: clients' explanatory models for their psychotic experiences, as expressed in their psychotherapy with me.

This research was qualitative, using recordings of therapy sessions (with participants having given consent, of course) as the basic raw data upon which analysis is made. Only clients who were already engaged with me in

psychological work were invited to participate in the research, and then only when the client's understanding of his or her own psychotic experiences had already become a significant part of our work together. It was made clear to potential participants that their ongoing therapeutic work with me would not be influenced by whether or not they participate in the research. A total of 15 clients agreed to participate in the research (age range 19–35 years; 11 males, 4 females; 12 European, 2 Maori, 3 Pacific Islander – with 2 identifying with more than one ethnicity). The findings reported here are based on analyses of transcriptions of 51 sessions with those participants.

The research methodology was based on grounded theory (Charmaz 1990; Glaser and Strauss 1967), an explicit methodology for analysing qualitative data, which specifies procedures aimed to ensure that the analysis is based on, or 'grounded' in, the data. This involves the researcher systematically coding and then categorizing the data with a view to establishing theoretical constructs that provide an explanation for the data.

The findings reported here are largely descriptive in nature. A more thorough analysis of this data, along with a detailed discussion of the importance of subjective experience, is presented in Geekie and Read (2009), where three theoretical – or explanatory – constructs ('fragmentation', 'invalidation' and 'spirituality') are proposed as expressing some of the essential features of the subjective experience of psychosis.

General findings

Participants in the study were very welcoming of the opportunity to discuss their experience in some depth and to explore the possible meanings of this experience. This is consistent with my clinical experience. Of course, no universal statements can be made about this being something that all clients of first-episode psychosis services will accept or benefit from. Using McGlashan and colleagues' (1975) terminology, we might expect that 'integrators' are more likely to benefit from such discussions than those who 'seal over' – although these tendencies are probably not independent of the kind of help the client is offered. However, for some clients at least, the opportunity to talk in depth about what their experience means to them is important, and as such should be offered as a routine part of clinical services.

A second general finding is that there is a great variety of ways in which clients understand their psychotic experiences. These differences are manifest in both style and content. Stylistically, participants varied from adopting concrete, literal terms to using more poetic, metaphorical idioms for expressing understandings of their experiences. At times, participants' uses of metaphor conveyed a depth of meaning largely absent from typical textbook descriptions of psychotic phenomena. For example, one participant described his thought processes while psychotic as follows: 'It's kind of like going out to the mail box, and it's full of mail, but I feel I can't sort out the junk mail from the mail I want to

read.’ In terms of content, there was even greater variety in participants’ notions about their psychotic experiences. It is worth pointing out that this variety existed not only between participants, but also for individual participants, some of whom had more than one possible explanation for their experiences.

Specific findings

The framework I developed for conceptualizing the range of ways clients spoke about their psychotic experiences is illustrated in Figure 13.1. As can be seen, I have developed three broad categories, each containing sub-categories. ‘The nature of the psychotic experience’ and ‘the personal meaning of the experience’ relate to the psychotic experience itself, whereas ‘narrating experience’ relates to the more general issue of the telling of one’s story or, in Shotter’s (1981) terms, ‘authoring’. I discuss these three broad categories, providing examples to illustrate.

The nature of the psychotic experience

This category covers participants’ thoughts about the causes of psychotic experience, as well as attempts to explain what is happening when they have a psychotic experience. I have found it helpful to distinguish between participants’ thoughts about *why* they have psychotic experiences (*causes*) and ideas about what processes might be involved when they have a psychotic experience (*explanations*). These two domains are closely related, overlapping at times, and as both pertain primarily to the nature of the experience, I have found it useful to categorize them together.

Because of space constraints, I will not discuss in depth participants’ explanations for how psychotic experiences arise, but I give a few illustrations. One participant, who spoke of the causes of his psychotic experiences as including constitutional shyness and difficult experiences as a child, explained his tendency to manic thinking (which included ‘delusional’ ideas) as follows:

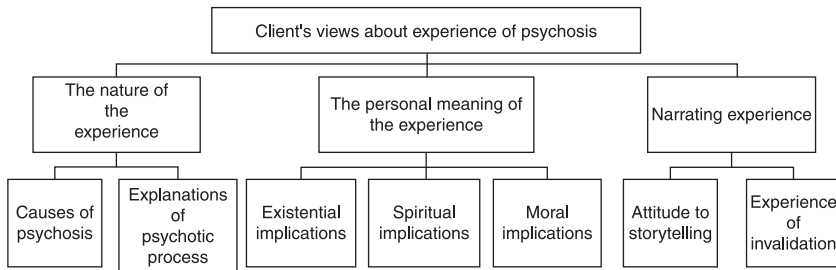


Figure 13.1 General categorization of clients’ views of experience of psychosis.

I suppose maybe that was my attempt to crush my inferior feelings. Maybe I thought if I just push myself enough to be overly confident then I wouldn't get those feelings, because I was thinking that when I was in the manic state.

Another participant saw impoverished communication at home when growing up as having caused his tendency to have psychotic thoughts and suggested that the process involved consisted of him filling in gaps where verbal information was missing:

It happened especially when there was very little verbal communication. When there was verbal communication, it was great, just like normal chatting. But, because I had nothing to say, I took in the world non-verbally, body language and that sort of thing, that's all I had. And, because it could be many things, as it's not quite concrete, I would interpret it in a certain way.

Causes of psychotic experiences

Possible causes of psychosis were of great interest to participants. A wide range of factors was considered. Participants did not appear troubled by multifactorial accounts of psychotic experiences, even when these contained apparent contradictions. Although most had no difficulty in expressing and exploring ideas about possible causes of psychosis, there were times when participants said quite simply that they had no idea why they had psychotic experiences. This was sometimes associated with distress, as with K, who stated, 'I can't figure it out. But, I mean, I think about it, about why it's happening, but I just don't know and that makes me feel bad'. Or with curiosity, as with M, who explained 'I don't understand much, but I am curious about it'. Statements such as the above were relatively rare, with participants (including the two who made the comments above) mostly having fairly well formed but flexible ideas about causality.

Although participants' notions of causality were, indeed, varied, it was possible to classify these into psychological, social/interpersonal and biological accounts. Within each of these three main classes of causality, I have made further sub-categorizations. Figure 13.2 illustrates the range of factors contained within each of the three main classes of causation.

Psychological causes

Psychological causes include accounts where an internal psychological process of some sort was implicated in the genesis of psychosis. There was considerable variety in the range of psychological factors considered by participants as possible causes of psychosis.

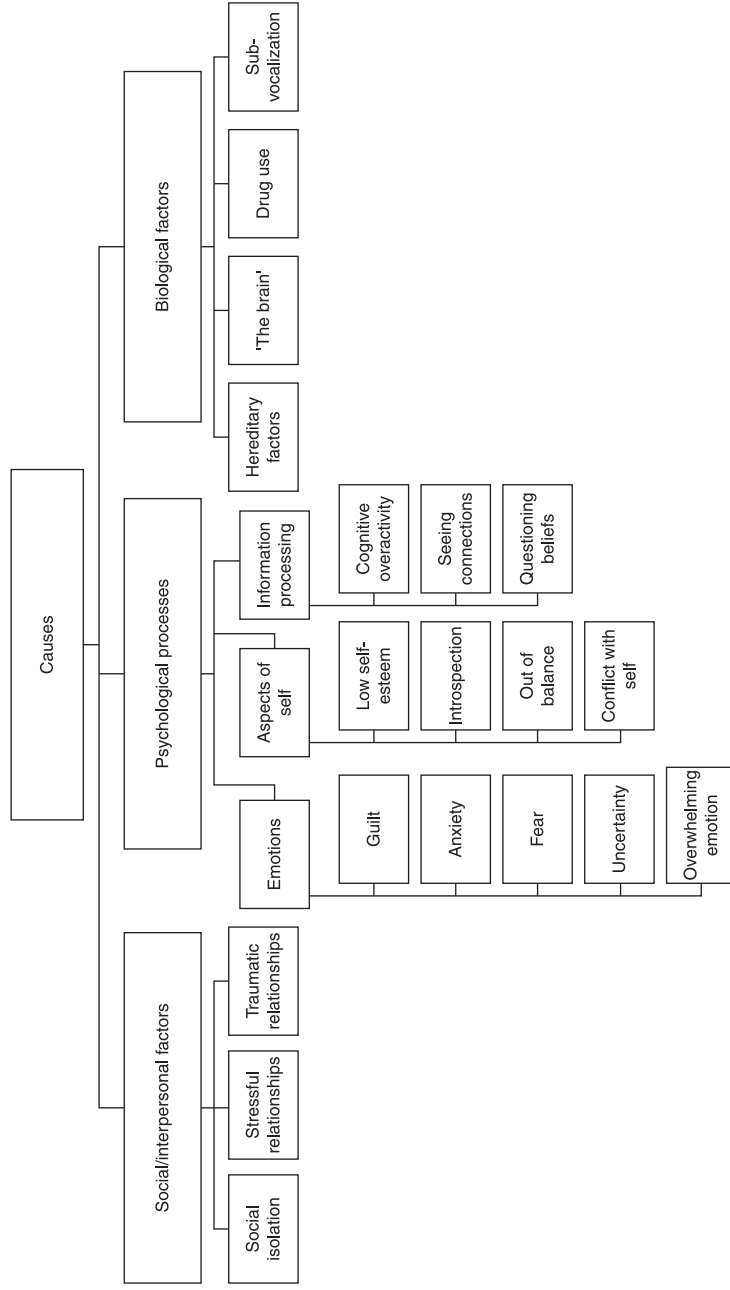


Figure 13.2 The wide range of variables viewed by participants as possible causes of their psychotic experiences.

Emotional explanations covered an array of specific emotions, including guilt, anxiety, fear and uncertainty, as well as the more general notion of overwhelming emotional arousal (with the specific emotion considered of less import). Within 'aspects of self', further sub-categorizations included low self-esteem, a tendency to introspection, being 'out of balance' with the self and conflict between different aspects of the self. 'Information processing' factors implicated by participants included those relating to cognitive overactivity ('too many thoughts'), information processing styles (such as tending to see connections between events) and the questioning of fundamental beliefs.

The notion of psychosis having been caused by the questioning of one's own beliefs is illustrated by A, who believed he had become psychotic because:

I questioned too many things all at once and without having an answer drew assumptions. It was just like cutting out my own foundation, and I didn't have much to stand on in the end, so I made myself vulnerable. I think we all have values in our life and if we take those away or we shatter them, we can be very vulnerable to us. I think that is what I did.

Social/interpersonal causes

This class of causal explanation for psychotic experiences included boredom, social isolation, stressful relationships and traumatic experiences. 'Traumatic experiences' refers largely to experiences from childhood, including abuse, as well as events such as teasing at school and parental divorce. A very clear and persuasive argument for the causal role of abuse was provided by B, a 30-year-old White female, who stated:

When I think of the way my father used to behave around me and that, well, I think to myself 'it's no bloody wonder I've got a mind like that'. When my father exposed himself in front of me and things like that, it's no wonder I've got this shit going on inside of me.

The same person explained *how* the abuse played a causal role: 'I can see what's happening is the lack of boundaries, and that was the cause of all this in the first place.'

Biological causes

Participants suggested a range of biological factors that they considered causal, including hereditary factors, the brain, and use of illicit drugs. One participant attributed his hallucinations to sensations in the 'voice box' – an idea that bears a striking resemblance to Frith's (1992) notion of sub-vocalization, of which the participant had no prior knowledge. Examples of quotes from participants who saw the use of drugs as implicated in the onset

of psychosis include D, who argued: 'I think it's because I started taking the drugs and it just changed my state and everything.' An example of an hereditary explanation was given by one woman who saw her psychosis as caused by 'having been born with something not right that I inherited'.

The personal meaning of the experience

The kinds of themes that emerge in this area include *existential*, *spiritual* and *moral* aspects or implications of the experience as seen by the participants, who found themselves faced with some profound questions about the nature of life and the nature of self. These questions, and the kinds of answers participants came up with, were clearly influenced by the recent experience of psychosis. For example, one participant found herself repeatedly wondering whether the content of her psychotic experiences means that she is a 'bad' person. Similarly, questions about the nature of one's relationship with God were also commonly provoked by the psychotic experience. Issues such as the above were clearly of great concern to participants (Geekie 2012). This is an area where the professional literature on psychosis is particularly deficient.

Narrating experience

In addition to expressing thoughts about the nature and personal meaning of psychotic experiences, participants also spoke about their experiences of telling their story to others, clearly seeing this as an important aspect of their psychotic episode (Geekie and Read 2008).

Comments on story telling could be fitted into two sub-categories: *attitude to storytelling* and the *experience of invalidation*. Some clients spoke passionately about the importance of telling their story. S explained: 'I've *got* to tell this story, got to summarize it' and proceeded to spend a number of sessions going over his story in great detail, with great enthusiasm and with apparent benefit. F expressed eagerness to explore her story because 'I want to find out what has happened and why it has happened to me.' And, finally, M asked to talk in greater depth about his experience because 'I just hope that I will be able to make sense of the whole lot.' Thus, participants made clear pronouncements that they see making sense of their experience, including sharing it with someone else, as of considerable importance to them. Furthermore, they viewed being actively involved in this sense-making as a vital part of this process (as opposed to being told what their experience meant). The importance of active participation was expressed succinctly by one participant who commented on the contents of a psycho-educational brochure explaining the nature of psychosis: 'That is very interesting and it makes good sense, but I want to make my *own* sense of it.'

The importance of being author of one's own experience relates closely to the second sub-category within *narrating experience*. *Invalidation* refers to

comments on the experience of feeling that their attempts to narrate experience have been undermined, undervalued or overlooked. This feeling was noted in connection both with making sense of psychosis and, at times, with the feeling that, as a consequence of having been psychotic, one's viewpoint on other issues was also invalidated. For example, M, whose spiritual interpretation of her experience had been dismissed by associates in her church, complained, with an obvious sense of loss, 'I had a lot of very profound experiences, but they have all been put down to "No, it's all in your own mind, you're talking to yourself".' She later went on to say that she felt that her viewpoint on other, unrelated matters was also dismissed by friends: 'All my close friends, including a very close friend, have written me off. They don't even bother with me now. It's like they see me as a lost cause, as cracked.'

CONCLUSIONS AND CLINICAL IMPLICATIONS

I hope I have demonstrated that clients of a first-episode psychosis service have much to contribute to the understanding of psychosis. Participants in my research, and others with whom I have worked clinically, welcome the opportunity to discuss their experience in depth and to develop an understanding of the experience that is personally meaningful to them. As the examples above indicate, clients can do this with considerable sophistication, providing accounts that cover both aetiology and process, shedding light on the experience that is often lacking in the professional literature.

Given that a psychotic episode can sometimes be an overwhelming experience, it is not surprising that clients are eager to make sense of it. Clients' explanatory models may not always correspond to professional understandings of psychosis, and it may, therefore, be necessary for some form of discussion and negotiation to take place if a shared understanding between clinician and client is to be established. (If this does not occur, therapeutic work will be more difficult, if not impossible.) It is important to recognize that clients want to be active participants in this process rather than passive recipients of the clinician's model. Clinicians who insist on 'telling' clients what their experience is or means run the risk of further engendering a sense of invalidation in the client, at the same time as missing out on an opportunity to learn something.

Following a first episode of psychosis, clients are no doubt in the process of building up their understanding of the experience, and it is at this point that clinicians may be able to contribute most to the client's efforts to develop a useful explanatory model. This contribution may involve challenges to some of the client's notions about the experience. It will need some sensitivity on the part of the clinician to be able to do this without leaving the client feeling invalidated. I am not arguing that we should abandon our professional notions and explanations for psychotic experience. However, if

we remind ourselves that there is not one but many professional perspectives on psychosis, and if we resist the temptation to adopt wholeheartedly one or other of these perspectives, this may make it easier for us to work with clients whose views do not correspond with our own. It is, of course, important that we provide clients with whatever information we may have that relates to their experience. But we should also be cognizant of the importance for the client to be actively participating in the authoring of their own experience.

A report on psychotic experiences by the British Psychological Society (2000) recommends that ‘Service users should be acknowledged as experts on their own experiences.’ Such an acknowledgement would, I believe, have important implications for research and clinical practice. I hope the research findings presented here show that this is both possible and desirable. Acknowledging service users as experts need not be viewed as undermining the expertise that clinicians and researchers bring to the subject. Rather, recognizing that there are multiple perspectives – or multiple ‘knowledges’ – on schizophrenia may enhance the quality and usefulness of the debate, illuminate the nature of the experience and lead to improved clinical services, even if it brings us no closer to finding the Holy Grail.

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PSYCHOSIS, POVERTY AND ETHNICITY

John Read, Lucy Johnstone and Melissa Taitimu

One of the many issues minimized by overemphasis of bio-genetic factors is the impact of belonging to a disempowered group. There is strong evidence that poverty and racism are causative factors for psychosis. Nevertheless biological psychiatry has on the whole either ignored this evidence or tried to interpret it in ways that are compatible with bio-genetic causal theories.

This chapter does not cover all groups that experience discrimination or disadvantage. Clearly, being female in a patriarchal society (Chapter 15) or lesbian or gay in a homophobic society (Gilman *et al.* 2001) can have psychological consequences. The common threads, besides discrimination, are relative lack of power and social status and the concomitant lack of access to the physical and social resources necessary to establish and maintain emotional wellbeing (Chapter 20). There is strong evidence that these threads, and the personal meaning they have for the individual, are mediating factors in the pathways that lead to mental distress.

One of the difficulties in unpicking the causal threads is the uncritical use of the term 'schizophrenia' throughout the research. As discussed in Chapter 5, the use of a non-valid category as a starting point undermines the conclusions that can be drawn from any given study. For these reasons, the term 'schizophrenia' is used in speech marks in this chapter. However, we argue that the collective findings are consistent enough to show that both poverty and racism play a causal role in 'schizophrenia' if that construct is taken as a proxy for severe forms of mental distress, including psychosis.

POVERTY

The facts

The public, all over the world, believe that poverty is among the leading causes of psychosis (Chapter 11). It seems they are right. Mental hospitals have always been filled primarily with poor people. In seventeenth-century

France the people confined in the Hôpital General were 'the poor of Paris' (Chapter 2). The first systematic study found that the deprived central areas of Chicago had far higher psychiatric admission rates than the wealthier suburbs (Faris and Dunham 1939). Contrary to the notion that only the less severe mental health problems are socially caused, the difference was particularly high for 'schizophrenia'. This relationship between poverty and 'schizophrenia' was soon replicated in nine other American cities (Clark 1949).

The famous New Haven study (Hollingshead and Redlich 1958) found that the poorest class (V: 'unskilled, manual') was three times more likely than the wealthiest two classes (I and II: 'business, professional and managerial') to be treated for psychiatric problems and eight times more likely to be diagnosed 'schizophrenic'. By 1976, there had been over 50 studies across Europe and North America showing that 'schizophrenia' occurred most frequently among the lower classes of urban society. (Kohn 1976: 177)

By 1980, the relationship between schizophrenia and poverty was described as 'one of the most consistent findings in the field of psychiatric epidemiology' (Eaton 1980). Studies have continued, with some exceptions (Bresnahan and Susser 2003), to confirm the earlier findings, in England, Wales, Finland, New Zealand, Canada and Nigeria. More often than not the strongest relationship with poverty is for 'schizophrenia' (Read 2010). A British study found that deprived children are four times more likely than non-deprived children to develop 'non-schizophrenic psychotic illness' but eight times more likely to grow-up to be 'schizophrenic' (Harrison *et al.* 2001). Even among children with no family history of psychosis, the deprived children were seven times more likely to develop 'schizophrenia'. A recent population-based longitudinal study in Israel found that education level of parents at birth, and being born in a poor neighbourhood, were significant risk factors for 'schizophrenia' (Werner *et al.* 2007). A study of 811 counties across 13 states in the United States found that rates of hospitalization for 'schizophrenia' correlated with percentage of residents living in poverty and with percentage unemployed (Fortney *et al.* 2009). Researchers measuring the more reliable constructs of hallucinations and delusions have found that they are more common among people who grew up in poverty (Brown *et al.* 2000).

Psychiatric admission rates increase during economic declines, with the relationship being, again, particularly strong for psychosis in general and 'schizophrenia' in particular (Warner 2004).

Relative poverty

It seems that *relative* poverty may be even more damaging than poverty *per se*. A study of 17 wards in South London found that high level of inequality was associated with incidence of 'schizophrenia', after adjusting for age, sex, absolute deprivation, and ethnicity, but only in the group of the most deprived wards (Boydell *et al.* 2004). A similar study comparing seven municipalities

in South Africa found a significant relationship between first episodes of psychosis and inequality, after controlling for gender, age, ethnicity, urbanicity and employment status (Burns and Esterhuizen 2008).

Powerful additional evidence comes from ‘The Spirit Level’ (Figure 14.1), in which two British epidemiologists summarize multiple studies to demonstrate the powerful relationship between relative poverty in societies and a whole range of social, health and mental health outcomes (Wilkinson and Pickett 2009: 181). They found that ‘across whole populations, rates of mental illness are five times higher in the most unequal compared to the least unequal societies’. Currently, the United Kingdom and the United States both feature in the top four most unequal societies of the 23 that they surveyed. Wilkinson and Pickett suggest that ‘if Britain became as equal as the 4 most equal societies (Japan, Norway, Sweden and Finland), mental illness might be more than halved’, while if the United States did the same, ‘mental illness . . . might . . . be cut by almost two-thirds’ (p. 261). However, the trend for both countries in the last 30 years has been in the opposite direction. Although they did not single out ‘schizophrenia’

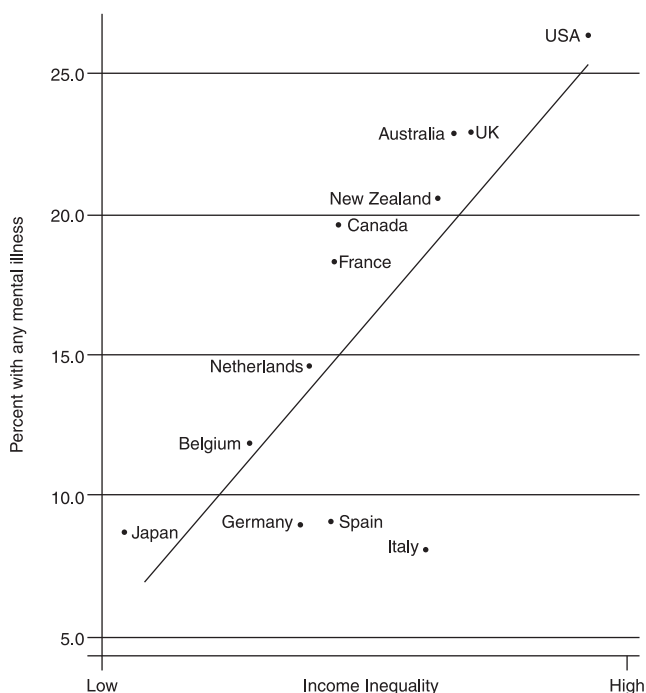


Figure 14.1 The prevalence of mental illness is higher in more unequal rich countries.

Source: Wilkinson & Pickett, *The Spirit Level* (2009)

for investigation, they argue that 'severe mental illness' is 'strongly correlated with inequality; mood disorders less so.' Their work is given added impetus by a recent World Health Organization report (WHO 2009), which found that as countries get richer, rates of mental illness increase, and that this needs to be understood and treated less in terms of individual pathology and more as a response to the complex consequences of inequality.

Important as this evidence is, we lack a full understanding of the pathways by which inequality exerts such devastating effects. Wilkinson and Pickett suggest that we are fundamentally social creatures, hard-wired to be deeply conscious of our status and standing in relation to others, and to suffer 'social pain' when poverty excludes us from community life and a sense of belonging and being valued. Like Warner (2004), they argue that inequality has its impact at least partially through the personal meaning it has for the individual, in terms of feeling shamed, worthless and excluded.

Urbanicity

The relationship between social class and psychosis is strongest in urban settings, as confirmed by over 20 studies that have found a correlation between urban living and psychosis (van Os *et al.* 2009). The specific causal factors in urban living have been identified as stressful life-events, social isolation, overcrowding, overstimulation, crime levels, pollution and, again, poverty (Sharples *et al.* 2001). The relationship holds even more strongly for 'schizophrenia' than for other psychoses (Lewis *et al.* 1992; Mortensen *et al.* 1999). It remains even after controlling for family history of psychiatric disorder in general (Lewis *et al.* 1992), or of 'schizophrenia specifically' (Mortensen *et al.* 1999). The 'attributable risk' from urban birth is four times greater than that from having a mother with 'schizophrenia'. Moreover, the greater the numbers of years lived in urban areas the greater the risk of 'schizophrenia' (Pedersen and Mortensen 2001.)

Biological psychiatry's explanations

In spite of the strong evidence that 'schizophrenia', like other expressions of human distress, is, at least partly, a consequence of growing up with the disadvantages and stresses associated with poverty (Read 2010), the majority of research has been on biological factors. The percentage of all 'schizophrenia' studies investigating biological causes or medication increased from 33.0% before 2000 to 50.7% thereafter, while the percentages investigating socioeconomic status was 1.0% and 0.9%, respectively (Read *et al.* 2009). The 4,500-page *Comprehensive Textbook of Psychiatry* makes no reference to the relationship between poverty and psychosis or 'schizophrenia' (Saddock *et al.* 2009). When not ignored all together, the research findings about social factors tend to be interpreted in ways that maintain the dominance of bio-genetic theories.

Madness causes poverty: 'social drift' and 'social selection'

One response to the evidence linking poverty with psychosis is the argument that 'schizophrenia' afflicts all classes equally, but those at the top of the economic pyramid drift to the bottom as a result of their illness. This is the 'social drift' theory. A weaker variation of the same theme is the 'social selection' theory (Eaton 1980), which suggests that although 'schizophrenics' have not actually drifted downwards themselves, their 'illness' has caused them to be of a lower social status than their parents.

The New Haven study tested the 'social drift' theory by investigating whether 'Class V' patients had drifted to the slums in the course of their lives' and whether 'schizophrenics' were socially downward mobile. No evidence of such 'social drift' was found. The study also rejected the weaker 'social selection' version of the theory because 91% of the 'schizophrenics' were in the same social class as their parents – rather than a lower class, as predicted (Hollingshead and Redlich 1958). Other studies confirmed these findings (e.g. Wheaton 1978). The relationship between urban life and schizophrenia has also been shown to be a consequence not of social drift (Dauncey *et al.* 1993), but of the extra stresses of growing up in the city (Lewis *et al.* 1992; Warner 2004).

More recently it has been shown that as well as predicting who will develop 'schizophrenia' in the first place, low 'social class of origin' (at birth or during childhood), which 'cannot be caused by schizophrenia', even predicts negative outcome *among* people with severe psychosis (Samele *et al.* 2001).

Of course, there *are* social and economic consequences to being extremely distressed, alienated or disoriented. Therefore it would be surprising if the social drift hypothesis is entirely without foundation. Indeed, psychiatrist Richard Warner has assembled an impressive amount of evidence demonstrating that the relationship between poverty and 'schizophrenia' can be attributed to unemployment, which is a frequent consequence of a diagnosis of psychosis in Western societies, and specifically the impact and meaning of the associated poverty – exclusion and lack of social power and status (Warner 2004.) In addition, you may suffer consequences such as prejudice and discrimination from having your response to adversity explained in terms of having a 'mental illness' (Chapter 12), and these, in turn, can add to the trap of poverty and stress. A reasonable conclusion to draw, therefore, is that both social causation and social drift are at play, the former in the causation of madness and the latter in its maintenance (Read 2010). In the words of a WHO report, 'Poor mental health is both a cause and a consequence of social, economic and environmental inequalities' (WHO 2009).

Poor people don't face more stress, they just find it harder to deal with

Being poor means that you are more likely to suffer a range of other disadvantages and stressors. Even those with enough to eat and somewhere to live are

more likely to experience powerlessness, isolation, lack of self-respect, physical ill-health, etc (Tizon in press). They are also many times more likely to suffer neglect and abuse as children (van der Kolk *et al.* 2001), which are strongly predictive of psychosis later in life (Chapter 18).

Some researchers see it differently. Those who found ten-fold differences in schizophrenia between the poorest and wealthiest parts of Nottingham saw 'no clues' about causes: 'There was no particular suggestion of unusually high rates of stressful events' (Giggs and Cooper 1987: 633). Others have claimed 'there is little evidence that lower-status individuals suffer from more situationally induced stress' (Rushing and Ortega 1979: 1192) and concluded that 'at any given level of stress, people of lower social class position are more likely to become mentally disturbed than are people of higher social class position'. It was even seriously proposed that no further research be undertaken on the topic and that researchers 'look for class-constant stresses, not stressors that are more frequent in the lower class, such as events related to the economy' (Eaton 1980).

More recently the *New Oxford Textbook of Psychiatry*, while acknowledging that poor people do have a harder life, argues that these stressors lead to psychosis only in 'genetically vulnerable individuals who have a restricted capacity to cope with complex or stressful situations' (Jablensky 2009: 549)

What happens to poor people when they enter the psychiatric system?

Diagnostic bias

A complicating factor in the relationship between poverty and 'schizophrenia' is diagnostic bias. Ten studies have found that more severe diagnoses are applied to poorer people than wealthier people with the same symptoms (Read 2004, 2010). One of them found that psychiatrists assigning severe diagnoses on the basis of class genuinely believed they were basing diagnoses on patient behaviour rather than occupation and education. Thus there was 'intellectual denial of social status effects and a subconscious utilization of status information' (Lebedun and Collins 1976: 206).

Treatment bias

Class bias also operates in treatment selection (Johnstone 2000). Starting with the well-known Hollingshead and Redlich (1958) study, it has repeatedly been demonstrated that poor people are more likely to receive physical treatments such as drugs, electro-convulsive therapy (ECT) and lobotomy, to be hospitalized, and to remain in hospital for longer periods (Read 2004, 2010). A survey of nearly 900,000 American children and adolescents showed that youths whose families are eligible for Medicaid (i.e. from poor families) are

5.8 times more likely to be on antipsychotic drugs than those from families with private health insurance (Zito *et al.* 2003). A recent study of 3,359 outpatient visits by individuals diagnosed 'schizophrenic' found that those on Medicaid were 44 times more likely than those with private insurance to be hospitalized (Rost *et al.* 2011). Poverty has also been found to be the 'strongest and most consistent predictor' of *compulsory* admission (Bindman *et al.* 2002).

The cycle of oppression

Psychiatric diagnosis and admission can lead to a further circle of disadvantage in which poor people are even less likely to receive appropriate help based on an acknowledgement of the role their difficult social circumstances have played in causing their distress and confusion. The label 'schizophrenia' is likely to further lower their self-esteem, and the effects of medication may reduce motivation and energy. They will find it even harder to gain employment, partly because of prejudice about 'schizophrenia' (Chapter 12). They will be less likely to appreciate the social origins of their distress, since this has been attributed to the illness of 'schizophrenia'. They may, as a result, be even more powerless to change the circumstances that have resulted in entering the psychiatric system in the first place.

RACE AND ETHNICITY

The facts

High rates of Black inpatients diagnosed psychotic were first identified in the United States towards the end of the nineteenth century (Fernando 2003: 32). As was the case for poverty, the first systematic study appeared about 80 years ago (Ødegaard 1932). It has since been repeatedly demonstrated that 'ethnic minorities' – whether immigrants or colonized indigenous peoples – are significantly more likely to be diagnosed with 'schizophrenia' than members of the dominant culture. By 1974, 24 of 25 US studies had found higher rates of admission for 'schizophrenia' among foreign-born people compared to the 'native' population (Rosenthal 1974). A recent longitudinal study found that African-Americans are 3.3 times more likely than White Americans to be treated for 'schizophrenia' (Bresnahan *et al.* 2007). Higher rates of 'schizophrenia' in ethnic minorities have been found in 10 other countries: Australia, Belgium, Denmark, Germany, Greenland, Israel, the Netherlands, New Zealand, Sweden and the United Kingdom (Fernando 2003; Read 2004). The largest incidence study in Britain recently confirmed that, compared to the White British group, Black Africans were nine times more likely to have received this diagnosis. Rates for other ethnic minority groups were also elevated, but to smaller extents (Fearon *et al.* 2006). Studies focusing on the

specific phenomenon of hearing voices also find higher rates among ethnic minority groups (Beavan *et al.* 2011).

Most studies have focused on migrants. For example, Moroccan immigrants are 5.8 times more likely to be diagnosed 'schizophrenic' than native Dutch people (Veling *et al.* 2006).

A meta-analysis of 18 studies (1977–2003) found that migrants were 2.9 times more likely to be diagnosed with 'schizophrenia', with even higher ratios for second-generation migrants (4.5), migrants from developing countries (3.3) and Black migrants (4.8) (Cantor-Graae and Selten 2005). The most recent meta-analysis found the rate of psychotic disorders for migrants to be more than double that of 'natives', with, again, much higher ratios for Black migrants (Bourque *et al.* 2011). Colonized, indigenous peoples, such as Māori in New Zealand, are also significantly more likely to be diagnosed 'schizophrenic' than Europeans, in both inpatient (Wheeler *et al.* 2005) and outpatient (Wheeler 2007) settings.

Biological psychiatry's explanations

As is the case for poverty, this is an under-researched area. A 1999 review reported that only one in six research studies about 'schizophrenia' gives the ethnic composition of the sample, and only one in 36 actually analyses data by ethnicity (Lewine and Caudle 1999). None of the studies in the most recent meta-analysis are from the United States (Bourque *et al.* 2011). Australian psychiatry is similarly remiss in relation to native Australians. Clearly this is a highly sensitive area, and the possibility that higher rates of mental distress in minority ethnic groups and colonized people are at least partly the result of discrimination raises very challenging questions about the roles of psychiatry and of the racism of the dominant culture. This may explain why a number of other hypotheses have received greater attention, as discussed next.

Selective migration

The predominant research focus for the past three decades has been migration. The 'selective migration' theory proposed that individuals who immigrate are predisposed to, or already have, 'schizophrenia' (Cochrane and Bal 1987). The collective version of this theory argues that there are higher rates of 'schizophrenia' in the countries from which people have emigrated. This comes dangerously close, however, to the racist argument that certain ethnic groups are genetically inferior. Nevertheless it *has* been argued that 'genetic as well as social factors have contributed' (Harrison *et al.* 1988: 654). Others have even claimed to have identified a relationship between a particular gene and 'schizophrenia' in African-Americans that was absent in their White counterparts (Nimgaonkar *et al.* 1992).

It was subsequently acknowledged that there is no evidence to support either the individual or the collective version of this theory, in any ethnic group (McKenzie and Murray 1999). For example, the incidences of 'schizophrenia' in Jamaica, Trinidad and Barbados are all 'comparable to the rate for the UK White population and significantly lower than the comparable rate for the UK-Caribbean population' (Fearon *et al.* 2006: 1546). Unfortunately, some mental health professionals continue to hold such views privately. When New Zealand psychiatrists were asked 'Why do you think Maori are over-represented in psychiatric institutions?' most identified issues related to colonization, loss of land and language, and poverty. However, 18% of the male psychiatrists stated that it was because Maori are biologically or genetically more predisposed to mental illness than Europeans. Examples included 'genetic loading' and 'Maori are biologically predetermined to mental illness, especially psychosis' (Johnstone and Read 2000: 142).

Migratory stress

The notion of 'migratory stress', referring to the general stress of moving countries, has also been deployed. However, it is several years after arriving that the incidence of 'schizophrenia' increases, and rates are even higher in 'second-generation' descendants of immigrants than in those who migrated (Cantor-Graae and Selten 2005; Harrison *et al.* 1988). The specific stressors associated with the migration process have been shown to be unrelated to 'schizophrenia' (Zolkowska *et al.* 2001). A large British community survey showed that hallucinations are more common in non-migrants than in migrants (South Asia) or are equally common (Caribbean) (Johns *et al.* 2002). The most recent meta-analysis concludes that 'post-migration factors play a more important role than pre-migration factors of migration per se' (Bourque *et al.* 2011: 897).

None of these unsubstantiated migration theories can be applied to African-Americans (Bresnahan *et al.* 2007; Cowan *et al.* 2011), who can no longer reasonably be considered migrants, or to colonized indigenous peoples, who never were.

'Perceptions' of racism

Other explanations have focused on the supposedly flawed perceptions of members of ethnic minorities. Some British researchers have argued that it is not increased exposure to adverse life events that differentiates African Caribbeans from other patients so much as their greater likelihood to attribute adverse events to discrimination (Gilvarry *et al.* 1999).

In fact, the research suggests that *actual* racism plays a greater role in psychosis than perceived racism. Members of ethnic minorities who believed that British employers were racist were 1.6 times more likely to be experiencing psychotic

symptoms than their counterparts. However, those who had actually experienced racist verbal abuse in the past year were 2.9 times more likely than others to be experiencing psychotic symptoms, while those who had suffered a racist physical attack were 4.8 times more likely to have such symptoms (Karlsen and Nazroo 2002). Another British study found that having experienced racially motivated verbal abuse or physical assault in the last year was associated with between a 2.0- and 3.5-fold increased risk of psychosis for Caribbean, Indian, Pakistani and Irish people and an almost 8-fold increased risk for Bangladeshi people. A recent Norwegian study found that even *within* a sample of people diagnosed psychotic, both 'feeling unjustly treated or insulted because of one's cultural background', and actual experiences of 'discrimination in housing and denial of employment due to immigrant status', were related to positive symptoms of 'schizophrenia' (Berg *et al.* 2011).

Black schizophrenia is different from White schizophrenia

It has been suggested that what is often mistaken for 'schizophrenia' among African Caribbeans is actually an 'acute psychotic reaction' or a 'psychogenic psychosis' – that is, madness brought on as a psychological reaction to unpleasant life events (Lewis *et al.* 1990), and that this 'misdiagnosis' may partly explain the apparently higher prevalence.

Studies have indeed suggested that African Caribbeans with 'schizophrenia' have a shorter 'illness' with a better outcome (McKenzie *et al.* 1995) and are more likely to have experienced a stressful event shortly before admission (Littlewood and Lipsedge 1981). This comes close to admitting, contrary to the narrow biomedical model, that psychotic breakdown can be a result of life events.

In their classic book *Aliens and Alienists*, Littlewood and Lipsedge (1982) outlined a more radical version of this position in their argument that symptoms of 'schizophrenia' in ethnic minority groups can be an intelligible response to racism and disadvantage. As an example, they present evidence that the greater incidence of paranoid reactions among immigrants may be 'merely a strong reiteration of the experience of discrimination' (p. 246). While this is a welcome perspective, the authors stopped short of exploring the possibility that 'schizophrenia' may be an understandable reaction to social stresses in the dominant White culture as well.

Social causation: racism can help drive you crazy

An encouraging recent development is the increase in attention by researchers (mostly European) to the most obvious likely causes of higher rates of madness in minority ethnic groups. Their studies have found that the main causal factors are racism, poverty, unemployment, urbanicity, isolation and a range of other measures of social disadvantage (Read 2004).

In the Western world, there is an association between minority ethnic status and poverty. Minority ethnic groups, asylum seekers and refugees are more likely to be unemployed and poor and to suffer all the consequences of lack of power and social status discussed in relation to class. In other words, they lack access to 'social capital' (McKenzie *et al.* 2002.) Along with this, they will inevitably experience the damaging effects of social inequality and the personal meanings that result from it – humiliation, worthlessness and entrapment.

The factors most consistently linking ethnicity to 'schizophrenia' are financial disadvantage, unemployment, social isolation and racist discrimination (Gilvarry *et al.* 1999; Sproston and Nazroo 2002). One reviewer found that the high rates of schizophrenia among British African Caribbeans are entirely explicable by the 'sociodemographic factors prevalent in this community', such as poverty (Sashidharan 1993). Unemployment and early loss of a parent have been found to be two of the social factors related to the high rate of 'schizophrenia' among Black Caribbeans in England (Morgan *et al.* 2008). A US study found that the ratio, for 'schizophrenia', of 3.3 for African-Americans compared to White Americans is reduced to 1.9 by controlling for family socioeconomic status, showing that 'causal pathways involving family SES played some role' (Bresnahan *et al.* 2007).

Different levels of disadvantage may help explain the differing rates of psychosis among different ethnic groups. Rates of psychosis and schizophrenia among Asians, while higher than for White British, are not as high as for African Caribbeans or Black Africans, who are more likely to be poor and unemployed (Brugha *et al.* 2004). In the Netherlands, different levels of discrimination experienced by different immigrant groups map onto odds ratios for schizophrenic disorders relative to the Dutch native population: low discrimination (1.6) – Turkey; medium (2.0) – Netherlands-Antilles, Surinam, 'other non-Western countries; and high (4.0) – Morocco (Veling *et al.* 2006). Janssen *et al.* (2003) recorded experiences of discrimination in 4,067 Dutch people who had shown no signs of psychosis and, three years later, the frequency of delusions and hallucinations. Those who had reported discrimination in two or more domains (skin colour/ethnicity, gender, age etc.) were three times more likely to be experiencing psychotic hallucinations than those who had reported no discrimination, and five times more likely to be experiencing psychotic delusions.

Interestingly, psychiatric admission rates are lower in areas where there is 'clustering' of an ethnic minority group (Boydell *et al.* 2001). It has been suggested that the protective element is '... reduced exposure to direct prejudice and increased social support' (Halpern and Nazroo 2000: 34). Similarly, a Dutch study found that psychosis rates were elevated only for immigrants living in neighbourhoods where their own ethnic group comprised a small proportion of the population (Veling *et al.* 2006). In what may perhaps be a related finding, the same research team found that 'negative identification'

with one's own ethnic group is 'a risk factor for schizophrenia in immigrants living in a context of social adversity' (Veling *et al.* 2008). This is consistent with a prospective birth cohort study in New Zealand, which found that elevated rates of mental disorder among Maori involve an interaction between socioeconomic disadvantage and childhood adversity, with secure cultural identity operating as a protective factor (Marie *et al.* 2008).

In summary, all the social adversities that are related to poverty and unemployment have been found to contribute to the elevated rates of psychosis among ethnic minorities and colonized peoples, who are exposed to particularly high levels of these factors. An ingenious analysis recently calculated that 'Up to 22% of all psychoses in England could be prevented if exposures associated with increased incidence in ethnic minority populations could be removed; this is equivalent to 66.9% within ethnic minority groups themselves' (Kirkbride *et al.* 2010)

What happens when members of ethnic minorities enter the psychiatric system?

Diagnostic bias

As was the case for poverty, there is evidence that high rates of 'schizophrenia' among ethnic minorities are partially explained by clinician bias in the diagnostic process. Historically this has even resulted in inventing new 'psychiatric illnesses' for ethnic minorities, starting with the nineteenth-century 'Drapetomania', which was said to cause slaves to run away (Cartwright 1851). A more recent British example is 'cannabis psychosis', which in one study was diagnosed in African Caribbean males 95 times more frequently than in White males (McGovern and Cope 1987).

Present-day bias in diagnosis is suggested by studies such as a survey that found that despite much higher rates (up to 9 times) of diagnosed 'schizophrenia' among British African Caribbeans in clinical settings, the rate is only twice as high in the community (Sproston and Nazroo 2002). In New Zealand, Maori are not only admitted to hospital at more than twice the rate than would be expected from their share of the general population but, once in hospital, are significantly more likely than Europeans to be diagnosed with psychotic disorders (as are Pacific Islanders and Asians) (Wheeler *et al.* 2005).

These diagnostic discrepancies in clinical samples might be partly explained by ethnic minorities entering mental health services later and, therefore, in more disturbed states – perhaps because of mistrust of systems run by the dominant culture. Evidence of diagnostic bias, however, has been found using various methodologies, including studies showing that when structured diagnostic protocols are used, fewer ethnic minority members are diagnosed than in clinical settings (Whaley 2001). The bias can occur both at the level of what constitutes a symptom (e.g. hallucinations being more readily perceived by psychiatrists in

African-Americans than in other patients) and at the level of deciding that a given symptom is indicative of 'schizophrenia' (Trierweiler *et al.* 2000).

There have been a few attempts to modify the diagnostic system in relation to ethnic minority groups. For example, the 'cultural formulation' is a supplement to the *DSM* categories, and it includes consideration of the influence of cultural factors and identity on the person's difficulties (Lewis-Fernandez and Diaz 2002). A recent Dutch study used a 'culturally sensitive' approach to diagnosis, in which hearing voices, seeing things or dead persons and being influenced by an outside force were deemed to be indicative of a possession state rather than symptoms of psychosis. Using this approach, there was no difference in the psychosis rates of Moroccan immigrants and Dutch natives (Zandi *et al.* 2010). While these developments can be seen as a step in the right direction, they are modifications of, not replacements for, the diagnostic system and are thus only a partial answer to the flaws and limitations of *DSM*. In addition, they are not widely used. In New Zealand, only 1% of psychiatrists and 13% of psychologists agreed that 'European diagnostic systems should not be used with Maori' (Johnstone and Read 2000.)

Treatment bias

Immigrants, colonized indigenous peoples and other ethnic minority groups are more likely, compared with their representation in the general population or in 'clinical samples' of inpatients or psychotic patients, to be forcibly admitted to, and detained in, hospital, to have the police involved in their admission, to be admitted to secure/forensic/special hospitals and to public rather than private hospitals, to receive antipsychotic drugs or electroconvulsive therapy and to not be offered psychotherapy (including therapy for psychosis: McKenzie *et al.* 2002). One or more of these examples of discrimination has been documented in Australia, Belgium, Germany, Israel, New Zealand, the United Kingdom and the United States (Read 2004). For example, the recent study of 3,359 outpatient visits by individuals diagnosed 'schizophrenic' found, after controlling for any clinical differences, that Black Americans were 1.7 times more likely to be medicated with antipsychotic drugs and 3.5 times more likely to be hospitalized, than White Americans (Rost *et al.* 2011). Even if therapy is offered, it is likely to be based on Western assumptions, which are not shared in all cultures. (For a discussion of racism in the theory and practice of psychotherapy, see Fernando 2003.)

In New Zealand, Maori are far more likely to be readmitted, particularly if diagnosed with 'schizophrenia' (Ellis and Collings 1997: 93). Another New Zealand study found that, compared to Europeans, Maori are given higher doses of antipsychotic drugs and are more likely to be given two or more at the same time (Wheeler *et al.* 2008). They are also more likely to be injected rather than given the drugs orally, as is the case for African-Americans (Arnold *et al.* 2004.)

Similar findings come from the United Kingdom, where Black and ethnic minority people are over-represented in psychiatric services, and more likely to be judged to be dangerous by professionals as dangerous (Ferns 2005). In 2010, 'Black and other ethnic minority groups' were six times more likely to be admitted to English psychiatric hospital (Care Quality Commission 2011). Black people in the United Kingdom are also more likely to be compulsorily detained or sent to locked wards, are more often given physical treatments such as medication and ECT (in higher doses) and are less often referred for counselling and therapy (Ferns 2005). Price *et al.* (2004) found that on psychiatric wards, seclusion is used more with Asian and Black patients than with White patients, even though there was no race-related difference in the episodes of aggressiveness against others.

It is not surprising that ethnic minority groups often experience services as 'inhumane, unhelpful and inappropriate' and say that their worldviews are not understood or even acknowledged by services (Sainsbury Centre for Mental Health 2002: 9).

Ethnicity and mental health: summary

Psychiatrists who themselves are members of ethnic minorities are often at the forefront of efforts to address the issues raised here, especially in Europe (e.g. Bhugra and Bahl 1999; Fernando 2003, 2011; Sashidharan 1993). As they have noted, the implications are far-reaching for the practice of psychiatry across all ethnic groups:

Establishing the role of such social factors would not only raise possibilities of prevention within the Afro-Caribbean community but could also provide an important pointer towards a possible causal role, rather than a precipitating role, for social factors in psychosis, a role that is currently discounted by much orthodox psychiatric research.

(McKenzie and Murray 1999: 57)

In fact, transcultural psychiatrists have argued that since all psychiatric practice is based on Western values and assumptions, it cannot be meaningfully applied to a non-Western culture (Fernando 2003, 2011). Western medicine, for example, assumes a split between mind and body that is absent in traditional African, Asian and Native American worldviews. Other cultures may see what we would call psychiatric disorders in religious, spiritual, philosophical or ethical terms. Fernando describes the imposition of Western models of medicines as a form of 'psychiatric imperialism' which is more subtle, but no less damaging, than more overt forms of colonialism.

CONCLUSIONS

Establishing causal relationships is always difficult, and simple cause-effect links are rare. The pathways from poverty and ethnicity to psychosis are complex. Nevertheless, there is ample evidence that inequality, deprivation and discrimination, filtered through their social and personal meanings, are key causal factors in psychosis. As a recent WHO report states, 'levels of mental distress among communities need to be understood less in terms of individual pathology and more as a response to inequalities involving relative deprivation across society' (WHO 2009). There is strong evidence that intervening at these levels can reduce the impact of the damage and hence avoid later psychotic breakdown (Chapter 20). Some progress has been made in recognizing these links and targeting interventions at a broader social level. In the United Kingdom, for example, there have been a number of reports into the role of social inequalities in the mental health of Black and ethnic minority groups (e.g. Department of Health 2003; Sainsbury Centre 2002). However, the biomedical model of mental distress continues to be a barrier to acknowledgement of the full implications of these findings. Indeed, critics have argued that this is part of the purpose of psychiatry: to sweep up the social casualties of Western industrialized societies by re-defining their difficulties as individual, biologically based 'illnesses' (Johnstone 2000; Warner 2004) and applying 'treatments' that silence and disable people even further. The alternative – to accept the fundamental challenge to biomedical psychiatry and to 're-structure our provisions for all of the poor' (Warner 2004) – is very costly and conflicts with too many vested interests. The struggle for full recognition of the impact of poverty and discrimination, and the implications for psychiatry as currently practiced, is ongoing.

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GENDER AND PSYCHOSIS

John Read and Vanessa Beavan

Depending on age and the diagnostic criteria employed, some researchers find that more men than women have 'schizophrenia' while others find there is no difference (Bogren *et al.* 2010; Jablensky 2009; Kirkbride *et al.* 2006). More importantly, there are definitely some clear differences *between* men and women with this diagnosis. These differences are so pronounced that they have been summarized in terms of men having 'typical schizophrenia' and women 'atypical schizophrenia' (Lewine 1981). Recently, reviewers reported that 'Sex differences in schizophrenia are one of the most consistently reported aspects of the disease' (Abel *et al.* 2010: 417). These differences remain something of a mystery. In this chapter we suggest that a psychosocial approach might provide some answers in an area where the bio-genetic paradigm has been particularly unproductive.

THE DIFFERENCES

Age of onset

The best-known difference is that men have an earlier onset. Kraepelin actually characterized the illness he invented (Chapter 3) as a disorder of young men. The peak period of onset for men is at the ages of 18–24 years, but for women it is 25–30 years (Castle 2000; Kirkbride *et al.* 2006; Salem and Kring 1998). This difference is 'specific to schizophrenia, not an artefact of admission practices, and similar across cultures' (Goldstein and Lewine 2000: 112). Sub-clinical psychotic experiences are also found more frequently in 17- to 21-year-old males than in females of this age group (Spauwen *et al.* 2003).

However, gender differences in rates of new cases of 'schizophrenia' change over the lifespan (Kirkbride *et al.* 2006). Childhood-onset schizophrenia is between two and three times more common in boys than girls. This difference lessens (Goldstein and Lewine 2000) or disappears (Orr and Castle 2003) in early adolescence, and then the male preponderance reappears again in late adolescence and continues through the twenties (Orr and Castle 2003). In the

thirties, rates are roughly the same, but women over 40 are nearly twice as likely as men to be diagnosed 'schizophrenic' for the first time. The female predominance becomes even greater after the age of 60 (Castle 2000; Kirkbride *et al.* 2006).

Behaviours, thoughts and feelings

Men and women diagnosed 'schizophrenic' tend to have different experiences and behaviours. Among the more consistent findings have been negative symptoms in men and affective symptoms in women (Abel *et al.* 2010; Goldstein and Lewine 2000; Morgan *et al.* 2008). The preponderance of negative symptoms in men has been confirmed in a large general population study (Maric *et al.* 2003).

Men have more cognitive deficits than women. Women tend to be more depressed (Abel *et al.* 2010) and more emotionally expressive in other ways, including being more angry and explosive. Some studies find positive symptoms, overall, distributed equally, while others find a higher overall rate in women (Abel *et al.* 2010; Lewine 1981; Ragland *et al.* 1999). Women tend to experience more auditory hallucinations, paranoia and persecutory delusions (Morgan *et al.* 2008), while men have more grandiose delusions and ideas of reference (believing that events are related to oneself). Women's delusions tend to focus more on spiritual issues and interpersonal wrongs (Goldstein and Lewine 2000; Salem and Kring 1998). Men are far more likely to be experiencing symptoms associated with abuse of drugs or alcohol (Barnett *et al.* 2007), with cannabis use especially common (Abel *et al.* 2010; Cotton *et al.* 2009).

In summary men tend to be emotionally and socially closed down, have problems thinking clearly, self-medicate with drugs and alcohol and sometimes believe themselves to be very important. Women tend to have strong feelings, to dislike themselves, to hear voices and to be frightened that others will hurt them. These tendencies have also been found in non-clinical populations (Beavan *et al.* 2011).

Brain functioning

Men diagnosed 'schizophrenic' not only have more cognitive deficits but also have more structural brain abnormalities, particularly enlarged ventricles and a reduction in hippocampus volume (Jablensky 2009; Salem and Kring 1998).

Social functioning before psychosis

Gender differences have also been documented before the onset of psychosis (Abel *et al.* 2010; Cotton *et al.* 2009). Young girls who later become 'schizophrenic' are more shy and sensitive than other girls, while 'pre-schizophrenic' boys act out more than other boys, sometimes to the point of aggression.

(This is a partial explanation for the higher rates of young boys in mental health services in general, since ‘externalizing’ behaviours are more likely to gain attention and referral.) By adolescence, however, ‘pre-schizophrenic’ male youths, compared with their female counterparts, are more socially isolated and withdrawn and perform less adequately in all social and occupational functions (Morgan *et al.* 2008; Salem and Kring 1998). During late adolescence, the best predictors of ‘schizophrenia’ for heterosexual boys include feeling sensitive, having fewer than two friends, and not having a girlfriend (Malmberg *et al.* 1998). These social difficulties continue after the men become unwell. In the woefully under researched area of sexuality, young heterosexual men diagnosed with ‘schizophrenia’ are significantly less likely to have dated, to have had sex, or be living with a partner than their female counterparts (Test *et al.* 1990).

Course and outcome

Apart from being in better shape before becoming unwell, women also fare better than men *after* being diagnosed ‘schizophrenic’ (Jablensky 2009). Women’s symptoms remit more quickly and they have fewer and shorter hospitalizations (Angermeyer *et al.* 1989). Antipsychotics are even less effective for men than they are for women (Abel *et al.* 2010; Salem and Kring 1998) – although some of the ‘atypical’ drugs are more likely to cause obesity and metabolic and cardiovascular problems in women (Abel *et al.* 2010).

A 20-year study found significantly better outcomes for women on both global outcome scores (work functioning, life adjustment, self-support and relapses) and psychotic symptoms (Grossman *et al.* 2008). Men attending a New Zealand early intervention psychosis service were less engaged with the service than the women, and their symptoms, particularly negative symptoms, were slower to remit (Theuma *et al.* 2007). Men are more likely to kill themselves (which is consistent with the general population) and to self-harm (which is not) (Harvey *et al.* 2008; Test *et al.* 1990).

THE EXPLANATIONS

Biology

As was the case for poverty and ethnicity, the predominant response of biological psychiatry to all these gender differences is to ignore them. ‘The examination of gender differences in schizophrenia and other chronic mental illnesses has not kept pace with the literature on depression’ (Sparks 2002: 306). Only about 1% of the 450-page text *The Epidemiology of Schizophrenia* (Murray *et al.* 2003) deals with gender. The 4,500 page *Comprehensive Textbook of Psychiatry* makes no mention at all of any gender differences (Saddock *et al.*

2009). A *Medline* analysis estimated that only about half of all articles on 'schizophrenia' up to 2010 even record gender, and a mere 2.5% analyse research findings by gender (Barker-Collo and Read 2011).

The few researchers who do address gender differences tend to consider only biological factors as possible causes. Even in a book specifically on *Women and Schizophrenia* (Castle 2000: 24, 28), any psychosocial factors that might explain gender differences are portrayed, via terms like 'artifacts due to psychosocial differences' or 'social confounding factors', as getting in the way of understanding the real, biological causes: hormones and reproductive functions. This is a long-standing tradition, reaching back to the male priest-physicians of Ancient Greece (Chapter 2) and maintained by British and German male psychiatrists of the nineteenth century and beyond, including such revered 'fathers' of modern psychiatry as Henry Maudsley and Thomas Clouston (Masson; 1986; Showalter 1985).

We offer just a few examples of the type of possible explanations for gender differences that come into view once bio-genetic blinkers are removed. None is intended as a singular explanation for any of the gender differences, because none will apply to all women or all men diagnosed 'schizophrenic'.

Gender roles: exaggerated roles as causal factor or diagnostic punishment for non-compliance with roles?

Most of the gender differences in the symptoms of people diagnosed 'schizophrenic' identified above can be found in the general population (Lewine 1981), leading to the conclusion that 'Sex differences in schizophrenic symptomatology may represent an exaggerated reflection of the small but consistent sex differences that are seen in the normal population' (Raine 1992). For example, in two clinical populations of young men, an extreme masculine role (particularly 'restrictive emotionality') was correlated with psychosis (Good *et al.* 1996). So, one possible explanation for the gender differences in 'symptoms' is that severely disturbed young men and women are falling back on an exaggerated version of their learned gender role in an attempt to find solid ground. Alternatively, perhaps adopting an extreme version of the traditional masculine or feminine roles contributes to the development of psychological disturbance. Both could be true, with the two interacting in a downward reciprocal spiral towards either intense helplessness, despair and paranoia for women or, for men, total emotional shutdown and social withdrawal into an internal fantasy world where one believes oneself to be very important.

In *Women and Madness*, however, Chesler (1972) suggests that it is people who do *not* conform to traditional gender roles that are particularly likely to be diagnosed 'schizophrenic'. She cites studies showing that female 'schizophrenics' are more dominant and aggressive than either male 'schizophrenics' or 'normal' women, while 'schizophrenic' men are more passive than 'normal

males'. Clear examples of broken gender roles were among Kraepelin's original symptom list (Chapter 3). Subsequent research has provided additional support, with 'schizophrenics' (male and female) more often demonstrating a 'confused gender role' or choosing opposite gender roles than either 'normals' or other psychiatric patients (LaTorre and Piper 1979). Two interpretations are possible here. One is that 'inappropriate' gender roles are causal factors in mental health problems and require treatment (La Torre and Piper 1979). Chesler, however, proposes that women and men are being punished diagnostically for rejecting societal norms.

There is certainly evidence that men and women *are* treated differently diagnostically. Studies have found that more severe diagnostic evaluations are made of 'physically unattractive' women (Cash *et al.* 1975) and (among 'non-liberal' clinicians) women with 'left' political values, especially if they *act* on those values (Abramowitz *et al.* 1973). Sadly, research into diagnostic bias in relation to gender waned after the 1970s. The few studies that were conducted continue to find bias (e.g. Beckwith 1993). Diagnostic bias, and the disproportionate use of drugs and electroconvulsive therapy on women, have been discussed elsewhere (Busfield; 1996; Johnstone 2000; Russell 1995; Williams 1999). Women continue, internationally, to be twice as likely to receive ECT (Chapter 8).

The stressors of adolescence and early adulthood

One of the defining characteristics of Bleuler and Kraepelin's illness was that it is something that develops early in life. Because of the bio-genetic ideology that spawned the construct, little attention has been given, ever since, to what might be going on in late adolescence and early adulthood that could cause young people to go mad, and how these life events and challenges might help explain the even earlier onset of psychosis in young men. In 1981, however, it was suggested that 'Tasks such as leaving home, establishing independent quarters, obtaining a job, and competing with others seem to be especially associated with schizophrenia in men, whereas childbirth and family conflict are associated with schizophrenia in women' (Lewine 1981: 441).

Harrop and Trower (2001) highlight the blocking of the normal adolescent tasks of separating from and de-idealizing parents, and establishing peer relationships. They also put the hormonal theories about gender differences into perspective by showing that age of puberty and hormonal levels are themselves strongly influenced by the social environment.

Poorer pre-morbid social functioning is associated with earlier onset of psychosis (Gureje and Bamidele 1998), suggesting a relationship between hindered developmental achievements in the interpersonal domain and risk of developing psychosis. While this association between social functioning and early onset is found across genders, men tend to fare worse than women on these factors.

It is possible that another component of traditional masculinity, high expectations for autonomy and independence, may expose young men to more stressors outside home at an earlier age. Expectations to take the initiative in sexual interactions may be an additional burden. Not having a girlfriend is very common for young heterosexual men who later become psychotic (Harrop and Trower 2001; Malmberg *et al.* 1998).

In addition, the finding that at first signs of 'mental illness' more men than women are still living at home (Hafner *et al.* 1998) suggests that a significant proportion of the men may be 'blocked' at the first of the two stages – separation from parents – identified by Harrop and Trower (2001). Men and women are equally damaged by living with highly critical or over-involved parents, but men are more likely than women to be living in such a situation (Bebbington and Kuipers 1995). Men living at home are exposed to more criticism than their female counterparts, perhaps because of gender role expectations regarding autonomy and independence. Parents show less tolerance and greater rejection and criticism of 'schizophrenic' sons than of 'schizophrenic' daughters even when their 'illness severity' is the same (Goldstein and Kreisman 1988; Haas *et al.* 1990). This may partly explain the earlier hospitalization of males (Lewine 1981).

Regardless of the severity of the illness, the parents will be more protective and tolerant of daughters than of sons, while sons will more often be sent to hospital for care. The implicit assumption here is that sons are expected to function in a self-sufficient manner, an expectation that does not exist in the same way for daughters.

(Goldstein and Kreisman 1988: 862)

'Schizophrenics' have childhoods too

Many researchers conclude from the gender differences that men are more prone to a 'poor-prognosis, neurodevelopmental subtype of schizophrenia, for which early environmental brain insults play an important etiological role' (Salem and Kring 1998: 795). However, the only brain insults ever mentioned are *medical* complications, usually around pregnancy and birth. Such complications are more common in men and are related to early onset of 'schizophrenia' in both men and women. This may indeed be a partial explanation for the greater neurocognitive deficits in men and for men having an earlier onset of 'schizophrenia'.

Other possible sources of brain insult have, until recently, been ignored. Child abuse is strongly predictive not only of the symptoms of schizophrenia but of earlier onset and earlier admission to hospital (Chapter 18). One inpatient study found that while only 9% of non-abused men were first admitted before age 18 years, this was the case for 60% of those who had been physically abused as children ($p < .005$) – and 44% of those sexually abused

($p < .05$). The differences for the women were in the same direction but were not statistically significant (Read 1998).

Why might child abuse in men be related to an earlier onset and worse outcome than in women? It has been suggested that: 'The greater vulnerability of the male brain to environmental insults is presumed to underlie the sex differences in severity, age of onset and other aspects of schizophrenia' (Harrison 1995: 6). Traumatic brain injury is significantly correlated with 'schizophrenia' (Malaspina *et al.* 2001). It seems that the male pathway to greater cognitive deficits, more negative symptoms and earlier onset of 'schizophrenia' includes, for some men, structural brain damage (Sachdev *et al.* 2001) resulting from the kind of physical trauma to the head to which boys are exposed at higher rates than girls (Boney-McCoy and Finkelhor 1995).

Another hypothesis is that because men are less likely to disclose child abuse (physical or sexual) (Alaggia 2005), to be asked about abuse (Read and Fraser 1998) or to have abuse disclosures properly responded to (Agar and Read 2002; Sleath and Bull 2010), they are less likely to receive the kind of help that might prevent deterioration into psychosis. A recent study, which confirmed that both men and women who had been abused as children were more likely to experience psychosis as adults, found that the men used the coping style 'seeking guidance and support' significantly less often than the women (Barker-Collo and Read 2011).

The effects of psychological trauma, such as sexual abuse, on the developing brain of the child are remarkably similar to the brain dysfunction and structural abnormalities found in many adult 'schizophrenics' (Read *et al.* 2001). The preponderance of negative symptoms in men and positive symptoms in women might be partially explained by the differential response of boys and girls to trauma. Boys tend to react to psychological trauma with hyperarousal, while girls typically respond with dissociation (Perry 1994; Read *et al.* 2001). Both the dissociative response to trauma (Perry *et al.* 1995) and the positive symptoms of schizophrenia (Andreasen and Olsen 1982) are primarily dopamine-mediated. Meanwhile, the hyperarousal trauma response and negative symptoms, which are both more common in males, are more related to structural brain changes such as cerebral atrophy and ventricular enlargement. Ventricular enlargement is more common in male 'schizophrenics' and is correlated with negative symptoms (Andreasen *et al.* 1990). It is plausible, then, that the typically male hyperarousal response to childhood trauma leads to more profound disturbance, mediated by cerebral atrophy and marked more by negative symptoms, both of which are more common in men. One study even found that the correlation between negative symptoms and early onset is true for men but not women (Gureje and Bamidele 1998). This hypothesis would also explain why men respond less to antipsychotic medication aimed at biochemical imbalance, since their brain disturbance is, because of birth complications, brain injury or

atrophy resulting from their hyperarousal response to trauma, more often of a structural nature rather than biochemically mediated.

CONCLUSIONS

These hypotheses, along with the many other variables potentially mediating the relationship between gender and psychosis, are clearly worthy targets of future research. In particular, they may provide clues about preventable causal factors and helpful clinical interventions. It is a wonder that they have been so neglected to date.

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UNDERSTANDING PSYCHOTIC SYMPTOMS

Cognitive and integrative models

Richard Bentall

STUDYING SPECIFIC COMPLAINTS RATHER THAN 'SCHIZOPHRENIA'

Attempts to understand the biological and psychological causes of mental illness have been confounded by the absence of an agreed and scientifically valid method of classifying the phenomena of interest. Throughout much of the past century, Kraepelin's (1899/1990)'s concept of *dementia praecox*, reformulated by Bleuler (1911/1950) as *schizophrenia* (Chapter 2), was the central organizing concept around which theories of severe psychiatric symptoms were constructed. However, many now agree that this concept has outlived its usefulness (Bentall 2003, 2009). The concept collects together symptoms so heterogeneous that two people with the diagnosis may have no symptoms in common, its borderlines with other diagnoses such as bipolar disorder are disputed and it fails to predict either the course of illness or response to treatment (Chapters 5–7).

But whereas many researchers now agree that the concept of schizophrenia is an impediment to progress in psychopathology, there is much less agreement about how to replace it. One approach has been to argue that there is but a single psychosis syndrome, accompanying the categories of schizophrenia, schizoaffective disorder and bipolar disorder contained in traditional classification manuals (Berrios and Beer 1995). Another, more widely discussed approach advocates dimensional models, which are apparently supported by a phenomenological continuum between psychotic symptoms such as hallucinations and delusions and normal mental states (Linscott and van Os 2010). For example, it has been argued that the symptoms attributed to schizophrenia, schizoaffective disorder and bipolar disorder can all be encompassed within a five-dimensional model consisting of positive symptoms, negative symptoms, symptoms of cognitive disorganization, depression and mania (van Os and Kapur 2009). Statistical analyses of covariations

between the symptoms reported by psychiatric patients suggest that there is some merit in each of these approaches, so that, at the highest level of generality, there seems to be a psychosis syndrome, but also variation between patients on the five dimensions (Reininghaus *et al.* 2012).

Whether these novel classificatory proposals can inform research into the aetiology and mechanisms responsible for severe mental illness remains to be seen. A further approach, which makes absolutely minimal assumptions about the structure of mental illness, involves attempting to explain the actual experiences and complaints of psychiatric patients. After all, unless educated to do so by well-meaning but possibly misguided mental health professionals, patients rarely come to the psychiatric clinic complaining of schizophrenia. Instead, they or their carers complain that they hear voices, have strange beliefs, lack motivation, or suffer from some other variety of phenomena that psychiatrists think of as symptoms. In the past two decades, research focusing on some of these specific phenomena has led to enormous progress in understanding their causes. Arguably, a complete and satisfying science of psychopathology can be constructed in this way. (For a detailed exposition see Bentall 2003.) In this chapter, I illustrate this approach by discussing recent research on auditory-verbal hallucinations (AVHs) and paranoid beliefs.

AUDITORY-VERBAL HALLUCINATIONS (AVHS)

Some observations about auditory hallucinations

AVHs are one of the most common complaints reported by people with a diagnosis of schizophrenia, but are also reported by patients diagnosed as suffering from severe affective disorders, such as bipolar disorder (Goodwin and Jamison 2007) and major depression (Lattuada *et al.* 1999) and by patients diagnosed as suffering from dissociative disorders (Ross *et al.* 1989) and borderline personality disorder (Slotema *et al.* 2012). One or many voices may be heard (Nayani and David 1996), consisting of a voice or voices speaking the individual's thoughts aloud, carrying out a running commentary on the person's behaviour, or speaking about the individual in the third person (Hamilton 1983). Voices issuing commands are common and are often a source of concern if the commands include injunctions to hurt the self or others, although, fortunately, obedience to such commands is rare (Juninger and McGuire 2001; McNeil *et al.* 2000). In the majority of cases, hearers believe that they know the identity of the voice or voices (Leudar and Thomas 2000), a finding confirmed by a study of 50 voice-hearers from the general population, which also found that often the content of the voices is personally meaningful, the person has a relationship with the voices, the experience has a significant impact on the voice-hearer's life and the experience has a compelling sense of reality (Beavan 2011).

A distinction has sometimes been drawn between 'true hallucinations' and 'pseudohallucinations' defined as perceived as occurring within the body (Jaspers 1913/1963) or because they are known to be not real by the perceiver (Sedman 1966). *DSM-IV-TR* acknowledges that 'One person with auditory hallucinations may recognize that he or she is having a false sensory experience, whereas another may be convinced that the source of the stimulus has an independent physical reality. . . . Some clinicians and investigators would not include [as hallucinations] those experienced as coming from inside the head' (American Psychiatric Association 2000: 767), but then goes on to argue that neither of these distinctions has any clinical utility. There is no research-based reason for disputing this conclusion, and clinical experience would tend to support it, although attempts to identify mechanisms specific to different kinds of hallucination have so far been rare and inconclusive.

Early psychological theories suggested that AVHs might be the result of conditioning processes, or some kind of intrusion of unconscious material into consciousness (perhaps because of some kind of dysfunction of mechanisms regulating wakefulness and sleeping), or abnormal mental imagery, but each of these theories has been shown to be inadequate (Bentall 1990), either on conceptual or empirical grounds. (For example, it is not obvious how to measure mental imagery objectively, and therefore unsurprising that attempts to compare imagery vividness in hallucinating and non-hallucinating individuals has led to completely inconsistent findings.) More recent theories have been informed by five crucial observations.

First, hallucinations are *much more common than was once thought*. Although fewer than 1% of the population of developed nations are diagnosed as suffering from schizophrenia (McGrath 2005), epidemiological studies have consistently reported that 5–10% of the population experience hallucinations at some point in their lives (Tien 1991; van Os *et al.* 2000), with a recent review of 17 studies of voice-hearing estimating a median (the mid-point of the findings) of 13.2% (Beavan *et al.* 2011). It appears that many people experience hallucinations, often over long periods, without feeling the need to seek psychiatric help (Beavan *et al.* 2011; Romme and Escher 1993). Comparisons between people who cope with voices and psychiatric patients suggest that it is differences in attitudes towards hallucinations that are important, rather than differences in phenomenology. Copers typically feel more powerful than their voices, whereas patients often feel submissive and subordinate towards them (Honig *et al.* 1998). Causal explanations about voices (e.g. 'I'm mad', 'It's a gift') and differences in the stress generated by various explanations are also predictive of who becomes disturbed following the onset of voices (Morrison *et al.* 2004).

Second, the tendency to experience *hallucinations appear to vary with culture*. A variety of studies (Al-Issa 1995), including large-scale cross-cultural investigations (Jablensky *et al.* 1992), have reported that visual hallucinations are more commonly experienced in the developing than the industrialized

world. There is an unresolved question of whether shamanic experiences reported in some cultures are equivalent to the hallucinatory experiences of psychotic patients in the developed countries. Whether or not this is true, there is no question that, in some cultures, some kinds of hallucinatory experiences are positively valued and shaped by specific cultural practices and local theories about mental processes (Lambrecht and Taitimu 2012; Luhrmann 2011).

Third, hallucinatory experiences are *affected by ongoing environmental stimulation*: individuals with a disposition towards auditory hallucinations are especially likely to experience them during periods of reduced sensory stimulation or when stimulation is unpatterned (Gallagher *et al.* 1994).

Fourth, it has long been known that auditory hallucinations are associated with activations of the speech musculature (a phenomenon known as *subvocalization*) (Gould 1948; McGuigan 1966) and brain activations that, although complex, involve both speech production and speech perception areas in the brain (Jones and Fernyhough 2007). Consistent with these observations, when individuals experiencing auditory hallucinations engage in verbal tasks such as reading, naming objects or humming, this temporarily suppresses their voices (Gallagher *et al.* 1995); when the person is unable to generate inner speech, the voices abate.

The fifth, more recent observation, the significance of which is only just beginning to be appreciated, is that hearing voices is associated with a history of childhood trauma. Consistent with models of psychosis that emphasize the role of early experience (e.g. Read *et al.* 2001), a very large volume of research conducted over the last decade has convincingly demonstrated that childhood adversities, varying from different kinds of abuse, through bullying, to the sudden loss of a parent, can increase the risk of psychosis in later life (Varese *et al.* 2012a). Moreover, a number of studies have reported a specific association between childhood sexual abuse and hallucinations (Hammersley *et al.* 2003; Read *et al.* 2003; Shevlin *et al.* 2007). Elevated rates of childhood sexual abuse have been reported not only in patients who hear voices, but also in 'healthy' hallucinators (Daalman *et al.* 2012). Recently, I tested this association in an epidemiological dataset of more than 7,000 UK citizens, finding that reports of childhood rape were specifically linked to hallucinations in adulthood but not to paranoid symptoms (Bentall *et al.* 2012).

Some theories about auditory hallucinations

Confusing external and internal stimuli ('source monitoring')

The findings on subvocalization point to a close relationship between auditory hallucinations and inner speech (roughly, verbal thinking, the process by which we talk covertly to ourselves). This develops during early childhood (Vygotsky 1962) and, in ordinary people, is accompanied by subvocalization

(McGuigan 1978). Many of the theories that try to account for this relationship (e.g. Bentall 1990; Frith 1992; Laroi and Woodward 2007) amount to the proposition that people hear hallucinated voices when they misattribute their own inner speech to a source that is external or alien to themselves. Hallucinations in other modalities can presumably be explained in the same way. Visual hallucinations, for example, may be the result of failing to discriminate between mental imagery and visual perceptions.

This theory has stimulated attempts to investigate the ability of hallucinating persons to discriminate between their own verbal thoughts and other stimuli (a process known as source monitoring or reality monitoring). Three paradigms have been employed. In signal detection studies (e.g. Barkus *et al.* 2007; Bentall and Slade 1985; Varese *et al.* 2011) participants listen to brief bursts of white noise and try to identify those on which a 'signal' (typically a voice) is superimposed. The general finding is that both hallucinating patients, and people scoring high on measures of hallucination-proneness, compared to controls, show an abnormal bias to detect signals (under conditions of uncertainty they assume that signals are present) but no decrement in perceptual sensitivity. In source monitoring experiments (e.g. Bentall *et al.* 1991; Brebion *et al.* 2005; Rankin and O'Carroll 1995), participants are asked to generate words to cues and also to listen to word lists. Later they are asked to identify which words had been self-generated and which other-generated. The general finding (with a few exceptions) has been that individuals who hallucinate, compared to appropriate controls, are more likely to mistake self-generated words for words that had been presented to them.

Finally, in a paradigm developed by Louise Johns (Johns and McGuire 1999; Johns *et al.* 2001, 2006), individuals read into a microphone, receiving feedback through headphones. Sometimes the feedback is distorted, and at other times it is replaced entirely by someone else's voice. Consistent with the source monitoring hypothesis, the general finding (with some exceptions; Vermissen *et al.* 2007) is that people who hear voices are poor at detecting when their own voice is replaced. Overall, the findings from these and similar studies suggest that people who hear voices are poor at recognizing their own voice and therefore more likely to attribute it to someone else (Waters *et al.* 2010).

These findings help to explain the increased tendency to hear voices when exposed to unpatterned stimulation (e.g. when in a noisy crowd). This presumably reflects the fact that judging the source of a stimulus is more difficult in these circumstances. However, it is less easy to explain *why* hallucinating patients are more prone to make these errors. One possible explanation is some kind of malfunction in the neuropsychological mechanisms responsible for judging the source of a stimulus. In a series of electrophysiological and fMRI studies, Judith Ford and colleagues have shown that the generation of inner speech results in an efference copy signal that causes reduced responsiveness in the auditory cortex (in effect, the frontal speech-generating parts of the brain tells the auditory cortex not to pay attention) and that this

process is impaired in people who hear voices (Ford and Mathalon 2004; Whitford *et al.* 2011).

Trauma and dissociation

The specific association observed between childhood sexual abuse and hallucinations has also been examined within the source monitoring framework. It is possible that childhood trauma directly affects source monitoring ability, but there is currently no evidence of this. It is therefore more likely that early trauma leads to other abnormal psychological processes that, combined with source monitoring deficits, lead to voices. There is currently considerable interest in the role of dissociative processes in this regard (Moskowitz *et al.* 2008). These processes involve disruption of the integration of conscious experiences with ongoing activity (for example, leading to feelings of derealization and, in extreme cases, fugue states) and are widely recognized to be consequences of childhood adversity (Kisiel and Lyons 2001; Sanders and Giolas 1991). Several studies have reported high levels of dissociation in patients with psychosis (Schafer *et al.* 2011), especially in those patients who suffer from AVHs (Perona-Garcelan *et al.* 2012a). In two recent studies dissociative experiences appeared to mediate between childhood trauma and voice hearing – trauma was associated with high levels of dissociation, which, in turn, were associated with current voice hearing (Perona-Garcelan *et al.* 2012b; Varese *et al.* 2011b). However, in one of these studies source monitoring (measured using the signal detection paradigm) was found to be equally impaired in patients with current hallucinations and those who had previously experienced hallucinations. Also, source monitoring was not related to dissociation or childhood trauma (Varese *et al.* 2011b). These findings suggest a two-factor model in which voice hearing results from a combination of poor source monitoring (a vulnerability factor) and trauma-related emotional processes.

Despite the tangible progress in understanding the social, psychological and neurobiological mechanisms that lead to hallucinations, many questions remain unanswered. For example, it is not clear why the phenomenology of hallucinatory experiences (e.g. the number and function of voices) varies between individuals. To some extent this reflects our incomplete knowledge of the process of inner speech itself, which has not been sufficiently studied by investigators of normal cognition (Ferryhough 2004).

PARANOID DELUSIONS

Until recently, delusions were often dismissed by researchers in psychology and psychiatry as ‘empty speech acts, whose informational content refers to neither world or self’ (Berrios 1991). This flies in the face of clinical

observations that patients' abnormal beliefs nearly always reflect concerns about their position in the social universe (Bentall 1994) or core existential anxieties (Musalek *et al.* 1989). For example, surveys of psychiatric patients have shown that the most common type of delusional belief is persecutory (paranoid) (Moutoussis *et al.* 2007), followed by extreme negative beliefs about the self and grandiose beliefs of unusual powers, special identity or secret wealth (Garety and Hemsley 1987). These findings seem to have some cross-cultural validity (Stompe *et al.* 1999).

Paranoia as exaggerated but understandable wariness

Paranoid beliefs appear to lie on a dimension that encompasses ordinary feelings of suspiciousness and mistrust, suggesting that extreme persecutory beliefs reflect an exaggeration of healthy wariness about the intentions of others (Freeman *et al.* 2005). This observation is concordant with the suggestion that paranoia is an adaptive process under some circumstances, so that patients see their paranoia as a logical, valued strategy for avoiding threatening events (Morrison *et al.* 2011). Consistent with this idea, there is evidence from epidemiological studies that the onset of paranoid beliefs is often preceded by actual experiences of victimization (Janssen *et al.* 2003; Mirowsky and Ross 1983) and also by early life experiences that disrupt the formation of normal attachment relationships (Bentall *et al.* 2012). This last observation is significant because attachment researchers regard the child's early bond with a caregiver as providing the psychological template upon which secure and trusting relationships can be built in the future (Read and Gumley 2008). Hence, clinical paranoia most likely reflects an extreme sensitization of normal mechanisms that are involved in the detection and avoidance of social threats. These mechanisms are thought to involve neural circuits in the basal ganglia that utilize dopamine (Moutoussis *et al.* 2008), which provides an obvious point of contact with biological models of psychosis (Moutoussis *et al.* 2007).

Despite the apparent continuum between clinical and sub-clinical paranoia, some differences between the beliefs of psychiatric patients and the paranoid fears of non-patients are important. Clinical paranoia tends to be relatively unaffected by social context, fluctuating relatively independently of circumstances, whereas non-clinical paranoia tends to worsen under stress or in the presence of unfamiliar others (Collip *et al.* 2011). Based on detailed clinical observation, Trower and Chadwick (1995) have argued that the paranoid beliefs of patients can be divided into two main types: 'poor-me', in which patients believe themselves to be the innocent victims of unfair persecution, and 'bad-me', in which patients feel that they are being persecuted for good reason (for example, because they have done terrible things). Not surprisingly, poor-me patients typically have higher self-esteem than bad-me patients (Chadwick *et al.* 2005). Paranoia in non-clinical samples is almost always bad-me and associated with very low self-esteem

(Melo *et al.* 2006) whereas, in patients, it is much more often poor-me (Fornells-Ambrojo and Garety 2005), or fluctuates between poor-me and bad-me (Melo *et al.* 2006; Udachina *et al.* 2012).

Explaining anomalous experiences

Various psychological models of delusion formation have been proposed. One of the earliest suggested that the apparently irrational beliefs of psychotic patients might reflect rational attempts to explain anomalous experiences (Maher 1974). For example, delusions might be the consequence of rational attempts to explain hallucinatory experiences. Note that, according to this model, reasoning is unimpaired in deluded patients. In fact, empirical studies of the association between delusional thinking and anomalous experiences have produced results that have been mixed. Some rare delusional systems (most notably the Capgras delusion, in which the individual believes that a loved one has been replaced by an impostor) can be explained in this way (Capgras patients suffer from a subtle impairment in their ability to tell whether faces are familiar; Ellis and Lewis 2001). Nonetheless, the evidence for the role of anomalous experiences in more common delusional systems is less clear-cut.

Freeman and colleagues (2009, 2012) for example, have shown that clinical paranoia is closely associated with insomnia, which potentially might be explained in the framework of the anomalous experience model. (It could be argued that chronic sleep deprivation leads to subtle changes in conscious experience, which might provoke delusional interpretations, although an obvious alternative is that there is a direct effect on reasoning ability.) Using experience sampling (a method by which participants are cued to make repeated reports of their immediate experiences, e.g. by beeps from digital devices), Ben-Zeev *et al.* (2011) found no association between the occurrence of anomalous experiences and the onset of paranoid episodes. There is more support for the idea that the slow onset of deafness can provoke paranoid thinking (especially if the individual remains unaware of his or her deafness and hence begins to wonder why people are no longer being communicative). Although early tests of this hypothesis were inconclusive (Cooper *et al.* 1976; Watt 1985), recent epidemiological studies have tended to support the association (Thewissen *et al.* 2005; van der Werf *et al.* 2010).

Biased reasoning

Jumping to conclusions

Several kinds of reasoning bias have been associated with paranoid thinking. Hemsley and Garety (1986) first suggested that people might develop abnormal beliefs as a consequence of being unable to weigh up evidence that either

favoured their theories or were at variance with them. In an influential series of studies, they set out to measure probabilistic reasoning in deluded patients (Garety *et al.* 1991; Huq *et al.* 1988). Participants were shown two jars, in each of which there were beads of two colours. In one jar, one colour predominated; the beads in the other jar were of the opposite ratio. They were then shown a sequence of beads apparently drawn from one of the two jars and were told to ask for beads until they felt able to decide which jar they were being taken from. Deluded patients, in comparison with others, ‘jumped to conclusions’, guessing more quickly. This observation has been replicated many times, sometimes with patients specifically suffering from paranoid delusions (Corcoran *et al.* 2008), sometimes with different tasks (John and Dodgson 1994), and sometimes with non-patients who score highly on a measure of delusional beliefs (Linney *et al.* 1998). The bias appears to be specifically associated with delusional thinking rather than psychosis in general (Bentall *et al.* 2009; Moritz and Woodward 2005) and worsens during emotional stress (Lincoln *et al.* 2010). However, the underlying psychological mechanisms responsible are unclear: a recent Bayesian analysis of beads in the jar data from psychotic patients and controls concluded that the deluded patients’ early responses occurred because they responded more randomly, presumably because they lacked the executive function abilities to integrate sequential information across time and respond appropriately (Moutoussis *et al.* 2011).

Difficulty reading other people (‘theory of mind’)

A second type of reasoning abnormality with respect to paranoia that has been investigated concerns the ability to understand the mental states of others. It has long been understood that a lack of ‘theory of mind’ (ToM) skills is a core feature of autism (Baron-Cohen 1995). Frith (1994) argued that the temporary loss of ToM skills during an acute psychotic episode might lead to misunderstandings about the intentions of others and, if these intentions were believed to be concealed, to paranoid beliefs. Early studies indeed supported the idea that paranoid patients were impaired on ToM tasks (Corcoran *et al.* 1997; Frith and Corcoran 1996), although later studies implicated ToM in other symptoms of psychosis, such as thought disorder (Sarfati *et al.* 1997). A large transdiagnostic study comparing paranoid and non-paranoid patients found that ToM impairment was specifically related to paranoia but also appeared to be related to more general executive dysfunction (Bentall *et al.* 2009).

Attributions: defending against low self-esteem?

The rich psychological data on paranoid thinking that have emerged over the past two decades has prompted several attempts to formulate integrative models. A model proposed by the present author and revised several times in the light of new evidence (Bentall *et al.* 1994, 2001) was originally prompted

by the results of a study in which paranoid patients were found to exhibit a specific attributional style (style of explaining events) in which negative experiences are attributed to causes that are global (likely to affect all areas of life), stable (likely to persist) and external (the cause is something to do with the actions of other people or circumstances, rather than the self) (Kaney and Bentall 1989). A subsequent study found that paranoid patients tended to make specifically personal attributions (the cause is the actions of others) rather than situational/circumstance attributions for negative events (Kinderman and Bentall 1997). These findings led to the idea that paranoid beliefs serve a self-protective function, so that the tendency to attribute negative experiences to the actions of others, while feeding a paranoid worldview, prevents awareness of negative beliefs about the self. The model, which bears some similarity to earlier psychodynamic models (e.g. Colby 1977), therefore proposes that paranoia arises from dysfunctional attempts to defend against low self-esteem.

Emotional vulnerability

A somewhat simpler model has been advanced by Daniel Freeman and colleagues (2002, 2008), who have argued that, rather than being a form of defence, paranoid delusions are a direct extension of emotional concerns of the individual. They have argued that, at the delusion formation stage, stressful life events, drug use or a number of other factors (perhaps provoking the kind of anomalous experiences that Maher hypothesized to be important) combine with pre-existing emotional vulnerabilities to drive a search for meaning that culminates in a paranoid hypothesis about the world. They point out that, during this search for meaning, individuals are likely to draw upon pre-existing beliefs about the self, others, and the world. On this view (and consistent with the evidence on the social origins of paranoia discussed above), individuals will be more likely to develop a paranoid explanation if their earlier experiences (such as trauma) have provoked concerns about personal vulnerability. Pre-existing anxiety is considered to be particularly important in this model, as it reflects expectations about impending danger. However, other emotions such as anger and depression may also add to the contents of the delusional system. Cognitive biases such as the tendency to jump to conclusions, theory of mind impairments and attributional biases are considered to play an important role in shaping the way that individuals try to understand the meaning of their experiences.

Testing and integrating two models

For reasons of space, it is impossible to discuss in detail the considerable efforts that have been directed towards adjudicating between the external attributions as defence against low-self-esteem model and the emotional

vulnerability model. Freeman (Freeman *et al.* 1998) pointed out that patients with paranoid beliefs tend to have low self-esteem, which has subsequently been confirmed (Bentall *et al.* 2008). This is arguably inconsistent with a defensive model.

Some studies (McKay *et al.* 2007; Moritz *et al.* 2006; Valiente *et al.* 2011) – but not all (Vazquez *et al.* 2008) – that have attempted to compare explicit (consciously reported) and implicit (revealed by indirect measures) self-esteem, find that paranoia is associated with a discrepancy between the two, which might be thought consistent with the idea of defensive processes. Other studies find that self-esteem in paranoid patients is highly unstable over both short (Thewissen *et al.* 2008) and long periods (Thewissen *et al.* 2007), which also might be thought to reflect abnormal strategies for regulating self-esteem. In patients, dramatic shifts in self-esteem seem to be associated with shifts between poor-me and bad-me delusional thinking (Melo and Bentall 2012). More importantly, and significantly for the defensive hypothesis, in an experience sampling study (in which patients recorded their symptoms and emotions many times a day, cued by electronic devices), Udachina *et al.* (2012) found that, in patients who currently held bad-me beliefs, an exacerbation of paranoid fears led to a drop in self-esteem, whereas in poor-me patients the onset of severe paranoid thoughts was followed by an improvement in self-esteem.

A second focus has been on the attributions made by paranoid patients. Studies in this area have been highly inconsistent, some (e.g. Fear *et al.* 1996) supporting the original observations of Kaney and Bentall (1989), others (e.g. Diez-Alegria *et al.* 2006; Humphreys and Barrowclough 2006) giving contrary results. A complication when interpreting these findings is the selection of participants. Several research groups report that attributional abnormalities are only found in paranoid patients, and not in people with sub-clinical paranoia (Janssen *et al.* 2006; McKay *et al.* 2005). Some find attributional biases only in patients with poor-me delusions (Fornells-Ambrojo and Garety 2009; Melo *et al.* 2006).

Overall, the evidence in favour of defensive processes is mixed. If these processes are involved, they are almost certainly only operative when patients suffer from acute poor-me delusions. As sub-clinical paranoia almost always tends to be of the bad-me variety, it is likely that, during the initial stages of the development of paranoid delusions, the emergence of paranoid fears is directly stimulated by the pre-existing emotional concerns of the individual as suggested by Freeman and colleagues (2002, 2008). However, it is also worth noting that the two models described here have much in common, emphasizing the role of early adversity, specific cognitive biases and particularly the key role of self-esteem related processes in the development of paranoid thinking.

Given the substantial consensus that is emerging about the kinds of processes that are important in paranoia, future research needs to examine how these processes change longitudinally, during the lifetime of the individual, within a developmental framework.

EXPLANATIONS AND CAUSES

I hope that this brief summary of what is known about hallucinations and delusions is sufficient to convince the reader that research on specific complaints constitutes a viable alternative to the Kraepelinian paradigm. Positive symptoms have been most extensively investigated in this way, but there has also been considerable work on psychotic speech disorder and, in recent years, an emerging research literature on specific negative symptoms such as anhedonia. If this programme of research is successful, then, once the full range of psychotic complaints has been explained, there will be no ghostly 'schizophrenia' left behind also requiring an explanation.

It has been argued that the complaint-orientated approach does not reveal the aetiology of severe mental illness (Mojtabai and Rieder 1998). This objection is no longer valid – if, indeed, it has ever been true – as research on individual complaints has increasingly focused, not only on the cognitive and emotional processes that are immediate precursors of the complaints, but also on biological and environmental influences on those processes. In this chapter, I have given examples of research on the role of neurobiological concomitants and environmental determinants of both voices and paranoia.

Although there is insufficient space to fully develop the claim (see Bentall 2003, 2009), I believe that complaint-orientated research is leading to a much richer understanding of the aetiology of psychotic experiences than has been possible within the framework of the Kraepelinian paradigm. It therefore offers us a better chance of developing more effective ways of helping people to deal with their complaints (Chapter 22).

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PSYCHODYNAMIC APPROACHES TO UNDERSTANDING PSYCHOSIS

Defenses against terror

Brian Koehler, Ann-Louise Silver and Bertram Karon

As psychoanalysts who work with people diagnosed as having a ‘schizophrenic’ disturbance, our contribution to understanding causation comes in the form of trying to understand the personal meaning of psychosis in the context of the particular lives of individual human beings. In this chapter we present a selective review of how psychodynamic (psychoanalytic) ideas have been applied to psychosis over the last century, highlighting the work of some individual clinicians whose work we have found of particular value in building our own understanding. We then outline the views we have come to on the basis of our reading of the literature and our own clinical experience. We offer first our view of how psychosis may be understood from a psychodynamic perspective, and then our perspective on some key aspects of psychodynamic therapy.

A CENTURY OF PSYCHODYNAMIC THEORY AND PRACTICE

Psychodynamic therapies follow in the centuries-old tradition of moral treatments – that is, treatments that include striving to understand each unique patient, developing a full and accurate history, encouraging work and socializing, and never humiliating patients. Moral treatment became eclipsed by somatic treatments, though without these being adequately shown to be better (McGovern 1985; Warner 2007; Whitaker 2002, 2010).

At the beginning of the last century Freud thought that sufferers from psychosis could not be analyzed, believing that they could not form a transference relationship that allowed analytic work. He told his younger colleague,

Richard Sterba, that schizophrenic patients scared him (B. Karon, personal communication). Nonetheless, he expressed hope that future modifications in technique would make psychoanalytic therapy for psychosis possible (Freud 1905). Carl Jung, however, asserted that psychoanalytic approaches were very relevant in psychosis, published this view in *The Psychology of Dementia Praecox* (1907) and was reiterating it five decades later. Eugen Bleuler, the psychiatrist responsible for introducing the term 'schizophrenia', worked with Jung and applied psychoanalytic ideas at the Burghölzli Clinic in Switzerland, apparently with promising results (Federn 1943). Other prominent early analysts also worked with psychosis, including Ferenczi, the charismatic founder of psychoanalysis in Hungary, who launched the interpersonalist branch of psychoanalysis (Dupont 1988; Silver 1993). Other early analysts who worked with psychosis included Karl Abraham, Robert Waelder, Paul Federn and Gustav Bychowski, the founder of psychoanalysis in Poland.

In Britain, Melanie Klein, a student of Abraham and Ferenczi, developed her theory of internalized object relations focusing on conflicts over hostile and sadistic fantasies (Klein 1948), which many therapists working with people with psychosis have found to be of crucial importance. Neo-Kleinian analysts such as Bion, Rosenfeld and Segal treated borderline and psychotic individuals, using approaches that included interpreting their patients' positive and negative feelings about the analyst and the underlying sources of manifest and latent anxiety. Kleinian analysts understand 'schizophrenic' phenomenology as a defense against overwhelming annihilation and persecutory anxieties. They argue for interpretation of unconscious material at the level of greatest anxiety, to develop awareness of the links between fantasy and reality (Jackson 2001; Jackson and Williams 1994). Winnicott (1971), who developed a profound interest in play, also proposed that what we see in psychotic phenomena is a defense against primitive agonies. Fairbairn (1954) and Guntrip (1969) developed object relations theory, stressing the universal human need for a good object (person), and that we all feel compelled to preserve the fantasy of the parents' goodness, even at the cost of maintaining the badness of the self. They suggested that fearing that one's love will destroy the object ('schizoid anxiety') is more debilitating than fearing that one's anger will destroy the object ('depressive anxiety'). Guntrip also described other deep anxieties of relevance for some individuals with psychosis: the fear of never finding one's way back to a human relationship, as well as the dread of losing oneself in the merger experience, being merged into oblivion.

Herbert Rosenfeld (1987) was amongst those applying Kleinian theory to treating psychoses. He suggested that during the acute state, many patients diagnosed with 'schizophrenia' unconsciously place fragmented parts of themselves completely into objects (animate or inanimate, hated or loved). These objects are then felt to so dominate the patients that there is very little experience of a separate autonomous self left. Rosenfeld believed that a precise interpretation, offered sensitively, can let a psychotic patient feel held together

by the analyst's words. He suggested that patients also often respond to interpretations as reflections of the analyst's state of mind, in particular his or her capacity to retain quietness while focusing on the patient's conscious and unconscious anxieties. Severely traumatized patients may compulsively repeat past traumatic interactions and situations within the therapy, projecting terrifying experiences into the analyst. Rosenfeld suggested that frequently work with psychotic patients activates the analyst's own psychotic anxieties, most frequently the dread that the patient will drive one mad. He suggested that the analyst must avoid the extremes of, on the one hand, over-involvement and entanglement or, on the other, of emotionless detachment and objectification, and that the violence of the patient's projections diminishes gradually if the analyst correctly interprets the patient's anxieties and their need to share these experiences.

In the United States in the first decades of the twentieth century, psychodynamic approaches were part of the treatment offered for severe mental illness by prominent figures, including Adolf Meyer, the first head of psychiatry at Johns Hopkins University, William Alanson White and Smith Ely Jelliffe in the Binghamton State Hospital and the St. Elizabeth's Federal Hospital in Washington. St. Elizabeth's developed into a national center for the psychodynamic treatment of severe mental illness (Noble and Burnham 1969). Psychoanalyst Harry Stack Sullivan worked there for a time. Sullivan's emphasis on the interpersonal roots and treatment of schizophrenic disturbances was an important influence on psychodynamic approaches in schizophrenia (Evans 1996). In 1935, Frieda Fromm-Reichmann, another psychoanalyst who made major contributions in psychosis, escaped Nazi Germany and came to Washington. Her work with psychosis is perhaps best portrayed in the book *I Never Promised You a Rose Garden*, in which Joanne Greenberg, who was diagnosed with psychosis and was severely disturbed for several years, describes the recovery she achieved with the help of therapy with Fromm-Reichmann in Chestnut Lodge (Greenberg 1964). Fromm-Reichmann and Sullivan called for analytically oriented treatments for psychosis and through their students influenced practice worldwide (Hornstein 2000; Silver 1989, 1993). The most prolific and influential of these, Harold Searles (1967), along with Boyer and Giovacchini (1967), Volkan (1976) and Kernberg (1986), integrated insights from British object-relations with American ego-psychology.

Harold Searles also worked at Chestnut Lodge in the United States, which became widely known for its psychoanalytic treatment of psychosis and set a standard for brutally honest self-awareness (Searles 1967, 1979). Searles, like Semrad (1969), Robbins (1993), Garfield (1995) and others, understands 'schizophrenic' experience as frequently being a defense against intense emotions. He suggests that the most extreme defense is failure to maintain a human identity. Searles describes a phase of therapy, which he terms 'therapeutic symbiosis', where there is a state of emotional oneness, an extremely

close attachment between patient and therapist, which he sees as crucial for all patients, psychotic or otherwise. He sees this as leading on to a process of mutual re-individuation and re-humanization once each person has accepted his or her feeling of having developed non-human defensive features and lets these come into play in the patient–therapist relationship (Searles 1965, 1979). Searles suggests that the patient's illness gradually serves as a security blanket for the patient and also comes to take on this meaning for the analyst, so that the eventual resolution of patients' symptoms causes both satisfaction and loss.

Gaetano Benedetti joined the staff at Burghölzli Clinic, where Bleuler had worked with Jung, in 1947 (two years before Searles came to Chestnut Lodge). As the psychopharmacologic era began, Benedetti continued to emphasize psychodynamics, working closely with Gustav Bally, Medard Boss, Marguerite Sechehaye and Christian Müller. In 1956, with Müller, Benedetti founded the International Symposium for the Psychotherapy of Schizophrenia, which evolved into the currently thriving International Society for Psychological and Social Approaches to Psychosis (www.isps.org). Benedetti proposes a model of schizophrenia in which he sees as the core psychological problem two incompatible nuclei of the self, each of which is non-self with respect to the other. One is characterized by excessive symbiotic needs, the other by excessive autistic needs for separation. 'The two nuclei of the self, symbiotic and autistic, fragment each other every time separative needs or symbiotic needs are both contemporaneously present and therefore intensely felt because they are not integrated' (Peciccia and Benedetti 1998). Patients react to the danger of fusion by autistic retreat. However, this withdrawal does not provide safety or protection, since the patient feels alone in his isolated world and risks loss of self (Koehler 2003). This model incorporates the primary dialectic between autonomy and interdependency, identified by many as key determinants in personality development and psychopathology in general (Blatt and Ford 1994). In relation to therapy with people with psychosis, Benedetti comments on the kind of therapeutic relationship that therapists need to offer. He suggests that therapists must provide the patient's ego 'with that amount of narcissism [self-love] it needs to integrate and understand itself' (Benedetti 1990) and that when we partially identify with patients' suffering, we encourage them to 'dare to seek out the places in our unconscious where human existence comes into contact with death' (Benedetti and Peciccia and Benedetti 1998). He also cites the risk to the therapeutic relationship in psychosis when interpretation 'meets a nothingness into whose vacuum the therapist may feel himself being dragged (from which) springs up fear of, and social aggression towards, psychotics' (Benedetti 1987: 187).

Bertram Karon, working in the United States, made valuable contributions to research into the effectiveness of psychodynamic therapy. His study compared, and blindly evaluated, chronic (previously unmedicated) inner-city patients suffering from schizophrenia randomly assigned to one of three

groups: (a) an average of 70 sessions of five-times-weekly psychoanalytic therapy without medication; (b) three-times-weekly psychoanalytic therapy, combined with medication; or (c) medication and support. Psychoanalytic therapy with continued medication was concluded to be better than medication alone, but not as helpful as psychoanalytic therapy without medication (or with initial medication withdrawn as the patient could tolerate it). While there are greater improvements in symptoms, work, affect, relations with spouses and children, discharge and less rehospitalization, in this research the most striking advantage of psychoanalytic therapy compared with medication is the greater improvement in thought disorder – that is, the ability to think rationally when you want to (Karon and VandenBos 1981). There were, however, problems in the study design which make it difficult to draw firm conclusions. The control group was transferred to a long-term hospital, and the group receiving only therapy had more frequent therapy sessions than the therapy-plus-medication group.

In Europe in the closing decades of the twentieth century, developments emerged both in practice and in research. Attention was focused on integrating therapy with other elements of multidisciplinary care and on developing approaches that explicitly emphasize supportive rather than exploratory techniques, particularly in the early stages of therapy, and that attend explicitly to the changing needs of patients at different stages of illness and recovery. There has been renewed interest in research into the effectiveness of contemporary models of therapy – for example, in Denmark, where a supportive approach has been studied in semi-randomized trials (Rosenbaum *et al.* 2012). There have also been exciting findings regarding the outcomes of ‘need-adapted’ care, where services offering psychodynamic therapy as one option in a program of care carefully adapted to the individual appear to produce excellent outcomes (Aaltonen *et al.* 2011; Seikkula *et al.* 2011).

Across continents, new thinking is emerging from cross-fertilization of psychodynamic approaches and work in different fields. For example, cognitive behavior therapy has important commonalities and important differences from psychodynamic therapy, and therapists are beginning to describe ways in which the two approaches may each enhance the other (e.g. Garrett and Turkington 2011). Developments in understanding the role of attachment disorganization, mentalization deficits, trauma and dissociation offer opportunities for enriching our understanding of psychosis and are beginning to be used to enhance psychodynamic approaches (e.g. Liotti and Gumley 2008).

A PSYCHODYNAMIC UNDERSTANDING OF PSYCHOSIS

To understand psychosis, it is essential to understand the unique history of the individual experiencing the psychosis and the meanings that they consciously and unconsciously assign to their experiences. The interest in

unconscious meanings is a crucial feature of the psychodynamic approach. Observed through a psychoanalytic lens, the seemingly meaningless utterances and acts of people suffering psychotic states often have clear meanings.

Although people diagnosed with 'schizophrenia' are very different from each other, their symptoms can often be understood as protecting, or defending against unbearable experience (Martindale and Summers 2013). Such experience includes chronic terror (Fromm-Reichmann 1950; Karon and VandenBos 1981; Sullivan 1962) and the threat of annihilation or unrelatedness (Will 1987). What is particular to defenses in psychosis is that they protect the sufferer by changing their experience of reality.

Thus, it is the stress of the sufferer's life, as subjectively experienced, that has driven that person into psychosis. The severity of the stress is determined by the conscious and unconscious meanings of experience and the quality of social support. Conscious and unconscious meanings are a product of a person's inner world, which has itself been shaped by life experiences, particularly those that happen early on when a sense of self and identity are developing. The personal meaning of experiences determine the ways people perceive themselves and others, experience anxieties and emotions and develop habitual defenses against these and their capacities to cope with stress.

In summary, difficulties with emotion are intrinsic to psychosis and can best be understood within the context of social developmental pathways (Birchwood 2003). Two further points are worth making. First, a psychodynamic view sees mental health and psychosis on a continuum. Second, a psychodynamic understanding of psychotic symptoms is entirely compatible with the view that at least some psychoses have an important neurodevelopmental component.

A psychodynamic understanding of psychosis is supported by contemporary research in other disciplines. Abnormalities in the broad domains of genetics, neurobiology, cognition, emotion, life experience and interpersonal relationships in severe mental disorders do not exist in isolation, and comprehending their interrelationships is important (Cicchetti 2010). Until recently, a radical reductionistic paradigm has guided much research and treatment, often working on the assumption that there is single direction of causality with brain changes leading to changes in subjective experience. However, emerging research is increasingly highlighting that life experience is very important in shaping brain structure and function – for example, through influencing gene expression (epigenetic changes) or changing the pattern of the brain's response to stress. Many aspects of life experience have been shown to be important in this respect, including attachment relationships, varieties of trauma and adverse social experience, social isolation and defeat, poverty and urban living. It is important to note that these include experiences early in life, long before psychosis emerges. In 1997 Koehler proposed that the neuroscience of schizophrenia and bipolar disorder overlapped significantly with the neuroscience of chronic stress, relational trauma and social isolation.

Since then it has been repeatedly demonstrated that the neuroscience findings in severe mental disorders (Read *et al.* 2001; Strakowski 2011; Whitford *et al.* 2011) overlap significantly with the neuroscience findings in early life trauma and social adversity (De Bellis 2010; Mill 2011; Read *et al.* 2009; Teicher *et al.* 2010). Suomi (1997), working with non-human primates (with whom we share 98–99% of our genome), demonstrated that the most dramatic effects of early experience occur later in life, in the context of stress or challenge. These findings are clearly consistent with the psychodynamic view that early life experience is a crucial factor in psychosis.

Liotti and Gumley (2008) propose the central importance of attachment disorganization and trauma – which can be transgenerational (Davoine and Gaudilliere 2004) – and dissociation in understanding the roots of psychosis, with attachment disorganization leading to problems in affect regulation, both directly and through facilitation of dissociative responses to traumas and mentalization deficits.

Psychodynamic explanations have also been proposed for particular psychotic symptoms. For example, most paranoid delusions may be seen as transferences (see below) to the world at large, or attempts to make organized sense out of one's self and world, despite unusual experiences. Hallucinations may be seen as waking dreams, akin to night dreams. Catatonia may be seen as a terror response evolved to cope with predators (Moskowitz 2004).

A PSYCHODYNAMIC APPROACH TO THERAPY OF PSYCHOSIS

Psychodynamic therapy can seem like a romantic heroic quest, with clinician and patient joining forces to quell terror and create a secure place for creativity. Playful shared activity is integral to the 'serious work.' Shared understanding of psychotic experience inspires hope. A strengthening mutual trust provides the glue to repair shattered egos.

Psychodynamic work involves listening to the symbolic meaning in a patient's words, symptoms and actions and how, also, disturbing meaning may have been disguised by psychotic defense mechanisms that change too painful realities. It involves learning about the person from the transference – that is, the way in which the therapy relationship becomes distorted in line with the person's habitual way of perceiving and responding to others. In contrast to Freud's ideas, experience has informed many clinicians that patients diagnosed 'schizophrenic' can form inherently analysable transference reactions. We monitor our own responses – that is, our countertransference – trying to stay attuned to our countertransference experience and are curious about how this may relate to the patient's conscious and unconscious inner world and in turn to their developmental and social experience. This provides clues to understanding, and thus containing, the patient's anxieties.

If we share our observations with the patient with humility and not professional authority, our patient will often correct us. Our aim is together to gradually discover the self-defeating (but in their development 'preservative') aspects of the patient's worldview. The relationship between therapist and patient is crucial. 'The various modes of actually being with another person are more important than any method' (Benedetti 1987). As therapists, we need to be open to the patient's communications without responding defensively, because limitations in therapy may lie in the therapist's capacity to 'take the transference' (Mitrani 2001), by which we mean the patients' subjective view of ourselves. To become more effective in working with people struggling with psychosis and develop the confidence necessary to stay with patients over the long haul, one therefore needs a deep knowledge of oneself, including one's underlying terrors and traumas. One also needs to develop a playful acceptance of one's own delicate balance. This is a lifelong task, supported by one's own psychotherapy, ongoing supervision, and reading of works of senior clinicians.

Therapists need to adapt what they do to the needs and capacities of the patient at any given moment. This is especially important with patients who experience psychosis or who readily resort to psychotic defenses, and in whom developing respect for each patient's stubbornly fixed defenses is therapeutic (Fromm-Reichmann 1950). Thus at some points supportive approaches that reinforce the patient's adaptive defenses will be essential, and at others it will be possible to use more exploratory approaches, challenging defenses and enabling growth.

Some patients may respond to insight, but psychotic symptoms are frequently a defense against insight, which is therefore resisted, so attempts to offer it may sometimes just heighten the psychotic defenses. What is often more important is internalizing a tolerant, kind, confused and stubborn therapist, who is there to help, can tolerate negative affect as well as not understanding and who doesn't quit. We match our stubbornness against the patients' stubborn symptoms. We need confidence in our unique subjectivities, our most powerful healing instruments. Attachment assists in regulating emotion, overriding extremes, and individual psychotherapy maximizes this potential. Although a rich array of theory and technique has evolved, providing security for clinicians, most effective therapists are more concerned with helpfulness than with theoretical consistency.

This relational transition from I-it to I-Thou (Buber 1958) marks the essence of spirituality. A good metaphor for psychotherapy of psychosis could be that it is a form of prayer: striving to bring order out of chaos, helping patients recover confidence in their humanness, seeking something of a resurrection, returning the patient to emotional life from a position of deadness. Even with patients others consider incurable, a very good outcome can sometimes result (Benedetti and Furlan 1993; Karon and VandenBos 1981; Robbins 1993; Siani & Siciliani 2000; and Chapter 23).

CONCLUSION

We hope that we have been able to show that a psychodynamic approach to understanding psychosis and conducting therapy with people with psychosis is compatible with emerging research, that it covers a range of approaches with different emphases and that it is a field that is continuing to develop. We end by reiterating that in all their forms the essence of psychodynamic approaches to psychoses is in attending to people's individual stories, in recognizing psychotic symptoms as potentially meaningful responses to intolerable mental experience, in considering unconscious as well as conscious levels of meaning and in dealing with these in present relationships, including the relationship with a therapist.

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CHILDHOOD ADVERSITY AND PSYCHOSIS

From heresy to certainty

John Read

Mental health researchers and clinicians have been slow to consider childhood adversities at all, let alone its relationship to psychosis (Read and Bentall 2012). It is now widely acknowledged, however, that childhood adversity plays a causal role in most mental health problems, including, in childhood, conduct disorder, attention-deficit hyperactivity disorder and oppositional defiant disorder; and, in adulthood, depression, anxiety disorders, eating disorders, sexual dysfunction, personality disorder, dissociative disorder, PTSD and substance misuse (Kessler *et al.* 2010). Childhood adversity is also related to *severity* of disturbance, regardless of how one defines severity. People subjected to childhood physical or sexual abuse have earlier, longer and more frequent admissions to psychiatric hospital; are more likely to try to kill themselves; and have higher global symptom severity (Read *et al.* 2004, 2009). Adults scoring highly on the *Adverse Childhood Experiences* scale are 10 times more likely to be prescribed antipsychotics and 17 times more likely to be prescribed antidepressants (Anda *et al.* 2007).

Nevertheless, 'Until very recently the hypothesis that abuse in childhood has a causal role in psychosis was regarded by many biologically oriented psychiatrists as heresy' (Read and Bentall 2012: 90). In the past decade or so, however, the deafening silence on the issue has been broken by an avalanche of studies. Numerous reviews have now been published (Bebbington 2009; Bendall *et al.* 2008; Larkin and Read 2008, 2012; Matheson *et al.* 2012; McCarthy-Jones 2011; Morgan and Fisher 2007; Morrison 2009; Read 1997; Read *et al.* 2005, 2008, 2009; Skehan *et al.* 2012; Varese *et al.* 2012). Our 2005 review, summarizing the first signs of the avalanche, concluded:

Symptoms considered indicative of psychosis and schizophrenia, particularly hallucinations, are at least as strongly related to childhood abuse and neglect as many other mental health problems.

Recent large-scale general population studies indicate the relationship is a causal one, with a dose-effect.

(Read *et al.* 2005: 330)

This review turned out to be a turning point in the history of research into the causes of ‘schizophrenia’ and psychosis. Apart from being cited over 500 times in the scientific literature, it received international media coverage. The UK’s *Guardian* newspaper described it as ‘tectonic plate-shifting evidence’ and predicted that ‘The psychiatric establishment is about to experience an earthquake that will shake its intellectual foundations. . . it will have to rethink many of its most cherished assumptions’ (James 2005). In the United States, *Newsweek* portrayed the review as ‘the most definitive look at schizophrenia to date’ and suggested that ‘The cumulative impact of this research has swayed opinion in the profession’s highest echelons’ (Pepper 2005).

‘That would be nice’ we thought. Nevertheless someone at the Institute of Psychiatry in London *was* sufficiently swayed to invite me, in 2006, to propose the motion ‘This house believes that child abuse is a cause of schizophrenia’ at a public debate. Following a rebuttal from geneticist Peter McGuffin, and lengthy debate, the motion was carried by 114 to 52 votes (www.iop.kcl.ac.uk/podcast/rss/Maudsley_Debates).

In the same year the first book devoted to psychosis and trauma was published (Larkin and Morrison 2006), and the second soon followed (Moskowitz *et al.* 2008). The first conference on trauma and psychosis, organized by Andrew Gumley at the University of Glasgow, had taken place in 2005.

Two subsequent reviews critiqued the 2005 review, argued that it was too early to reach conclusions and called for more research (Bendall *et al.* 2008; Morgan and Fisher 2007). The call was more than adequately responded to. Subsequent reviews (Read *et al.* 2008, 2009) identified 11 large-scale general population studies. All but one found, after controlling for a range of other factors – including family history of psychosis (thereby negating the ‘only in the genetically predisposed’ argument) – that child maltreatment is significantly related to psychosis. The one exception subsequently corrected a flaw in the original study and found the same as the other ten (Cutajar *et al.* 2010). Nine of the 11 tested for a ‘dose-response’ (a relationship between the *degree* of maltreatment and the probability of psychosis), and all nine found this to be the case. Another reviewer concluded that: ‘There is now considerable evidence of an association between child sexual abuse and psychosis. This relationship is at least as strong as, and may be stronger than, that with other mental disorders’ (Bebbington 2009).

Another sign of lessening hostility to the idea that bad things happen and can drive you crazy was the publication in February 2012 by the *British Journal of Psychiatry* of an invited Editorial on the topic (Read and Bentall 2012). Later that year the research teams led by Richard Bentall in Liverpool and Jim van Os in Maastricht combined to conduct the first meta-analysis of

this rapidly expanding body of research (Varese *et al.* 2012). A second meta-analysis soon followed (Matheson *et al.* 2012). I report the first meta-analysis in some detail here, because it used the broader inclusion criteria of ‘psychosis’ or ‘psychosis symptoms’, netting 41 studies, whereas the second focused on the narrower and problematic construct ‘schizophrenia’, reporting on 25 studies. We will see later, however, that the findings were remarkably similar.

The 2012 psychosis meta-analysis

A meta-analysis transcends ordinary literature reviews by adopting rigorous inclusion criteria and allowing for differences in sample sizes and methodologies when conducting analyses. In the 2012 meta-analysis Filippo Varese, Feikje Smeets and colleagues focused on sexual abuse, physical abuse, emotional abuse, neglect, bullying and parental death. They included only the 41 best designed studies of the 736 screened. The 41 comprised 18 comparisons of psychotic ‘patients’ (2,048 in total) with non-psychiatric controls (1,856), ten prospective studies (following people over time) (41,803), and eight general population surveys conducted at a given point in time (35,546). Analysing the 41 studies together revealed that people who had suffered childhood adversity were 2.8 times more likely to develop psychosis than were those who had not ($p < .001$).

Nine of the ten studies that tested for a dose-response found it. For example, a UK survey of 8,580 people found that those subjected to one form of adverse event (sexual abuse, bullying etc.) were 1.7 times more likely to be diagnosed with a psychotic disorder, compared to 18 times more likely for three adverse events and 193 times for five types (Shevlin *et al.* 2007a). A prospective study found (after controlling for family history of psychosis) that adults abused as children were 9.3 times more likely to have ‘pathology-level psychosis’. The odds ratios for ‘mild’ and ‘severe’ abuse were 2.0 and 48.4, respectively (Janssen *et al.* 2004). A subsequent study, of female prisoners, has also found a dose-response (Kennedy *et al.* 2013).

The meta-analysis generated odds ratios for each type of adversity: sexual abuse, 2.4; physical abuse, 2.9; emotional abuse, 3.4; neglect, 2.9; bullying, 2.4; parental death, 1.7.

Other evidence

In summarizing the research, I focus primarily on those studies considered sufficiently rigorous to be included in the 2012 meta-analysis. However, a far greater number of studies, using a range of methodologies (Read *et al.* 2004, 2005, 2008, 2009), including qualitative interviews (Read 2012; Romme *et al.* 2009), confirm the findings of the meta-analysis.

Even *within* samples diagnosed psychotic or ‘schizophrenic’, child abuse is related to poor premorbid functioning, poor cognitive and intellectual

functioning, substance abuse, unemployment, stigma, social withdrawal, anger (at self and others), poor engagement with services, low satisfaction with diagnosis and treatment, anxiety, depression and suicidality (Bae *et al.* 2010; Campbell *et al.* 2013; Connor and Birchwood 2012; Lothian and Read 2002; Outcalt and Lysaker 2012; Shannon *et al.* 2011). Similar findings are emerging for children experiencing psychosis (Hainsworth *et al.* 2011).

These relationships between adverse events in childhood and negative outcomes are further confirmed by studies of first-episode psychosis (Conus *et al.* 2010), and of young people considered at risk for developing psychosis (Addington *et al.* 2013; Sahin *et al.* 2013; Thompson *et al.* 2013). Child abuse has also been shown to be related to bi-polar disorder (Hammersley *et al.* 2003; Maniglio 2013). Child abuse may even help explain the high use of cannabis by people who experience psychosis, many of whom may be using the drug to self-medicate for the effects of the abuse (Alemany *et al.* 2013; Konings *et al.* 2012; Shevlin *et al.* 2009).

The six types of adversity covered in the meta-analysis are by no means an exhaustive list. Other social factors have a causal role in or are significant risk factors for psychosis. These include: mother's health, nutrition and stress during pregnancy; being the product of an unwanted pregnancy; early loss of parents via abandonment; separation of parents; witnessing inter-parental violence; dysfunctional parenting (often intergenerational); war trauma; torture; rape or physical assault as an adult; racist or other forms of discrimination; and heavy marijuana use in early adolescence (Larkin and Morrison 2006; Read and Bentall 2012). A study of over 10,000 Australians found that 'delusional experiences' were related to all ten of the traumatic events studied: combat; life-threatening accident; fire, flood or other natural disaster; witnessed someone being injured or killed; rape; sexually molested; physically attacked; threatened with a weapon, kidnapped, tortured or terrorism; other extremely stressful or upsetting event; and someone close to you traumatized (Scott *et al.* 2007). Combat and rape were the most strongly related.

Controversial?

Not everyone is pleased with all this research. The five most common negative responses, other than ignoring the evidence altogether, are:

- 1 active denial of the evidence;
- 2 arguing that the research is bad because it is 'family blaming' (Chapters 19 and 26);
- 3 accepting that there is some evidence but arguing that it only applies to those with a genetic predisposition to 'schizophrenia';
- 4 claiming that you can't believe abuse disclosures made by mad people;
- 5 suggesting that adults who experience psychosis exhibited early signs of psychosis as children, of the kind that somehow made them more

vulnerable to being abused (Daly 2011) – without wondering whether these ‘early signs’ are themselves caused by abuse or neglect; six pointing out that childhood adversity causes other mental health problems besides psychosis, which is true and irrelevant (see p. 262).

For example, the *New Scientist’s* (2006) coverage of the London debate reported that ‘Mainstream psychiatrists are not impressed’: ‘There are no methodologically robust studies showing that schizophrenia is caused by child abuse’, says Robin Murray of the Institute of Psychiatry. ‘The strongest predictor of schizophrenia is a family history of the disorder.’ Peter McGuffin, also at the Institute, warns that refocusing on abuse risks a return to the 1960s ‘when it was fashionable to blame the parents for causing schizophrenia’ (*New Scientist* 17 June 2006, p. 66).

While some psychiatry textbooks simply ignore all this research, one acknowledges it but portrays it as ‘very controversial’ (Murray and Dean 2008: 305). The idea that bad things happening in childhood can drive you crazy is not, however, controversial in the real world beyond biological psychiatry. The public, all over the world (including ‘patients’ and their families) believe that ‘schizophrenia’ is caused more by things like abuse, neglect, stress, poverty etc. than by chemical imbalances and genetic predispositions (Chapter 11). A British High Court judge has ruled: ‘The terrible abuse to which A was subjected led to both his suffering PTSD and that disorder of the mind which is symptomatic of schizophrenia. . . . What is important is that his adult psychiatric problems, however they are classified, were caused by his childhood sexual abuse’ (*A v The Archbishop of Birmingham* 2005).

RELIABILITY OF ABUSE DISCLOSURES

Perhaps the most damaging of the five objections identified above is the notion that when psychiatric patients, especially those labelled ‘schizophrenic’, tell you they were abused, you should not believe them. This argument has been deployed for decades to discredit people brave enough to tell mental health professionals what has happened to them. It is also used in efforts to discredit research studies of the kind described in this chapter. Our 2012 meta-analysis was accompanied by a ‘commentary’ making precisely this argument (Susser and Widom 2012). This is not an evidence-based argument (Goodman *et al.* 1999; Meyer *et al.* 1996). Despite the enforced secrecy surrounding the sexual abuse of children, evidence to corroborate disclosures of patients has been found in 74% (Herman and Schatzow 1987) and 82% (Read *et al.* 2003) of cases. One study found that ‘The problem of incorrect allegations of sexual assaults was no different for schizophrenics than the general population’ (Darves-Bornoz *et al.* 1995: 82). A recent study found

that disclosures of child sexual or physical abuse, or neglect, by adults diagnosed with psychotic disorders had high concurrent validity (i.e. they correlated with responses to a questionnaire about early relationship with parents), and good convergent validity (with case notes), were stable over time and were *not* related to severity of psychotic symptoms (Fisher *et al.* 2011).

SEXUAL ABUSE

A review of 59 studies (52 inpatient samples and seven outpatient samples in which at least 50% were diagnosed psychotic) found that 47% of the women and 29% of the men had been sexually abused in childhood (Read *et al.* 2008). Another review, of samples diagnosed psychotic, produced sexual abuse rates of 42% for women and 28% for men (Morgan and Fisher 2007). A later study of schizophrenia spectrum disorders found child sexual abuse rates of 75% for women and 36% for men (Outcalt and Lysaker 2012). Another recent (mixed gender) study found a rate of 39% (Connor and Birchwood 2012).

All 20 of the relevant meta-analysis studies found that people who had been sexually abused as children were more likely to be psychotic later in life, with odds ratios varying from 1.4 to 9.7 (Varese *et al.* 2012). In 17 of the 20 the difference was statistically significant. For example, in a prospective study of 1,129 adolescents, those sexually abused by age 13/14 were 3.3 times more likely than others to report psychotic experiences two years later (De Loore *et al.* 2007). In a subsequent study, of 233 young people considered at 'ultra high risk' of developing psychosis, those who had been sexually abused as children (but not those who had suffered other types of abuse) were significantly more likely to become psychotic (Thompson *et al.* 2013).

Incest

There has been little research into the relationship of incest to psychosis. One might assume, however, that the additional betrayal involved, combined with the fact that the betrayal is committed by the person to whom the child would normally turn for support when betrayed, might render incest an obvious candidate when searching for events that can drive us mad. The review of inpatient samples and outpatient samples where at least 50% were diagnosed psychotic found that 29% of women and 7% of men had been subjected to incest (Read *et al.* 2005). Another review of inpatient studies found that 62% of the sexual abuse cases were incest (mostly by fathers), compared to 29% in community samples (Read *et al.* 2004). New Zealand studies found that both inpatients and outpatients who had suffered incest exhibited significantly more 'schizophrenia' symptoms than those who suffered extra-familial sexual abuse (Read and Argyle 1999; Read *et al.* 2003).

Content

One of the many aspects of psychosis buried by the 'medical model' is the *meaning* of the 'symptoms'. A study of severely maltreated children found that the content of their hallucinations or illusions was strongly reminiscent of episodes of traumatic victimization (Famularo *et al.* 1992). The same has been found for many adolescents (Heins *et al.* 1990) and adults (Honig *et al.* 1998). In a recent study of young people considered at high risk of developing psychosis, childhood trauma was related to feeling watched or followed (Falukozzi and Addington 2012). Four studies have found that the content of at least half of the hallucinations and/or delusions of adults who were abused as children are related to the abuse (Bowe 2002; Hardy *et al.* 2005; Read and Argyle 1999; Reiff *et al.* 2012).

Examples from studies of incest survivors include: 'one believed that her body was covered with ejaculate and another that she had had sexual relations with public figures', and 'a man who had been raped several times by an uncle at age seven heard voices telling him he was "sleazy" and should kill himself; a woman who had been sexually assaulted by her father from a very young age, and raped as a teenager, had the delusion that people were watching her as they thought she was "a sexual pervert" and had auditory hallucinations accusing her of doing "dirty sexy things"' (Heins *et al.* 1990: 563–4).

Examples from another study include: 'Another's chart read "Sexual abuse from an early age. . . . Raped several times by strangers and violent partners."' This person believes that he is being tortured by people getting into body, for example, the Devil and the Beast and "had bleeding secondary to inserting a bathroom hose into self, stating "wanting to wash self as people are trying to put aliens into my body".' (Read *et al.* 2003: 12).

PHYSICAL ABUSE

In the review of 59 studies of inpatient samples and outpatient samples in which at least 50% were diagnosed psychotic, 47% of the women and 49% of the men had been physically abused in childhood (Read *et al.* 2008). The review of samples diagnosed psychotic produced physical abuse rates of 35% for women and 38% for men (Morgan and Fisher 2007). A subsequent mixed-gender 'schizophrenia' study found a rate of 47% (Connor and Birchwood 2012). Again, the majority of the abuse is perpetrated by family members; again, predominantly fathers (Read *et al.* 2004).

All 13 of the studies included in the meta-analysis found that people physically abused as children were more likely to be psychotic as adults; with odds ratios from 1.5 to 9.4. In 12 of the 13 the difference was significant (Varese *et al.* 2012).

For example, a survey of 2,353 adults in the United States found that those 'badly beaten up' by a parent were 4.5 times more likely to have experienced

auditory hallucinations (Shevlin *et al.* 2011). In a study of over 2,000 British children those who had been physically maltreated after age 7 were 2.4 times more likely to exhibit psychotic symptoms by age 12, with the odds ratio increasing to 3.5 if the abuse started *before* age 7. The difference between abused and non-abused remained significant after controlling for gender, poverty, IQ and, again, 'genetic liability' for psychosis (Arseneault *et al.* 2011).

EMOTIONAL ABUSE

Psychosis researchers have only recently begun to study adverse events other than physical and sexual abuse. A 2008 review identified only six studies on the prevalence of emotional abuse in 'schizophrenia' (Read *et al.* 2008). They ranged from 32% to 94%, with a weighted average of 47%. A recent study found a rate of 62% (Connor and Birchwood 2012). Only six emotional abuse studies met the inclusion criteria for the 2012 meta-analysis. All six found that emotional abuse was more common in people who later became psychotic; with odds ratios from 1.4 to 12.2. In five of the six the difference was statistically significant.

For example, an Italian study found that people diagnosed 'schizophrenic' were 12 times more likely to have been emotionally abused than the general population, and twice as likely as people diagnosed with depression (Rubino *et al.* 2009). Is it possible that these ongoing, more subtle, types of maltreatment may, for some, be even more damaging than discrete instances of physical or sexual abuse? The 'schizophrenia' group in the Italian study were only six times more likely than the general population to have been physically abused and only twice as likely to have been sexually abused. Several other studies have found that emotional abuse and/or neglect are even more strongly related to psychosis than sexual and physical abuse (e.g. Connor and Birchwood 2012; Daalman *et al.* 2012). Indeed, the overall odds ratio in the meta-analysis for emotional abuse (3.4) was the largest of any of the six types of adversities examined. However, it is possible that *severity* of the adversities (age of onset, duration, frequency, number of perpetrators and types of abuse, intra vs. extra-familial) is more important than the *type* of adversity.

Two other bodies of research confirm the relationship between emotional or psychological maltreatment and psychosis. Expressed Emotion (EE) is a euphemism for parental hostility, criticism and intrusiveness (Chapters 19 and 26). A US study followed adolescents for 15 years and found that 36% of those whose parents had both scored high on EE were now diagnosed with schizophrenia, compared to 0% if only one, or neither, parent was high EE (Goldstein 1987).

The Parental Bonding Instrument retrospectively measures parent-child relationships. The *Caring* scale is a dimension from empathy and affection to neglect and emotional coldness. The *Over-protection* scale ranges from intrusion, control, and infantilization, to autonomy and allowance of

independence. High *over-protection* and low *care* is called 'affectionless control'. Studies of psychosis and schizophrenia consistently find high rates of affectionless control (Berry *et al.* 2007; Read and Gumley 2008).

NEGLECT

The prevalence of childhood emotional neglect among people diagnosed with 'schizophrenia' has been found to range (depending partly on definition) from 19% to 91%, with an average of 51%; the range for physical neglect was 19–89%, with an average of 41% (Read *et al.* 2008). A recent study found rates of 77% for emotional neglect and 65% for physical neglect (Connor and Birchwood 2012).

The 2012 meta-analysis combined physical and emotional neglect and found that in all seven studies psychosis was more common in those who had been neglected; with odds ratios from 1.1 to 8.4 (Varese *et al.* 2012). Five of the seven differences were significant. For example, of 5,877 adults in the United States those who had been 'seriously neglected' as children were 8.4 times more likely to meet diagnostic criteria for psychotic disorders (Shevlin *et al.* 2007b).

BULLYING

Another kind of maltreatment that has only recently received the attention it deserves is bullying. All six of the relevant studies in the meta-analysis found that people who had been bullied as children were more likely to be psychotic later in life, with odds ratios ranging from 1.2 to 4.4. Four of the six differences were significant. For example, in a study of 4,561 Japanese teenagers, those who had been bullied were 2.4 times more likely to be experiencing 'psychotic-like experiences' such as hearing voices and believing that they receive personal messages through the television (Nishida *et al.* 2008). Another meta-analysis, focused exclusively on bullying, found that combining seven general population studies produced an odds ratio of 2.7 (van Dam 2012).

PARENTAL LOSS

Loss of parents, a less taboo topic than child abuse, has been researched for far longer. Eight of 13 studies between 1943 and 1963 had found that 'schizophrenics' have significantly higher rates of parental death in childhood than comparison groups (Granville-Grossman 1966). Four of the five studies about maternal death were significant, compared to only one of the four on paternal death. The remarkable book *The Schizophrenic Disorders*, by Eugen Bleuler's son Manfred, includes five of his own studies, of 932 people diagnosed

'schizophrenic'. These produced a rate of parental loss before age 15, from death and other causes, of 31% – significantly greater than in the general population (Bleuler 1978). A later study found that loss of mother during childhood (but not loss of father) was significantly higher (55%) in 'schizophrenics' than for other diagnoses (23%) (Friedman *et al.* 2002).

The meta-analysis focused on parental death, rather than loss for other reasons. In seven of the eight studies adults experiencing psychosis were more likely to have had a parent die during childhood (Varese *et al.* 2012). However, only two of these seven differences were statistically significant. Overall, people who had had one or both parents die were only 1.7 times more likely to experience psychosis. This was the only odds ratio of the six types of adversity investigated not to be statistically significant.

An Italian study, however, suggests that losing a parent for other reasons may be more predictive of psychosis than having a parent die. People diagnosed with 'schizophrenia' were only 1.8 times more likely to have had a parent die but were 5.5 times more likely to have had a parent permanently leave home, to have been adopted or to never have had the father live with the family (Rubino *et al.* 2009).

HOW DO ABUSE, NEGLECT AND LOSS LEAD TO PSYCHOSIS?

Many researchers, accepting that there *is* a causal relationship, are now investigating the processes by which, for instance, being regularly beaten or humiliated in childhood leads to, for instance, paranoia or hearing derogatory voices later in life. In our *British Journal of Psychiatry* Editorial, Richard Bentall and I wrote:

At a psychological level, researchers have focused on mechanisms that might mediate between childhood adversity and later mental health problems, including attachment, dissociation, psychodynamic defences, coping responses, impaired access to social support, and revictimisation. This research has the potential to uncover specificities in the effects of adversity which may be difficult to see when only broad diagnoses are considered.

(Read and Bentall 2012: 90)

Space does not permit a full account of these rapidly growing bodies of research. Table 18.1 provides a few examples of accounts of the most researched mechanisms. The two books mentioned earlier cover these in more detail (Larkin and Morrison 2006; Moskowitz *et al.* 2008). For summaries see Read and Bentall (2013), Read and Sanders (2010), Bebbington (2009) and Read *et al.* (2008, 2009).

Table 18.1 Examples of works describing some of the psychological processes that explain the link between childhood adversity and subsequent psychosis

<i>Psychological process</i>	<i>Work</i>
<i>Cognitive processes</i> e.g. source monitoring difficulties; jumping to conclusions; scanning for, and oversensitivity to, adverse events; negative self-schema; mentalization (difficulty understanding others' feelings and intentions)	Garety <i>et al.</i> 2007 Bentall and Fernyhough 2008 Morrison 2009 Garrett and Turkington 2011 Bentall <i>et al.</i> 2012 Connor and Birchwood 2012 and Chapters 16 and 22
<i>Dissociation</i>	Kilcommons <i>et al.</i> 2008 Moskowitz <i>et al.</i> 2008 Varese <i>et al.</i> 2011 Laddis and Dell 2012 Longden <i>et al.</i> 2012
<i>Psychodynamic defences</i>	Abraham [1907] 1955 Ferenczi 1933 Garrett and Turkington 2011 and Chapter 17
<i>Attachment</i>	Berry <i>et al.</i> 2007 Gumley <i>et al.</i> 2008 Read and Gumley 2008 Connor and Birchwood 2012 Benamer 2010
<i>Coping responses</i>	Barker-Collo and Read 2003, 2011 Horan <i>et al.</i> 2007 Bebbington 2009
<i>Social withdrawal</i>	Murphy <i>et al.</i> 2013

All the different approaches describe the effects of adverse childhood events as ways of *responding to*, or *coping with*, those events. It is unlikely that any of the mechanisms or models can explain the link between childhood adversity and psychosis in isolation from the others. They all interact with one another. One attempt to capture some of these interactions is reproduced in Figure 18.1. The different models may have different uses at different stages of therapy (Garrett and Turkington 2011).

Sometimes the models are describing the same process through different lenses. For example, experiencing an intrusive abuse memory as occurring outside oneself might be described by a cognitive researcher in terms of problematic 'source monitoring' but in terms of projection by a psychoanalyst.

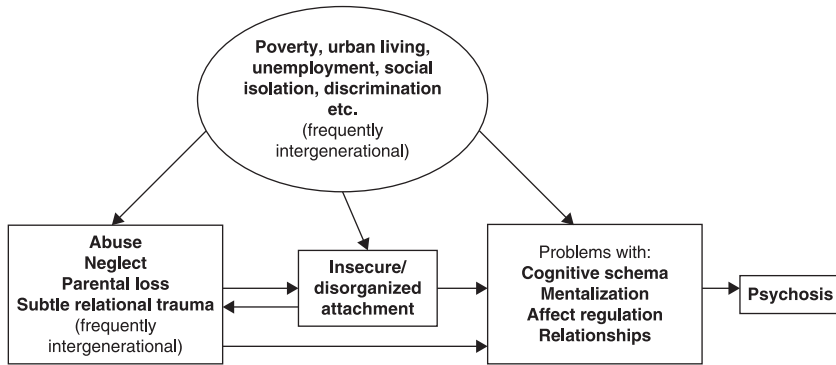


Figure 18.1 A diagram of the role attachment may play in the relationship between childhood adversity and psychosis (from Read and Gumley 2008). Two-directional arrows indicate that insecure/disorganised attachment resulting from childhood adversity can increase probability of further abuse and neglect, e.g. in adolescence.

Each model will have varying levels of importance (including none) for each person. This becomes apparent when reading the growing number of first-person accounts that have recently been published (Geekie *et al.* 2012; Romme *et al.* 2009). The ISPS journal *Psychosis* regularly publishes first-person accounts, most of which describe the role of adverse events in the development of psychosis (e.g. Bullimore 2009; Dillon 2009; Lampshire 2009; Longden 2010).

Elsewhere, another ‘expert by experience’ wrote:

I don’t think that abuse itself is a strong cause for psychosis. It hurts, but it is rather simple. I think that the threat and the betrayal that come with it feed psychosis. The betrayal of the family that says, ‘you must have asked for it’, instead of standing up for you. That excuses the offender and accuses the victim. And forces the child to accept the reality of the adults. . . . You are forced to betray yourself. That is what causes the twilight zone. What makes you vulnerable for psychosis.

(Boevink 2006: 17–19).

Revictimization

In 1907 German psychoanalyst Karl Abraham noted some dementia praecox cases in which ‘the illness followed upon trauma’ and some ‘in which a subsequent experience similar to the traumatic one was the precipitating factor’ ([1907] 1955: 17). A study of people diagnosed with schizophrenia or dissociative disorder (85% of whom suffered child abuse or neglect) found: ‘In most patients, the onset of auditory hallucinations was preceded by either a

traumatic event or an event that activated the memory of earlier trauma' (Honig *et al.* 1998: 646).

Most research on violence and psychosis is concerned with violence *by* 'psychiatric patients'. They are, however, far more likely to be assaulted than to assault others. In addition to the high levels of abuse in childhood, the majority of psychiatric patients suffer physical assaults as adults. The majority of the women (and about one in four of the men) suffer sexual assaults (Read *et al.* 2004, 2005).

In a study of female psychiatric patients, physical assault was found to be significantly related to only one diagnostic group: 'nonmanic psychotic disorders (e.g. schizophrenia, psychosis not otherwise specified)' (Briere *et al.* 1997). Another study of female inpatients found that adult sexual assault was more strongly related to a diagnosis of schizophrenia than to mania, depression, substance abuse or borderline personality disorder (Cloitre *et al.* 1996). A recent study found that women attending a centre for rape victims in Denmark in 2003 were (after controlling for a previous psychosis diagnosis) ten times more likely than the public to be diagnosed with 'schizophrenia, schizotypal or delusional disorder' in the following four years (Elklit and Shevlin 2011).

Very few studies have examined the cumulative effects of being abused in both childhood and later life. In one study a significant relationship between adult sexual assault and 'schizophrenia' was no longer significant when child abuse was controlled for (Cloitre *et al.* 1996). In a New Zealand study only two types of 'symptoms' (tactile hallucinations and voices commenting) were predicted by child abuse if no adult abuse followed. However, hallucinations (in general), delusions and thought disorder were all predicted by child abuse if the child abuse was followed by adulthood abuse (Read *et al.* 2003). These studies suggest that being abused in childhood *and* adulthood is, unsurprisingly, more predictive of psychosis than either alone.

Specificity

As argued earlier, 'this research has the potential to uncover specificities in the effects of adversity which may be difficult to see when only broad diagnoses are considered'.

For example, childhood sexual trauma appears to have a specific effect on the risk of hallucinations, which may reflect a long-term impact on the processes underlying source monitoring (the ability to differentiate internal and external stimuli); whereas attachment difficulties and more chronic victimisation, for example bullying, may increase the risk of paranoid delusions by affecting the way that individuals appraise unpleasant experiences.

(Read and Bentall 2012: 90)

Much of the research in this area has been conducted by Richard Bentall and colleagues (Bentall and Fernyhough 2008; and Chapter 16). For example, when analysing data on a general population sample of over 7,000, all five types of childhood adversity studied were significantly related to both paranoia and hearing voices. However, after controlling for the co-occurrence of the other type of psychotic symptom, childhood rape was associated with hallucinations but not paranoia, and institutional care was related to paranoia but not hallucinations (Bentall *et al.* 2012).

The other specificity issue is whether 'schizophrenia' or psychosis in general are more or less related to childhood adversity than other mental health problems. The second meta-analysis to be published was explicitly designed to address this (Matheson *et al.* 2012). The 25 'schizophrenia' studies that met inclusion criteria showed that people with this diagnosis were 3.6 times more likely than the general population to have suffered childhood adversities ($p < .0001$). 'Schizophrenia' was found to be just as related to childhood adversity as affective psychosis, depression and personality disorders, and more strongly related than anxiety disorders. Dissociative disorders and PTSD were even more strongly related than schizophrenia. These sorts of findings can be used to try to minimize the importance of the relationship between adversity and madness, by highlighting that the relationship is not specific. What they actually show, of course, is that most or all mental health problems have their roots in childhood and that 'schizophrenia' and psychosis, contrary to decades of claims that they are bio-genetically based, are no exception.

WHAT ABOUT GENES AND THE BRAIN?

The traumagenic neurodevelopmental model

'Schizophrenia' has repeatedly been portrayed as a 'brain disease'. The evidence for this is, supposedly, that the brains of 'schizophrenics' are different from those of 'normal' people. Chapter 6 provides evidence that this claim is poorly researched and wildly exaggerated. However, what would it mean if there were some differences?

Of course there are biological processes accompanying the psychological processes by which trauma leads to psychosis. All mental processes have neurological and biochemical correlates. Some researchers, however, believe that if they can identify these correlates, they have discovered a cause. This is akin to assuming that because the brain operates differently when we are grieving, it is the brain that caused our sadness rather than the loss. Brain researchers studying schizophrenia often operate as if the brain exists in a social vacuum, seemingly unaware of the obvious fact that a primary function of the brain is to react to the environment (Read *et al.* 2001, 2008).

An important recent development, however, is that this simplistic thinking has been effectively challenged. In 2001, along with the renowned psychiatrist and neuroscientist Bruce Perry (see www.childtrauma.org) and psychologist Andrew Moskowitz (an expert in trauma, dissociation and psychosis), I published a paper introducing the *Traumagenic Neurodevelopmental Model* of psychosis (Read *et al.* 2001).

We pointed out that the heightened sensitivity to stress found in ‘schizophrenia’ consists, for many patients, of neurodevelopmental changes to the brain caused by trauma in the early years of life. The evidence is that all the structural and functional differences between the brains of ‘schizophrenics’ and ‘normal’ adults (which are cited to support the ‘brain-disease’ hypothesis) are the same differences found between young children who have been traumatized and those who have not. These include: overactivity of the hypothalamic-adrenal-pituitary axis; dopamine, serotonin and norepinephrine abnormalities; and structural differences such as hippo-campal damage, cerebral atrophy, ventricular enlargements and reversed cerebral asymmetry (Read *et al.* 2001).

We also summarized the research linking child abuse and neglect to those psychological deficits in childhood – such as reduced intellectual and cognitive functioning (Campbell *et al.* 2013) – that are typically interpreted as early signs of ‘schizophrenia’ rather than as the result of adverse events in childhood. The main conclusion we drew, however, is that the heightened sensitivity to stress evidenced by dysregulation of stress regulation mechanisms is not necessarily inherited. It can be caused by childhood trauma and neglect.

An explicit test of the Traumagenic Neurodevelopmental Model subsequently found that within a sample of ‘schizophrenia patients’ those abused as children (especially those emotionally abused) had greater HPA axis dysregulation, measured by cortisol levels, than their non-abused counterparts (Braehler *et al.* 2005).

Many studies have now demonstrated the relationship between psychosis and overreactivity to stress of the HPA axis, and the dopaminergic system (Holtzman *et al.* 2013; Read *et al.* 2009; Teicher *et al.* 2012; van Winkel *et al.* 2008). A recent Egyptian study found that low levels of brain-derived neurotrophic factor (BDNF), ‘fully mediated the associations between cumulative trauma and overall schizophrenia psychopathology’ (Fawzi *et al.* 2013: 23). The role of stress (Holtzman *et al.* 2013), including childhood trauma is finally being recognized. A review of the neurobiological mechanisms underlying psychosis suggested:

A heuristically useful framework to study the underlying mechanisms is the concept of “behavioral sensitization” that stipulates that exposure to psychosocial stress – such as life events, childhood trauma, or discriminatory experiences – may progressively increase the behavioral and biological response to subsequent exposures’.

(van Winkel *et al.* 2008: 1095).

Our 2001 paper seems to have had some impact, in academia at least, having been cited over 300 times. A 2007 review concluded that ‘the traumagenic neurodevelopmental model is an example of a genuine integration between social, psychological and biological factors’ (Yui *et al.* 2007: 279).

Some readers may worry that the fact that child abuse can alter the brain suggests that these changes might be irreversible, and therefore does support the notion that ‘schizophrenia’, is, in a way, a brain disorder. Others might be concerned that all this also supports the notion that medication is the answer. However, just as negative experiences can damage our brains’ ability to respond to stress, so positive experiences can repair that damage (although the earlier those positive experiences occur the better). Recent studies show that psychotherapy can enhance the ability of the HPA axis to respond to stress, in both children (Schuengel *et al.* 2009) and adults (Moller 2011; Olff *et al.* 2007).

In 2012, leading Harvard brain researcher Martin Teicher and his colleagues published the largest of their extensive studies demonstrating that childhood maltreatment has long-lasting effects on the brain. The finding was not new. What was new is that the disorders for which this is considered relevant no longer exclude psychosis:

Childhood maltreatment increases risk for an array of psychiatric disorders and is associated with an earlier age of onset, more severe course, and poorer response to treatment. This is likely attributable, at least in part, to early stress-induced alterations in trajectories of brain development. . . . Childhood maltreatment or abuse is a major risk factor for mood, anxiety, substance abuse, *psychotic*, and personality disorders, and it is associated with reduced adult hippocampal volume.

(Teicher *et al.* 2012: 563, 569; *italics added*)

Epigenetics

In an article reporting Teicher’s findings, *Newsweek* reported that ‘Childhood abuse or poverty can alter which genes are active in the developing brain through a process called epigenesis. These changes can lead to diseases such as schizophrenia and bipolar disorder’ (Coghlan 2012).

Epigenetics involves some complex processes (Champagne and Curley 2009; Read *et al.* 2009) but the main point is that rather than inheriting predispositions to, say, psychosis or depression, which inevitably and invariably affect our chances of developing those ‘disorders’, our genes are actually turned on and off by the environment.

An array of processes in the chromatin in the cell nucleus turn gene transcription on and off through mechanisms that are highly influenced by the individual’s socio-environmental experiences. . . . In the

central nervous system, these processes are specifically sensitive to an individual's psycho-social experience; hence neuropsychological development from birth to adulthood is governed by epigenetic mechanisms which are shaped by interactions with the environment (Champagne & Curley 2009). . . . This means that, at the cellular level, a new picture is emerging of how our biological heritage interacts with the environment, and in this picture, the environment seems to play a much more crucial role than hitherto thought.

(Read *et al.* 2009: 300)

Most epigenetic research is conducted on rats:

This research shows that the frequency of licking and grooming by the dam in early life governs the development of brain and behaviour in the offspring with effects that last into adult life. Poor and absent maternal care directly affects and shapes the activity of the hypothalamus-pituitary-adrenal (HPA) axis, the main stress system of the brain. . . . Behaviourally, the exposed pups exhibited increased anxiety, fear and startle responses, passivity, and reduced explorative behaviour in novel situations in adult life, and *females tended to exhibit the same behaviour towards their own pups as they had experienced postnatally.*

(Read *et al.* 2009: 304; italics added)

So it is the behaviour of the parent that determines whether things 'run in families' rather than some unalterable genetic structure. We concluded:

This emerging view also suggests that the idea that specific DNA polymorphisms provide a vulnerability to mental health problems may be seriously flawed. . . . A lot of money has been wasted on well-intentioned but ultimately useless genetic research, with similar amounts poured into context-less brain research. The genuinely integrated socio-bio-psychological approach proposed in this paper, and increasingly being adopted internationally, will lead to genuinely productive research programmes, at both the theoretical and clinical levels.

(Read *et al.* 2009: 306)

CLINICAL IMPLICATIONS

Asking

None of these impressive research developments will matter if we don't act on them. Every assessment of every person who comes into contact with mental health services should include a full 'psychosocial history', without which a

comprehensive formulation of their problems and an effective treatment plan are impossible (Read and Sanders 2010; Read *et al.* 2007). That history must include questions about abuse, bullying neglect and loss.

Inpatient studies in the United States and the United Kingdom in the 1990s found that clinicians identify fewer than half of the cases of abuse that are reported to researchers (Read *et al.* 2007). For example, a study of 30 'heavy users' of acute inpatient and emergency services who disclosed to researchers that they had been sexually or physically abused as children found that none had ever been asked about abuse before (Rose *et al.* 1991). A more recent survey, of New Zealand women who had been sexually abused during childhood and were later treated by mental health services, found that only 22% had ever been asked about sexual abuse by those services (Read *et al.* 2006). A recent review, of 11 studies, found 'little evidence of widespread routine enquiry during mental health assessments in acute settings' (Hepworth and McGowan 2012).

An early New Zealand inpatient study had found that even when abuse questions were added to the admission form, they remained unasked 68% of the time. When people *were* asked about abuse, 59% disclosed either physical or sexual abuse in childhood, but of those who were *not* asked only 6% disclosed such abuse at any point while in hospital (Read and Fraser 1998a). This is consistent with multiple studies showing that many adults are reluctant to spontaneously disclose child abuse (Read *et al.* 2007) – especially if they have never disclosed it before or did so but were not believed (Reiff *et al.* 2012). This is yet another reason why it is incumbent upon us to ask.

Moreover, it seems that males, old people and people with diagnoses indicative of psychosis, such as 'schizophrenia', are particularly unlikely to be asked; and that male clinicians, psychiatrists, and staff with strong bio-genetic causal beliefs are particularly unlikely to ask (Read *et al.* 2007; Sampson 2012; Young *et al.* 2001).

A rare study of what users of mental health services think about being asked about childhood abuse found that although the majority (64%) had experienced abuse, 78% had not been asked about this (Lothian and Read 2002). Their comments included:

'There were so many doctors and registrars and nurses and social workers and psychiatric district nurses in your life asking you about the same thing, mental, mental, mental, but not asking you why.'

'I think there was an assumption that I had a mental illness and, you know, because I wasn't saying anything about the abuse I'd suffered no one knew.'

'I just wish they would have said "What happened to you? What happened?" But they didn't' (Lothian and Read 2002).

Responding

There has been little research on what mental health professionals do *after* a client discloses abuse. Studies, in New Zealand (Agar and Read 2002; Read

and Fraser 1998b) and the United States (Eilenberg *et al.* 1996; Posner *et al.* 2008) found low levels of response in terms of offering information or support, documenting the abuse in files, asking about previous disclosure or treatment, reporting to legal or protection authorities, including the abuse in summary formulations or treatment plans, or referring for abuse-related therapy.

Progress?

Only two studies have evaluated progress in this crucially important aspect of clinical practice. US psychiatrist Julia Eilenberg and her colleagues compared the quality and usage of trauma histories at a psychiatric clinic in 1996 and 2008 (Eilenberg *et al.* 1996; Posner *et al.* 2008). They found, in the files of those for whom trauma was recorded, a significant improvement (from 15% to 56%) in the recording of the frequency and severity of the traumatic events. Recording of posttraumatic symptoms in the trauma cases was still absent for 65%. There had been no improvement in the extent to which the trauma histories were included in either formulations (14%) or treatment plans (9%).

A New Zealand study points to the importance of training staff. In 2000 Auckland District Health Board introduced a policy that all mental health clients should be asked about childhood trauma *and* that all staff should attend a one-day training on how to do this (Cavanagh *et al.* 2004; Read *et al.* 2007); this was led initially by myself, with significant input from Debra Lampshire, who provided a service-user perspective. The programme specifically addressed barriers to asking, identified in a national survey of mental health professionals (Young *et al.* 2001). In 2010, 250 charts from the DHB's four community mental health centres (Sampson 2012) were compared to 200 charts from one of those CMHCs prior to the training (Agar *et al.* 2002).

By 2010, the recorded rate for childhood physical abuse had doubled, from 17% to 36%, and that for child sexual abuse had increased from 20% to 32% (with similar increases for assaults in adulthood). The percentage of the child sexual abuse cases that were mentioned in formulations increased from 22% to 57%, and, in treatment plans from 20% to 44%; with increases for childhood physical abuse of 12% to 47% for formulations and 12% to 24% for treatment plans. However, the proportion of child sexual abuse cases referred for abuse-related treatment remained static (22/23% for sexual abuse; 18/20% for physical abuse).

This was the first study to include childhood neglect. Of the recorded cases of emotional neglect, only 33% were mentioned in formulations and 15% in treatment plans, with figures for physical neglect of 23% and 14%, respectively (Sampson 2012).

While some of the progress identified in this study may be attributed to the training programme, a recent study has highlighted, as well, the importance of a supportive service culture (Toner *et al.* 2012).

Some progress *is* being made. In 2008 the National Health Service in the United Kingdom introduced guidelines (NHS Confederation 2008) similar to those of Auckland DHB. An interviewer recently asked me to name my two most satisfying accomplishments. After describing the 2006 London debate, I talked about having been invited to lead the first training programme, for 100 psychiatrists, on how to implement the NHS guidelines (i.e. how to ask about abuse) at the *Royal College of Psychiatrists Training Day on Trauma and Violence* in London. Again I made sure I was accompanied by an ‘expert by experience’, this time Jacqui Dillon, Chair of the UK *Hearing Voices Network*. If I had been asked to name a third accomplishment, I would probably have said being invited to publish a description of our New Zealand training programme in the professional journal of the *Royal College of Psychiatrists* (Read *et al.* 2007). It seems that an increasing number of psychiatrists, including some in positions of authority, do understand the need for change.

In our *British Journal of Psychiatry* editorial we concluded that ‘The most important implication is in the domain of primary prevention’ (Read and Bentall 2012: 90). The 2012 meta-analysis calculated that the ‘estimated population attributable risk’ was 33%. This means that if the six childhood adversities reviewed were eliminated, the number of people with psychosis would be reduced by a third (Varese *et al.* 2012).

While waiting for politicians to fund programmes designed to prevent child abuse and neglect (Chapter 20), we can perhaps at least provide some decent support for those already abused or neglected. Strong advocates of any particular model of therapy for psychosis should remember, as is the case for all mental health service users, that the quality of the therapeutic relationship (regardless of models or techniques) is a strong predictor of client satisfaction (Lothian and Read 2002) and of objective outcome measures (Dunn and Bentall 2007; Priebe *et al.* 2011). It is also important to be aware that having been abused can render the development of a therapeutic alliance more difficult (Lysaker *et al.* 2011).

Hopefully, in the coming years, more mental health services *will* offer people who experience psychosis and who have suffered abuse or neglect, or significant losses, the kind of support and therapy that addresses their life histories (Chapters 21–26). If not, all the wonderful research efforts of the last decade will have been a waste of time.

In the first edition of this book, this chapter ended with 37 research questions (Read *et al.* 2004: 239–244). Many remain unanswered: for example, in the area of gender differences in pathways from childhood abuse to psychosis (Barker-Collo and Read 2011; and Chapter 15). The two most pressing research questions now, are: ‘What are the psychological, social, political and economic barriers to making mental health services trauma-informed?’ and ‘How do we remove them?’

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PSYCHOSIS AND FAMILIES

Intergenerational parenting problems

John Read and Fred Seymour

If we look at some experience and behaviour without reference to family interactions, they may appear comparatively socially senseless, but if we look at the same experience and behaviour in their original family context they are liable to make more sense.

(Laing and Esterson 1970: 26)

To cause – even inadvertently – such terrible suffering in someone you love seems unbelievable and insulting. One of the reasons for emphasizing heredity and disturbed physiology is to avoid the guilt engendered by focusing on the possible mistakes of the parents. But all parents make mistakes. No one knows how to be a perfect parent. Parents generally believe that they have a great deal to do with how their children turn out; when children do not turn out well, parents have to cope with the question, ‘What did I do?’ Because our knowledge is so inexact, the answers are apt to be wrong. A parent guided by a wrong theory makes the situation worse rather than better.

(Karon and Widener 1994: 47–8)

FEAR OF ‘FAMILY-BLAMING’

Few people doubt that our emotional wellbeing as adults has something to do with how we were raised as children. For most parents (including the authors), the first reaction when something goes wrong for our sons or daughters is to wonder whether we might be somehow responsible. However, the possible role of families in the causation of ‘schizophrenia’ has become a taboo subject. Relatives are often told, in ‘psycho-education’ programmes, that they couldn’t possibly have had anything to do with the development of the ‘illness’. Research investigating the notion that families have the capacity to drive us

mad has often been denigrated as ‘family-blaming’ (Johnstone 1999; Martindale 2008). British psychologist Mary Boyle (2011: 39) recently wrote: ‘The less subtle aspects of this censorship range from accusations of blaming the parents, or being old fashioned and out of touch with evidence, to being told bluntly that you cannot say that’.

It is assumed that relationships between relatives and staff are important (which is true) and that these relationships will inevitably be damaged if we discuss whether families might have contributed to the problems of the ‘patient’ (which is not true).

What is missing from the ‘family-blaming’ debate is discussion of what the word ‘blame’ means. This complex issue is often reduced to the simplistic notion that if family factors are identified, then the parents must be bad people. Our efforts as parents, however, are best understood not from a simple moral notion of ‘badness’, but in terms of multiple interacting factors, including stressors on the whole family (such as poverty), conflicts within the parental relationship, isolation, lack of education about parenting skills, and intergenerational cycles of unhelpful parenting (Diamond and Doane 1994; Karon and Widener 1994).

Although every family will be different, the research discussed in this chapter may help families consider what might have gone wrong in their own family and what might usefully be tried to make things better for *all* concerned.

FAMILY MEMBERS’ CAUSAL BELIEFS

Contrary to the views of drug-company-funded organizations claiming to represent families, like SANE in the United Kingdom, NAMI (National Alliance on Mental Illness) in the United States, and the Schizophrenia Society of Canada, the majority of family members believe that their relatives’ difficulties are caused more by life events and circumstances, including family problems, rather than by genetics or chemical imbalances (Read and Magliano 2012). Out of 14 studies (in 11 countries) of family members of people diagnosed ‘schizophrenic’, 13 found that they endorse psychosocial causes more than medical/biological ones (Chapter 11).

For example, relatives in Turkey cited stressful events (50%) and family conflicts (40%) more often than biological/genetic factors (23%) (Karanci 1995). Some studies suggest that relatives reject the ‘medical model’ even more strongly than other people. In Italy 68% of relatives believe ‘schizophrenia’ is caused entirely by psychosocial factors, compared to 34% of the public and 20% of professionals (Magliano *et al.* 2004). An Austrian study found that relatives were twice as likely as the general public to believe that their loved ones’ difficulties were caused by ‘occupational stress’ and ‘unhappy

family situation' (Grausgruber *et al.* 2007). In the United States, family members were more than twice as likely as clinicians (43% vs. 18%) to cite 'the way he was raised' (van Dorn *et al.* 2005).

FAMILY INTERACTIONS AND RELAPSE

Although there has been a marked increase in research into child physical and sexual abuse, more subtle processes, such as emotional abuse and neglect, have received less attention from researchers (Chapter 18). However, the largest body of research explicitly relating family issues to 'schizophrenia' for the last 40 years *has* focused on these less overt processes, by studying the relationship between 'expressed emotion' and relapse (Tienari and Wahlberg 2008). The term 'expressed emotion' is a euphemism, designed to create an impression of non-judgmental neutrality. What it measures is far from neutral. Expressed emotion (EE) has three components: the *hostility* component measures global criticisms and rejection; the *criticism* component measures negative comments about what the 'patient' does, thinks or feels; and the *emotional over-involvement* component measures over-protectiveness, intrusiveness, emotional display, excessive praise, self-sacrifice and preoccupation (Vaughn and Leff 1976).

By 1992, a review reported on 23 studies (in 12 countries) of EE and relapse over 9 or 12 months. Of the 23, 20 found higher relapse rates in high-EE than in low-EE families. In four studies with a follow-up period of two years, post-discharge the median relapse rates were 27% for low-EE and 66% for high-EE families (Kavanagh 1992). Two studies have found similar differences at five years and one at eight years (Weardon *et al.* 2000). As the number of hours per week spent with high-EE families increases, so does the relapse rate (Kavanagh 1992; Vaughn and Leff 1976). The negative effects of the hostility, criticism and over-involvement occur at the psychological level, in terms of level of stress experienced (Cutting *et al.* 2006), and the biological level, in terms of activation of brain regions that process aversive social information (Rylands *et al.* 2011).

Further evidence that living in a difficult family environment is not good for one's sanity is provided by evidence that programmes that *decrease* parental EE reduce relapse rates. An early review of six studies calculated that while normal treatment (primarily medication) led to a 24-month relapse rate of 71%, EE-targeted interventions led to a rate of 33% (Kavanagh 1992). Subsequent reviews have confirmed these findings (Pharoah *et al.* 2006), including for first-episode psychosis (Askey *et al.* 2007).

EE levels in families of people diagnosed 'schizophrenic' are lower in 'developing' countries. This might help explain their higher recovery rates (Chapter 5). The relationships between EE, relapse and culture, however, are complex (Bhugra and McKenzie 2010).

FAMILY INTERACTIONS BEFORE ONSET

Expressed emotion

EE cannot be considered causative unless it precedes the 'illness'. This possibility has received far less attention than the role of families in relapse. When not ignored altogether, it is usually couched in terms of family factors merely hastening the timing of, or 'precipitating', the first episode rather than helping to cause the disorder itself.

Evidence that EE precedes, and is predictive of, 'schizophrenia' comes from the UCLA Family Project, a 15-year prospective study. The researchers noted that: 'It is sometimes implied that current interest in the family's role in the course of schizophrenia has little bearing on etiological issues', adding, however, that it is difficult 'to view family life as so discontinuous across the life-span that those attributes of the family environment related to the onset of schizophrenia do not overlap with those associated with differential course after an initial episode' (Goldstein and Doane 1982: 693). Their thinking is consistent with earlier findings by the pioneers of research into the role of stressful events, including EE, in both relapse *and* onset. Birley and Brown (1970) had found that in the three weeks before admission, most 'schizophrenics' had experienced significant life events that were beyond their control. Leff and Vaughn (1980) found that episodes of schizophrenia (but not depression) are *predicted* by living in high-EE families.

In the UCLA study the parents of 64 troubled adolescents, none of whom had symptoms of psychosis, were tested for EE. Fifteen years later, the diagnoses for the adolescents (now adults) whose parents had both been low-EE were: broad-spectrum schizophrenia, 6%; definite diagnoses of schizophrenia, 0%. Where one parent had been high in EE, the results were: broad-spectrum, 37.5%; definite schizophrenia, 0%. Where both parents had been high in EE, the rates were: broad-spectrum, 73%; definite schizophrenia, 36% (Goldstein 1987). The finding that having one parent who was *not* high-EE led to no definite diagnosis of schizophrenia is important: it confirms other studies documenting the protective nature of having one supportive attachment figure in one's early years (Fergusson *et al.* 2003).

Few studies have pursued this research avenue. One found that the parents of children with schizophrenia-spectrum disorders 'were more likely to direct harsh criticism toward the child than were parents of depressed children or parents of normal controls' (Hamilton *et al.* 1999). Another found that the parents of young Italian adults at 'ultra-high risk' for psychosis exhibited EE levels as high as the parents of young adults with actual schizophrenia-spectrum diagnoses (Meneghelli *et al.* 2011).

Affective style

The UCLA study also measured 'affective style' and 'communication deviance'. Affective Style (AS) has three sub-categories: personal criticism (e.g. 'You have an ugly, arrogant attitude'), guilt induction ('Most of our fights are because of you') and intrusiveness (I know damn well that you agree with me though you refuse to admit it') (Diamond and Doane 1994).

After 15 years, the adolescents (now adults) from 'negative AS' families were twice as likely to have a diagnosable mental disorder as those from 'benign AS' families. The difference was even greater for schizophrenia. Of those from benign AS families, 4% had schizophrenic spectrum diagnoses, compared with 56% of those from negative AS families. None from the benign AS families but 16% from the negative AS families had a definite diagnosis of schizophrenia (Goldstein 1987).

Communication deviance

Communication deviance (CD) measures the tendency to speak in a way that makes shared meaning difficult or impossible. Like EE and AS, parental CD is related to probability of relapse (Velligan *et al.* 1996).

At the 15-year follow-up, the figures for the broader schizophrenia spectrum diagnoses were: low CD, 9%; intermediate CD, 26%; high CD, 50%. None from the low CD group but 11% of the intermediate- and 10% of the high-CD group had definite diagnoses of schizophrenia. When siblings were included, 70% of high CD families had an offspring with a broad-spectrum diagnosis and 15% had a definite schizophrenia diagnosis (Goldstein 1987).

A strong interaction between CD and AS emerged. After 15 years, none of the sons and daughters of families rated as benign on AS *and* low on CD had any diagnosable mental disorders. All of the offspring of families rated as negative for AS and high on CD had at least one diagnosis, and 89% of these were broad-spectrum schizophrenia diagnoses, with 22% having definite diagnoses of schizophrenia. CD is also highly related to EE (Kymalainen and Weismani 2008).

Parents with High CD may be experienced by their adolescent as a chronic source of vague and confusing messages. If, in this setting, the adolescent also experiences critical, intrusive, or guilt-inducing affect from his parents, he would have little recourse through communication with them for exploring, clarifying or correcting his feelings of unworthiness, rejection, or isolation. A second adolescent would have parents poorly focussed in their communication but providing compensatory emotional support. A third teenager, one whose parents were hostile but clear thinking, would at least have the

avenue of functional communication as a means of understanding and adapting to their negative posture.

(Doane *et al.* 1981: 685)

The early studies

Goldstein and Doane were not the first to study families. In 1956, Gregory Bateson, Don Jackson, Jay Haley and John Weakland (the 'MRI Group') published 'Toward a Theory of Schizophrenia'. Dismissive of both biological psychiatry and intrapsychic psychology, they focused on social context. Their aim was to understand the behaviour of 'schizophrenics' in terms of communication patterns learned within their families and thereby develop a model of family therapy. They described the 'double-bind' situation whereby conflicting messages render an effective response, or clarification, impossible (Bateson *et al.* 1956). Their case studies are complex, thoughtful and convincing (e.g. Haley 1959).

They were not blaming anyone: 'The group prefers an emphasis on circular systems of interpersonal relations to a more conventional emphasis upon the behavior of individuals alone or single sequences in the interaction' (Bateson *et al.* 1962: 157). They were among the first to focus on intergenerational cycles of family difficulties.

In 1949, Ruth and Theodore Lidz had already found that most parental relationships (61%) in a sample of families of 'schizophrenics' were 'marked by strife', including 42% that were either 'seriously incompatible' or had led to separation or divorce. Child-rearing patterns were rated 'bizarre or extremely faulty according to conventional standards' in 41% of the families. In 23%, at least one parent had rejected the patient 'to an extreme degree'. In 48%, at least one was 'either psychotic or chronically and seriously neurotic or psychopathic'. Only 11% of the patients had 'come from reasonably stable homes in which they had been raised by two stable and compatible parents according to fairly acceptable principles of child rearing' (p. 343).

Later studies by Theodore Lidz and his Yale colleagues (Lidz *et al.* 1986) identified two predominant patterns of parental relationships. The 'schismatic' relationship was characterized by differences, competition and hostility. In the 'skewed' relationship, 'an apparent harmony or pseudo-mutuality was maintained because the aberrant ideas and ways of childrearing of one parent were not countered by the passive spouse' (Lidz 1969). Far from blaming parents, the focus was, again, on the intergenerational spiral of damaging interpersonal patterns. Much later (1990: 121) Lidz was to bemoan the distortions, by others, of his focus and intentions: 'I also find it very distressing that because parents' attitudes and interactions are important determinants of schizophrenic disorders, some therapists and family caseworkers treat parents as villains who have ruined the lives of their patients.'

By 1969, Lidz concluded that the problems found in nearly all the families studied 'had started prior to the birth of the patient and were continuing when the patient became overtly psychotic'. Reviewing his own studies, those of the MRI Group, the work of Alanen (1960) in Finland, Delay *et al.* (1957) in France and a group of Maryland researchers (see below), he found that 'they reveal essentially the same difficulties'.

The first to use the term 'communication deviance' were Lyman Wynne and Margaret Singer. In their research, in Maryland for the National Institute of Mental Health, they identified 'pseudo-mutuality' as a valuable concept for understanding schizophrenia (Wynne *et al.* 1958). Pseudo-mutuality is one attempt to solve the essential human conflict between needing to be related to others and needing one's own identity. This 'absorption in fitting together at the expense of the differentiation of the identities of the persons in the relation', they hypothesize, 'contributes significantly to the family experience of people who later, if other factors are also present, develop acute schizophrenic episodes'. Wynne and Singer stressed that some pseudo-mutuality occurs in most families, but if the available roles for family members were rigid and few, and were all based on the need to preserve the illusion of sameness between members, individuation could only be achieved by withdrawal into an isolated, distorted world of madness.

They published convincing case studies. Their first empirical demonstration that parents of 'schizophrenics' score significantly higher than 'normal' controls on measures of CD (Wynne and Singer 1963) was repeatedly replicated over the next 40 years. For instance, Wild *et al.* (1965) found that in 58% of the families of 'schizophrenics', both parents were high in CD compared to 12% in a control group. (Again, there was evidence that having one parent who communicates satisfactorily is a protective factor).

By 1984, the Wynne and Singer group had conducted five studies showing high CD in the families of 'schizophrenics' and could point to five reviews of research concluding that 'parental CD is an aspect of family interaction that has consistently been found to characterize families of schizophrenics' (Sass *et al.* 1984). By 1992, 12 studies had found higher CD among the parents of schizophrenia patients than among the parents of other patients or 'normal' individuals (Miklowitz and Stackman 1992). A later study found high CD in 73% of a sample of families of 'schizophrenics' (Nugter *et al.* 1997). Norwegian studies have even linked specific subtypes of parental CD to the same subtypes in their offspring (Rund and Bakar 1986).

The UCLA study had shown that parental CD precedes and predicts the onset of schizophrenia. Parental CD is also related to 'pre-schizophrenic' behaviours in young children (Velligan *et al.* 1988). In one inpatient sample, 84% of children with 'schizophrenia-spectrum' diagnoses were living in families with high CD, compared with 52% of depressed children (Asarnow *et al.* 1988).

WHAT CAUSES WHAT? WHO NEEDS WHAT?

It is clear from the research reviewed above that negative feelings and intrusiveness (expressed emotion, affective style) and poor communication (communication deviance) are not just related to relapse, but are present well before the first signs of the 'illness'. This is consistent with the fast growing literature on child abuse (most, but not all, of which happens at home) and neglect (almost all of which happens at home). But are the 'symptoms' caused by the parents' behaviour? Isn't it possible that the parents' behaviour is a *reaction* to the difficult behaviour of children who later become 'schizophrenic'?

This question can be addressed by researching the stability of EE, AS and CD. If these behaviours are stable over time, their causal role in the problems of the offspring is more plausible. Whereas if they are unstable, the assumption that the parents' are reacting to the behaviour of the offspring is more likely (Nugter *et al.* 1997).

CD is indeed a stable and enduring trait (Velligan *et al.* 1988), even in the face of a programme specifically designed to reduce it (Nugter *et al.* 1997). The Finnish Adoptive Family Study, led by Pekka Tienari (and including Lyman Wynne) found CD to be 'a traitlike feature, stabilizing after adolescence, as with certain personality traits' (Wahlberg *et al.* 2001). They also found that the CD of the adopting parents during the adoptees' adolescence *predicted* thought disorder, 11 years later, regardless of whether adoptees did or did not have a 'schizophrenic' parent (Wahlberg *et al.* 2000).

In contrast to the trait-like stability of CD, several studies have shown that EE and AS are, for some parents, quite variable over time and can be reduced by targeted programmes (Kavanagh 1992; Miklowitz 1994). This is consistent with the hypothesis that parental criticism and hostility might be explained partly in terms of parents reacting to the annoying 'symptoms' of their offspring.

Parents high in EE do, however, have other traits that *are* stable, including intolerance, lack of empathy and inflexibility (Hooley and Hiller 2001). Goldstein *et al.* (1992) found that 100% of parents whose high EE *was* stable over time had themselves experienced a major psychiatric disorder. The percentage of the relatives of 'schizophrenics' who have a 'psychiatric disorder' at the time of onset of their offspring's disorder ranges from 57% to 79% in studies from Norway (Bentsen *et al.* 1997), England (Barrowclough and Parle 1997) and Australia (Winefield and Harvey 1993), with high-EE relatives consistently having more disorders than low-EE relatives.

There is evidence that high EE is, in many cases, *not* a reaction to the offspring's 'symptoms'. One study found that parents high in EE have low global acceptance of the patient (Dorian *et al.* 2008). A review cited four studies showing that 'EE does not appear to be correlated with any obvious clinical or demographic characteristics of patients' (Hooley and Hiller 2001), which was later confirmed in the families of first-episode patients (Meneghelli

et al. 2011). Another review found seven studies showing that the relationship between EE and relapse 'is not explained by patient characteristics such as severity of symptoms or premorbid adjustment' (Kavanagh 1992: 610). Another reviewer concludes that the relationship between EE and relapse is 'independent of patient symptom severity, history of illness or medication regime' (Miklowitz 1994). Even after the 'patient' improves, EE remains high for many parents, especially those with a history of mental health problems (Goldstein *et al.* 1992). Negative AS has even been found to increase after the offspring has improved (Doane *et al.* 1986).

Perhaps the search for an answer to the question 'what causes what' that would apply to *all* families should have been called off 60 years ago. In 1962, George Brown, whose work had inspired the EE approach to schizophrenia research, found, unsurprisingly, that there were different answers in different families:

In about one-third of the interviews it seemed that the past or current behavior of the patient was a sufficient cause of the 'high emotional involvement'. . . . In roughly another third the 'high emotional involvement' was clearly due to unusual behavior of the relative, and could be independent of the severity of the illness of the patient. . . . The remaining families could be included in neither group . . . it is possible that an interaction between the behavior of the patient and that of the relative was responsible for the 'high emotion'.

(Brown *et al.* 1962: 56)

Of course there is reciprocal interaction between parents and children. The interactions will often have been going on for years. However, some 'interactional' models tend to consider only interactions *after* the onset of schizophrenia and still insist on calling the behaviours of some family members 'reactions' and the behaviour of other family members 'symptoms' (e.g. Kavanagh 1992). So there remains a danger of replacing a simplistic unidirectional *causal* model blaming parents with an equally simplistic unidirectional *reactive* model blaming the sons and daughters. There is little room in such a model for the possibility that the disturbing behaviour of the sons and daughters might be a reaction to years of parental criticism and hostility, perhaps from an early age. Kavanagh is far from alone in encouraging us to avoid the 'temptation to label people', meaning the labelling of parents as being high in EE, while being comfortable with the equally pejorative labelling of other family members as 'schizophrenic', 'patients', and 'ill'. This is unhelpful to everyone. It merely switches responsibility for the families' problems from one family member to another while pretending that no blaming is going on at all.

Others have returned to Bateson's 'circular systems of interpersonal relations' (1962: 157) and developed more genuinely interactional models (Hahlweg *et al.* 1989; Wichstrom and Holte 1992). Miklowitz (1994)

presents findings that the offspring frequently 'countercriticize' their parents, thereby reciprocating the high-EE or negative AS behaviours, and coins the term 'long chains of negative reciprocity'. Weardon *et al.* (2000) prefer 'complex circular relationship' to describe a pattern of escalating mutual negativity.

BEHIND THE ARGUMENTS

Sensitive sons and daughters

Some researchers have taken the valuable step of probing behind what family members do and say to study the *perceptions* they have of one another. It seems the patients' experience of their parents' attitudes and feelings towards them is an even more powerful predictor of negative outcomes than the actual parental attitudes and feelings (Scott *et al.* 1993; Tompson *et al.* 1995). 'Parents in the relapse group were not in touch with how the patient saw them, whereas the patients were sensitively aware of how they were seen' (Scott *et al.* 1993). This suggests that the patients might be paranoid; but in one study only the views of the patients who relapsed were accurate (Scott *et al.* 1993).

Troubled parents

One study found that: 'The more a relative perceives his or her own life to be unpredictable, and thus exposed to uncontrollable dangers, the greater the likelihood that he or she will overprotect the patient' (Bentsen *et al.* 1997: 562). Emotionally over-involved relatives are more prone to feeling guilty in general, while critical and hostile relatives have low 'hostility-guilt', meaning they have weak internal inhibitors against being aggressive. These researchers stressed the need to work with relatives' guilt feelings, whether deficient or excessive (see Martindale 2008; Martindale and Smith 2011). This is taboo for EE-based family interventions that adhere to the 'it's not your fault, because it's an illness' line.

A particularly important study found that critical relatives experienced their offspring's behaviour as more upsetting and felt less able to cope with, or control it, than other relatives. When all the patient and relative variables were analysed, the extent to which the relatives *perceived* the offspring's symptoms as upsetting, but not the actual severity of the symptoms, predicted whether the relatives were rated as a psychiatric 'case' (Barrowclough and Parle 1997). Compared with low-EE parents, high-EE parents more often explain their offspring's difficulties in 'illness' terms (Barrowclough *et al.* 1994) and estimate that the difficulties have gone on for longer (Mintz *et al.* 1989).

Thus some relatives are less critical, hostile, intrusive and overprotective than others. These relatives are more flexible, tolerant and empathic and have fewer 'psychiatric disorders'. This group is slower to believe an 'illness' has started in a distressed relative than relatives who are more critical, hostile, intrusive, overprotective, inflexible, non-empathic, intolerant and who often have their own mental health problems. This second group's hostility and criticism has been described as an understandable but futile attempt to reduce their anxiety by deploying 'an intense interpersonal social control coping response' (Greenley 1986).

So the most disturbed relatives are the most critical of, and hostile to, their sons and daughters. Their own poor mental health is sustained by their experiencing unusual behaviours as upsetting and, therefore, being easily distressed by them. There is often more than one person who needs help if the family is to get better.

MOVING BEYOND BLAME

In 1994, Karon and Widener described a series of studies that measured 'the degree to which the parent, when there is a potential conflict between the needs of the child and the needs of the parent, unconsciously acts in terms of the parent's needs without regard to the potentially conflicting needs of the child'. The parents of 'schizophrenics' did this more often than the parents of 'normal' people. Far from blaming anyone, the researchers stress: 'This is not a conscious choice by the parent: the parent is typically unaware of the potential conflict', adding that parenting problems 'are better understood as symptoms or coping devices developed in order to come to terms with the parent's own childhood' (pp. 52, 53). This issue of intergenerational cycles of parenting difficulties (Diamond and Doane 1994) is very important if we are ever to develop effective prevention programmes to reduce the amount of suffering that is going on in so many families (Chapter 20).

Instead of relying on pre-packaged 'psycho-education' programmes, we would be more helpful to family members if we focused on acknowledging, normalizing and discussing the inevitable confusion and ambivalence about causes (Chapter 26). This is more useful than trying to suppress or deny relatives' feelings by forcing them to accept either a simplistic illness-blaming model that says it's *not at all* their fault, or an equally simplistic approach that says it's *all* their fault. Everyone would learn more and produce better outcomes if we listened to relatives' thoughts and feelings (Read and Magliano 2012; Neugeboren 2003, 2009) and offered support rather than 'educating' them to think like biological psychiatrists.

Helping families is hard using a manual based on studies that have measured just one or two 'variables' from the complicated, ongoing, circular, reciprocal and dynamic processes occurring in all families. Just as the world of

a very distressed or disorientated individual is neither described nor understood by the word 'schizophrenic', so the efforts of a parent or family to do their best cannot be encapsulated by terms like 'high EE'. Both are equally simplistic and do more damage than good to the people concerned. Both convey permanence where there is variation over time and circumstances. Both convey categories where there are dimensions. Both convey something within the person that is somehow separate from the world in which they are doing their best to cope, where there is really a myriad of forces operating on the families and on all the individual family members, in the present and from the past.

A recent Editorial in the *British Journal of Psychiatry* (the publication of which suggests that the issues discussed in this chapter are no longer taboo) summarized the relationship between a range of childhood adversities and psychosis and argued:

Some of these adversities have been shown to be intergenerational, so that parents who themselves suffered in childhood struggle to provide an optimum environment for their own children. This finding can be used to counter the argument that research into intrafamilial causes of mental health problems is undesirable because it is 'family blaming'. On the contrary, the findings should encourage us to identify the needs not only of the 'identified patient' but of parents and other family members, whose problems – often originating in their own childhoods – tend to go unnoticed.

(Read and Bentall 2012: 89)

Families used to be ignored by mental health professionals. We now make more contact with them. But we still often make no real contact with what has been going on in the family. Until we are ready to ask parents and other relatives how they are dealing with the 'What did I do wrong?' question, and to listen properly without claiming to know the answer, we will remain unhelpful. Propagating our own pet theory is no substitute for responding to the different fears, worries, anger and guilt experienced by different families, and by different family members within each family (Martindale 2008). New understandings can bring new solutions, for all members of a family. This in turn might decrease the chances of psychosis in the next generation of the family.

A family member, who also happens to be a professor of clinical psychology, wrote:

I can understand why it is so easy to slip into medicalizing because, to be honest, I have found my brothers at times to be incredibly difficult, a strain and even infuriating. I too have sometimes wanted a quick answer and a quick fix to their troubled and troublesome states.

But deep down, like many family members, I wanted professionals to understand our lives and traumas and help them and us to find support and work – to help build a life for them, not just offer a slow death through medication.

(Dallos 2011: vii)

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Part III

SOCIAL AND
PSYCHOLOGICAL
APPROACHES TO
RESPONDING TO MADNESS

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PREVENTION OF PSYCHOSIS

Creating societies where more people
flourish

Judi Clements and Emma Davies

The dominant bio-genetic model of mental health argues that psychosocial factors have little to do with causing what mental health professionals call 'schizophrenia'. It is, therefore, widely assumed that little can be done to prevent it. Indeed, a PsycINFO search in January 2012 revealed that of 111,674 research papers ever published on 'schizophrenia' or 'psychosis', only 96 (0.09%) focus in any way on 'primary prevention', and only about 20 of these 96 actually do address primary prevention. Many deal, instead, with young people already identified as being 'at high risk' of developing psychosis, which is usually defined as 'indicated', 'targeted' or 'secondary' prevention (Chapter 24). The most commonly identified risk factors identified as possible targets for *primary* prevention are obstetric complications or mother's health during pregnancy (9) and substance abuse (5). Only one identifies childhood abuse (Bebbington *et al.* 2011), and only one focuses on poverty (Read 2010).

To the best of our knowledge only one study of the effectiveness of a primary prevention programme in relation to psychosis has ever been published. A two-year environmental enrichment programme at age 3–5 years, focusing on nutrition, physical exercise and education (multimodal stimulation, use of toys, art, handicrafts, drama and music), was found to reduce schizotypal traits in early adulthood (Raine *et al.* 2003).

Research into the social aetiology of 'schizophrenia', buried under bio-genetic ideologies for several decades, is now receiving the attention it deserves (Chapters 14, 18 and 19). The evidence indicates that the social factors that lead to other mental health problems are similar to those that lead to the behaviours and experiences described as 'schizophrenic'. What lays the foundation for the prevention of 'schizophrenia' will, therefore, also help prevent PTSD, depression, suicidality etc.

Indeed, Kessler *et al.* (2010) analysed data from 21 countries and concluded that ‘childhood adversities have strong associations with all classes of disorders’ and that ‘childhood adversities associated with maladaptive family functioning (e.g. parental mental illness, child abuse, neglect) were the strongest predictors of disorders’. This is consistent with our understanding of ‘high-risk children’:

What distinguishes the high risk child from other children is not so much exposure to a specific risk factor but rather a life history that is characterised by multiple familial disadvantages that span social and economic disadvantages; impaired parenting; neglectful and abusive home environment; marital conflict, family instability; family violence and high exposure to adverse family life events.

(Fergusson and Horwood 2003: 131)

In this chapter, therefore, we are not advocating specialized programmes to prevent ‘schizophrenia’ *per se*. Instead, we outline some paths to create the conditions for societies where more people flourish. This requires reframing mental health as something much broader than the absence of illness.

MENTAL HEALTH IS A WHOLE POPULATION ISSUE

Mental Health is a state of wellbeing in which every individual realises his or her own potential, can cope with the normal stresses of life, can work productively and fruitfully, and is able to make a contribution to her or his community.

(WHO 2011)

The World Health Organization’s understanding of mental health is broader than governments’ mental health policies, which remain almost entirely focused on services and response to diagnosed ‘mental illnesses’. This deficit focus extends from the health sector into the legal system, with mental health law having little or nothing to do with developing, promoting or sustaining mental health and everything to do with assessing, treating and containing people with ‘mental disorder’. Many citizens interpret the term ‘mental health’ as related to problems and diagnoses. Many see ‘mental health’ as something to be avoided rather than embraced, nurtured and valued.

To achieve gains for the population, the social and interpersonal conditions that support mental health and wellbeing need to be the focus of comprehensive and integrated public policy. To create societies where more people flourish, the discourse needs to change and the paradigms be reframed. Positive mental health goes far beyond the absence of a diagnosis. It is a different realm of functioning (Ryff *et al.* 2003)

An individual's mental capital and mental wellbeing crucially affect their path through life. Moreover they are vitally important for the healthy functioning of families, communities and society. Together they fundamentally affect behaviour, social cohesion, social inclusion and our prosperity.

(Foresight Project: Government Office for Science 2008: 10)

The Foresight Project goes on to describe mental wellbeing as a dynamic state that enables people to develop potential, work productively, create positive relations and contribute to community. Whilst not suggesting that governments cease addressing the particular needs of those diagnosed with 'mental disorders', the Foresight report urges that policies and choices have to nurture mental capital and wellbeing in the population so that 'everyone can flourish throughout their lives' (p. 10).

Nurturing mental capital and wellbeing is the *raison d'être* of mental health promotion as a public health discipline. The ideas are not new. Albee's 1985 model offers a conceptual framework for thinking about how to prevent madness.

$$\text{Incidence of emotional unwellness} = \frac{\text{organic factors} + \text{stress} + \text{exploitation}}{\text{coping skills} + \text{self-esteem} + \text{support groups}}$$

The framework draws attention to the preconditions necessary for 'a life worth living'. It includes the social determinants of health. To develop prevention approaches using Albee's model, the numerator needs to be decreased and the denominator increased.

There are strong links between mental health outcomes and relative poverty (Wilkinson and Pickett 2010). Decreasing the systemic factors in Albee's numerator is about reducing inequality, improving people's access to opportunities, security and dignity. It includes reducing debt, unemployment, inadequate housing and abuse. The question of how to do this is beyond the scope of this chapter. Recent developments in Positive Psychology, however, help us work out how to increase the denominator in Albee's equation. A first step is sharpening definitions and measurement of wellbeing.

MEASURING MENTAL WELLBEING

Measurement is made possible by the growing international (albeit Western) consensus that is developing around core measurable components of sustainable mental wellbeing. For nations with indigenous populations, like New Zealand, work is needed to identify and include indigenous measures (Durie 2006).

The generic definitions that are emerging include how people feel (pleasure, enjoyment, and satisfaction) and how they function (sense of autonomy, competence, meaning and purpose, pro-social behaviour). They include social, psychological and emotional dimensions (Keyes 2007). The term wellbeing is

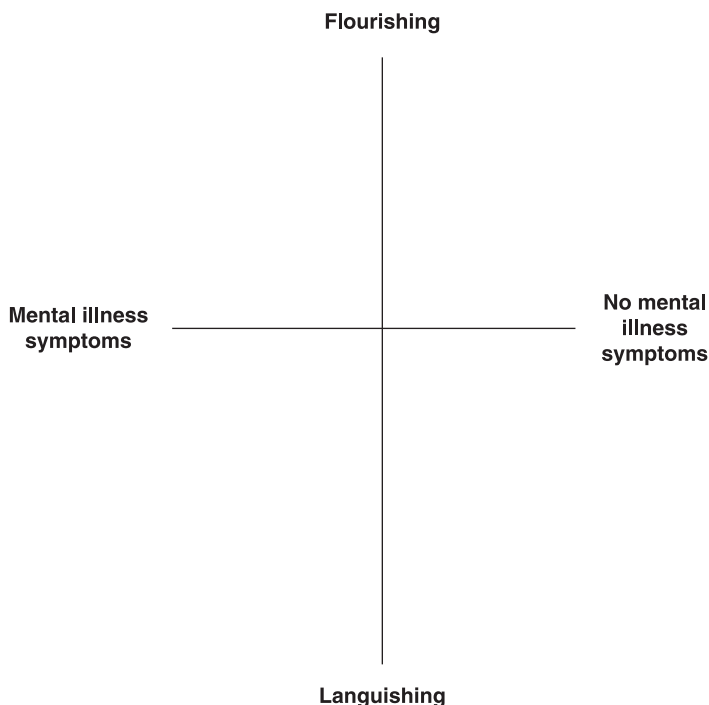


Figure 20.1 Flourishing and mental illness (adapted from Keyes 2007).

to encapsulate a state of mental wellbeing, is starting to be defined in ways that can be measured. Keyes conceptualizes this dimension as separate from, and interacting with, the presence or absence of mental illness symptoms – as illustrated in Figure 20.1.

These two continua give a broader, whole population view and also a model that encompasses the reality that people diagnosed with ‘mental illness’ can flourish, which aligns with the service user/consumer definition of recovery and influences the best anti-stigma campaigns. Fredrickson has defined a tipping point into flourishing with a balance of three positive to one negative emotional states (outlined in Table 20.1). On the other hand ‘languishing’ includes states of experience where people describe their lives as ‘hollow’ or ‘empty’ (Fredrickson and Lasoda 2005). It is experienced by both people who have and those who have not been diagnosed with a ‘mental illness’ (Keyes *et al.* 2010).

Using epidemiological data from the United States, and emerging definitions of mental health and flourishing, Keyes (2007) found that fewer than 20% of the US population is experiencing optimal mental health, or is flourishing. Using data from the European Union Survey, the Wellbeing Institute at Cambridge University, UK, has established indications of the level of

Table 20.1 Emotional states

<i>Positive</i>	<i>Negative</i>
Joy	Anger
Gratitude	Shame
Serenity	Contempt
Interest	Disgust
Hope	Embarrassment
Pride	Guilt
Amusement	Hate
Inspiration	Fear
Love	Stress

flourishing in Europe. The highest measured national level of flourishing was in Denmark at around 33%. The United Kingdom (like the United States) was just under 20% (Huppert and So 2009).

PRIORITIZING AREAS FOR ACTION

Accentuating the positive: emotions as a key resource

There are emerging strategies and practices that can increase positive emotions at the individual level. These include the maintenance of gratitude diaries (Lyubomirsky 2008), mindfulness practice, social connection, random acts of kindness (Sheldon and Lyubomirsky 2004), exercise and diet. There are also emerging public health messages regarding how to promote and maintain positive mental health, developed as part of the UK Foresight Project. The underpinning report ‘Five Ways to Wellbeing: The Evidence’ provided five key messages that reflect the behaviour that people can undertake and that evidence supports can lead to improvements in mental health and wellbeing. The Five Ways are simplified to Connect, Give, Take Notice, Learn, Be Active. The five messages help make the point that individuals, groups and populations as a whole can influence their mental health and wellbeing.

There are also emerging evidence-informed guidelines for mindfulness practice in clinical practice (National Institute for Clinical Excellence [NICE], UK). The UK Mental Health Foundation has developed a strategy to establish a national network of information and practitioners (www.be-mindful.co.uk)

Strengthening the foundations of human development

New science has transformed our understanding of child development, learning and learning difficulties. We need to capitalise on

these developments for all children by placing more emphasis on starting early . . . improving parenting, and home and nursery care environments and combining interventions in different settings – the family, nursery care and school.

(Government Office for Science 2008: 15)

The argument for intervening early in the life of the child, early in the life of problems, has been well made by scientists, governments, the World Health Organization and the OECD (Allen 2011; OECD 2009; WHO 2008). The OECD (2009) concludes that early investment in children is vital, particularly for the most marginalized and vulnerable children in society. The specifics of programmes that work to improve outcomes are more elusive.

Recent developments in neurobiology indicate that neglect, stress and trauma in early childhood can lead to compromised brain development and heightened response to threat throughout life (Perry 1997; Perry and Pollard 1998; Shonkoff and Phillips 2000). This has been explicitly linked to the development of psychosis (Read *et al.* 2001; and Chapter 18).

This makes it critical that we identify abuse and neglect as early as possible. Good quality universal screening, referrals, targeted assessments leading to quality services are all important components of a protective system. A recent systematic review of reviews of child maltreatment prevention concluded: ‘. . . home-visiting, parent education, abusive head trauma prevention and multi-component interventions show promise in preventing actual child maltreatment (Mikton and Butchart 2009).

Targeted intensive home visiting programmes are the most widely studied strategy for preventing abuse and neglect. While there is mixed evidence on the efficacy of these programmes overall, the Nurse–Family Partnerships and Early Start show particularly good results (Donelan-McCall *et al.* 2009; Mikton and Butchart 2009; Scribano 2010). Five common features of these programmes could explain their success: (a) developed as practice and research programmes simultaneously; (b) trained professional staff; (c) focus on high-risk groups; (d) relatively high involvement, and (e) substantial investment in ensuring the fidelity of programme delivery (Donelan-McCall *et al.* 2009; MacMillan *et al.* 2009).

Early Start is a home-visiting service for families based on a social-learning-model approach. Its goals are to improve child health, reduce the risk of child abuse and neglect, improve parenting skills, encourage family socio-economic and material wellbeing and encourage stable partnerships. Crucial elements include assessment of family needs and resources; development of a positive partnership between client and family support worker; collaborative problem-solving; and provision of support, advice, and mentoring to mobilize families’ resources. The programme has shown positive effects in a trial in Christchurch, New Zealand, and requires evaluation elsewhere.

Early Start has reduced related outcomes such as injuries and hospital admissions for child abuse and neglect, although rates of child protection reports did not differ between the intervention and control groups (MacMillan *et al.* 2009).

Nurse–Family Partnerships seek to improve prenatal health and birth outcomes; improve child health, development, and safety; and improve maternal life-course outcomes, including assisting women to plan for subsequent pregnancies and employment. The programme is targeted at low-income, unmarried, first-time mothers. Entry is before late pregnancy and exit at two years. Home visits are conducted by trained nurses, with six to nine visits during pregnancy and 20 during the first two years. There is evidence, from many high-income countries, of reduced child physical abuse and neglect, measured by official child protection reports, and related outcomes such as injuries in children of first-time disadvantaged mothers (MacMillan *et al.* 2009).

Creating secure attachment

Attachment-based interventions might also improve insensitive parenting and infant attachment insecurity, but there is not yet any direct evidence that these interventions prevent severe psychological neglect (MacMillan *et al.* 2009).

There is limited evidence to suggest that the ‘Circle of Security’ group protocol intervention, based on attachment theory, can change disorganized and insecure attachment in high-risk toddlers and pre-school children into secure attachments immediately post-intervention (Hoffman *et al.* 2006). By using edited versions of videotaped interactions, carers are encouraged to reflect on their own histories and interactions with their children to increase the sensitivity and appropriateness of their behaviours (Marvin *et al.* 2002). Longer-term effects are unknown. However, the sound theoretical underpinning, relationship-focus and early results are encouraging well-established multi-component programmes (e.g. Early Head Start) to incorporate these focused interventions into their broader programme development (Weatherston 2007).

MENTAL HEALTH PROMOTION PROGRAMMES

There is also emerging evidence that sustained mental health promotion programmes in schools can produce long-term benefits for children and young people, including improved emotional and social functioning as well as academic achievement. The most effective programmes appear to be those that take a whole school approach and are aimed at promotion of mental health combined with more targeted cognitive-behavioural therapy (Maxwell *et al.* 2008).

CONCLUSION

Recent publications in prominent psychiatry journals, on both sides of the Atlantic, suggest that the recent surge in research about childhood adversity and psychosis (Chapter 18) is beginning to be matched by greater attention to the need for a broader societal-level perspective. A 2012 invited Editorial in the *British Journal of Psychiatry*, having summarized the research in trauma and psychosis, concluded that ‘the implications of our having finally taken seriously the causal role of childhood adversity are profound. . . . The most important implication is in the domain of primary prevention’ (Read and Bentall 2012). Meanwhile, in the United States, *Archives of General Psychiatry* (a prominent bio-genetically oriented journal) has published several important papers on the relationship between childhood adversity and mental health outcomes. For example Green *et al.* (2010) not only confirmed Kessler’s (2010) findings that childhood adversity, particularly when occurring within the family, is predictive of all disorders, but went on to calculate a ‘population-attributable risk’ of 32%. This means that if all the adversities included in their study were eliminated from society the number of people with the mental health problems studied would be reduced by a third. However, the data used by Kessler and Green excluded ‘psychosis’ (probably a by-product of the now discredited assumption that psychosis is unrelated to childhood adversity). The first ever meta-analysis of research linking childhood adversity to psychosis also calculated a ‘population-attributable risk’, based on 41 studies. It was 33%. The authors concluded: ‘The implications of our findings for primary prevention are obvious and urgently in need of greater attention’ (Varese, Smeets *et al.* 2012).

It is particularly encouraging that *Archives of General Psychiatry* published a commentary by Australian psychiatrist James Scott and colleagues:

Interventions to improve the safety of children in their homes have been relatively poorly studied. This is even more disappointing when one consider that family functioning is intergenerational. . . . With such a large proportion of mental illness attributable to childhood adversity and the evidence of transgenerational transmission of childhood adversity, part of any future national policy initiatives regarding mental health must address the welfare of families and children. . . . We must guard against this field becoming trapped in uninformative circular epidemiology (i.e., the continuation of epidemiologic studies beyond the point of reasonable doubt). Instead, we need to focus on research related to prevention and intervention.

(Scott *et al.* 2010: 112)

George Albee put it this way:

Primary prevention research inevitably will make clear the relationship between social pathology and psychopathology and then will work to change social and political structures in the interests of social justice. It is as simple and as difficult as that!

(Albee 1996: 1133)

The task will be a little less difficult if we abandon a narrow 'mental illness' framework, pay greater attention to reducing adversity in early childhood and accept that 'mental health' is, indeed, a whole-population issue.

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THE WORK OF EXPERIENCE-BASED EXPERTS

*Jacqui Dillon, Peter Bullimore, Debra Lampshire and
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USER-LED SERVICES IN NORTH AMERICA [JC]

One of the outgrowths of the consumer-survivor movement has been the development of projects and programmes run by service users themselves. These take many forms, including support groups, drop-in centres, advocacy programmes, housing programmes, phone lines, skill teaching services, and others. These programmes adhere to certain basic principles:

- run and managed by people who have themselves received mental health services;
- based on a model of peer support rather than professional expertise;
- striving to minimize hierarchy and to value the contributions of every participant.

When consumers/survivors come together, they often identify critical needs, which are largely unmet in traditional mental health programmes. This occurs whether the participants have a critical analysis of the mental health service delivery system, or whether they are accepting of its basic tenets. Although critics of the consumer/survivor movement often label it 'anti-psychiatry' this is an over-simplification. In fact, there is a wide range of opinion within the movement on many questions, including the value of psychiatric treatment and the validity of the medical model of mental illness. However, even those consumer/survivors who adhere closely to traditional psychiatric theories are critical of many features of the mental health system, particularly the lack of respect people encounter within the system and the inability of people to get their self-defined needs met.

In general, mental health services for people identified as long-term clients are focused on maintenance and medication. There are limited opportunities, in most cases, for access to ongoing emotional support, educational programmes, long-term housing in integrated settings, or routes out of the mental health system. Many people end up living in mental health housing, attending dead-end day programmes, surviving on meagre public benefits and, in general, leading emotionally impoverished lives.

Many user-run services originate when consumer/survivors come together and articulate their unmet needs. Often, people tell stories about how they have tried to make the system meet their needs: for example, gathering in the waiting room of a programme to socialize with their peers, only to be told that this is an 'inappropriate' use of the facility. Consumer/survivors often express frustration at the fact that when they express the wish to lead more independent lives, to complete their educations, or to have satisfying careers, they are told they are being 'unrealistic,' 'grandiose,' or 'non-compliant.' It becomes clear that in order to get the help they want, they will have to do it themselves. This, for example, is how the Mental Patients' Association (MPA) started in Vancouver, Canada.

In early 1971 a group of ex-patients in Vancouver called a public meeting to discuss dissatisfaction with the psychiatric system. The organizers had been patients together in a day hospital, a supposedly progressive arrangement that treats patients during the day and permits them to go home evenings and weekends. This group of patients, however, did not find the setup supportive, since they discovered that crises frequently arose precisely during the times when staff were unavailable (evenings and weekends), and it was against the rules for patients to see one another or talk on the phone outside the institution. One Monday morning the patients arrived on the ward and learned that over the weekend one of them had committed suicide. Strong emotions were expressed, and one immediate result was the clandestine circulation of a patients' phone list. Over time, some of the patients discovered that they were relying on these 'illegal' phone calls far more than on the therapy they were receiving during the day hospital hours, and they began to talk among themselves about the kinds of 'help' that were truly useful. By this time all of them had been discharged from the hospital, and they felt that their informal network was the one real support they had. They decided to try to find more people who had similar feelings about psychiatric treatment and to discuss with this larger group what could be done (Chamberlin 2001).

MPA's experience is typical. The group identified a problem: the lack of ongoing peer support, and the fact that the professionally run programme overtly discouraged such support. They came up with a solution, the development of the phone list and the utilization of it by members for support. They recognized that this support was extremely helpful to the members. They decided to develop a more permanent solution, and, as a result of the public meeting, started their first drop-in centre. Starting out with one small volunteer

drop-in centre, the organization grew. They wrote grant applications, hired staff, and enlarged the programme to include cooperative housing. MPA become incorporated, and gradually became an established service-providing agency that utilized a self-help model to serve hundreds of members.

Another example is the Ruby Rogers Advocacy and Drop-In Center (RRC) in Somerville, Massachusetts, which this author helped to start in 1985. RRC developed from an older consumer/survivor organization, the Mental Patients' Liberation Front (MPLF), founded in Boston in 1971. MPLF was a political action group that organized meetings and demonstrations aimed at bringing about radical change in the mental health system. MPLF published a handbook, *Your Rights as a Mental Patient in Massachusetts* (1973), which provided legal and practical information for people who wanted to challenge decisions made about their care, particularly involving involuntary commitment and forced medication, and distributed the booklet directly to patients in hospitals and programmes. MPLF was also instrumental in a court case, *Rogers v. Commissioner*, that established a limited right to refuse medication in Massachusetts.

Many MPLF members wanted more companionship and support than a weekly business meeting and committee work could provide. This led to the decision to seek funding from the Massachusetts Department of Mental Health (DMH) to establish RRC. The decision to seek DMH funding was controversial. Some members objected to the idea of the agency of which they were so critical also being a funder, but the application was submitted and accepted. RRC was operated as a totally peer-run programme. Like MPA, programmatic decisions were made by members through business meetings open to all.

The basic philosophy of MPA, RRC, and many similar programmes is that people who have been labelled mentally ill have the capability to help themselves and one another. In the traditional mental health system, there is a strong division between professionals and clients. Professionals are there to give help, clients to receive it, and these roles are seen as immutable. In self-help programmes roles are fluid. People may come into the group seeking help, or because they see themselves as wanting to help others, but it is common and expected that members will fulfil both roles at different times. One of the most important things that happens to people within self-help groups is that they get to experience themselves filling a different role. A person who has been seen, and has always seen him/herself, as needy and dependent responds dramatically when he or she has the opportunity to help another member. Often, a person's entire self-image begins to change when he or she experiences another member's appreciation for some advice or support. Whether in a formal support group setting or in the more informal membership interaction that is the heart of the self-help experience, words of comfort and empathy often have dramatic results. Knowing that another person has had similar experiences, and that they have overcome them, provides valuable insights. Labelling and diagnostic terminology is

discouraged, although many people, because of years of experience in the mental health system, have learned to use labels to describe their own behaviour and the behaviour of others. Members are encouraged to use common sense terminology instead. This is one of many ways members are encouraged to see their thoughts, feelings, and actions, and those of their peers, as normal reactions to real-life stresses, rather than as mysterious 'symptoms' over which they have no control.

In self-help programmes, members are free to engage in whatever degree of contact and participation they choose. There is no single right way to utilize a self-help programme. One member may attend on a daily basis, another only when he or she is feeling troubled or needy. This allows people to tailor the service to meet their self-defined needs, rather than having to fit into a programme or structure designed by others.

Self-help groups are structured with the minimal number of rules. Often, groups start with the idea that no rules are necessary. This is a common reaction when people try to structure something that they intend to be totally different from the coercive mental health system. In time, however, most groups realize that a few basic rules are necessary: the same rules that govern civil behaviour generally. This author has seen posters in many client-run programmes, in many countries, which list the rules that the group has arrived at cooperatively. Although independently arrived at, these rules are remarkably similar. They usually include prohibitions against stealing, weapons, violence, illegal drugs, and verbal harassment of others. Usually a process is put into place to deal with violations of the rules, in which every attempt is made to deal with the rule-breaker in a caring and respectful manner.

Another important feature of these groups is that they take a pragmatic stance about such questions as the nature (or existence) of 'mental illness,' or the utility of psychiatric medication, with members holding various positions. Although critics frequently accuse self-help groups of condemning psychiatric treatment or medication, in fact, many members may also be seeing mental health professionals and using psychiatric drugs. These are seen as matters of individual choice.

EFFECTIVENESS OF USER-LED SERVICES [JD, PB, DL]

Throughout history, people who have had their experiences defined as mad have protested at having their realities redefined by others and at the treatments that result (Porter 1987). With issues of power, choice and control being central to all mental health crises, and responses to those crises, many users have felt compelled to work together to improve their individual and collective experiences and have benefited from the mutual support of their peers.

An early attempt at collective action in pursuit of positive change was the Alleged Lunatics' Friend Society, in England (1845–63), often cited as

the first user-led advocacy group (Hervey 1986). It was formed by, amongst others, John Perceval (the son of an MP), who was admitted to asylums because of his religious beliefs. It was not until the 1970s that the first signs of coherent independent action by users and their organizations were integrated into collaborative attempts to improve mental health services (Campbell 1996), creating a burgeoning affirmation of the potential of individuals, groups and communities to support each other, outside and alongside the contributions of professional disciplines.

Within contemporary mental health services, it has become increasingly recognized that input from users regarding decision making and planning in mental health services is essential if the needs of users are to be met (Perkins 2001). People who use services are progressively being seen as the experts on how those services should be developed and delivered; they are the ones who know exactly what they need, what works well, how things should be delivered and what improvements need to be made. For example, many Mental Health NHS Trusts in the United Kingdom are introducing Peer Workers into their staff teams, building on innovative work that has been taking place in the voluntary sector and peer-led mental health organizations for many years.

Several studies worldwide have found that user-run peer support services produce positive outcomes, including for people who experience psychosis (Corrigan 2006; Doughty and Tse 2005, 2011). Improvements have included increases in wellbeing generally and in satisfaction with mental health services (Doughty and Tse 2005). Participation in peer support programmes has been shown to yield decreases in levels of distress and hospitalization (Galanter 1988), larger social networks (Rappaport *et al.* 1992), enhanced self-esteem and social functioning (DeMasi *et al.* 1997), decreased lengths of hospital stays, and lower services costs overall (Dumont and Jones 2002).

Most studies have been descriptive or qualitative, seeking to identify the characteristics of people who choose to participate in these programmes, the processes that lead to change, and the users' perspective on benefits of participation (Kaufmann *et al.* 1994; Mowbray and Tan 1992; Segal *et al.* 1995). Doughty and Tse (2005) found that users of user-run services in New Zealand reported higher levels of satisfaction compared to traditional mental health services, and reported increases in their quality of life. A cross-sectional study in America, of 1,824 individuals with diagnoses of bipolar, schizophrenia or major depressive disorder, found that being involved in user-run peer services was positively associated with nine out of ten indices of recovery and empowerment (Corrigan 2006).

Strategies for Living (Faulkner 2000), a qualitative study designed and executed by service users, involving interviews with 71 mental health service users, explored people's strategies for living and coping with mental distress. Peer support, the support of others in similar circumstances and the value of self-help received warm and grateful praise. The first experience of meeting

others with similar problems, in a group or day centre, was often a significant turning point in people's lives, emphasizing the value of acceptance and belonging against a background of stigma and discrimination. This suggests that practitioners should pay more attention to the role of self-help and peer support in overcoming stigma and discrimination (Faulkner and Thomas 2002).

Wallcraft and Michaelson (2001) have stated 'it is now possible to discern a clear alternative discourse relating to the phenomena of severe distress, non-rational behaviour and beliefs' (2001: 179). This is as a result of years of user action to generate our own discursive practice and to create our own discourse about mental distress. The 'self-advocacy' discourse offers greater hope of recovery and less social rejection as it promotes self-knowledge, self-advocacy, self-empowerment and self-determination. Wallcraft and Michaelson (2001) have outlined a number of ways in which the self-advocacy movement is challenging the biomedical discourse, including:

- providing alternative concepts to describe mental distress, for example defining severe mental illness as a breakdown;
- emphasizing life histories to contextualize the complex origins of breakdown and crisis;
- developing self-assessment schedules of needs and wants;
- developing self-management techniques for specific problems;
- setting up alternatives to hospital care, i.e. self-help projects;
- designing and carrying out user-led research;
- enabling service users to become trainers and designing training for mental health workers.

In light of the current global financial crisis where mental health care resources are limited, investing in user-run services for mental distress makes good sense. More individuals experiencing mental distress could be better supported on their recovery journeys, more cost-effectively. However, as psychiatrists Pat Bracken and Phil Thomas (2003) note, professionals 'need to move beyond the modernist technical frameworks and models of twentieth-century psychiatry if we are to keep faith with the growing user movement'.

THE INTERNATIONAL HEARING VOICES MOVEMENT [JD, PB, DL]

The Hearing Voices Movement was founded over 25 years ago, following the ground-breaking research of Marius Romme and Sandra Escher, who advocated for a radical shift in the way we understand the phenomenon of Hearing Voices. In contrast to traditional, biomedical psychiatry, which views voices as an aberrant by-product of genetic, brain and cognitive faults, Romme and

Escher's research firmly established that voices make sense if the traumatic circumstances that frequently provoke them are taken into account (Escher 1993; Romme and Escher 1989, 2000, 2011). While 'auditory hallucinations' is the preferred jargon within psychiatric literature, the term 'hearing voices', which uses ordinary, non-pathologizing language framed subjectively, has been reclaimed. This is part of a wider aim within the mental health user movement to decolonize medicalized language of human experience.

Romme and Escher's research shows that the majority of people who hear voices have had some traumatic experience that they connect with hearing voices. Subsequent research has confirmed their findings (Chapter 18) and attests to what many of us with first-hand experience of madness have always known – bad things that happen to us can drive us crazy.

Launched in 1988, inspired by the work of Romme and Escher and the Dutch self-help organization, Stichting Weerklank, the national Hearing Voices Network, England (HVN), a registered charity, has earned a reputation as an innovative, prominent and influential grass-roots movement, which openly critiques the traditional psychiatric relationship of dominant-expert clinician and passive-recipient patient (www.hearing-voices.org).

HVN is a network of people who hear voices, their friends and relatives, carers, allies, support workers, social workers, nurses, psychologists, psychiatrists and others. We work together to gain a better understanding of hearing voices, seeing visions, unusual tactile sensations and other sensory experiences. We work to reduce ignorance and anxiety about these phenomena. The aims of the network are to raise awareness of these experiences, to give men, women and children who have them an opportunity to talk freely about them together and to support anyone with these experiences seeking to understand, grow and learn from them in their own way. This is important to counteract the invalidation so often experienced when people disclose their experiences (Geekie and Read 2009; Romme *et al.* 2009).

HVN creates sanctuary; safe spaces to share taboo experiences, where there are real possibilities for healing and growth. People are free to share and explore their experiences in detail, including the content of what their voices say (Beavan and Read 2010), without the threat of censorship, loss of liberty or forced medication, which are a common feature of disclosure in traditional psychiatric settings.

Although HVN has existed for many years on limited funding with a small team of paid staff and volunteers, the organization has achieved a great deal. We have established and supported more than 200 Hearing Voices groups across England. Judi Chamberlin mentioned, earlier, the difficulty that many people have moving on from the mental health system. Our groups have led to many members safely ending reliance on statutory services. Groups are run mainly in the community but also within statutory and voluntary sector services. These include acute settings, prisons and low-, medium- and

high-secure settings. In addition a number of specialist groups for women, people from Black and minority ethnic communities, prisoners, children, parents and young people have also been established. A recent review of Hearing Voices groups states:

Group treatment specifically for voice-hearers can increase group homogeneity and therefore aid normalization and treatment specificity. Such groups are currently being offered in both state and voluntary sectors internationally. The longest standing provider is the Hearing Voices Network. . . . A key aim of the Network's Hearing Voices Groups is to create a wider acceptance that voice hearing is a valid experience with a variety of explanations and not an indication of mental illness.

(Ruddle *et al.* 2011: 758)

The review identified only one evaluation of an HVN group, a single group pre-post study (Meddings *et al.* 2004). For the 12 participants with pre- and post data self-esteem and coping strategies increased and duration of admissions and voice frequency and power decreased. This is consistent with numerous unpublished evaluations. For example, six-week groups at a Community Mental Health Centre in New Zealand, led by an experience-based expert with a clinician, led on average to a 50% (self-reported) reduction in both frequency of, and distress from, voices. The majority (88%) reported that they found the groups 'very helpful', with 77% reporting increased understanding of the voices, 70% increased ability to influence the voices, and 73% reduced negative impact of the voices (Lampshire 2011). However, there is an urgent need for further studies, with control groups.

As well as our group work, we raise awareness of the benefits of talking about voices, and the value of peer support. We promote this innovative approach to voice hearers, family members and mental health professionals. We regularly provide training to staff, support on-going research into hearing voices and allied subjects, and liaise with the media, which has resulted in a number of positive broadcasts and publications about HVN and voice hearing, both nationally and internationally (Adams 2008; Dillon 2010a; Hilpern 2007; Kirsch 2007; Lakhani 2009; Smith 2007). We also offer a confidential helpline for voice hearers, publish a quarterly magazine and have produced a range of publications (e.g. Baker 1989; Downs 2005a, 2005b; James 2001; Thomas and May 2005).

Hearing Voices Groups and the 'Hearing Voices Approach' are now accepted, commonplace and embedded within mainstream services. A recent Healthcare Commission report (2008) commends mental health trusts that provide Hearing Voices groups as offering 'appropriate and safe interventions' in inpatient settings.

There are now Hearing Voices networks in Australia, Austria, Belgium, Canada, Denmark, England, Finland, Germany, Greece, Holland, Ireland, Italy, Japan, Kenya, Malaysia, New Zealand, Norway, Palestine, Scotland, South Africa, Spain, Sweden, Switzerland, Uganda, United States and Wales. International developments are co-ordinated via INTERVOICE – the International Network for Training, Education and Research into Hearing Voices – an international organization dedicated to spreading positive and hopeful messages about the experience of hearing voices across the world (www.intervoiceonline.org).

Representatives and members of the international networks meet annually, at the World Hearing Voices Congresses, to promote new developments, innovative practice and translation of materials, to support activities in burgeoning countries and to share a sense of community, solidarity and camaraderie. One of the fundamental strengths of the Hearing Voices Movement, which has led to its extraordinary success, is that it is based on mutually respectful relationships – authentic partnerships between experts by experience and experts by profession, working together to bring about the emancipation of voice hearers.

HVN's starting point is that the crises that people experience are real and that they are happening for a reason that is directly connected to the person's life. We endeavour to support people to make sense of the real events in their lives that may have precipitated their crisis. We show a genuine interest in the range of people's inner, subjective experiences. When people describe experiences that are deemed 'psychotic', we look for the meaning in their madness. Sometimes people use metaphorical or symbolic language to convey their realities and sometimes they are talking literally about things that have happened to them. However crazy someone appears, we believe that they are making a meaningful attempt to survive maddening experiences.

Contrary to traditional approaches, HVN sees voice-hearing as significant, decipherable and intimately connected to a person's life story. Consequently, we encourage and support people to listen to their voices and attest to their reality in order to better understand their meaning. We acknowledge that people are having normal reactions to abnormal stress. Instead of asking people 'WHAT is wrong with you?', we ask them 'What has happened to you?' On a daily basis we hear stories of physical, sexual and emotional abuse, the impact of neglect and poverty, as well as of racism, sexism and classism. Those people, whose experiences do not fit so neatly into a category of trauma as it is currently understood, raise the question of developing our understanding of the huge range of painful and damaging experiences that can be inflicted on children and adults.

By listening to people's stories and looking at the rapidly emerging evidence in the field, it is clear that the impact of trauma, particularly experiences like child sexual abuse, is frequently at the root cause of 'psychosis'

(Chapter 18). However, other more mundane and less obvious experiences such as serious attachment difficulties, enmeshment in families, the consequences of intergenerational trauma, the long-term impact of poverty, racism and other social inequalities can also be profoundly affecting and are all significant factors in the development of 'psychosis' (Morgan *et al.* 2006; Read 2010; van Os 2004; and Chapters 14 and 18).

We show respect for the reality of the trauma and suffering that people have experienced and a keen awareness of how these may limit their expression of feelings, ability to think clearly or capacity to connect. A key part of our role is to magnify the voices of people who are not normally listened to, by promoting the belief that each person has a deep wisdom and expertise about managing and dealing with their own problems. We validate and support people's resilience, creativity, stamina and emotional strengths, even when they themselves doubt the existence of these qualities.

Acknowledgment enables people to develop true insight into their own distress and suffering, which leads to an increasing sense of meaning and purpose in their lives. We are interested in people's subjective experiences – including their altered states of consciousness, unusual perceptions, ideas and ways of seeing and experiencing the world. When your own feelings, thoughts, experiences and bodily sensations begin to make sense to you, insight is a natural consequence. When you understand your own 'symptoms' as meaningful and essential survival strategies, a more respectful and loving acceptance of yourself begins to emerge (Bullimore 2009; Dillon 2010b; Lampshire 2009; Longden 2010; McNamara 2011).

We recognize that recovery is an on-going process with no fixed end point and that each person's recovery is unique. HVN recognizes that crises may occur again because recovery is an evolving process – an expansive process, not a reductive one that seeks to control and maintain people. We have faith in people's inherent right and capacity to heal, to make mistakes, to learn and to grow. We know that there is much about human experience that we do not understand and we remain humble and curious and open to new ways of seeing the world. We are not interested in complying with social control or in servicing normality. 'Instead of being a list of symptoms, with side effects on top, we are people who hear voices and see visions, have unusual thoughts, passionate feelings, intense experiences' (Dillon and May 2002). We celebrate our differences.

For many voice hearers and psychiatric survivors, active participation and social action supports and enhances the recovery process; having a shared survivor mission and becoming part of a collective voice creating change in the world, both inside and out, is healing, empowering and liberating. This is also true for many mental health workers, who have become increasingly disillusioned with an inadequate biomedical model, disturbed by the collusion demanded of staff in a mental health system driven by fear, control and bureaucracy. For them, becoming part of the Hearing Voices Movement also

enables a recovery of meaning, purpose and optimism, a renewal of important values, a rediscovering of a sense of self. For all of us, the possibility of reconnecting to ourselves and each other as unique and equal human beings is life affirming.

From Romme's perspective, psychiatry must change its attitudes toward voice-hearing in the same way it changed its attitudes toward homosexuality: accepting it as a normal variation of human experience, rather than pathologizing it and attempting to cure it. Within the current context there is often a literal denial of the experience of hearing voices and the injustices that may have caused the voices in the first place. Instead, intelligent people are expected to accept discredited diagnoses for fear of being labelled as 'lacking in insight', are coerced into taking medication that they don't want and which frequently does more harm than good and are incarcerated against their will – 'for their own good'. Fighting for the rights of people deemed mad, who have already suffered more than enough, is the last great civil rights movement.

NOTE

Judi Chamberlin died in January, 2010 (Hevesi 2010). This chapter consists of a shortened version of Judi's chapter in the first edition of *Models of Madness*, followed by a summary on the effectiveness of user-led services and an account of the Hearing Voices Movement by Jacqui Dillon, Peter Bullimore and Debra Lampshire.

Intervoice: www.intervoiceonline.org

Hearing Voices Network, England: www.hearing-voices.org

EXAMPLES OF OTHER GROUPS RUN BY EXPERIENCE-BASED EXPERTS

The Bonkers Institute: www.bonkersinstitute.org

Coming off Psychiatric Medication: www.comingoff.com

The Freedom Centre: www.freedom-center.org

The Icarus Project: <http://theicarusproject.net>

International Network Toward Alternatives and Recovery: <http://intar.org>

Mad Chicks: www.mad-chicks.org.uk

Mad Pride: www.ctono.freemove.co.uk

MindFreedom International: www.mindfreedom.org

UK Advocacy Network: www.u-kan.co.uk

World Network of Users and Survivors of Psychiatry: <http://wnusp.rafus.dk>

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COGNITIVE THERAPY FOR PEOPLE EXPERIENCING PSYCHOSIS

Anthony Morrison

Over the last two decades, the cognitive model originally outlined by Beck (1976) has been applied to the understanding and treatment of psychosis. This model suggests that the way we interpret events has consequences for how we feel and behave, and that such interpretations are often maintained by unhelpful thinking biases and behavioural responses. Beck's cognitive model also suggests that these interpretations are influenced by our core beliefs, which are formed as a result of life experience. There have been several recent attempts to develop cognitive models of psychosis and psychotic symptoms or experiences (Bentall *et al.* 2001; Chadwick and Birchwood 1994; Garety *et al.* 2001; Morrison 1998, 2001; and Chapter 16). Many of these models suggest that it is the way that people interpret psychotic phenomena that accounts for distress and disability, rather than the psychotic experiences themselves.

Cognitive therapy (CT) for psychosis (Hagen *et al.* 2010; Morrison 2002) is based on the same principles that were outlined for standard cognitive therapy for emotional disorders (Beck 1976). CT is time-limited and based on a cognitive model of the disorder in question and on individual case formulations derived from the model.

There is an increasing body of evidence regarding the efficacy of cognitive therapy for psychosis, including for first episodes (Morrison 2009). Several randomized controlled trials have shown cognitive behavioural approaches to the treatment of psychosis to be superior to both routine care and treatment-as-usual (Kuipers *et al.* 1997, 1998; Tarrier *et al.* 1998), in addition to demonstrating benefits in comparison with befriending or supportive psychotherapy (Sensky *et al.* 2000; Tarrier *et al.* 1998). Meta-analyses have concluded that cognitive behavioural treatment of psychosis is effective and should be delivered routinely as part of the treatment package that is offered to people with a diagnosis of schizophrenia (Gould *et al.* 2001; Sarin *et al.* 2011; Wykes *et al.* 2008). This is now recommended by the guidelines for the

treatment of schizophrenia in several countries, including the United Kingdom (NICE 2009).

The first study to evaluate CT in the absence of medication found that 20 people diagnosed with schizophrenia spectrum disorders had significantly reduced symptoms (positive and negative) and improved social functioning at end of treatment and at six months follow-up (Morrison *et al.* 2012a).

The primary aims of CT for psychosis are to reduce the distress experienced by people with psychosis and to improve their quality of life. The purpose of CT is not necessarily to reduce the frequency of psychotic symptoms but, rather, to help patients to achieve their own specific goals in relation to the problems that they have identified. CT for panic disorder does not aim to eliminate body sensations; instead, it aims to help people generate less upsetting appraisals of these experiences. CT for psychosis works similarly, focusing on generating less distressing explanations for psychotic experiences, rather than attempting to eliminate these experiences. Indeed, CT should recognize that psychotic experiences may well serve a function for the person.

The process of CT for psychosis begins with assessment and the identification of problems and goals. It then involves the development of a cognitive behavioural formulation that is used to guide selection of treatment strategies. These strategies are implemented and evaluated, and the outcomes of interventions are used as data with which to modify the formulation, if indicated. The process ends with relapse prevention in an attempt to consolidate and maintain treatment gains.

ASSESSMENT

Assessment of people with psychosis is very similar to a cognitive behavioural assessment of a non-psychotic patient. After setting the scene, explaining confidentiality and the practicalities of therapy, it is often helpful to begin with an analysis of a recent problematic incident. The aim of this is to generate information that will be useful in understanding the development and maintenance of problems and in suggesting change strategies.

Problem maintenance

It is common to begin with an assessment of problem maintenance. This can be done by examining a recent incident in terms of events, thoughts, feelings and behaviours, and collaboratively searching for meaningful links between these factors. For example, Chadwick and Birchwood (1994) recommend viewing auditory hallucinations as an antecedent event within this kind of framework. This analysis of a recent time when someone was distressed by his or her voices is likely to provide information about how people interpret their voices and how this makes them feel. If they believe that the voice belongs to

the Devil, who is trying to make them do bad things, this is likely to make them feel scared, angry or hopeless (depending on their belief system), whereas if they interpret the voice as being a sign that they are special, or as a dead friend who is advising them, then they may feel happier about the experience. The interpretation of the voice, combined with the emotional response, is likely to determine what the person does in relation to the voices. This framework is also applicable to delusional beliefs, as illustrated below:

<i>Event</i>	→ <i>Interpretation</i>	→ <i>Feeling</i>	→ <i>Behaviour</i>
sees man in suit	→ they're following me	→ scared	→ evasive action
	they're going to hurt me	→ paranoid	→ avoid going out

The main purpose of assessment is to generate information that can be used to develop a formulation. The cognitive model of psychosis should therefore guide the process of assessment. Questions should be asked to identify problematic events or intrusions, the subsequent interpretation of these, and emotional, behavioural, cognitive and physiological responses. Specific factors include culturally unacceptable interpretations, selective attention, control strategies, positive beliefs about psychotic phenomena, imagery in relation to psychotic phenomena and metacognitive beliefs (thoughts about thoughts and thought processes). It is also important to examine environmental factors that may be involved in the maintenance of the problem, such as the kind of neighbourhood a person lives in, their housing, (un)employment and financial circumstances.

Problem history and development

It is important to assess early experiences that may have contributed to the development of current difficulties. Life events in childhood contribute to the development of core beliefs and dysfunctional assumptions or rules that guide behaviour and the selection of information-processing strategies (Wells and Matthews 1994). Particular events that are worth assessing include childhood sexual and physical abuse, emotional abuse and neglect, social isolation and bullying. This is especially important in patients with psychosis, given the high prevalence of such experiences in this population in both childhood and adulthood (Larkin and Read 2011; Read *et al.* 2005 Varese, Smeets *et al.* 2012). A general assessment of family life, cultural and spiritual upbringing, school experiences and friendships should also be performed.

The influence of life experiences on the development of self and social knowledge should be considered. Assessment should include an analysis of core beliefs. These are unconditional statements about oneself, the world and other people (e.g. 'I am bad', 'Other people cannot be trusted', 'The world is dangerous'). The conditional beliefs or rules that people adopt to compensate for these core beliefs should also be assessed. These often occur in the form of 'If ... then ...'

statements, such as, 'If I am perfect, then I will be lovable.' The compensatory strategies that are the behavioural expressions of these rules should also be identified (such as striving for perfection or subjugating personal needs).

Procedural beliefs, which guide the selection of information-processing strategies, should also be assessed (Wells and Matthews 1994). Procedural beliefs that are particularly relevant to people with psychosis include beliefs about the utility of paranoia and suspiciousness (e.g. 'Staying on your toes keeps you safe') or beliefs about unusual perceptual experiences (such as 'Having odd experiences can make life more interesting' or 'Hearing voices can help me cope').

PROBLEM AND GOAL LIST

The development of a shared list of problems and goals, which can then be collaboratively prioritized, is a central part of CT and is invaluable in engaging patients (whether they are psychotic or not). Problems should start out quite general and be phrased in a more specific manner after additional information is gained. The goals set in relation to the problems should then be made as SMART as possible. This means making the goals Specific, Measurable, Achievable, Realistic and Time-limited. Here are some examples:

- *Problem:* Hearing distressing voices that interfere with relationships.
Goal: Within six weeks, to reduce distress from 70% to 40%, or to reduce frequency from about 6 hours per day to 4 hours per day, or to increase belief that 'I can have a conversation despite my voices' from 0 to 25%.
- *Problem:* Feeling paranoid in social situations.
Goal: Within four weeks, to reduce belief that 'Other people are trying to harm me' from 90% to 50%, or to reduce anxiety when in social situations from a weekly average of 85% to a weekly average of 50%.
- *Problem:* Uncertainty about the link between childhood abuse and my current difficulties.
Goal: Within one week, to find out more about traumatic life events and the development of unusual experiences, and to increase belief that 'I know what research says about childhood trauma and hearing voices' from 10% to 80%.
- *Problem:* Having no close friends to confide in.
Goal: Within four weeks, to find one person that I would feel comfortable telling some personal details to.

FORMULATIONS

Once assessment has been conducted and problems and goals agreed upon, the next step is the development of a shared formulation. There are several

levels at which a person's difficulties can be formulated. Basic formulations can be easily constructed by summarizing recent incidents in the format of event→thought→feeling→behaviour cycles, as mentioned earlier. More detailed maintenance formulations that focus on specific symptoms can also be useful. For example, one model of the maintenance of auditory hallucinations has suggested that safety behaviours (strategies designed to avoid feared outcomes in relation to voices) can prevent disconfirmation of the interpretation and also increase the frequency of voices (Morrison 1998). It also emphasizes the role of affect and physiology in the maintenance cycle and incorporates normalizing triggers such as sleep deprivation, drug use, stressful life events and biochemical changes. This model can be easily translated into a case conceptualization if a patient prioritizes voices on their problem list. An example of this type of formulation is given in Figure 22.1.

Another level of formulation is the development of an historical case conceptualization. This type of formulation incorporates early experiences and life events and the impact that these have had on core beliefs, procedural beliefs, dysfunctional assumptions and compensatory strategies, in addition to information regarding the current maintaining factors. One model of psychotic symptoms (Morrison 2001) has suggested that it is the cultural acceptability of interpretations that determines whether someone is viewed as psychotic or not, and that these interpretations are

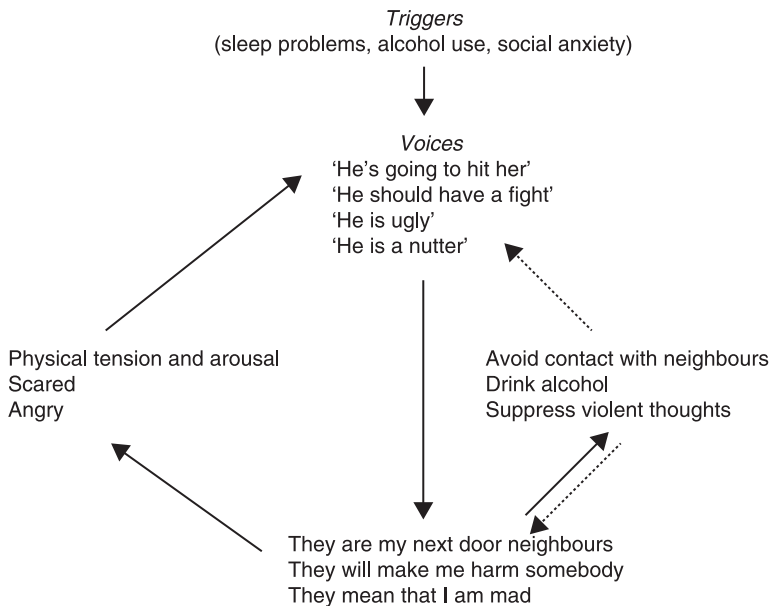


Figure 22.1 A formulation of the maintenance of voices.

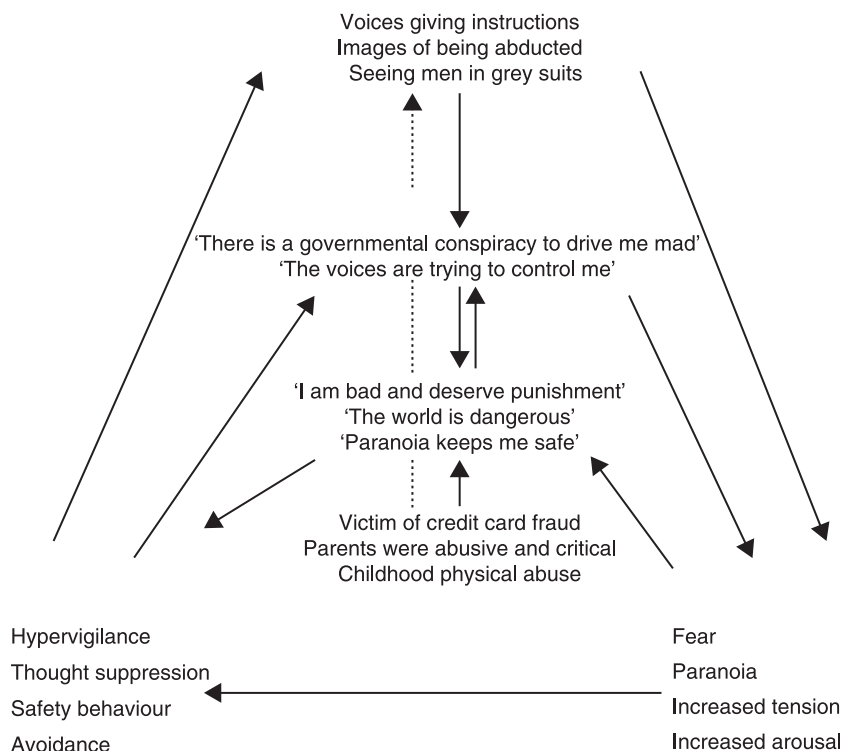


Figure 22.2 An idiosyncratic formulation of the cognitive model of psychosis.

influenced by life experiences and beliefs. It also suggests that it is the initial interpretation of psychotic experiences, and the way in which people respond to such experiences, that determines whether or not the experience will cause distress and recur. This model of psychosis is also easily translated into an idiosyncratic case conceptualization that can explain the development and maintenance of psychosis (see Figure 22.2 for an example).

INTERVENTIONS

Once a case formulation has been collaboratively developed, strategies for change can be chosen on the basis of what is likely to achieve quick success or what will have the most significant effect on the person's quality of life. Most change strategies can be described as verbal reattribution methods or behavioural reattribution methods.

Verbal reattribution

Advantages and disadvantages

It is important to consider the advantages and disadvantages of a particular belief or experience. This should be done before attempting to change a belief, even if the belief is distressing. Positive beliefs about psychotic experiences can be identified through this process. These can include the advantages of being paranoid, such as safety and excitement, or beliefs about the benefits of hearing voices. Such beliefs are important, as they have been shown to be implicated in the development and maintenance of psychotic experiences (Morrison *et al.* 2004a). Common positive beliefs about hallucinations include the benevolence of the voices (Chadwick and Birchwood 1994) and perceiving the voices as providing companionship or relaxation (Miller *et al.* 1993). The origins of psychotic experiences should also be explored within this context, as they may have been functional at some stage, as coping or survival strategies. For example, paranoia is helpful if you are in a dangerous or abusive environment. Voices frequently evolve as a coping response to trauma (Larkin and Read 2011; Romme *et al.* 2009). Some negative symptoms can also be conceptualized in a similar way (see below).

If significant advantages are identified, then these should be evaluated with regard to current utility, alternative potential sources of the benefits and the relative weighting of advantages and disadvantages. It may be that work is required to provide an alternative way of achieving the benefits before the problematic aspects of the psychotic experience can be dealt with. For example, if someone has their self-esteem maintained by a belief that they are special and chosen by God (but is experiencing difficulties as a result of behaviour arising from this), then it is important to provide alternative sources of self-esteem. Similarly, if someone views the company that their voices provide as very significant, then it is important to help them develop social networks that will address this need. At times, it is necessary to recognize that the benefits of a psychotic experience outweigh the disadvantages, and that other ways of proving the advantages are not achievable or realistic. In such cases, the patient should be enabled to make their own decision and be supported as best as possible in relation to their environment (familial, social or psychiatric). Facilitating contact with support groups or interest groups with similar experiences or beliefs can be helpful.

Usually, however, it is possible to provide an alternative way of achieving the benefits, or the advantages are outweighed by the disadvantages. In addition, many positive beliefs about psychosis are inaccurate. Also, survival strategies that have been helpful in the past may no longer be necessary if the environment has changed. If a decision is reached to proceed, then therapist and patient can continue to evaluate, and possibly change, the belief or reduce the frequency of, or distress associated with, an experience. Strategies that will facilitate this process are described below.

Evidential analysis

A standard procedure in CT is to examine the evidence for and against a particular thought or interpretation of events (Wells 1997, 2000). This process begins with the identification of a specific thought or belief to be considered, and some discussion of what can be construed as evidence (e.g. a feeling that something is true is unlikely to be viewed as evidence in a courtroom). Two columns can then be used to collate evidence for and against the particular belief (Table 22.1). The evidence should be elicited Socratically, using questions such as ‘What makes you think that this is true?’ and ‘Is there anything that is incompatible with your belief?’ Consideration of modulating factors can also be helpful in generating evidence, as can questions related to the content of the belief.

Generation of alternative explanations

Generating alternative explanations is another useful strategy to help reduce the distress associated with psychotic experiences. Delusional ideas and beliefs about voices are open to examination, and it is important to help people to consider a wide variety of possible explanations for their experiences if they are distressed by their current explanation. Brainstorming exercises, and the consideration of what other people have suggested or would suggest, can facilitate this process. These alternative explanations should be rated for degree of belief or conviction on an ongoing basis, and recent problematic situations and moderators can be evaluated in relation to consistency with each explanation. This process can be applied to general beliefs or specific incidents. For example, one piece of evidence in the analysis above was the rowdy noises from next door. The person in this example should be

Table 22.1 Evidence for and against a belief

<i>‘The neighbours are going to attack me’</i>	
<i>Evidence for</i>	<i>Evidence against</i>
There are rowdy noises from next door.	I have seen them three times this week and they haven’t attacked me.
I have been assaulted by other people in the past.	I have never been assaulted by anyone from my street.
They can read my mind.	I have never seen the neighbours be violent to anyone.
	I don’t think they are going to attack me when I am drunk or when I am with other people.

encouraged to generate as many possible explanations for such an incident as possible. It can be helpful to include some highly unlikely alternatives, as humour can help develop the therapeutic relationship and lighten the tone in therapy. This process may yield a list such as:

<i>The rowdy noises from next door are due to:</i>	
	<i>Conviction</i>
<i>Initial belief: 'The neighbours want to attack me':</i>	80%
<i>'The neighbours are having a party':</i>	25%
<i>'The neighbours are having an argument':</i>	50%
<i>'The neighbours are making noises to wind me up':</i>	50%
<i>'I am imagining the noises':</i>	10%
<i>'The noises are being beamed into the house from outer space':</i>	0%

Normalization

The use of normalizing information can combat the negative effects of stigma, reduce distress and provide information to facilitate the generation of alternatives. Such information can include facts about the prevalence of psychotic experiences and beliefs in the general population. For example, over 10% of the population hear voices (Beavan *et al.* 2011), and up to 70% of the general population endorse beliefs that could be labelled as delusional (Verdoux *et al.* 1998). When providing such information, calculating the implications for the prevalence in your own country can increase the impact that results. For example, in the United Kingdom, over 5 million people will hear voices, many of whom will never have contact with psychiatric services. Such information can be liberating for someone who believed that they were the only person with such experiences, or that such experiences automatically meant they were mad.

Information about the links between life events and specific psychotic experiences can also be helpful in reducing distress and providing an alternative explanation for hallucinations and delusions. For example, there is evidence that auditory hallucinations are linked to childhood trauma (Read *et al.* 2005), bereavement (Grimby 1993) and drug abuse (Shevlin *et al.* 2009), and that psychotic experiences are related to urban living (van Os *et al.* 2001). A summary of the normalizing approach to psychosis is provided in Kingdon and Turkington's (2008) book, which can be helpful for patients to read, as can another book, *Think You're Crazy? Think Again* (Morrison *et al.* 2008).

Imagery

People with psychotic experiences often have recurrent images associated with them (Morrison *et al.* 2002). For example, people who hear voices will frequently have an image of the perceived source of their voice (e.g. God, the

Devil, their next-door neighbour, a celebrity) or an image that is associated with the content of the voices. Patients with persecutory ideas often have images of their persecutors or images that are associated with their feared outcomes (e.g. going mad or being assaulted or killed). Such images can be a useful way of accessing personal meaning and core beliefs and can be modified to become less distressing or powerful (Hackmann 1997).

Core beliefs and content of voices

Core beliefs, such as 'I am bad' or 'Other people cannot be trusted', can be evaluated and changed using the techniques outlined above. They are also amenable to other strategies such as historical tests (seeking information from any time in the past that is inconsistent with the belief), the use of criteria continua (operationalizing factors such as badness, and rating self or others on the resulting continua) and positive data logs (Padesky 1994).

Trauma-based strategies

Given the high prevalence of trauma in patients with psychosis (Read *et al.* 2005; Varese, Smeets *et al.* 2012), it is likely that treatment strategies that have been developed for patients with post-traumatic stress disorder or dissociative disorders may be helpful. Such strategies include examining the idiosyncratic meaning of the trauma or the sequelae of the trauma and developing an explanatory narrative for the trauma to aid contextualization in memory (Ehlers and Clark 2000). In addition, strategies for dissociation, such as the use of grounding and distraction techniques, refocusing and cognitive restructuring, can be useful (Kennerley 1996).

Behavioural reattribution

The use of behavioural strategies within a cognitive framework can be an extremely effective way of achieving belief change and reducing distress (Chadwick and Lowe 1990). Behavioural experiments are a vital component of CT. It can be particularly important to address safety behaviours that are used to prevent feared outcomes (especially in relation to negative symptoms).

Behavioural experiments

Behavioural experiments are central to CT for psychosis. Beliefs about voices and delusional ideas are often translatable into testable hypotheses that can be collaboratively investigated by patient and therapist. Behavioural experiments should be designed carefully to ensure a 'no-lose' outcome. Predictions should be stated in a concrete way, and possible results should be reviewed in

advance so that that the outcome is meaningful and will not be dismissed or accommodated within the problematic belief system. Examples of behavioural experiments include the use of activity scheduling to evaluate beliefs about the consequences of activity or the lack of pleasure (helpful for negative symptoms), or exposure to feared situations to evaluate beliefs about voices or paranoid ideas.

An example illustrates the process.

Jim, a 19-year-old man, was certain that his thoughts were being transmitted via telepathy. He cited evidence from several examples in which it appeared that people around him had made comments about the topic he was thinking about, or reacted in a way that suggested that they were aware of his thoughts. An experiment was devised in which he was asked to operationalize the kind of response that would occur if people were able to read thoughts of imminent personal attack. He decided that people would flee a building by the nearest exit. He then decided, after considerable time had been spent using verbal reattribution methods to allow experimentation to feel safe, that an appropriate test of this belief would be to try to broadcast thoughts that implicated himself as a terrorist who was about to set off a bomb. He did this in several different buildings, and noticed no sudden rush for the exit (Although he did notice a few people leaving, this had been discussed in advance, and it had been agreed that the majority of people would have to leave quickly to confirm his belief). This resulted in a sustained decrease in both conviction and distress associated with the beliefs.

Safety behaviours

Safety behaviours, first identified in relation to anxiety disorders, are behaviours adopted to prevent a feared outcome. They are problematic when they prevent disconfirmation of the catastrophic beliefs about the feared outcome. People with psychotic experiences frequently use safety behaviours (Morrison 2001). Studies have demonstrated the presence of such behaviours in people with distressing auditory hallucinations (Nothard *et al.* 2008) and persecutory delusions (Freeman *et al.* 2001). Examples of such safety behaviours include avoidance of particular places, hypervigilance and selective attention, and idiosyncratic strategies to prevent psychosis-related fears. For example, if someone believes that their voices will make them go mad, then they will adopt behaviours designed to maintain mental wellbeing (e.g. thought suppression, monitoring for signs of impending madness, staying in the company of others or trying to relax), which prevent disconfirmation of this belief and may actually increase the frequency or intensity of the voices. If someone believes they are being persecuted, they may develop strategies such

as taking different routes to the shops, being hypervigilant for men in suits, suppressing signs of arousal or avoiding eye contact, which can be similarly problematic. Such safety behaviours can be manipulated to demonstrate their counterproductive effects and are amenable to being dropped in behavioural experiments to evaluate the relevant beliefs.

Negative symptoms

Negative symptoms include apathy, withdrawal, flat or blunted affect, anhedonia and poverty of speech. Such symptoms are frequently assumed to be the result of a biological syndrome or deficit state. However, many of these experiences can be understood, from a CT perspective, as safety behaviours. Some patients report having developed flat affect as a deliberate survival or coping strategy to avoid the feared outcome of physical punishment or humiliation in childhood. Others report having developed a similar strategy to manage their contact with the psychiatric system, since strong expressions of emotion frequently lead to increases in medication or admission to hospital. Poverty of speech can develop for similar reasons, and inactivity is frequently adopted to avoid exacerbations in positive symptoms. Other possible causes of negative symptoms include over-medication, depression, anxiety and substance abuse.

ADJUSTMENT AND RECOVERY

Once people have recovered from the distress associated with their psychotic experiences, there are other factors that are potential targets for CT. Such difficulties should also be considered at the beginning of CT, as many people will prioritize problems that are traditionally considered 'co-morbid' as more distressing than their experience of psychosis. For example, there is a significant literature that suggests that post-traumatic stress disorder is common in people with psychosis (Mueser *et al.* 1998), which is unsurprising given the prevalence of traumatic life events in people with psychosis (Read *et al.* 2005). It is also worth considering that people with psychosis frequently develop post-traumatic stress disorder in response to their treatment experiences or the psychotic symptoms themselves (Frame and Morrison 2001; McGorry *et al.* 1991). Depression and hopelessness are also common responses to an episode of psychosis (Birchwood *et al.* 2000). Such problems are clearly appropriate targets for CT.

RELAPSE PREVENTION

Core beliefs and conditional assumptions, as well as positive beliefs about psychotic experiences, can be conceptualized as vulnerability factors for

relapse. These can be addressed at this stage of therapy, providing that the patient consents to this, using the strategies outlined earlier. A more formal approach to relapse prevention has been described by Max Birchwood and colleagues (Birchwood *et al.* 1989). This approach involves the identification and subsequent monitoring of early warning signs such as sleep difficulties, anxiety, depression, unusual thoughts, irritability and social withdrawal.

Monitoring systems, often involving case managers and relatives or friends, are then developed, and action plans are collaboratively generated. There is evidence to suggest that CT targeted at people exhibiting such signs is feasible and can reduce relapse rates by 50% (Gumley *et al.* 2003). It is also important to help people to distinguish between a lapse and a relapse, and to ensure that they do not overly catastrophize the emergence of early signs, as this could potentially fuel the development of a relapse (Gumley *et al.* 1999). It is worth noting that a recent trial of CBT for relapse prevention in people with psychosis was unsuccessful (Garety *et al.* 2008); however, it is likely that this result was affected by both the selection of participants who were not distressed or help-seeking and by the failure of the specific model of therapy to change the hypothesized mediators such as core beliefs and problematic appraisals.

MONITORING OUTCOMES

The most useful method of monitoring outcomes is to develop idiosyncratic measures that are related to the SMART goals that have been set. These could include regular ratings of belief, distress, frequency, severity and impairment in relation to specific psychotic experiences or other problems, using methods such as Likert scales, visual analog scales or percentage ratings. It can also be useful to use standardized interview-based or self-report measures such as the PSYRATS (Haddock *et al.* 1999) to assess dimensions of hallucinations and delusions. There are other scales that can be used to monitor appraisals of voices (Morrison *et al.* 2004a) and paranoia (Gumley *et al.* 2011). It is also important to monitor social and occupational outcomes. All measures should reflect user-defined outcomes (Byrne *et al.* 2010; Neil *et al.* 2009).

SUMMARY AND FUTURE DIRECTIONS

Many studies have clearly shown CT for psychosis to be effective. Future research should include consideration of the possible side-effects of CT. Normalizing CT hopefully minimizes side-effects, but problems such as stigmatization, pathologizing normal variations of human experience and increased short-term distress remain possibilities. There are several areas that are showing promise in the psychological treatment of psychosis that require further development and evaluation. The development of trauma-based interventions for

people with psychosis is in its infancy (Larkin and Morrison 2006). The role of beliefs and safety behaviours in the development and maintenance of negative symptoms is beginning to be demonstrated (Grant and Beck 2009), and the efficacy of treatment strategies based on such a conceptualization is currently being evaluated. The use of CT for people with psychosis is also being extended to specific populations such as those with command hallucinations (Trower *et al.* 2004) and those with persecutory ideas (Foster *et al.* 2010).

Another development is the use of CT for the prevention of psychosis in populations defined to be at high risk of developing a psychotic disorder within the subsequent year. The first study of people from such a high-risk population, in the absence of medication, found that they are significantly less likely to develop psychosis over a 12-month period if they receive cognitive therapy, in comparison with a monitoring control condition (Morrison *et al.* 2004a). The second, much larger, study found that the 144 who received CT and 'monitoring of mental state' had significantly reduced symptoms, in terms of both severity and frequency, compared to the 144 who received only monitoring (Morrison *et al.* 2012b). Overall, only 8% transitioned to psychosis (CBT: 7%; controls: 9%) and depression, anxiety and satisfaction with life all improved for both groups, but there were no significant differences between the two groups on these measures, or on level of distress. We concluded:

It is also possible that the therapeutic effects of our monitoring control were greater than expected, since it provided regular contact with non-judgmental, warm, empathic, and accepting individuals who provided a non-stigmatising, non-catastrophising reaction to disclosures of unusual experiences and beliefs, which may have been beneficial. . . . Cognitive therapy reduces the severity of psychotic experiences without the use of antipsychotic drugs, which should provide the benefits of symptom based improvement without the associated risks of serious side effects. However, the low rate of transition to psychosis in both groups and the recovery from psychosis and affective symptoms is clearly an important and optimistic message to convey. . . . It should encourage a normalising, non-catastrophic perspective on their psychotic experiences, and any treatment should largely be needs driven on the basis of what problems are presented and prioritised by service users; this is something that cognitive therapy is suited to, given its collaborative, problem oriented nature. . . . On the basis of low transition rates, high responsiveness to simple interventions such as monitoring, a specific effect of cognitive therapy on the severity of psychotic symptoms, and the toxicity associated with antipsychotic drugs, we would suggest that antipsychotics are not delivered as a first line treatment to people meeting the criteria for being in an at-risk mental state.

(Morrison *et al.* 2012b: 6, 7)

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PSYCHODYNAMIC PSYCHOTHERAPY FOR PSYCHOSIS

Empirical evidence

Alison Summers and Bent Rosenbaum

To those of us who believe that psychodynamic approaches have much to offer people who experience psychosis, it is a great concern if the empirical evidence is unconvincing to others. In particular, if the evidence is not persuasive to those making decisions about funding health services for psychosis, and research, then we can expect a continuing decline in psychodynamic treatment, training and research, and a vicious circle where the decline is consolidated by the diminishing number of practitioners and researchers with experience, or understanding, of psychodynamic approaches in psychosis.

Sackett *et al.* (1996) describes evidence-based practice as the ‘conscientious, explicit and judicious use of current best evidence’. With more or less the same body of evidence, previous reviewers have found it possible to conclude (as we ourselves have done) that psychodynamic approaches can be valuable (Gottdiener 2004; Karon 1989), or that they should be dispensed with (Lehman *et al.* 2003; Mueser and Berenbaum 1990) or that there is insufficient evidence to draw firm conclusions (Malmberg *et al.* 2010; NICE 2009). These discrepancies seem to relate principally to the question of what constitutes adequate evidence. We discuss what evidence is needed, then review what is available.

WHAT KIND OF EMPIRICAL EVIDENCE IS NEEDED?

Study design

In Westernized health care systems it is widely accepted that a positivist view of evidence should guide decision making, and that randomized controlled trials (RCTs), as the study design most likely to minimize bias, are the gold standard test of efficacy. In this context, it seems highly unlikely that

disagreements over efficacy will be resolved without RCTs, and without these RCTs also meeting additional quality standards. This seems so regardless of how we may feel about underlying assumptions about what constitutes knowledge or truth and about political, economic and cultural influences that limit objectiveness of RCT evidence (Sturdee 2001).

Non-RCT evidence can be arranged in hierarchies of strength; for example, well-designed non-randomized controlled trials, then case series, then expert opinion. However, relying solely on non-RCT evidence is likely to remain contentious, as there is no agreement about what is sufficient.

Interventions

Psychodynamic interventions can be defined as those where unconscious factors are taken to be important and where therapist–patient relationships and interactions are regarded as crucial sources of information.

Details of the therapy have important effects on outcome. Contemporary psychodynamic therapies include significant modifications of classical psychoanalytic approaches and emphasize supportive elements (Cullberg and Johannessen 2004). Each psychodynamic therapy is unique, as the therapist's intervention at any particular moment is determined by their formulation of the patient's needs in that moment (e.g. for a supportive intervention or an interpretation). In addition, the therapist's response is determined by their experience, theoretical understanding, personality and confidence in the psychodynamic approach; and these factors thus become integral aspects of the treatment offered. Structural factors also affect outcome, particularly length of therapy, with longer therapies, though off-putting to funders, offering more possibility of good recovery (Leichsenring and Rabung 2011).

The treatment context will have its own dynamic effects, may support or inhibit a patient's ability to make use of therapy and needs to be considered. Employing psychodynamically trained staff may have effects that accrue only over lengthy time periods, through effects on organizational culture and capacities. The research context may have its own impact – for example, when randomization leads to 'artificial' matching of therapist and patient or when use of a manual or recording affects the therapist's work.

Population studied

The outcome of all treatment trials is likely to be greatly affected by the population studied. Psychosis, even when more tightly specified as 'schizophrenia', is a heterogenous condition with marked differences between individuals in the relative contributions, for example, of neurodevelopmental, traumatic and affective factors, in the stage of psychosis (McGorry *et al.* 2008; and Chapter 24), and in personality, non-psychotic defences, and treatment goals. All of these may affect treatment response. Suitability for therapy is not

necessarily best determined by diagnosis in any case (Valbak *et al.* 2003). Interest in self-understanding, for example, may be a better predictor of benefit (Watzke *et al.* 2010).

Outcomes assessed

Trial outcomes will also be affected by the outcome measures chosen. This choice is value-driven rather than objective. Focusing on variables that are easier to measure privileges certain types of outcome. Trials of psychosis treatments have frequently taken measures of function and symptoms as main outcomes, though the ‘recovery’ movement is beginning to challenge this narrow focus (Slade 2009). Other benefits, such as finding meaning, remoralization or better relationships, may be equally important to patients (Coursey *et al.* 1995). With the most commonly used outcome measures, those psychodynamic therapies where the primary aim of which is internal change are likely to appear to perform less well than therapies that directly target the chosen measures, even when they are achieving other outcomes valued by patients. Adverse outcomes are also important to consider, as any group outcome may conceal subgroups with significant benefit and significant harm. Length of follow-up may affect both the benefits and adverse effects identified.

Conclusion: convincing empirical evidence

We believe that for there to be some consensus about the efficacy of psychodynamic approaches in psychosis, two conditions need to be fulfilled. First, there need to be well-designed RCTs. Second, these RCTs need to study psychodynamic approaches that meet contemporary standards. The likelihood of a study demonstrating efficacy will also be determined by choice of study population and outcome measures.

WHAT KIND OF EMPIRICAL EVIDENCE DO WE HAVE?

In the first edition of this book, Gottdiener (2004) reported on a meta-analysis (Gottdiener and Haslam 2002) of individual therapy in schizophrenia. He identified 37 studies where an effect size could be calculated and found that overall, among patients who had individual therapy, 66% improved, compared with 35% with no therapy. He concluded that psychodynamic therapy had a similar effect size to cognitive behaviour therapy, and that both were more effective than supportive therapy. This meta-analysis relied mainly on non-RCT evidence and may therefore not have convinced critics of the value of psychodynamic therapy.

The most recent review we have found is by the English Schizophrenia Guideline Group (NICE 2009). The reviewers state that in the absence of RCTs, other types of controlled studies may be considered (p. 37). However,

in the discussion on psychodynamic therapies, although none of the RCTs discussed use contemporary forms of therapy, they do not consider other types of study. They found no clear evidence to support the use of psychodynamic therapies as discrete interventions but concluded that further well-conducted research is warranted. They suggested that health care professionals might consider using psychodynamic principles ‘to help them understand the experiences of people with schizophrenia and their interpersonal relationships’.

Methods for our review

We attempted to build on previous reviews by considering:

- studies of any format of psychodynamic therapy, not just individual therapy;
- RCTs published at any date, plus contemporary studies of any design that collected data on a prospectively identified cohort;
- studies in any language, not just English.

We searched relevant databases from 1990 to May 2011 and also read review articles and reference lists and contacted expert colleagues. This approach may not have identified studies with non-specific titles (e.g. psychotherapy for schizophrenia) and is likely to have missed studies where psychodynamic work is one component of a service.

Individual therapy

We found only six RCTs of psychodynamic therapy (Durham *et al.* 2003; Grinspoon *et al.* 1968; Gunderson *et al.* 1984; Karon and VandenBos 1981; May *et al.* 1968; O’Brien *et al.* 1972), all but one over 30 years old. None of them compared a contemporary psychodynamic therapy with a non-psychodynamic treatment using an adequate research design.

One study (Karon and VandenBos 1981) found that psychodynamic therapy patients achieved greater improvement in function and less hospitalization than controls. However, there were problems in the study design, including control patients being transferred to a different hospital. Two studies showed no benefit, or only a very limited one, from psychodynamic therapy compared with a control condition (Durham *et al.* 2003; May *et al.* 1968). May found that inpatients treated with psychotherapy alone did little better than controls and that adding psychotherapy to medication conferred only marginal additional benefit. Durham found that supportive psychodynamic therapy achieved less symptom reduction and less patient satisfaction than cognitive behaviour therapy (CBT), but that both psychodynamic therapy and CBT improved delusions more than treatment as usual. In both

of these studies, limited therapist experience is likely to have affected outcome and in one (Durham) the psychodynamic therapists were markedly less experienced than the comparison therapists.

Grinspoon *et al.* (1968) found that in 'chronic' male inpatients, psychotherapy plus medication improved adaptation whereas psychotherapy alone produced no demonstrable change. However, limitations in study design raise possibilities of bias, and the therapists lacked experience with schizophrenia.

Two studies are comparisons of different formats of psychodynamic therapy. Gunderson *et al.* (1984) found that 'reality-adapted' therapy led to better functional outcome and less hospitalization than 'exploratory insight-oriented' therapy, which led to greater improvements in ego-function. However, differences between therapies were less than intended and differences in outcomes were small. O'Brien *et al.* (1972) found that group therapy compared with individual therapy led to greater improvement and less hospitalization. However, the group therapy patients were also treated by more experienced therapists.

Within these RCTs there is additional empirical support for psychodynamic therapy. In one study, therapists whose sessions were rated as more skilful achieved better impact on negative symptoms (Glass *et al.* 1989). In another, experienced therapists achieved more rapid and extensive benefit (Karon 1989). The studies also identify predictors of good outcome, e.g. good premorbid functioning (Mintz *et al.* 1976) and early therapeutic alliance (Frank and Gunderson 1990).

We found reports of one other controlled trial (Rosenbaum *et al.* 2005, 2006, 2012) and one case series (Varvin 1991). Both suggest psychodynamic therapy can improve social functioning and symptoms.

There are also some retrospectively identified case series (Cullberg and Levander 1991; Rund 1990) and many individual case reports that attribute improvement to psychodynamic therapy (e.g. Garrett and Turkington 2011; Lysaker and Gumley 2009).

Group therapy

We found the one RCT mentioned above (O'Brien *et al.* 1972), which found group psychodynamic therapy more effective than individual psychodynamic therapy. We found no other controlled studies that met our criteria, although there are a number of case series and case reports (e.g. Pesek *et al.* 2010) that suggest that group psychodynamic therapy can have important effects on patients' lives.

Family therapy

We found no RCTs or other controlled studies of family therapy with a specifically psychodynamic approach, though again there are individual case reports

(e.g. Martindale and Smith 2011) that suggest that psychodynamic family work may benefit all family members.

Services including psychodynamic therapy as a treatment option.

Again we found no RCTs. In controlled studies of services which include the option of psychodynamic therapy for appropriate patients with psychosis, reported outcomes are as good as, or better than, comparable services (Alanen *et al.* 1994; Ciompi *et al.* 1992; Seikkula *et al.* 2006, 2011). A comprehensive service which includes psychodynamic therapy and which is offered to everyone with a first presentation to mental health services seems to reduce the incidence of schizophrenia (Aaltonen *et al.* 2011).

Additional supporting evidence

There is also an increasing empirical evidence base for psychodynamic conceptions of psychosis development (Dimaggio and Lysaker 2010; Liotti and Gumley 2008) and for some of the presumed mechanisms of treatment effect, for example enhancement of mentalizing, hope and self-understanding (García-Cabeza and González de Chávez 2009; Harder and Rosenbaum 2012; Lysaker and Gumley 2009). Lastly, case study evidence suggests that psychodynamic approaches may be usefully integrated with others such as cognitive-behavioural therapy (Garrett and Turkington 2011).

Summary of evidence

(a) There are no well-designed RCTs of psychodynamic individual or group therapy that use a contemporary approach to therapy (i.e. one flexibly adapted to the needs of the patient) and compare this with a 'no-therapy' condition. There are no RCTs at all of psychodynamic family therapy or of services that include psychodynamic therapy as one component of treatment. RCT evidence therefore is inconclusive about whether contemporary psychodynamic approaches are, or are not, effective.

Controlled studies and case series suggest that contemporary psychodynamic therapy may be more effective than 'treatment as usual' (Rosenbaum *et al.* 2005, 2006, 2012; Varvin 1991) but less effective than more intensive multimodal interventions. The one meta-analysis available supports this view (Gottdiener and Haslam 2002). Claims that psychodynamic therapy causes harm are not based on substantive evidence.

(b) Within controlled studies, there is evidence suggesting that, at least in some cohorts of patients, there are differences in effect between different forms of therapy. Therapies that help patients adapt to reality may have more impact

on adjustment than more exploratory therapies, which may have more impact on ego-strength and thinking (Gunderson *et al.* 1984). Therapies delivered by more skilled or experienced therapists may be more effective than others (Glass *et al.* 1989; Karon 1989). Group therapy may be more effective than individual therapy (O'Brien *et al.* 1972).

There is also evidence of differences between patient groups, for example, with greater improvement among patients not misusing substances (Rosenbaum *et al.* 2012), those with good premorbid function (Ciompi *et al.* 1992; Varvin 1991) and those where a good therapeutic alliance is established in the first six months (Frank and Gunderson 1990).

(c) Studies of services that offer psychodynamic therapies (Alanen *et al.* 1994; Ciompi *et al.* 1992; Seikkula *et al.* 2011) indicate that they produce outcomes at least as good as those for other services.

Conclusions

Empirical evidence suggests that to maximize the benefits of psychodynamic interventions, it is important to adapt to individual patient needs, both in selection of patients for therapy and in the delivery of this therapy. We need to know more about therapy and patient characteristics that affect outcomes, and about which outcomes are affected. Although various study designs may help with these questions, conflicts in views about efficacy of psychodynamic interventions are unlikely to be resolved without additional RCTs studying contemporary approaches. Nevertheless, the evidence currently available strongly supports the view that psychodynamic interventions can be an effective treatment for people who experience psychosis.

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THE DEVELOPMENT OF EARLY INTERVENTION SERVICES

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Were we to ask the question, 'for how long is it decent to let a young person suffering from psychosis go undiagnosed and untreated in our society?', the answer would be obvious to most people.

There seems to be, among some clinicians and researchers, an assumption that some psychotic disorders such as the schizophrenias are best compared with diseases such as Huntington's chorea or Creutzfeldt–Jakob's syndrome. They seem to think that there is no effective treatment, that the timing of the treatment is unimportant, and that both the patient and the relatives are best off not knowing what is going on, until it is inevitable due to the manifestation of a chronic disorder with a definite downhill course. And there is the notion that if the person recovers, then it was not 'a real case of schizophrenia' after all. That this fundamentalistic Kraepelinian view has been contradicted by well-designed outcome studies (Fenton 1997; and Chapter 5) has had limited impact in some circles.

The next step in this logic is that psychotic disorders must be of biological origin, and hence the only treatments must be those that work directly on a biochemical level. It is also believed that people with psychosis do not benefit from talking to, or relating to, therapists, as the disorders are endogenous brain diseases with no connection to significant people or interpersonal relationships. The final step in this line of thinking is that the only acceptable reason for trying to form an alliance with the patient is to improve medication compliance.

THE LOGIC OF EARLY INTERVENTION

People using early intervention strategies rely on a different logic. The most fundamental difference is the belief that psychotic disorders are dynamic,

psycho-bio-social, reversible processes, in which the psychotic breakdown is only one stage in the process, which in some cases can be prevented, delayed, modified and reversed.

The notion of psychotic disorders developing in stages (Figure 24.1) is not new. Many clinicians and researchers have held a corresponding view that the treatment of such disorders should be phase-specific and need-adapted. Both in the Anglo-American and the Scandinavian tradition, this has been a prevalent view for decades.

Sullivan pointed out in 1927 that ‘the psychiatrist sees too many end-states and deals professionally with too few of the pre-psychotic’, and ‘the great number of our patients have shown for years before the breakdown, clear signs of coming trouble’. Like Cameron (1938) and Meares (1959), Sullivan suggested intervening at an early stage of illness development, preventing a chronic course. Docherty *et al.* (1978) have outlined the process of decompensation (Table 24.1).

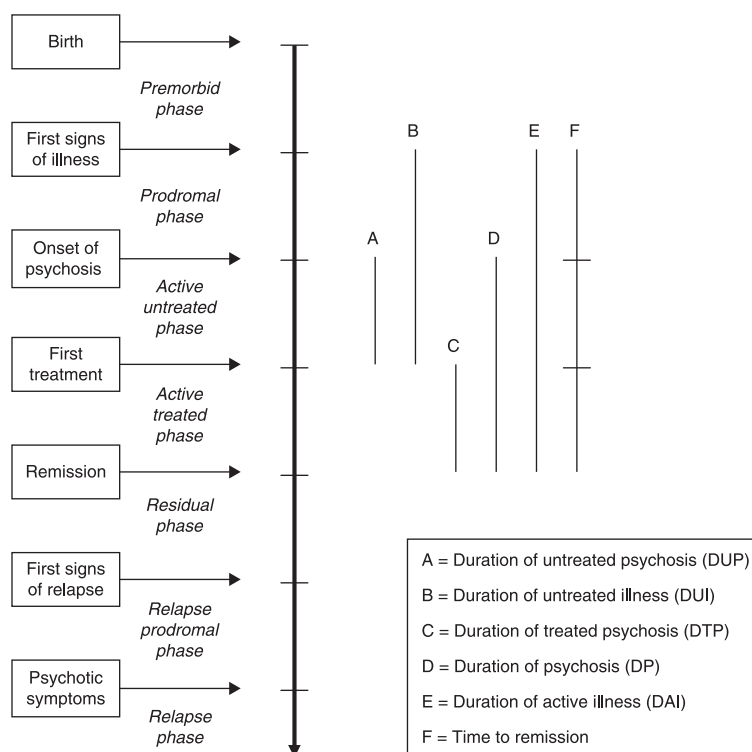


Figure 24.1 Early course of schizophrenia: phases and definitions (adapted from McGlashan & Johannessen 1996).

Table 24.1 Five stages of decompensation

0	Equilibrium
I	Overextension
II	Restricted consciousness
III	Disinhibition
IV	Psychotic disorganization
	1 destructuring of the external world
	2 destructuring of the self
	3 total fragmentation
V	Psychotic resolution

There are three possible levels of prevention. Primary prevention aims to reduce the incidence of a disorder (Chapter 20). Secondary aims to prevent existing disorders from causing symptoms and dysfunction. Tertiary decreases the morbidity of the disorder – that is, it improves the prognosis. In Figure 24.2, the different types of prevention are related to the different stages of psychosis development.

Regarding tertiary prevention, it has been shown that structured long-term treatment with psychosocial interventions involving family treatment, supportive psychotherapy, cognitive behaviour therapy and the monitoring of

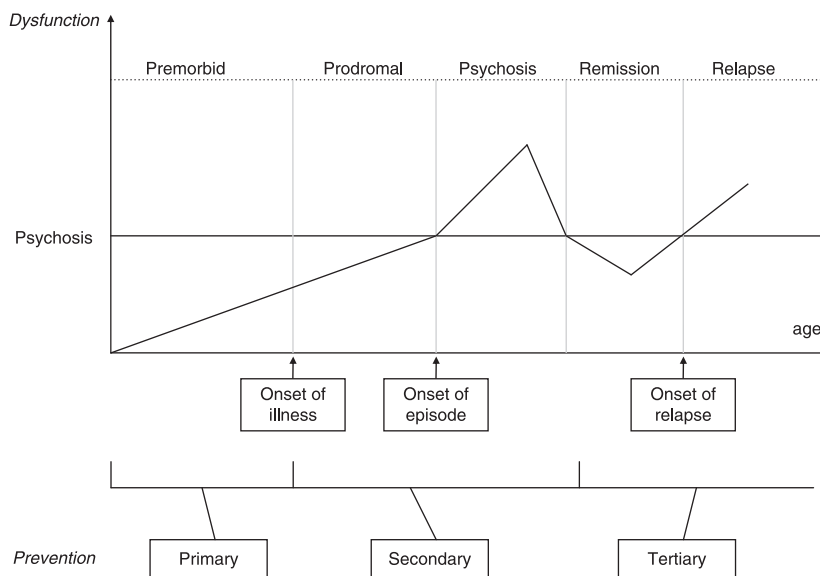


Figure 24.2 Stages of schizophrenia and prevention types (after Larsen et al. 1998).

early symptoms of relapse can improve long-term outcome (Hogarty *et al.* 1986; Wyatt 1991; and Chapters 22, 23, 25, 26). Many practical models for the identification of the prodromal symptoms of relapse in schizophrenia, or 'relapse signature' (Birchwood 1992), have been developed.

It has been clearly demonstrated that relapse rates can be significantly reduced through a systematized mapping of early warning signs. Many family studies have shown a marked reduction in relapse rates (Leff 1994; and Chapter 26). In principle, there is no difference between relapse prevention and the prevention of the first outbreak of psychosis. High expressed emotion is an expression of high stress in the milieu of the patient (Chapter 19). Thus, the capacity to handle additional stress will be reduced. Stress reduction and easy access to services upon early warning signs are important features of most psychoeducational family programmes. This is exactly what one wants to implement for patients who have not yet experienced their first psychotic breakdown, using early intervention programmes.

Other strong indicators of the importance of environmental factors include the large variance in the incidence of schizophrenia between different parts of the world, and between those raised in cities versus rural areas (Mulvany *et al.* 2001; Schelin *et al.* 2000; and Chapter 6). The high incidence of schizophrenia among second-generation immigrants in European countries points in the same direction. These functional psychotic disorders are the result of person–environment interaction and hence could be prevented.

Mrazek and Haggerty (1994) have developed another classification of preventive interventions. Their terms – 'universal', 'selective' and 'indicated' preventive interventions – reflect interventions focused, respectively, on the whole population, on asymptomatic subgroups and on individuals at high risk for developing a specific disorder. Early intervention in first-episode psychosis reflects indicated prevention and includes the possibility for primary prevention when applied in the prodromal phase.

INTERVENTION IN THE PRE-PSYCHOTIC PHASE (THE 'PRODROMAL' PERIOD)

The term 'prodromal psychosis' is, by definition, a retrospective concept and, as such, is of limited value (Johannessen and McGorry 2010). Furthermore, to talk about 'prodromal schizophrenia' in non-established cases raises difficult ethical questions. To our mind it is better to address a suspected case of a possible psychotic development in terms of the presenting symptom or symptoms. Hafner *et al.* (1999) showed that in approximately 80–90% of cases the first symptom to appear is depression, with a mean duration of symptoms before psychosis of five years. Several studies have addressed efforts to prevent the conversion of suspected 'prodromal' cases into psychosis.

The Buckingham study

A pioneering project was carried out in England, between 1984 and 1988 (Falloon 1992). The mental health service systems were directly connected to the general practitioners (GPs) in the area (population 35,000). The GPs were trained in detecting early cases of psychosis, using the *DSM-III-R* prodromal signs. There was immediate access to the specialized services, located in the same buildings as the GPs. Comprehensive treatment, including assertive crisis treatment, carer-based stress management, cognitive-behavioural strategies, psycho-education and targeted minimal dose medication, were initiated within hours of referral. Treatments were tailored to patients' individual needs. The expected annual incidence of schizophrenia was reduced ten-fold.

The study highlights the potential gains of early detection and intervention by focusing on the prodromal phase. It suggests that deterioration may be less irreversible and resistant to intervention than expected. The main limitation of this study was the low number of patients. The strategy of prevention at the prodromal level is also problematic due to the non-specific nature of the symptoms and the danger of falsely labelling persons who would never have developed psychosis (Johannessen and McGorry 2010).

The PACE study

The Personal Assessment and Crisis Evaluation (PACE) clinic in Melbourne aims to identify individuals at high risk of becoming psychotic, and then to prevent the transition into psychosis. To identify possible prodromal cases, the Melbourne group developed a screening instrument called Comprehensive Assessment of At Risk Mental States (CAARMS) (Edwards and McGorry 2002). The rate of transition to psychosis was 26% at 12 months (Amminger *et al.* 2006). In a study conducted by PACE at the ORYGEN Youth Health (OYH) clinic, the transition to psychosis rate at two-year follow-up was 16% (Yung *et al.* 2008). The characteristics of the PACE study and some other early intervention programmes are outlined in Table 24.2.

The Bonn Early Recognition study

In Germany, early symptoms of schizophrenia, called 'basic symptoms', are measured by the Bonn Scale for Assessment of Basic Symptoms. Greater emphasis is put on subjective experiences of patients before they develop psychosis. A study at six early-detection centres in Germany, Finland, the Netherlands and England included 245 patients assessed as prodromal according to either ultra-high-risk (UHR) criteria or the basic symptom-based criterion cognitive disturbances (COGDIS). At 18-month follow-up, the incidence rate for transition to psychosis was 19% (Ruhrmann *et al.* 2010).

Table 24.2 Early intervention programmes

<i>Buckingham</i>	<i>PACE</i>	<i>EPPIC</i>	<i>TIPS</i>
Catchment area 35,000	Catchment area not defined	Catchment area 800,000	Catchment area 400,000
Primary health care and psychiatric specialists located together	Services situated at health promotion centre for adolescents	Ordinary psychiatric services, community- oriented, ambulatory teams	Ordinary outpatient unit, detection team, community- oriented
Immediate access to treatment, home based	No referral from primary care needed (i.e. direct access)	Direct access to services for users 7 days a week	Direct access to services for users 5 days a week
Comprehensive treatment focused on presenting problems	Psychosocial and medication-based treatment	Comprehensive treatment programmes	Comprehensive treatment programmes
Focus on prodromal/ early psychosis, preventing conversion into psychosis	Focus on prodromal period, preventing conversion into psychosis	Focus on early, first- episode, psychosis; reducing DUP	Focus on early, first-episode, psychosis; reducing DUP

Note

DUP = duration of untreated psychosis.

The North American Prodrome Longitudinal Study (NAPLS) study

This study is a multi-centre study in North America, originating at Yale Medical School. The screening instrument used was the Structured Interview for Prodromal States (SIPS) (McGlashan *et al.* 2001), which bears some similarity to CAARMS. The conversion to psychosis rate for 377 patients at 12-month follow-up was 22% (Woods *et al.* 2009).

The TOPP study

This project is situated at Rogaland Psychiatric Hospital in Norway and, in contrast to the other prodromal projects, covers a catchment area and thus does not have a selected sample. The acronym TOPP stands for early Treatment Of Pre-Psychosis and relates to the TIPS Project (early Treatment

and Intervention in Psychosis), which is a first-episode psychosis study. Using the SIPS, a one-year conversion rate of about 40% was observed for the first 15 patients.

Controlled clinical trials of treatment in the prodrome

Five clinical trials have tested whether treatment can delay, or prevent, onset of psychosis in patients meeting prodromal criteria. The first randomly assigned people meeting CAARMS criteria to six months of a combination of antipsychotics (low-dose Risperidone) and cognitive behavioural therapy (CBT) versus basic support; 10% in the intervention group converted to psychosis (McGorry *et al.* 2002). The second randomly assigned persons meeting CAARMS criteria to six months of CBT versus basic support; 6% of the CBT patients converted to psychosis during follow-up (Morrison *et al.* 2004). The third randomly and blindly assigned persons meeting the SIPS criteria to an active antipsychotic (Olanzapine) versus placebo for one year. During the one-year follow-up 16% of the patients treated with Olanzapine converted to psychosis, compared with 38% of placebo patients (McGlashan *et al.* 2006). The fourth, a recent study, explored the effects of long-chain omega-3 polyunsaturated fatty acids (PUFAs) in a randomized controlled trial. After 12 months, 5% in the PUFAs group had converted to a psychotic disorder (Amminger *et al.* 2010). The final study recruited people not taking antipsychotic medication. The 144 who received cognitive therapy and 'monitoring of mental state' had significantly reduced psychotic symptoms compared to the 144 who received only monitoring (Morrison *et al.* 2012), but the difference between the two groups in terms of transition to psychosis (CBT: 7%; controls: 9%) was not significant. These studies demonstrate that treatment (psychosocial, pharmacotherapeutic or both) might reduce the conversion rate to psychosis.

INTERVENTION IN THE PSYCHOTIC PHASE AND DURATION OF UNTREATED PSYCHOSIS

Early intervention means that treatment is given as soon as possible after the psychosis is present. The key concept here is duration of untreated psychosis (DUP), the time between the onset of first psychotic symptoms and first adequate treatment.

More than 10 studies have shown that the typical DUP average is 1–2 years (Johannessen *et al.* 2001a). In addition, the duration of untreated *prodromal* symptoms is between one and five years (Yung and McGorry 1996). This represents a major public health problem. Even if one were to presume that it makes no difference to the long-term prognosis whether one intervenes early or late in the development of a schizophrenic disorder, it represents an additional, and in most cases almost intolerable, extra burden to be

psychotic without receiving adequate help. People in active psychosis are vulnerable to initiating irrational, unpredictable and often permanently damaging behaviours, both to themselves and others (Lieberman and Fenton 2000).

The reasons for treatment delay include ignorance, denial, stigma, lack of motivation, absence of information about early psychosis and lack of access to appropriate interventions. Larsen *et al.* (1998) found that treatment delay was correlated with social withdrawal and poor social functioning, and that young males have particularly long DUP. Lincoln and McGorry (1999) found that lack of knowledge about mental illness and mental health services in the general community, the health delivery system and the educational system also contribute to delay.

Reducing the DUP: health promotion aspects

Effective health promotion programmes must include multi-level strategies targeting individuals, families, neighbourhoods, schools, workplaces and communities. The community component includes modifying the organization of health services to emphasize prevention and supplementary training for health personnel. Attitudes and knowledge about mental illness can be significantly improved if the information strategies use multiple access points: for example, media, health service personnel and schools (US Department of Health and Human Services 1994). It has been demonstrated that schoolteachers are capable of identifying individuals who later develop serious mental illness, when given adequate information about such disorders (Olin *et al.* 1998).

The TIPS study

The early Treatment and Intervention in Psychosis (TIPS) project is a prospective clinical trial designed to examine whether it is possible, within a sectorized catchment area, to reduce DUP, and whether earlier treatment in first-episode psychosis can improve the course of the disorder. We studied whether it is possible to establish an effective early detection system (ED) based on: (a) information programmes aimed at the general population, health professionals and schools (teachers and students), and (b) easy access to services through a low threshold and ED-teams.

The study involved three sites: Rogaland county, Norway (370,000 inhabitants), Ullevål sector, Oslo, Norway (190,000) and Roskilde county, Denmark (100,000). The patients were treated with the same psychosocial and medical protocol across all three sites. The inclusion period was 1997–2000, with a five-year follow-up period. Currently, a ten-year follow-up is under way. An extensive education and information system has been implemented in Rogaland, while Ullevål and Roskilde are control sectors that rely

on existing detection and referral systems for first-episode cases. Details are provided in Johannessen *et al.* (2001b) and Larsen *et al.* (2001).

The average DUP in Rogaland was reduced from 114 weeks in 1993–94 to 26 weeks in 1997–2000. The patients detected earlier were younger and had better premorbid adjustment, less severe psychosis, with more frequent drug abuse. (This latter characteristic probably reflects developments regarding drug abuse in the area.)

It thus appears that patients accessed the psychiatric services when the symptoms were less severe. The pattern of referral also changed significantly, with more than 50% of the referrals coming from family, schools and friends, as compared to pre-TIPS, when all referrals came through the general practitioners.

The TIPS study concluded that it is possible in a quasi-experimental study to reduce DUP, and it compared its effect on outcome (Larsen *et al.* 2001, 2006; Melle *et al.*, 2006). At the two-year follow-up Early Detection had a positive effect on course of symptoms and functioning, especially negative symptoms (Melle *et al.* 2006). At the five-year follow-up three-quarters of patients were in remission (Larsen *et al.* 2008; Melle *et al.* 2009). The early-detection group had better outcomes for negative, cognitive and depressive components on the PANSS-scale. Patients from the early-detection sector had more regular contact with friends; i.e. 57% vs. 36% saw friends at least monthly. The TIPS study showed that the combined use of Detection Teams and Information Campaigns was successful in bringing FEP patients early into treatment (Joa *et al.* 2008).

At the ten-year follow-up, the groups no longer differed on most of the symptom dimensions, but significantly more early-detection patients were considered recovered on the basis of social functioning, particularly employment (Hegelstad *et al.* 2012).

The treatment programme

The basic requisite of treatment is continuity of care – that is, that the same therapist should stay with the patient for at least two years. Preliminary data indicate that 50% of the patients have only one therapist in these two years, and 40% have two. Teams working with the patient also secure continuity of treatment. This team consists of the patient's individual psychotherapist and primary contact nurse. In some cases, the main focus will be individual psychotherapy, but this will not be the most relevant in all cases. The psychotherapist has the main responsibility for all aspects of the treatment.

This emphasis on continuity of care is based upon the psychological needs of patients experiencing psychological disintegration. We believe that this disintegration, pathognomic for psychosis, can be reduced by stable external structures and that the inherent anxiety and suffering requires time, stability and a stable human relationship to re-establish inner structure where

structure has been dissolved. What the patients with psychosis need the least are unstable and/or broken relationships.

Active outreach psychotherapy

The supportive, active outreach psychotherapy guarantees a minimum of one psychotherapeutic contact per week with an experienced psychotherapist. The main focus of the therapy is supportive and psychodynamic. This dynamic understanding forms the basis for all treatment efforts and emphasizes the active interaction between the individual's biological equipment, psychological factors and the social and physical environment. We aim to understand the psychological development of each individual, as well as factors contributing to the person's psychological breakdown, or psychosis, in this context. The psychosocial needs of the individual patient beyond the standardized treatment are identified and addressed.

The phase specificity of the treatment is fundamentally important when relating to psychotic patients. There is a strong Scandinavian tradition for 'needs-specific treatment', originating from Finland in the late 1970s. The Finnish psychoanalyst Veikko Tähkä (1979) introduced the concept of 'psychotherapy as phase-specific treatment'. Alanen (2009) and others have developed this approach further. The mean annual incidence of schizophrenia significantly declined in the five-year period following the introduction of this 'need-adapted' approach (Aaltonen *et al.* 2011).

People experiencing psychosis should be met flexibly and individually, based on both an individual and interactional interpretation of the situation. It is from these that the therapeutic needs should be defined. Patients' needs vary, sometimes from day to day. This must be reflected by the frequency of meetings between therapist and patient and the length of each session. The supportive elements will always be an important part of psychodynamically oriented psychotherapy for this group of patients (Alanen 1997: 144).

McGlashan and Keats (1989) stress the importance of a thorough evaluation of the different therapeutic steps when treating a person with psychosis, moving gradually from building a relation to a more active psychotherapeutic phase, when the patient's psychological apparatus has been gradually re-established. In some cases, the therapist will, for long periods, work primarily at establishing contact and/or getting the patients to attend sessions. Other patients are strongly motivated for psychological treatments and an insight-oriented process. The therapist must be flexible and offer practical help. A main focus will be helping the patient to develop internal coping strategies and to enhance the understanding of illness-related experiences and personal vulnerability factors. It is extremely important not to overstretch the patient's psychological capacity, including cognitive capacity, during recovery. Cognitive-behavioural therapy applied in the acute phase may actually make the short-term prognosis worse (Turkington 2001). There will also

be a focus on accepting loss in a way that avoids demoralization, preserves hope and encourages reintegration into society. Active outreach approaches are close to modern cognitive-behaviour treatment strategies (Chapter 22) and imply an explicit responsibility to contact patients if they do not arrive for appointments, or visit patients at home if necessary.

Psychoeducative family work

Few studies on family intervention have been conducted on first-episode samples. Most have been on patients with a more chronic course. We have modified the approach of McFarlane for first-episode patients and their families (Øxnevad *et al.* 2000). First-episode patients are different and the prognosis varies. We do not define psychosis, in the way that has been the tradition in most psycho-educational approaches, as a biological brain disease (Chapters 19 and 26). We leave the question open and emphasize the psychological vulnerability that will vary from case to case, within the framework of a stress/vulnerability model. It is our experience that this is well understood and received by both patients and families.

The family work consists of three elements: single family meetings, multi-family groups and a workshop for the families. In the single-family meetings, the family meets with the therapist at least three times during the first weeks after the patient has been admitted. These meetings can be with or without the patient, depending on the patient's condition. A multi-family group consists of five or six families. The sessions last for 90 minutes and have a supportive, non-threatening atmosphere. They meet every second week and focus on problem-solving, crisis management, communication training and education about psychiatric disorders. A one-day educational workshop is conducted for new participants, including family members, close friends, teachers and others the patients want to include. The workshop involves:

- information on psychiatric health services;
- information on psychosis;
- possible outcome/prognosis;
- stress/vulnerability model;
- experiences of being a relative;
- what relatives can do to help;
- treatment and coping.

In the TIPS Project, about 50% of the 300 families have participated in the multi-family groups. At five-year follow-up participants had significantly lower drop-out rate than patients who did not participate (Rossberg *et al.* 2010). However, the participants had significantly less improvement in PANSS positive and excitative symptoms and significantly longer duration of psychotic symptoms at follow-up.

Medication

Medication is based on treatment with olanzapine as first choice, risperidone as second and perphenazine as third. The recommended dose of olanzapine is 10 mg and that for risperidone 1–4 mg per day. Compliance with medication improved significantly when perphenazine was replaced by olanzapine as first choice after the first year (Opjordsmoen *et al.* 2009).

The EPPIC study

In Melbourne, Australia, the Early Psychosis Prevention and Intervention Centre (EPPIC) is established as a psychiatric service, including early detection and specialized treatment for early psychosis. The catchment area is the western metropolitan area of Melbourne with about 800,000 inhabitants. The model is outlined in Table 24.2.

The EPPIC model aims to reduce both primary and secondary morbidity in patients with early psychosis, through the dual strategy of identifying patients as early as possible after onset of illness and providing intensive phase-specific treatment for up to 18 months. The service has been evaluated by comparing a matched group of patients from 1989 to 1992 with patients included in the EPPIC programme in 1993 (Edwards and McGorry 2002). The EPPIC patients had a significantly better one-year outcome than the pre-EPPIC group, including better quality-of-life scores and fewer negative symptoms. The effect was strongest for patients with DUP of 1–6 months. The studies did not show that the EPPIC programme was able to reduce the duration of untreated psychosis. The better one-year outcome is probably due to an improved, more comprehensive, treatment.

The Birmingham Early Intervention Service (EIS)

The EIS in Birmingham, UK, is for first-episode psychosis. It is part of the Northern Birmingham Mental Health National Service Trust, which includes 24-hour psychiatric emergency and home treatment teams, primary care and liaison services, assertive outreach, and rehabilitation and recovery services. The core component is an assertive outreach team, operating seven days a week and staffed by 10 case managers with case loads of 15 patients. The treatment protocol includes the following:

- low-dose antipsychotic regimes;
- cognitive therapy for delusions and hallucinations;
- cognitive therapy aimed at improving adjustment and reducing co-morbidity;
- psychosocial intervention for problem drug and alcohol use;
- prevocational training.

Other studies and services for first-episode psychosis

Important projects are being carried out in Canada (London, Ontario and Calgary, Alberta). The OPUS (early detection and assertive community treatment of young persons with untreated psychosis) study in Denmark, the 'Parachute-project' in Sweden, the RAPP (recognition and prevention of psychological problems) project in New York, the Early Assessment Service for Young People with Psychosis programme (EASY) in Hong Kong, the Dutch Early Detection and Intervention Evaluation (EDIE-NL) trial, the Programma2000 in Italy, the Early Intervention Service in Bedfordshire, England, the Early Case Identification Programme in Canada (ECIP), the Early Psychosis Intervention Programme (EPIP) in Singapore, the Croydon Outreach and Assertive Team (COAST) project in London and the FETZ (early detection and treatment for psychosis) study in Cologne, Germany, are also aimed at both primary and secondary prevention. There are also Early Psychosis Intervention and First Episode Psychosis services in New Zealand. A study of 100 users of one New Zealand service found significant improvement, in positive and negative symptoms and in quality of life, in the first three months, followed by ongoing gradual improvement over the subsequent 21 months (Theuma *et al.* 2007). A comprehensive overview of early intervention programmes is provided by Edwards and McGorry (2002) in *Implementing Early Intervention in Psychosis*.

CONCLUSION

For decades, patients with schizophrenia and other psychotic disorders have been treated without recognizing their individuality and individual needs. Psychosis is better described as a continuum than in categories (van Os *et al.* 2000; and Chapter 5). The growing understanding of psychosis as something that develops gradually, in phases or stages, and which is therefore possible to prevent, delay, modify and reverse, has serious implications for how we structure and deliver our services.

Our basic beliefs about these disorders strongly influence the way we meet and treat people with psychosis, both in terms of the *content* of the treatment and the *way* in which we provide treatment modalities. In recent years, in many Western countries, the 'talking therapies' have often not been offered to the most seriously ill. This reflects an attitude that these patients will not profit from telling their story, and working through their experiences and subjective suffering. For those of us working with the early stages of illness development, and meeting people in the phase where they are 'losing their grip' – that is, going into a psychotic state of mind – we know that this simply is not true.

Today we know that it is possible to reduce the duration of untreated illness for people suffering from psychosis. We do not have conclusive evidence that

reducing the duration of untreated psychosis improves the long-term prognosis, although most of us assume that it does, out of simple logic.

Many groups around the world have now established clinical programmes and research focusing on early psychosis. In many countries, including the United Kingdom, Australia, Norway, Denmark, Germany and Canada, there are government-supported development programmes for a national implementation of early intervention services. Hopefully, this will secure a place for psychological treatments for these serious disorders in the future.

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NON-HOSPITAL, NON-MEDICATION INTERVENTIONS IN FIRST- EPISODE PSYCHOSIS

Loren Mosher and John Bola

Psychosocial approaches to 'schizophrenia' have largely been relegated to an adjunctive role by psychopharmacologic interventions. Pharmacological treatment of 'schizophrenia' began with the introduction of chlorpromazine (Thorazine) in the 1950s. After the serious side-effects of these medications were finally acknowledged, the 'atypical' antipsychotic medications were introduced. 'Atypical' medications were touted as being more effective and as having fewer side-effects like tardive dyskinesia and neuroleptic induced deficiency syndrome, and they quickly began to replace 'conventional' antipsychotic medications. These medications provided new sustenance for everyone's wish to find the magic bullet to 'cure' 'schizophrenia.' A less sanguine view of them is presented in Chapters 9 and 10.

Today, it is widely believed that people with 'schizophrenia' will not recover without antipsychotic medications and will usually 'relapse' when medications are discontinued. We present several lines of evidence contradicting the beliefs that: (a) schizophrenia is a life-long illness with a deteriorating course, and (b) medications are universally necessary for the treatment of psychotic disorders.

RECOVERY FROM SCHIZOPHRENIA?

In long-term follow-up studies during the pre-antipsychotic era, about two-thirds of people treated in enlightened settings had good social recoveries (Bleuler 1968; Ciompi 1980; Huber *et al.* 1980). A majority of 'back ward' Vermont State Hospital patients were functioning well in the community 30 years after discharge (Harding *et al.* 1987); the vast majority had stopped taking antipsychotics. Harrow and Jobe (2007) found that at 15-years recovery

from schizophrenia was correlated with discontinuing antipsychotic medications. However, outcomes for people with schizophrenia appear to have deteriorated in the late twentieth century, corresponding to increased use of antipsychotics (Hegarty *et al.* 1994).

THE TREATMENT OF FIRST-EPISODE SCHIZOPHRENIA

The World Health Organization conducted two multi-country studies of first-episode psychosis (Jablensky *et al.* 1992; World Health Organization 1973) that found better outcomes in developing countries. There are two competing explanations for this finding: more social support in collectivist cultures (Lefley 1987) and lower use of antipsychotic medications (Whitaker 2002). Whitaker (2010) points out that at the five-year follow-up 16% of developing country patients were maintained on antipsychotics as compared to 59% in developed countries. A prospective study controlling for medications and measuring social support would help address the important unanswered questions on the effectiveness of antipsychotics and the role of social support in recovery.

Despite agreement in published psychiatric guidelines to treat early episodes of psychosis with antipsychotics for 6–24 months (Gaebel *et al.* 2005), evidence supporting this approach is quite limited. A recent systematic review and meta-analysis found no evidence that immediate use of antipsychotics in early episodes of acute psychosis improves outcomes (Bola *et al.* 2011, 2012). While nearly universal, this practice is not currently evidence-based.

A meta-analysis of first-episode studies found no evidence of long-term harm from short periods (2–6 weeks) of antipsychotic postponement (Bola 2006a), corroborating results from an earlier RCT (Johnstone *et al.* 1999). This paper (Bola 2006a) has been credited with removing a 15-year ethical prohibition on carefully conducted studies that do not immediately medicate all participants (Gould 2006). A new study of psychosocial treatment with random assignment to antipsychotics or placebo is currently recruiting first-episode subjects (Francey *et al.* 2010).

A SUBGROUP OF ‘MEDICATION-FREE RESPONDERS’?

A large subgroup of people with an acute episode of schizophrenia-type psychosis will recover without antipsychotic treatment. The size of this subgroup has been estimated at 25% (Dixon *et al.* 1995), 37% (Cole *et al.* 1964) and 40% (Thornley *et al.* 2006). Reliably differentiating those needing from those not needing antipsychotics would result in several clinical and scientific benefits (Bola 2006b): (a) providing specifically indicated treatment, (b) sparing unnecessary exposure to medication-induced side effects, (c) helping discriminate clinically similar from etiologically different illnesses

(Cloninger 1989), leading to diagnostic refinement, and (d) development of etiological hypotheses applicable to newly identified subgroups.

A long, but often overlooked literature seeks to identify spontaneous remitters, medication-free responders or what was described as 'schizophreniform disorder' (i.e. pseudo-schizophrenia) patients (Langfeldt 1939). Characteristics associated with recovery (Langfeldt 1969; Vaillant 1962) evolved into the concept of 'good prognosis' (e.g., acute onset, later onset, identifiable stressor, no family history of schizophrenia, etc.) that was previously considered essential in discriminating schizophrenia from schizophreniform disorder (APA 1987; Carpenter and Stephens 1982).

Two post-hoc studies have sought to identify individuals not needing antipsychotics in early-episode psychosis. The first study used data from the Soteria project (Bola and Mosher 2002b). Out of 179 subjects, 129 (72%) were available at the two-year follow-up. Of experimental completers, 43% were not treated with antipsychotic medications between the end of the six-week medication control and the two-year follow-up (29/68; 35% of 'intent to treat', 29/82). Older age, better prior social functioning and fewer cardinal symptoms of schizophrenia were associated with medication-free treatment (Bola and Mosher 2002a, 2002b).

A second study used data from the Finnish Need-Adapted project (Bola *et al.* 2006). This quasi-experimental acute early-episode study of psychosocial treatment in the community, compared to antipsychotic treatment in hospital, began with 135 subjects, of which 106 (78.5%) were available at the two-year follow-up. There were 46% medication-free completers in the experimental group (31 of 67; 31/84 or 37% of intent-to-treat), with four predictors of medication-free response. A higher score on the GRIP scale (Rakkolainen *et al.* 1979) – measuring (a) retention of life goals, and (b) social relationships – and, contrary to expectation, a higher score on the CPRS item *Wahnstimmung*, a sense of meaning or inner significance derived from the psychotic experience, were positively associated with medication-free response. Family mental health treatment and the Comprehensive Psychiatric Rating Scale (CPRS) (Jacobsson *et al.* 1978) item for distorted speech were negatively associated with medication-free response. Neither in this study nor in the previous one (Bola and Mosher 2002b) was duration of symptoms (more or less than six months; schizophrenia vs. schizophreniform) helpful in discriminating medication-free responders from those needing medications.

PILOT STUDIES OF PSYCHOSOCIAL TREATMENT

Four quasi-experimental pilot studies in early-episode psychosis have compared hospital treatment with medications to psychosocial treatment using a time-limited three- to six-week antipsychotic postponement, with two- to five-year follow-up periods: Soteria (Bola and Mosher 2003),

Soteria-Bern (Ciompi *et al.* 1992), the Finnish Need-Adapted Project (Lehtinen *et al.* 2000), and the Swedish Parachute Project (Cullberg *et al.* 2006). In each project, two- to three-year outcomes were equal to or better than those for hospital-treated subjects, and in each study 42–46% of completers (and 35–43% of intent-to-treat subjects) were not receiving antipsychotic medications (Bola *et al.* 2009). We present a synopsis of each project.

SOTERIA

Soteria rejected the ‘doing to’ connotation of ‘treatment’ and did not accept the disease model of ‘schizophrenia’ (Mosher and Hendricks 2005). At its core was the idea that psychosis should be dealt with directly, face to face, without the external impediments of theory, artificial institutions, professionally acquired belief systems and practices, and without the chemically induced changes in awareness from antipsychotic drugs. It focused on finding shared meaning and understanding of the subjective experience of ‘schizophrenia’, including the experience of others, in an interactional process. Soteria eschewed the medical model. Its facility was a home in the community. It utilized non-professional staff trained to relate to and understand madness without preconceptions, labels, categories, judgments or the need ‘to do’ anything to change, control, suppress or invalidate the experience. Antipsychotics were minimized for an initial six weeks. Many reports of the Soteria Project are available (Bola and Mosher 2003; Matthews *et al.* 1979; Mosher and Menn 1978; Mosher *et al.* 1995; Thomas 2013).

Background

Soteria was loosely fashioned after the Philadelphia Association’s Kingsley Hall (1964–72) in London (Bola and Mosher 2005), combined with an existential/phenomenological–interpersonal psychotherapy orientation. It incorporated ideas from: moral treatment in American psychiatry (Bockhoven 1963), Sullivan’s interpersonal theory (Sullivan 1962), the specially designed milieu at Shepard-Pratt Hospital in the 1920s, and the idea that growth was possible from psychosis (Laing 1967; Menninger 1959; Perry 1974).

Based on one of the author’s (LM) Kingsley Hall experience, it seemed evident that when dealing with psychotic persons in an open community residential (nonmedical) setting, some *contextual constraints* (rules) were required:

- do no harm;
- treat everyone, and expect to be treated, with dignity and respect;
- guarantee sanctuary, quiet, safety, support, protection, containment, and interpersonal validation;

- ensure food and shelter;
- most importantly, the atmosphere must be imbued with hope – that recovery from psychosis is to be expected – without antipsychotic drugs.

Within this defined and predictable social environment *interpersonal phenomenology* was practiced. Its most basic tenet is 'being with' – an attentive but non-intrusive, gradual way of getting oneself 'into the other person's shoes' so that a *shared meaningfulness* of the subjective aspects of the psychotic experience can be established within a confiding relationship. This requires unconditional acceptance of the experience of others as valid and understandable within the historical context of their life – even when it cannot be consensually validated. The Soteria approach includes thoughtful attention to the *caregiver's* experience (not unlike the psychoanalytic concept of 'transference'). This represented a new emphasis on the *interpersonal*, bringing phenomenology more into relationship with the modern concepts of systems and interactive fields without sacrificing its basic open-minded, immediate, accepting, non-judgmental, non-categorizing, 'what you see is what you get' core principles.

The essential characteristics of Soteria are:

- 1 small and home-like, sleeping no more than 10 persons including staff;
- 2 two staff on duty, a man and a woman, in 24–48-hour shifts;
- 3 ideologically uncommitted staff (to avoid failures of 'fit'), with positive expectations of recovery;
- 4 peer/fraternal relationship orientation, to mute authority;
- 5 preservation of personal power and, with it, maintenance of autonomy;
- 6 open social system to allow easy access, departure and return if needed;
- 7 everyone shares day-to-day running of the house to the extent they can;
- 8 minimal role differentiation to encourage flexibility;
- 9 minimal hierarchy to allow relatively structureless functioning;
- 10 integrated into the local community;
- 11 post-discharge continuity of relationships encouraged;
- 12 no formal 'therapy'.

Research methods

Soteria used a quasi-experimental (consecutive admission) design in the first cohort (1971–76; $n = 79$) and an experimental design with random assignment in the second (1976–79; $n = 100$). Subjects with the following characteristics were recruited from two county hospital psychiatric emergency facilities in the San Francisco Bay Area:

- 1 initial diagnosis of schizophrenia by three independent clinicians using *DSM-II* (APA 1968) and judged in need of hospitalization;

- 2 no more than one previous hospitalization of four weeks or less with a diagnosis of schizophrenia;
- 3 ages 15–32;
- 4 not currently married.

These criteria selected a relatively poor prognosis group by excluding the favorable characteristics of older age and being married (Strauss and Carpenter 1978). Assessments were conducted by an independent research team trained to maintain good inter-rater reliability (*kappa* of 0.80 or better). Diagnostic assessments were conducted prior to entry and again at 72 hours (to exclude drug-induced psychoses).

Soteria opened in 1971 and a replication, Emanon, in 1974. Despite (or perhaps because of) consistently positive results (Matthews *et al.* 1979; Mosher and Menn 1978), the project ended in 1983.

Results

Admission characteristics

Combined cohort subjects ($n = 179$) were remarkably similar on 10 demographic, 5 psychopathology, 7 prognostic, and 7 psychosocial preadmission (independent) variables. They were young (mean age 21), mostly White (10% minority), relatively well educated (high school graduates) men and women raised in typical lower-middle-class, blue-collar American suburban families (Bola and Mosher 2003).

Six-week outcomes: global psychopathology

Global Psychopathology Scale (1 = normal; 3 = mildly ill; 5 = markedly ill; 7 = extremely ill) results were similar, indicating *significant and comparable improvement in both groups* [experimental group admission: mean = 5.2; control = 5.3; six-week group means: experimental = 3.5; control = 3.5; paired-*t*, $p = .01$ (Mosher and Menn 1978; Mosher *et al.* 1995)]. Soteria subjects, only 24% of whom received any antipsychotic treatment during the initial six weeks, improved as much as the antipsychotic-treated controls. Much to our surprise, the Soteria environment proved to be as effective as antipsychotics for acute symptom reduction.

Milieu Assessments (Mosher *et al.* 1995; Wendt *et al.* 1983)

(a) Systematic measures

We compared the characteristics of Soteria's milieu with the hospital psychiatric wards using the *Moos Ward Atmosphere Scale* (WAS) and the

Community Oriented Program Environment Scale (COPEs; Moos 1974). These instruments scored 10 variables: involvement, support, spontaneity, autonomy, practicality, personal problem orientation, tolerance of anger, order, programme clarity and staff control. The Soteria-hospital differences were significant on 8 of the 10 WAS/COPEs sub-scales with Soteria scores much higher on the three 'psychotherapy' variables: involvement, support and spontaneity. Soteria scores were also significantly higher on personal problem orientation and programme clarity, and lower on practicality and staff control.

(b) *Clinical description*

- milieu *characteristics*: quiet, stable, predictable, consistent, clear and accepting;
- early milieu *functions*: supportive relationships, control of stimulation, provision of respite or asylum, and personal validation;
- later *functions*: structure, involvement, socialization, collaboration, negotiation and planning (Mosher and Burti 1994).

Staff characteristics

Both Soteria and hospital staff were psychologically strong, independent, mature, warm, and empathic (Hirschfeld *et al.* 1977). Soteria staff were significantly more intuitive, introverted, flexible, and tolerant of altered states of consciousness.

Two-year outcomes (Bola and Mosher 2002b, 2003)

(a) *Data analysis*

Multivariate outcomes for completing individuals ($n = 129$) compared Soteria and hospital-treated groups as a whole, then separately for schizophrenia and schizophreniform groups, and then for the 43% of the Soteria group that received no antipsychotics. Eight outcome measures (representing five domains: rehospitalization, psychopathology, independent living, working, and social functioning) were used. A composite outcome scale was created by converting each to standardized (z) scores, oriented with positive values for better outcomes, summing the measures, and restandardizing. *DSM-II* schizophrenia subjects with symptoms for more or less than six months were re-diagnosed with schizophrenia [71 of 169, 42% (APA 1980, 1994); or schizophreniform disorder (98 of 169, 58%)].

(b) 'Completers' ($n = 129$)

Compared to individuals receiving usual treatment, Soteria subjects had:

- +0.47 standard deviation (SD) better composite outcome ($t = 2.20, p = .03$), corresponding to a small-medium effect size ' r ' of +.23 [Effect size ' d ' (SD) converted to ' r ' via Rosenthal and DiMatteo's (2001, p. 71) formula $r = (d^2/d^2 + 4)$];
- a 20% higher probability of being in the lowest psychopathology categories ($z = 2.7, p = .03$);
- One less readmission ($z = -2.37, p = .02$).

(c) DSM III 'Schizophrenia' subjects ($n = 49$):

Compared to individuals receiving usual treatment, Soteria subjects had:

- +0.81 SD better composite outcome ($t = 2.42, p = .02$), or a medium-large effect size ' r ' of +.38;
- 44% higher probability of no or little psychopathology ($z = -2.11, p = .04$);
- 48% higher probability of excellent or good improvement in psychopathology ($z = -2.68, p = .01$);
- 40% higher probability of working ($z = 2.3, p = .02$).

(d) DSM-III Schizophreniform subjects ($n = 80$)

Compared to individuals receiving usual treatment, Soteria subjects had:

- +0.34 SD better composite outcome ($t = 1.22$, not significant), an effect size ' r ' of +.17;
- 1.24 fewer readmissions ($z = -2.36, p = .02$).

(e) 'Medication-free responders' ($n = 29$):

Among Soteria treated individuals, 43% used no antipsychotic medications during the follow-up period and were designated 'medication-free responders.' At two-year follow-up this group was performing well above the overall group mean, with an effect size ' r ' advantage of +.38, or 38% better, on the composite outcome scale.

A SOTERIA REPLICATION: SOTERIA-BERN

In 1984 Luc Ciompi and colleagues established a replication of Soteria, called Soteria-Bern, that remains open today. Due to government regulations, the staff included a majority of nurses, making the setting both more professional

and more expensive than Soteria. Second, Soteria-Bern developed a four-phase model of recovery that emerged from Ciompi's efforts to develop an explanatory theory for 'schizophrenia' (Ciompi 1994). This differed from the interpersonal phenomenology of Soteria. Third, a majority of Soteria-Bern's residents (31 of 51) received low-dose antipsychotics (average daily dose of 172 mg. chlorpromazine equivalents), whereas in Soteria a majority was initially treated without antipsychotics.

Since a random assignment study could not be implemented, a matched hospitalized case-control study was conducted, drawing comparison subjects from nearby hospitals. Describing Soteria-Bern, Ciompi wrote:

The evaluative research was mainly done in two studies, the first one concerning immediate outcomes without a control group, and the second one comparing the outcomes of Soteria patients after two years with carefully matched controls coming from four different hospital settings in Switzerland and Germany. From these studies, the following findings are of particular interest: Firstly, by using a German version of the well-known 'Ward Atmosphere Scale' (WAS) by Moos (1974), it was verified that the therapeutic atmosphere in Soteria differed significantly from the atmosphere in four traditional control institutions (Ciompi *et al.* 1993). Main differences concern greater emotional closeness and more warmth and spontaneity of patient-staff relations in Soteria and less hierarchy, order and control. Secondly, immediate results on the four 'axes' of psychopathology, housing situation, work situation, and global outcome were very good or good in about 2/3 of the first 56 treated cases. Statistically, women and less medicated patients had a significantly better outcome than men did, respectively, than patients receiving higher doses of neuroleptics (Ciompi *et al.* 1991). The main result of this first study is, therefore, the confirmation that actual psychotic patients with schizophrenic spectrum disorders (full schizophrenia in 39 out of 56 cases according to *DSM III* criteria, 14 schizophreniform psychoses, 3 unclear) can in fact be successfully treated in a Soteria-like setting, as claimed by Mosher and colleagues (Mosher and Menn 1978; Mosher *et al.* 1995). Thirdly, our comparative study revealed no significant differences between Soteria patients and controls, two years after first admission, concerning the same four axes, and, in addition, the relapse rates. Significant differences existed, on the contrary, concerning daily and total doses of neuroleptic medication that were about 3–5 times lower in Soteria, and also concerning the average duration of institutional care which was 6 months in Soteria, versus 3 months in the control settings (Ciompi *et al.* 1993). Globally, our evaluation confirms that with much lower doses of neuroleptic medication and

without higher costs, similar 2-year results can be obtained with Soteria approach, as with traditional drug-centered hospital treatments. In addition, observations and follow-up information over up to 13 years suggest that many former Soteria patients have had a considerably better long-term evolution than usual.

(Ciompi 1997: 641–4).

Comment

Although Soteria-Bern found no statistically significant two-year advantages (perhaps due to the small sample), it appears that subjects who availed themselves of post-discharge individual or family psychotherapy – much like what is available in Scandinavia (see the Finnish API project below) – had better outcomes. Soteria-Bern demonstrated that the Soteria approach of using non-professional staff, while less expensive, is not necessary to achieve good results. In addition, Soteria-Bern was theory-based – in contrast to Soteria’s atheoretical position. Yet, both Soteria-Bern and Soteria were able to achieve very good results during the acute phase of psychosis with interpersonal methods and no or low-dose antipsychotic medications. The two social environments were similar on the Moos Scale, giving credence to the notion that these caring, supportive, humane, protective, interpersonally focused non-hospital environments can reduce the need for antipsychotic drug treatment during the most disorganized period of psychosis. In addition, a virtually identical 43% of experimental subjects (6 of 14) were not treated with antipsychotic medications (Ciompi *et al.* 1992). The Soteria-Bern replication provides support for the idea that a well-organized social environment can significantly reduce the need to rapidly initiate antipsychotic treatment in most early episodes of acute psychosis, resulting in comparable or better results with an important reduction in medication dependence and side effects. The Finnish ‘API’ project provides further support for this idea.

THE FINNISH COLLABORATIVE STUDY

Beginning in 1967, Yrjö Alanen and colleagues in Turku developed a psychosocial approach to first-episode psychosis that has come to be called the Finnish ‘need-adapted’ or ‘integrated’ system of care (Alanen 2009; Alanen *et al.* 2000; and Chapter 24). This approach evolved from hospital-based psychotherapy to an outpatient and in-home-oriented family crisis intervention system and spawned projects in other parts of Scandinavia (Aaltonen *et al.* 2011; Cullberg *et al.* 2002; Seikkula *et al.* 2011). Its principles are:

- 1 Therapeutic activities are planned and carried out flexibly and individually to meet the real needs of the patient and family. This occurs in a

context in which both the patient and significant others are present with the special psychosis team that collaboratively develops a shared understanding of the experiences, observations and hypotheses that emerge within the meetings.

- 2 Examination and treatment are dominated by a psychotherapeutic attitude—an attempt to understand what is going on with all parties involved.
- 3 Various therapeutic activities should complement, not compete with each other. Teamwork is crucial.
- 4 Continuity of involvement by the same team should be maintained for as long as required.

In contrast to Soteria, this was a very professional system, with staff trained in psychotherapy and family therapy; however the role of antipsychotic medications was not addressed until the *Acute Psychosis Integrated Treatment Project* ('API') (Lehtinen *et al.* 1996).

The 'API' project

Although intended to be a ten-year study, only two-year outcome data have been published (Lehtinen *et al.* 2000), although five-year outcomes have been presented at a conference (Lehtinen 2001).

Methods

Subjects were consecutive first-episode patients with non-affective psychosis, from six centres in 1992–93. Three centres used a no or low-dose antipsychotic strategy (an initial three week antipsychotic-free period with medications introduced only when 'absolutely necessary') while the others used antipsychotics as 'usual.' Both groups were treated with the 'need-adapted model.'

Results

At two years, 79% of subjects were available for follow-up (106/135). Overall, both groups were doing well, especially when compared with contemporary American community-based treatment: 69% had spent a month or less in the hospital. Just over 50% had no psychotic symptoms in the previous year, and only 20% had five or more psychotic symptoms; 40% had Global Assessment Scale (GAS) scores of 7 or more. About 60% had retained their 'Grip on Life', and 47% were working.

Forty-three per cent of the experimental group and 6% of the control subjects had never received antipsychotic medications. The experimental group received less hospital treatment ($p = .01$), had fewer psychotic symptoms during the previous year ($p = .08$), and a higher proportion had GAS scores

of 7 or better ($p = .01$). There was *no association* between time since first psychotic symptoms and outcome (the 'duration of untreated psychosis' variable).

At the five-year follow-up, 37% of experimental subjects were not taking antipsychotic medications; 88% had not been hospitalized between the two- and five-year follow-ups, compared to 68% of usual care subjects ($p = .01$). GRIP on life scores were higher in the experimental group ($p = .03$); GAS and BPRS scores were comparable (Lehtinen 2001).

As in the Soteria and Soteria-Bern projects, the Finnish API project enabled a surprisingly identical 43% of newly diagnosed persons with psychosis to remain antipsychotic free for two years.

SWEDISH PARACHUTE PROJECT

In Sweden, Cullberg's Parachute Project provided 'need-adapted treatment' using mobile crisis teams and a residential therapeutic milieu that was compared to university hospital treatment for people with a first episode of psychosis ($n = 175$) (Cullberg *et al.* 2002, 2006). There were six treatment principles:

- 1 intervention without delay, preferably in the patient's home;
- 2 initial intervention structured according to the patient's needs;
- 3 immediate and recurring family meetings, preferably including the patient;
- 4 follow-up continuity of treatment over a five-year period;
- 5 use of the lowest possible dose of antipsychotic medications, with an effort to avoid all use of antipsychotics during an initial 1–2 weeks;
- 6 access to small-scale overnight residential care when needed.

Results

About two-thirds of the Parachute Group did not receive antipsychotic medications during the first week of treatment. At the three-year follow-up, 42% of Parachute completers (25/59; 25/71 or 35% of intent to treat) were not being treated with antipsychotic medications (some had previously received brief, targeted antipsychotic treatment). This nearly identical 42% of medication-free completers at three years is strikingly similar to the three other pilot projects. Due to an initial medication postponement in both treatments, it is not possible to estimate a medication effect size. Readers are referred to Cullberg's comprehensive and thoughtful treatise on biopsychosocial treatment in schizophrenia (Cullberg 2006).

WHY DID THESE INTERVENTIONS WORK?

These four successful psychosocial approaches to first-time psychotic persons differ in theory, professional experience, treatment site, culture and in myriad

other ways. Yet there are many common elements that may help account for their effectiveness. In addition to developing therapeutic milieus for psychosis (Gale *et al.* 2008), each endeavoured to maximize Frank's (1972) five non-specific factors:

- 1 the presence of a *healing context*;
- 2 the development of a *confiding relationship with a helper*;
- 3 the gradual evolution of a *plausible causal* explanation for the reason the problem at hand developed;
- 4 *the therapist's personal qualities generate positive expectations*;
- 5 the therapeutic process provides *opportunities for successful experiences*.

Residential alternatives to hospitalization can provide healing contexts. In the Finnish project it would appear that the 'team' came to be seen this way. Each project was structured to remove institutional barriers to the growth of therapeutic relationships. Each programme sought 'meaningfulness' within psychosis ('a plausible causal explanation') and was imbued with a positive expectation of recovery. Literature on therapeutic communities for psychosis, Frank's principles, and these pilot projects provide important guidance for future development of psychosocial programmes for early episode acute psychosis.

THE TAKE-HOME MESSAGE

Psychosocial treatment of early episode psychosis may be easier to initiate with previously untreated people than with those previously hospitalized and treated with antipsychotics. These four pilot projects demonstrate that early psychosis can be successfully 'treated' with no, or low-dose, antipsychotic medications. The successfully treated will suffer few, if any, of the adverse effects of antipsychotics. Tardive dyskinesia, tardive dementia and neuroleptic-induced deficiency syndrome are serious iatrogenic diseases (see Chapter 9); avoiding them is a tremendous accomplishment. This is why attention to 'first-break' persons is critical, as iatrogenically induced adverse consequences for those not requiring antipsychotics can be prevented best at this juncture. It is our view that a large-scale randomized controlled trial should be mounted that compares: (a) initial acute psychosocial treatment using a time-limited medication postponement, to (b) usual medication treatment in the hospital and, also to (c) medication plus psychosocial treatment of first-episode psychosis – as is being developed in the NIMH Recovery after an Initial Schizophrenia Episode (RAISE) project. In addition to outcome comparisons, such a study should be designed to maximize the identification of subgroups responding best to each treatment. The current practice of medicating all first-episode patients, without regard to the type or severity of psychosis, or to probable response to medication (positive or negative), appears scientifically and ethically untenable.

NOTE

This is an updated version of the parallel chapter in the first edition of *Models of Madness* which was written with my friend and colleague Loren Mosher (who died in 2004).

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FAMILY THERAPY AND PSYCHOSIS

Replacing ideology with openness

Volkmar Aderhold and Evelin Gottwalz

HISTORICAL OVERVIEW

In the last few decades, family therapy has become a well-established treatment for many mental health problems. Psychotherapy and family therapy for people diagnosed with 'schizophrenia', however, have traditionally been applied only after the acute symptoms have subsided. More recently, family approaches emphasizing early and acute intervention and treatment (Bloch Thorsen *et al.* 2006; Bradley *et al.* 2006; Bressi *et al.* 2008; Hogarty *et al.* 1997; Leavey *et al.* 2004) have been researched, including treatment at home (Alanen 1997; Cullberg *et al.* 2006; Seikkula *et al.* 2006). A combination of individual and family therapeutic approaches has also successfully been introduced (Garety *et al.* 2008; Hogarty *et al.* 1997; Lehtinen *et al.* 2000; Seikkula *et al.* 2006).

Family therapy has a long history in the treatment of psychosis. In the 1930s and 1940s, Harry Stack Sullivan's interpersonal theory of schizophrenia was widely accepted in the United States. He stressed the interdependence between 'schizophrenics' and their families:

The aetiology of a schizophrenic illness is to be sought in events that involve the individual. The significant events seem to me to lie wholly within one category, events relating the individual with other individuals more or less highly significant to him.

(Sullivan 1962: 248)

In the 1950s and 1960s, researchers made important progress in understanding the role of families in the aetiology of psychosis, including the double-bind model (Bateson *et al.* 1956), the first findings of unusual thought and communication patterns in the families of schizophrenic patients (Wynne

and Singer 1963), and the description of disturbed interaction patterns in some families (Lidz *et al.* 1958).

The pioneers of family therapy worked with the families of very disturbed patients (Boszormenyi-Nagy and Framo 1965; Bowen 1960; Haley 1963; Minuchin 1967; Richter 1970; Satir 1964). Their methods became popular. However, interest in family therapy faded because of the lack of well-designed evaluation studies, routine treatment with neuroleptics since the 1960s, and the development of bio-genetic models of schizophrenia.

The situation changed again in the mid-1970s. When the big psychiatric institutions closed, many patients returned home. This was often too great a burden on all concerned. It became evident that neuroleptic treatment cannot solve psychosocial problems. Psychotic crises every one or two years became the rule, leading to the 'revolving-door' pattern of readmissions. When problems of living together were recognized as a major risk factor for relapse, it again became evident that families need to be included in treatment efforts.

'PSYCHOEDUCATIONAL' FAMILY TREATMENT

By the end of the 1970s, researchers had developed the concept of 'expressed emotion', with its three components of criticism, hostility and over-involvement (Chapter 19). During this period, however, the idea that families could help cause psychosis was largely abandoned. It became taboo to discuss families in terms of causation. Yet it became acceptable to talk about their role in preventing relapse. Expressed emotion research proved that unfavourable attitudes of close relatives towards the patient are related to a higher risk of relapse (Vaughn and Leff 1976). Therefore, family-oriented approaches began to try to improve the relational system within the family and to develop the family's own resources.

By the end of the 1970s, various 'psychoeducational' family treatment programmes for schizophrenia had been developed (Dixon *et al.* 2001; Goldstein and Miklowitz 1995). This was a very different way of working with families. These programmes are based on the vulnerability–stress (Zubin and Spring 1977), or VSBK (*Vulnerabilitäts–Stress–Bewältigungs–Kompetenz*; vulnerability–stress–coping–competence), model. Ignoring Zubin and Spring's original view that 'family experiences' could contribute to the vulnerability, most proponents of the psychoeducational approach assumed a biological disposition due to hereditary transmission and possibly intrauterine and/or perinatal damage, causing a later vulnerability to stress via neuronal developmental processes, leading to 'schizophrenic' symptoms only via additional stressors.

Thus the vulnerability–stress and VSBK models became an explicit exoneration of the family of origin. Conditions within the family during childhood

were now considered irrelevant (Read *et al.* 2008). Family dynamics were not considered as possible causal factors. Techniques of the 'traditional' (e.g. strategic or systemic) family therapies were therefore not applied. The treatment goals were no longer to understand how the family might have contributed to the problem and how that might be resolved but, rather, to prevent the recurrence of a psychotic episode and the reduction of the burden on family members. Therein lies an important contradiction. Families are told that they could not have played any role in the development of the problem, while simultaneously being asked to change their behaviour to help prevent the same problem from recurring.

Psychoeducational family programmes consist of two general ingredients:

- 'education' about schizophrenia, from the predominantly disease model offered by the vulnerability–stress model;
- active skills training in communication, problem solving, stress management and crisis intervention, primarily designed to reduce high expressed emotion.

Many studies have demonstrated that these psychoeducation approaches can indeed prevent, delay or shorten relapses (Goldstein and Strachan 1986; Hogarty *et al.* 1986) by reducing high 'expressed emotion' or the amount of contact between patient and parents (Falloon *et al.* 1982; Leff *et al.* 1982; and Chapter 19).

It is often assumed that the educational component (which would be more accurately named 'bio-education' than 'psycho-education') is an important contributor to reduced relapse rates. It is believed that it lowers parents' hostility towards their sons and daughters, by encouraging the view that they have no control over their behaviour (Falloon *et al.* 1982; Leff *et al.* 1982). However, studies have shown that interventions that contain only the 'educational' component are not effective in reducing relapse rates. Relapse is actually prevented simply by teaching parents communication and stress management skills, actively discouraging criticism, intrusiveness and over-protection, and reducing the amount of contact between parents and patients (Kavanagh 1992; Tarrier *et al.* 1989).

There isn't, in any case, much evidence that families are convinced by efforts to persuade them that their offspring have a biologically based illness waiting to be 'triggered' by stressors. One study found 'absolutely no change in the amount of knowledge between pretests and posttests' (Cozolino *et al.* 1988: 683). Another assessed the impact of the educational component of the 'integrated mental health care' approach (Falloon *et al.* 1982). Before the 'education', 11% of the relatives believed the problems were caused by a disordered brain, whereas 32% stated this three months later. Thus only 21% had changed their minds, or their statements, as a result of the teaching. The number who stated a belief in genetic inheritance increased from 11% to

15%. Only 3% of the patients adopted an illness model, either before or after the attempt to 'educate' them (McGill *et al.* 1983).

The fact that it is new skills, rather than new causal beliefs, that are helpful is further demonstrated by a survey of New Zealand mental health staff one year after being trained in Falloon's 'integrated mental health care' model. The components they reported to be useful were teaching patients and relatives to identify early warning signs and to use problem-solving techniques (Allen and Read 1997).

The 'psychoeducation' approach to working with families, with its narrow focus on 'expressed emotion', excludes or minimizes other important family dynamics, including conflicts between autonomy and attachment. Furthermore, an ideologically driven and manualized approach can fail to adapt treatment to the needs of individual families. It forfeits the chance of an open dialogue that recognizes the specific problems and needs of each family (Wynne 1994).

The ideology of biological reductionism permeating most 'psychoeducative' family programmes, with its intention of avoiding any 'family blaming' (Chapter 19), means a loss of basic neutrality and openness to the different situations and needs of individual families. Because psychosocial family causalities, intergenerational processes, circularities, and so on are, quite unnecessarily, equated to having to feel guilty, any aetiological relevance is denied. These issues have been made taboo topics of discussion (Chapter 19). One might even wonder whether the taboo, which must be obvious to most family members, might actually serve to create or exacerbate guilt ('We must have done something really terrible if the whole issue is so awful it cannot even be spoken about').

These manualized treatment programmes do not adequately address, therefore, feelings, attributions of guilt and related emotions within the family. The three components of expressed emotion – criticism, hostility and overprotectiveness – can be understandable defences against feelings, especially guilt (Martindale 2008). Where this is the case, talking about it, rather than superficial statements bestowing an illusion of absolution, is required. An open and honest dialogue about the guilt question, which will be different for different families, seems to be a better approach. The focus should be on the acceptance of responsibility, grieving and forgiveness.

Real family work requires a setting where it is possible to ask questions and a shared search for answers that make sense for all family members can occur (Wynne 1994). Therapists should not impose answers but should assist families to feel safe enough to search for their own.

PSYCHOANALYTIC FAMILY THERAPY

Psychoanalytical family treatment (Martindale 2008), sometimes with weekly sessions for one to three years, obviously requires high motivation on behalf of

the family. The focus is on long-standing conflicts, with or without a focus on interactions in childhood, depending on whether the family finds this helpful. Treatment may include autonomy–dependence conflicts if an adult patient is still strongly attached to the family (Schwarz 2000).

An important goal is for diffuse and unclear communication within the family to become more transparent and clear. From an analytical perspective, this communication pattern can be seen as a resistance against anxiety-inducing changes in relationships over time. The therapist becomes a mediator in a constructive dialogue between family members. 'It should be possible for all family members to converse freely on all topics, especially on those topics that had been avoided so far' (Boszormenyi-Nagy and Framo 1965). The reactions of transference and countertransference between family members, and with respect to the therapist, may be interpreted. Anxiety and shame released in this way may have to be 'worked through', sometimes repeatedly. Reconciliation and mourning for what cannot be changed may need to be discussed.

Work with several generations is sometimes possible. Grandparents can be particularly constructive in finding solutions (Sperling 1975). It can sometimes take at least two to three years for psychoanalytic treatment to result in a reduction of acute psychotic relapses and an improvement in social relationships, emotional vitality, self-esteem and the development of a clear identity separate from that of others (Schwarz 2000).

SYSTEMIC FAMILY TREATMENT

The conception of a systemic treatment of psychosis is based on suggestions from American researchers in the 1950s (e.g. Bateson *et al.* 1969; Lidz *et al.* 1958; Wynne *et al.* 1958). Systemic family treatment was further developed by three treatment and research centres: the Palo-Alto group (Bateson), the Milan group (Palazzoli) and the Heidelberg group (Stierlin, Retzer, Simon).

The double-bind hypothesis of Bateson *et al.* (1969) became well known, entering everyday language. It led to new therapeutic opportunities to resolve paradoxes of communication. Starting from there, Palazzoli *et al.* (1975) developed specific intervention strategies that represented a 'counterparadox' to paradoxically disturbed communication. Based on Stierlin's (1975) work, the Heidelberg group also developed specific approaches to finding solutions, especially in cases of psychosis.

Important basic models of systemic family treatment are cybernetics, information and communication theory, mathematical theories and chaos theory (Retzer 2000). They are mostly concerned with formal organizational processes that determine the origin and change of structures. They are applied to the question of how a complex family system maintains its structure. Retzer

and Simon (2001) consider psychotic symptoms to be a deviation from what is expected. In the case of positive symptoms such as delusions or hallucinations, patients behave in an unexpected way. In the case of negative symptoms, they do not behave in the expected way. As with the psychoanalytic approach, emphasis lies on the issues of autonomy and dependence, guilt and responsibility. A major problem of many families is the unclear position of the family members with respect to the boundaries between generations. It can also be helpful to talk about who is different, and in what ways, from other members.

Systemic family therapy is primarily concerned with reversing 'excommunication', the patient's exclusion from the 'tone-setting community' (Stierlin 1996). The therapist must take care not to take on family functions. This is possible by adopting a neutral position, supportive of all family members. The overarching aim is to create space for new opportunities.

Symptoms, initially seen as entirely negative by all family members, are discussed, to discover the function they play in the family. They can then be seen in a more positive light, increasing the chances of self-regulation and tolerance. The essential goal of this treatment is the softening of hardened and incompatible positions of the family members. Helping families to learn new ways to manage conflict between members is central.

Seven randomized controlled outcome studies of systemic family therapy for families with a member diagnosed 'schizophrenic' have been published since 2003 (for reviews see Retslaff *et al.* 2009; von Sydow *et al.* 2010). The study size varied from 10 to 68, the number of sessions from 3 to 28, and the duration from 3 to 30 months. Each study showed significant positive effects, including:

- relapse reduction (Bertrando *et al.* 2006; Bressi *et al.* 2008; M. Zhang *et al.* 2006a; S. Zhang *et al.* 2006b);
- symptom reduction (Cao and Lu 2007, Zhou *et al.* 2003, M. Zhang *et al.* 2006a);
- reduced expressed emotion (Bertrando *et al.* 2006);
- improved social functioning (M. Zhang *et al.* 2006a);
- improved quality of life (Cao and Lu 2007).

Studies with at least eight sessions showed significantly better follow-up results than shorter treatments, after one (Bressi *et al.* 2008) or two years (M. Zhang *et al.* 2006a; S. Zhang *et al.* 2006b).

None of these seven studies is mentioned in the 2009 U.S.A. PORT (Patient Outcomes Research Team) report, which uses a narrow definition of 'family intervention': 'Key elements of effective family interventions include illness education, crisis intervention, emotional support, and training in how to cope with illness symptoms and related problems (Dixon *et al.* 2010: 49). It also cites none of the many studies showing that 'illness education' is

correlated with negative attitudes (Chapter 12). In the United Kingdom, however, the 2009 National Institute for Health and Clinical Excellence Guideline (NICE 2010), which does not mention 'illness education', argues that: 'Family intervention in schizophrenia derives from behavioural and systemic ideas, adapted to the needs of families of those with psychosis', includes family therapy studies (e.g. Bressi *et al.* 2008) and states that 'Family intervention is normally complex and lengthy (usually more than ten sessions) but delivered in a structured format with the individual family, and tends to include the service user as much as possible' (NICE 2010: 290).

There are no controlled efficacy studies comparing psychoeducational approaches with systemic or other family therapy, and none comparing different approaches to family therapy. Shadish *et al.* (1995) concluded, in their meta-analysis of family therapies, 'that if all treatments were equally designed, implemented, measured, and reported, significant differences among orientations might not be found' (p. 350).

COMBINING ACUTE HOME TREATMENT WITH SYSTEMIC FAMILY THERAPY AND INDIVIDUAL THERAPY: THE 'NEED-ADAPTED' APPROACH

A comprehensive psychotherapeutic treatment of first-episode patients, designed to meet the specific needs of individual patients and families, has been developed in Finland (Chapter 24). Therapeutic meetings at the beginning of treatment between the treatment team and the family, including the patient, proved to be therapeutically significant and important in determining the therapeutic requirements in each individual case. The success of this treatment model has been demonstrated by follow-up results (Alanen 1997). Alanen and his co-workers found that individual psychotherapy was well suited to some patients, whereas others benefited from family treatment or both. These two kinds of treatment can be complementary. The decision to use one or both should be taken case by case.

Lehtinen (1994) distinguishes between different patient–family constellations, with different needs and outcomes:

- 1 Patients with psychotic reactions in acute stress situations: early intervention and psychodynamic re-definition of the trigger situation led to remission with favourable prognosis after two to five family sessions.
- 2 Patients with less psychosocial integration, an identified relationship between their life situation and psychotic symptoms, and whose rather rigid families were more opposed to changes so that therapists lost their emotional access: in this situation, women were often able to make progress in the longer term by using individual therapy. Men more often withdrew from treatment and developed psychotic symptoms.

- 3 If, at the beginning of treatment, patients had already been ill for many years and were socially isolated, a long-term rehabilitative approach to treatment, including psychoeducative family treatment, has proved successful (Alanen 1997).

The treatment meetings

In the 'need-adapted' approach the whole family is supported from the beginning and becomes or remains motivated for ongoing family-focused treatment. In an acute crisis the client, the family and others from the social network are invited to a 'treatment meeting' within 24 hours, if possible in the patient's home. These initial meetings are intended to build a common ground of knowledge, experience and human relations between the staff and all others. The treatment meetings are based on a central principle of the need-adapted approach that requires the patient's presence in all situations that concern her or himself. They are held whenever needed. This can be daily to start with and less frequent later. They are used for conjoint treatment planning and decision making. They enhance communication and problem solving. They can be focused on specific therapeutic topics. Staff sometimes serve as role models.

The Turku Schizophrenia Project compared 56 first-admission 'schizophrenia' patients from 1976–77 with 30 from 1983–84. The 'treatment meeting' approach had been implemented in the meantime. The study found significantly more people with no psychotic symptoms (61 vs. 38%) and in full-time employment (57 vs. 30%) in the later group (Alanen 1997).

The NIPS project on the treatment of first-admission 'schizophrenia' group patients, conducted jointly in Denmark, Finland, Norway and Sweden (Alanen *et al.* 1994), clearly showed that the need for pharmacotherapy declined with the progress of psychotherapeutic treatment. At the end of the five-year follow-up, 56% of the patients were not taking neuroleptic drugs; 49% of all patients, and 39% of those with a *DSM III R* diagnosis of 'schizophrenic disorder' showed no psychotic symptoms (Alanen 2009).

In the multicentre API Project from 1992–93, the group receiving treatment based on the open dialogue family approach (43% of whom received no neuroleptics) were three times less likely to experience psychotic symptoms at two-year follow-up than a control group, 94% of whom received neuroleptics (Lehtinen *et al.* 2000).

Open dialogue approach

Under the leadership of Seikkula, Alakare and Aaltonen, therapy meetings, based on the need-adapted treatment model, were further developed in Western Lapland (Finland). Flexible meetings with the patient and the social network became the core intervention. This treatment system was influenced

by social constructionism, which was introduced into family treatment by H. Anderson, H. Goolishian and T. Andersen. They describe the psyche as an essentially social phenomenon, and reality as something constructed through conversations. Thus it is essential that psychotherapy is an open dialogue because new understanding cannot be processed by a single consciousness, but only through a process of interaction between people. The treatment meetings offer the best setting for this kind of dialogical interaction. No individual idea is considered more correct than any other. Meaning is created in social relations or, later on, in the inner dialogue between different voices that contain our life experiences. If many different views are given a voice, a greater potential for understanding and change is created. All voices must be heard, and the professionals must try to build bridges between the different voices. Therapists don't try to control the system but try to process a dialogue, generating new meanings that become part of the inner dialogues of each individual. The method of 'reflecting teams', introduced by Tom Andersen (1991), is used. Continuity is crucial. The whole treatment process can last three meetings or many years.

Seven leading principles have been formulated on the basis of research and clinical experience:

- immediate help within 24 hours;
- social network meetings (instead of individual or family) as the first and ongoing intervention if possible;
- flexibility and mobility of the team;
- responsibility of the team for the treatment process;
- psychological continuity in professional relations and creation of meaning;
- tolerance of uncertainty is possible if the setting is safe enough and premature conclusions and decisions can be avoided;
- facilitation of dialogues within the social network.

Besides systemic interventions, individual therapy is offered and is accepted by about 50% of all cases.

A five-year outcome study of patients with non-affective first psychotic episodes (1994–97) showed the following results. In the group of 42 patients treated within the open dialogue model, 82% did not have any residual psychotic symptoms during the last year, 86% had returned to their studies or a full-time job, and 14% were on disability allowance; 17% had relapsed during the first two years and 19% during the next three years, and 29% had used neuroleptic medication in some phase of the treatment. Of all first-episode schizophrenic patients, 65% had never used neuroleptics over five years (Seikkula *et al.* 2006). These results were replicated for a later cohort of first-episode patients (2003–5) with further decline of the duration of untreated psychosis from 3.5 months to half a month revealing a positive change in the whole culture of using mental health services, decrease of

stigmatization and ongoing inclusion of the patients in the social context (Seikkula *et al.* 2011).

Comparing the two five-year periods before and after this treatment system was fully established (1985–89 and 1990–94) revealed that the mean annual incidence of schizophrenia decreased significantly, whereas the incidence of schizophreniform psychoses and prodromal positions did not change. The number of new long-stay ‘schizophrenic’ hospital patients fell to zero (Aaltonen *et al.* 2011).

PREVENTION OF PSYCHOSIS

Sadly, there have been no randomized controlled studies of family therapy within the highly debated area of ‘ultra-high-risk’ youth (Preti and Cella 2010). One study has used psycho-education in multiple-family groups, along with other treatment modalities (Nordentoft *et al.* 2006). The conversion to psychosis after two years was 25%, significantly less than the control group (48%). There would seem to be an urgent need for greater availability, and evaluation, of family therapy for families with teenagers at risk of psychosis. High ‘expressed emotion’ and ‘communication deviance’ on the part of the parents of troubled adolescents have been shown to be highly predictive of the adolescents later being diagnosed ‘schizophrenic’ (Goldstein 1987). Conflictual communication of high-risk adolescents is a predictor for later positive symptoms (O’Brien *et al.* 2009). Caregivers’ positive remarks are associated with improvement in negative and disorganized symptoms, and their expressed warmth with improved social functioning (O’Brien *et al.* 2006).

CONCLUSION

Relatives often bemoan the failure of services to listen to them (Read and Magliano 2012). Psychotic disorders are heterogeneous and are determined by many factors. So are families. Therapeutic work with families must be conceived accordingly. The origin and development of psychotic disorders and of the social systems within which they occur must be seen as a complex and interactive process, involving reciprocal determinism between the widely divergent levels of a bio-psychosocial self/environment system. Attention must be paid to the role that traumatic and stressful life events, both within and beyond the family, can play in the genesis of psychotic disorders. It is impossible to reliably predict the course of psychological disturbance. The course is not set by the disorder alone, but is influenced by many factors, especially psychosocial ones.

Families with a member diagnosed ‘schizophrenic’ are clearly not all the same. Each has its own structural organization and communication patterns.

Therefore, only tentative hypotheses and interactive explanations are possible or sincere in therapeutic work with families. Our own position is that of experts with an awareness of hypotheses that may or may not fit for a given family, and with a certain ability to initiate a constructive dialogue, with and within the families, on the origins of, and solutions to, the subjectively experienced problems.

We do not try to work through a ready-made education or treatment programme that might not be suited to the needs of the individual family. We first provide information about the various treatment components and options. We retain the greatest possible openness to all subjects that may arise. We are only mediators, if possible, and regulators, if necessary. We try to detect each family's individual access to problems (their phenomenology) and use it to activate their own specific resources. Because of our different training, we use a variety of methods, but only if they make sense to the family.

In the event of a true dialogue, new meanings are discovered in life events, and relationships become clearer, with clearer boundaries and greater independence. Then we feel we are on the right path. Guilt and exculpation are obsolete opposites. The subtle handling of this question is part of the discussion and not – either as an exoneration or as an accusation – an expression of morals. It is not for us to judge.

Evaluation of any treatment system must assess its flexibility, especially its ability to adapt itself to the specific needs of different individuals and different families. As the system is in a continuous process of development, any treatment programme falls short of a perfect outcome. This must be stated truthfully, to the reader of this chapter, and to the families with whom we work.

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CREATING EVIDENCE- BASED, EFFECTIVE AND HUMANE MENTAL HEALTH SERVICES

Overcoming barriers to a paradigm shift

John Read and Jacqui Dillon

Our final chapter begins by asking *why* mental health services continue to be dominated by an unsubstantiated, simplistic and unhelpful ‘medical model’, then summarizes the progress that has been made since the first edition of *Models of Madness* in 2004, before making some suggestions about what each of us might do to ensure that this progress is accelerated.

WHY THE MEDICAL MODEL REMAINS DOMINANT: BARRIERS TO FURTHER CHANGE

A prerequisite for any social change is understanding the causes of the problem being addressed. We need to understand the barriers to change so we can develop strategies to overcome them. We must ask the question: why, in the absence of scientific evidence to support it and despite repeated documentation of the damage it causes, do we collectively allow an unsubstantiated, simplistic, reductionist, pessimistic and often harmful model to dominate our attempts to understand madness and to alleviate the suffering often associated with it? This question applies to the entire mental health world, not just to psychosis. The proliferation of the ‘diagnose and drug’ approach has occurred across the board, from ‘depressive disorders’ to ‘attention deficit hyperactivity disorder’.

We have seen (in Chapters 2–4) examples, spanning centuries, of inhumane ‘treatments’, typically meted out to relatively powerless groups, based on spurious theories. So the idea that the way modern mental health services operate can be partially explained in terms of their social control

function should be unsurprising. Even the powerful and corrosive role of drug companies, which might be considered a uniquely modern development, is actually a continuation of the theme of control by the powerful, for their own self-interests, of the relatively powerless. Indeed, it can be argued that all of the answers offered below could be applied, to some extent, to previous centuries as much as to the twenty-first century. This may seem a pessimistic beginning to our discussion but, as we shall see later, progress *is* being made and there *are* things we can do to accelerate these changes, at each of the levels identified next in answer to the question: why?

Political: power and social control

One of the few people who *have* addressed this question is British psychologist Mary Boyle (2013). In 'Making the World Go Away, and How Psychology and Psychiatry Benefit' (Boyle 2011), her central answer is that mental health services function to deny the importance of life experiences and social context such as 'child abuse and neglect, school and workplace bullying, domestic and sexual violence, discrimination, poverty and unemployment' in the creation of human distress and misery. She adds, however:

It is not difficult to work out that these events involve relatively powerful groups – governments, corporations, men, white people, adults – damaging less powerful groups. In the case of emotional distress, then, context seems to include or even equate to the operation of power.

(p. 39)

Until we acknowledge that mental health services have as much to do with controlling undesirable behaviour as with alleviating distress, we cannot fully understand the various ways that mental health professionals operate that are detrimental to individuals and families. One obvious example of social control is that services, and psychiatrists in particular, are charged with 'managing risk'. This stressful task primarily involves doing whatever it takes to avoid deaths. Inpatient units seem to be driven by this imperative above all others. We all seem happy for psychiatrists to hold this responsibility on our behalf (until something goes wrong, when the knives come out). Actually there is no evidence that suicide or homicide is prevented by medication, ECT or hospitalization. Nevertheless, we pass responsibility for controlling distressing behaviours, from suicide to hearing scary voices, from homicide to believing red cars are following you, to professionals whose job is not supposed to be social control but the relief of emotional distress. Under these circumstances it is inevitable, as it has been throughout history, that the professionals will use techniques that pacify and tranquillize while trying to convince us, and

themselves, that the techniques are helping the person. One of the themes found in the historical examples in Chapter 2 was:

The experts of the day often camouflage, even from themselves, the socio-political function, and the damaging nature, of the ‘treatments’ behind theories that the people concerned have personal defects that are ameliorated by the ‘treatments’.

Today the theory doing the camouflaging of the social control is the ‘medical model’:

The moral complexity and ambiguity that is inherent in the enterprise of policing human conduct is neatly reduced to the morally neutral and more predictable activity of managing a bodily disease.
(Rapley *et al.* 2011: 4)

Societal: denial and distancing

At a societal level, mental health services and the ‘medical model’ are maintained not only because of the policing function they serve on our behalf but because of what they deny and what (and who) they distance us all from. When Boyle writes about ‘Making the world go away’, she is referring to the ‘context’ of people’s lives. It is hard to acknowledge that we collectively continue to tolerate high rates of abuse and neglect of children, violence towards women, poverty and so forth, and that these adversities are causing distress and misery to millions of people. Perhaps the ‘medical model’ is allowed to dominate *because* it helps us deny these unpalatable facts. As Herman (1992: 4) stated, in her classic text *Trauma and Recovery*, ‘the knowledge of horrible events periodically intrudes into public awareness but it is rarely retained for long. Denial, repression and dissociation operate on a social as well as an individual level’

The medical model facilitates distancing from people who have suffered the worst of what we humans can do to one another. Rather than draw closer to someone in emotional pain and listen to their story, the medicalization of distress encourages us to see them as having a context-less ‘illness’, to be frightened of them, to pull away, and to call in the doctor. Although most societies no longer lock away distressed/distressing people (and their stories of what happened to them) in huge institutions, Chapter 12 reports numerous studies showing that the more we accept illness explanations for distress, the more frightened we become and the more we distance ourselves.

The denial and distancing facilitated by decontextualizing human distress gives politicians a perfect excuse, as if they needed one, to do nothing about reducing the social problems underlying the distress and, instead, to placate

us with a slight increase in the traditional mental health budget targeted, as always, at 'the bottom of the cliff'.

Economic: the profit motive

The immense and grotesque power of the pharmaceutical industry to nurture the illness model of distress (Busfield 2013) was documented in Chapter 10. It may be the largest barrier to change. While we like to think our colleagues are open to being influenced by research and rational argument about how to make mental health services more humane, this is of little interest to drug company executives who have, understandably, an entirely different agenda. By 'understandably' we mean in terms of the morally blind economic imperative to increase profit for shareholders and bonuses for themselves. For example, in 2012 GlaxoSmithKline was fined US\$3 billion, 'after admitting a multi-year criminal scheme to hide unhelpful scientific evidence, manipulate articles in medical journals and lavish gifts on sympathetic doctors' in order to increase sales of psychiatric drugs to children (Foley 2012).

Personal: more denial and distancing

At an individual level, we all benefit from the denial and distancing outlined above. Caught up in our own lives, including our problems and distress, it can be just too hard to think or feel anything about the violence and injustices around us, let alone do anything about it. It is convenient, when people around us 'crack up', to see it as a biological weakness on their part.

There is, however, another form of denial and distancing going on. Deploying a model that portrays those who crack as being biologically (and irreversibly) different from us can protect us from our own fear of going mad, or from acknowledging that we are already, sometimes, a bit bonkers. Causal beliefs that not only create the impression of a categorically separate group (thereby denying the dimensionality of emotional distress) but also exaggerate the difference between the two groups by proposing genetic and biochemical abnormalities can fuel the reciprocal processes of distancing, fear, projection and scapegoating. When the suggested differences imply brain functioning so grossly abnormal that a person is denied responsibility for their actions, our fear is further fuelled by the belief that the person could totally lose control at any moment. This may be accompanied by the belief that this unpredictability, which may express itself violently, needs to be severely – even harshly – controlled. Thus the 'medical model' not only meets our need to see ourselves as entirely different from mad people – that is, as sane – but increases that need by making us scared of being seen and 'treated' as mad. This in turn increases the need for the model, and so on.

To compound matters, in the absence of systems that support mental health workers to bear witness to the pain of people who have endured terrible things, there is a risk of vicarious traumatization, contributing to professionals feeling overwhelmed, burned out and increasingly negative and cynical towards those they are trying to help. Workers may employ a number of self-protective strategies like ‘the silencing response’, if they feel unable, or unsupported, to hear distressing stories.

Some families’ and carers’ groups – but by no means all (see Chapter 10) – have an investment in the notion that their relatives’ distress is caused by an illness rather than by disturbing events that may have occurred in the distressed person’s life, absolving themselves of culpability or responsibility. Clearly, some parents do inflict *deliberate* cruelty, neglect and abuse on their children, and the issue of blaming families is unproblematic in those instances. However, there are many parents who themselves suffered in childhood, or have experienced serious adversity as adults, and consequently struggle to provide an optimum environment for their own children. Rather than blaming families or exacerbating their difficulties by pathologizing their loved ones’ distress, acknowledging the effects of intergenerational trauma, poverty and other social ills would help us identify, and respond to, the needs not only of the ‘identified patient’ but of family members, whose problems, and needs, often go unnoticed (Read and Magliano 2012; and see Chapter 26).

Service users themselves may also become embroiled in the medicalizing of their distress, in an attempt to protect themselves and their families from painful truths. For some, having to deal with the consequences of the deleterious effects of others’ mistreatment or failures is a bitter pill to swallow. Abdicating responsibility for their own distress and recovery to an expert figure that can ‘cure them’ may be an unconscious attempt to receive the ‘care’ that they were denied as children. This can result in traumatizing re-enactments of earlier abusive dynamics with service users being infantilized, deemed demanding and non-compliant, and mental health professionals playing the role of controlling, withholding, punitive parents.

Besides denial and distancing in relation to unpleasant *emotions*, there are some *cognitive* processes at work. One is the capacity to categorize everything. This is wired into the human brain, for good reason. This ability, which is actually a requirement for survival, serves to perpetuate our diagnostic, labelling approach to understanding human distress. Then there is ‘the fundamental attributional error’ – the tendency we all have to attribute stable, personality-based traits to the behaviours of others while using situational factors to account for our own behaviour (see Jones and Harris, 1967, for the original experiment). Wikipedia offers an example: ‘If Alice saw Bob trip over a rock and fall, Alice might consider Bob to be clumsy or careless. If Alice tripped over the same rock herself, she would be more likely to blame the placement of the rock.’ So while we might be able to see the role of adverse social events in the development of our *own* distress, we tend to ignore such

events when it comes to the distress of other people, preferring to see the distress as a characteristic of the person, something within them, such as in illness. Interestingly, collectivist cultures are less likely to make this fundamental error than individualistic cultures (Miller 1984).

Cultural: we are not all individuals

Explanations of distress that focus on the individual are consistent with, and reinforce, the worldview of 'Western' cultures that see most things from an individual perspective.

Strategies which avoid confronting the importance of context in the initiation and maintaining of distress also gain credibility from the symbiotic relationship between psychological and psychiatric theory and method and Western, particularly North American, cultural assumptions about the primacy of the autonomous individual.

(Boyle 2011: 34)

There are many examples of active suppression of indigenous healers, and their more holistic understanding that humans develop not as isolated units but as part of a collaborative group. The Tohunga Suppression Act of 1907 in New Zealand was designed to replace traditional Māori healers, with 'modern medicine'. 'The once highly-advanced systems of North American First Nations were illegal in Canada until the 1960s, and have only recently been allowed to function at all' (Fernando 2011: 52).

Professional: it's not just psychiatry

We draw heavily, here, on Boyle's analysis in the remarkable book *De-Medicalizing Misery* (Rapley *et al.* 2011). Having argued, convincingly, that a decontextualized model is dominant precisely *because* it ignores social context, she identifies the 'avoidance strategies' used. For psychiatry, these include converting distress into 'symptoms' of 'disorders' and, for psychology, include a focus on individual 'abnormalities' in thinking processes etc. 'without ever mentioning context and life experience'. Boyle offers, as an example of the consequences of these strategies to avoid context, the 2006 Layard initiative in the United Kingdom to increase access to psychological therapy, especially cognitive therapy, in order to return depressed and anxious people to work and reduce benefit payments (Marzillier and Hall 2009). As the LSE's (2006: 1) *Depression Report* puts it:

'crippling depression and chronic anxiety are the biggest causes of misery in Britain today'. Note that depression and anxiety have been given the status of first causes, rather like supernatural deities whose

own existence requires no explanation. Should we now ask what causes misery, the official answer is anxiety and depression.

(Boyle 2011: 30)

Boyle also identifies some more subtle 'safety behaviours' that deflect attention away from social context, such as arguing that adverse environments are consequences, rather than causes, of 'mental disorders', and the use of the vulnerability-stress model to argue that only genetically inferior people are affected by adverse environments, which supposedly have no effect on 'normal' people.

Addressing her own question 'What is the threat?' to psychiatry (of addressing the social causes of madness and misery), she first clarifies that 'I am not suggesting that strategies for avoiding context are used in a planned and conscious way. Like any defence mechanism they are a mixture of conscious and unconscious' (p. 34). She concludes:

The reasons this is so threatening is because the medical model which underlies both the modern *DSM* and hence psychiatry's authority, is explicitly justified on the basis that those behaviours and experiences said to constitute mental disorders are *not* meaningful and intelligible, that instead they are outward symptoms of an internal pathology and therefore not subject to the same rules as so-called normal behaviour or normal distress. . . . It is no wonder that psychiatry is fearful of context and has to devise so many strategies to avoid it, because context constantly threatens to make emotional and behavioural problems intelligible or, to put it another way, it threatens to abolish psychiatry's self-defined subject matter.

(pp. 35, 37).

John Clarke, a New Zealand nurse, brings the threat to life:

For over 10 years I have become increasingly frustrated at psychiatrists who vehemently denied any link between schizophrenia and sexual abuse or trauma. I wondered at how seemingly intelligent people could be so blind. That largely evaporated one day, after again mentioning your name and your theory that a client's illness seemed linked to a history of sexual abuse, a psychiatrist whom I greatly respect turned to me with an overwhelmingly tired look and let out an exhausted groan and angrily said 'Don't mention that name. I get sick and tired of being attacked all the time – from clients, families, professionals. So you would get rid of all psychiatrists would you?' The outburst was gone as quickly as it came but for the first time I realized how this seemingly helpful theory was a cause of major threat to an entire profession.

(John Clarke, personal communication to JR, 2012)

Boyle reminds us, however, that psychiatry is not the only profession whose status is dependent on denying unpalatable social realities. She argues that, although 'on the face of it psychology has nothing to fear from making distress psychologically and socially intelligible, indeed that would seem to be one of its major purposes', both clinical psychology and its parent discipline, psychology, 'use these avoidance strategies every bit as much as psychiatry' (p. 37). She explains this in terms of 'psychology's extreme insecurity about its academic and social acceptances as a science' which has led to an adoption of methodologies more suitable for addressing biological than social questions, to an avoidance of politically sensitive topics and, like psychiatry, to an almost exclusive focus on the de-contextualized individual.

In following psychology in minimizing context, clinical psychology gains the added advantage of placating psychiatry. . . . In fact, what we seem to have are three very insecure groups who have implicitly agreed not to expose the operation of power in return for academic and professional privileges.

(p. 39)

Some British clinical psychologists, concerned about individualising, pathologising and context-less CBT, wonder whether their profession is at risk of reducing itself to the role of 'the magician's assistant' (Coles *et al.* 2013).

Space does not permit full discussion of all the other mental health professions or the 'para-professionals' such as residential support workers, community workers etc. A comparison of 94 psychologists and psychiatrists with 289 'nurses, social workers, vocational workers, etc.' in Switzerland found that the latter group endorsed antipsychotics for psychosis and antidepressants for depression significantly less than the psychologists and psychiatrists (Lauber *et al.* 2005). However, an Italian study found that nurses had similarly biological causal beliefs about 'schizophrenia' to psychiatrists and believed even more strongly than psychiatrists that little could be done to help and that 'schizophrenics' should not marry or have children (Magliano *et al.* 2004).

Because of their numbers, the role of nurses in maintaining the status quo or supporting progress is crucial. The same is true of GPs. They are the first port of call for the majority of 'mental health cases'. There has been little research into their views about mental health services or beliefs about the causes of psychosis. A study of Welsh GPs about depression found that they believed significantly less than psychiatrists in 'biochemical abnormality' as a cause and were more than twice as likely to endorse the statement 'If psychotherapy was freely available, it would be more beneficial than antidepressants for most patients' (Kerr *et al.* 1995). A subsequent survey of New Zealand GPs found that nine social factors (e.g. recent divorce, domestic violence) were endorsed as risk factors for depression more often than biochemical or genetic factors. In keeping with their Welsh counterparts, 92% agreed that

‘psychological therapy should be as affordable as GP treatment’ (Wilson and Read 2001).

There are no doubt other reasons, beyond those suggested here, for the continuing dominance of the ‘medical model’ (Bentall and Varese 2012). A succinct way to summarize all this, before moving on to solutions, is:

The medicalization of suffering and difference thrives because it sanitizes and simplifies.

(Rapley *et al.* 2011: 4)

PROGRESS

Although it can sometimes seem that progress towards more humane and effective services is either non-existent or painfully slow, we should note, and celebrate, some important developments in recent years.

In terms of how ‘psychosis’ is understood, there has been, at least in academic and professional circles, increased acceptance that social factors, such as child abuse, neglect etc., play a significant role (Chapters 14, 15, 18, 19). This is remarkable given the often scathing response with which such ideas were initially received in some quarters.

It was gratifying to see the National Health Service in the United Kingdom introduce guidelines in 2008 recommending that all mental health clients be asked about childhood trauma (NHS Confederation 2008). Furthermore, multiple research teams have begun to identify the psychological mechanisms by which early adversity leads to specific ‘complaints’ such as hearing voices or paranoia (Chapter 16), thereby further normalizing these experiences as understandable reactions to life events while pointing the way to therapies targeted at specific complaints and specific psychological processes.

The claim that hearing voices and having unusual beliefs can be explained in terms of a disease process for which certain people have a genetic predisposition, the bedrock of the medical model of psychosis, has increasingly been exposed as false (Chapters 6 and 7). The role of epigenetic processes, by which the environment turns genes on and off, is finally receiving the attention it warrants.

We should acknowledge, however, that these realizations mean only that the academics and professionals are finally catching up with the lay public, who have always understood that madness is caused primarily by bad things happening (Chapter 11). Indeed, another cause for celebration is that the public, all over the world (with the possible exception of the United States), have held on to this belief (and their preference for psychosocial treatments over drugs and ECT) despite the constant propaganda from biological psychiatry and drug companies aimed at getting us to change our minds.

As the publication date for *DSM-5* approached, international debate intensified, not only about the contentious issue of introducing a ‘Psychosis

Risk Syndrome' (Chapter 5) but about the value of categorical diagnostic labels in general. Partially in recognition of the poor reliability and validity of the 'schizophrenia' label, but also because of the stigma and pessimism associated with it, some significant organizations, including the Schizophrenia Fellowship (in the United Kingdom and New Zealand), have removed the term from the names of their organizations. In 2012 ISPS did likewise, becoming the *International Society for Psychological and Social Approaches to Psychosis* (Martindale 2012). The new title not only gets rid of 'schizophrenia' but acknowledges that there are more ways of helping than 'psychological treatments', as implied by our old name.

ISPS has continued to grow since 2004. We now have 1,500 members, with branches in 19 countries (www.isps.org). ISPS has published 16 books and, in 2009, launched its own scientific journal, *Psychosis* (www.tandf.co.uk/journals/rpsy).

Meanwhile the evidence base for a whole range of approaches to helping has continued to accumulate (Chapters 21–26). The number of pockets of excellence in services around the world, where this evidence is actually acted on, has increased; but they remain, nevertheless, just pockets. Perhaps the most exciting development has been the remarkable expansion of the Hearing Voices Network (www.intervoiceonline.org). As documented in Chapter 21, there are now Hearing Voices networks in 20 countries, with nearly 200 support groups running in England alone. This is a major cause for celebration. If a relatively disempowered and often marginalized group of people can achieve such successes, perhaps those mental health staff trying to achieve progress in their own services will be inspired to maintain their efforts.

In the United States, perhaps the strongest bastion of the 'medical model' of mental health, psychiatry, in its current form, is becoming unattractive to the next generation of doctors. While 38% of practicing physicians are 55 or older, in psychiatry nearly 55% are in this age range. Meanwhile, the number of medical students applying to, and graduating from, psychiatry training programmes has been steadily declining (Balon *et al.* 2011). In an article entitled 'Are Psychiatrists an Endangered Species?' Austrian psychiatrist Heinz Katschnig (2010: 21) wrote:

The validity of psychiatry's diagnostic definitions and classification systems is increasingly questioned. . . . Confidence in the results of therapeutic intervention studies is waning. . . . Challenges from outside include mounting patient and carer criticism, and psychiatry's low status within medicine and in society in general. In psychiatric journals, the question is being discussed whether and how psychiatry will 'survive into the second half of the 21st century', and the presence of 'considerable pessimism and a sense of foreboding among psychiatrists' is being described. In many countries, a shortage

of psychiatrists is reported. The question has even been asked whether psychiatry should exist.

There have even been some chinks emerging in one of the biggest barriers to further progress, the unwholesome relationship between psychiatry and the drug industry. Some prominent psychiatrists have spoken out (Chapter 10), and in some countries psychiatry's professional organizations have taken small first steps to building a proper ethical boundary between themselves and the industry. An increasing number of journals now insist that authors declare drug company income (leading to some rather long lists!).

NEXT STEPS

Hope and optimism

Despite these successes, the 'medical model' retains a stranglehold on most mental health services, as evidenced by the ever-increasing prescription rates for Ritalin, antidepressants and antipsychotics. In most services around the world someone hearing voices or fearing that people are following them is virtually guaranteed to be given a prescription of antipsychotics but remains far less likely to be offered safer, more effective psychological approaches. Family therapy remains virtually taboo in most countries. The idea that psychosis can be prevented remains a mere whisper.

With change being so slow, and often at personal cost to those involved in what can be rather fierce battles, we might despair about ever winning the war. Perhaps identifying the barriers was a bad idea. It can make things seem hopeless. Antonio Gramsci (1891–1937), an Italian political theorist, wrote, while imprisoned by Mussolini's fascist regime, about keeping his faith by thinking of himself as 'a pessimist because of intelligence, but an optimist because of will' and urged us 'to live without illusions and without becoming disillusioned' (Gramsci [1929] 1971). This is not easy but is, it seems, necessary if we are to grasp the enormity of what we are up against, especially the unbridled power of the pharmaceutical industry, yet never give up trying to create effective, humane mental health services. It may help to remember that the public is on our side in this endeavour. Just as mental health workers try to inspire hope that things can and will get better (which increases the chances that they actually *will* get better), we must choose to continue to hope that mental health services can and will get better; because if we don't they won't. As academic and civil rights activist Cornel West said, hope calls for a leap of faith that goes 'beyond the evidence to create new possibilities based on visions that become contagious'. These visions allow people to engage in 'heroic actions, always against the odds, no guarantee whatsoever' (Devere Smith 2006).

Richard Bentall (2009: 288) wrote, after hearing a taxi driver's story of recovery:

Sometimes ordinary stories of triumph over adversity are inspiring. They remind us that hope is the fire that will guide us through darkness. Psychiatry's greatest sin has been to crush hope in those it has claimed to care for. Without hope the struggle for survival seems pointless. With hope, almost anything seems possible.

A paradigm shift?

One example of the optimism required here is the proposition that we are already on the verge of a paradigm shift. Psychologist Andrew Moskowitz reminds us that in the famous book, *The Structure of Scientific Revolutions* . . .

Kuhn (1970) argued that paradigms change and a *scientific revolution* ensues when three conditions are met: (a) a period of crisis develops in which the paradigm fails to adequately answer questions considered fundamental; (b) serious 'anomalies' occur in which phenomena not clearly compatible with the paradigm are observed; and, importantly (c) a suitable alternative paradigm that explains many of the previous findings and at least some of the observed anomalies comes to light. Kuhn saw scientific revolutions as taking time to resolve; he argued that changing such strongly held beliefs involved a process of persuasion and fundamental reorganization not unlike that of *religious conversion*: 'Conversions will occur a few at a time until, after the last holdouts have died, the whole profession will again be practicing under a single, but now different paradigm' (Kuhn 1970, p. 152).

(Moskowitz 2011: 351)

Andrew believes that the necessary new paradigm is finally emerging and that it includes a central role for trauma and dissociation. This is broadly consistent with Mary Boyle's answer to her own question 'What can be done?' (2011: 41):

Every presentation of distress, in theory, research, teaching, case discussions, media presentations and everyday conversations between service users and professionals would have at its centre, unless there is strong evidence to the contrary, the idea that people's feelings and actions are consonant with their past and present experiences.

Psychiatrist, Glenn Roberts, a long-time champion of the Recovery movement (Roberts *et al.* 2006), is as optimistic as Andrew about a paradigm shift, but sees the future this way:

In a Kuhnian sense, I wonder if, as we all strive to move ahead to more values-led, person-centred, socially inclusive, partnership-based patterns of practice in which we try to work together for the hoped-for outcome of recovery as 'a common purpose' (to cite the Royal College of Psychiatrists position paper of 2006) that the paradigm shift (revolution) may be to move beyond models? And particularly those attributed to single professional groups by others.

(Glenn Roberts, personal communication, 2012)

Meanwhile, in Norway, Ragnfrid Kogstad (2012) thinks that 'it is difficult to estimate as to what degree fundamental changes will take place and if so, whether the changes qualify for a paradigm shift, i.e. if there really is a scientific revolution taking place':

Arguments in favour of the position that fundamental changes will come are the long-lasting tensions between a medical and psychosocial discourse, the unsuccessful attempts at documenting the existence of genetic and biological markers, the dark side of history as represented by very negative and painful user experiences over the centuries, the humanistic and successful alternatives outlined, and not least, the international ratification of the Convention on the Rights of Peoples with Disabilities, which is based on an international consensus about the need for substituting the medical and charity model with a human rights and social model.

(Kogstad 2012: 528)

That there may be sudden changes instead of a rational and controlled process based on evidence and scientific documentation is indicated by the former lead editor of *DSM-IV*, Allan Frances. Frances, who earlier did not want to be 'a crusader of the world', said in an interview in 2010 that,

The idea of more kids getting unneeded antipsychotics that would make them gain 12 pounds in 12 weeks hit me in the gut. It was uniquely my job and my duty to protect them. If not me to correct it, who? . . . Frances fears that by use of the proposed diagnostic category, 'psychosis risk syndrome', as well as other newly constructed diagnoses, 'DSM-5 will take psychiatry off a cliff.'

(http://www.wired.com/magazine/2010/12/ff_dsmv/Inside
the Battle to Define Mental Illness)

There is no doubt that the mainstream thinking in psychiatric and mental health care is challenged from different sides and that changes will come, though in what way and how comprehensive are not easy to predict. Thus, we are facing exciting times.

Some practical suggestions

The following suggestions of things we can do to facilitate these changes, and to nurture our hope and optimism during these exciting times, are far from being an exhaustive list. We suspect that each reader would be readily able to add to the list. They are offered merely as something to refer to from time to time. Rest assured that thousands of people around the world are trying these things, and many other strategies, to achieve our common goal.

- Ensure service users are involved in the management of, and training about, mental health services.
- Ask service users about their past and how they think that relates to their current problems: reject the ‘can of worms’ fallacy.
- Ask service users what they need.
- Support services, and research, led by ‘experience-based experts’.
- Help your colleagues focus on recovery not pathology.
- Advocate for reduction of coercion, in services and legislation.
- Form links between progressive professional organizations and groups of service-users and family groups.
- Stop using scientifically meaning, stigmatizing words like ‘schizophrenia’ and medicalizing terms like ‘mental illness’.
- Publicly name attempts by drug companies to influence mental health services, research and policy.
- Refuse to accept drug company money.
- Support psychiatrists, psychologists and other mental health staff who are under pressure from their colleagues to conform.
- Help psychiatrists share the responsibility for managing risk.
- Initiate or support relevant training programmes in mental health services and medical schools.
- Inform yourself about, and use, the research literature about psychosocial causes and treatments, and about the limitations and dangers of anti-psychotics (carry this book at all times!).
- Lobby for change: governments (local and national), mental health service managers, etc., and in the media (write letters).
- Join a group of some kind with similar goals (e.g. Hearing Voices Network, ISPS) or form your own local group to connect with others who share your values (you will need the support!).

A major dilemma in pushing for these changes is how direct to be. Is it sometimes better to ‘soften the message’ so as not to put off ‘middle of the road’ psychiatrists who might be open to new ways of thinking? Of course. But in the end, if we don’t clearly name the problem (and the relevant evidence) and firmly identify some alternatives, little is likely to change. At the risk of ending with a cliché, it all comes down to ‘speaking truth to

power'. 'Otherwise we become bystanders – our silence gives consent. We become complicit, colluding in the further oppression of people who have already suffered enough meanwhile providing huge profits for pharmaceutical companies with their voracious appetite for the control of human experience' (Dillon 2011).

One remedy for bad practices is to challenge the assumptions that underlie them. This is why the attempt to make scientific sense of psychiatric disorders, although certainly an exhilarating intellectual enterprise that will never lose its fascination for me, is not merely an academic game. The point is not only to interpret the world, but to change it.

(Bentall 2011: 169)

In recognition of the many people who devoted their lives to reforming the mental health world but are no longer with us, we leave the very last words to psychiatrist Loren Mosher. Loren insisted that the first chapter of the first edition of *Models of Madness* ended with the assertion that the book, besides gathering together all the relevant research and arguments, is 'a straightforward, unashamed wake-up call. Everyone involved should act, in whatever way your circumstances allow, to end this madness.'

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